

QIAGEN

EZ2

Global Product Support EZ2 Service Manual SM-5000-0047-001, QFS, QTS, EZ2



Service Manual

SM-5000-0047-001, QFS, QTS, EZ2

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1 Introduction

1.1 General Information

QIAGEN Global Product Support (GPS) is responsible for the creation and maintenance of Instrument Service documentation. It is the policy of QIAGEN to improve products as new techniques and components become available. QIAGEN reserves the right to change specifications at any time. In an effort to produce useful and appropriate documentation, we welcome your comments and suggestions on this publication.

Contact GPS via E-mail: service@qiagen.com

INFORMATION 	If an error is found in this Service Manual please contact Global Product Support immediately (see contact details above).
---	--

1.2 EZ2 Service Manual Overview

This Service Manual represents an integral part of the QIAGEN EZ2.

This Service Manual is intended to be used by QIAGEN authorized representatives.

The instructions contained in the manual regarding operation, service and the test set-up are to be strictly observed. Pay particular attention to any safety information.

1.3 Intended Use

As this manual covers all three EZ2 systems, please refer to the appropriate User Manual for the relevant instrument Intended Use statements.

1.3.1 Product Expected Use

The EZ2 and EZ2 FX systems are designed to perform automated isolation and purification of nucleic acids. The EZ2 is intended to be used only in combination with QIAGEN kits indicated for use with the EZ2 instrument for the applications described in the kit handbooks. The EZ2 system is intended for use by professional operators, such as technicians and physicians trained in molecular biological techniques and the operation of the EZ2 system.

The EZ2 MDx system is designed to perform automated isolation and purification of nucleic acids in molecular diagnostic and/or molecular biology applications. The EZ2 MDx is intended to be used only in combination with QIAGEN kits indicated for use with the EZ2 MDx instrument for the applications described in the kit handbooks. The EZ2 MDx system is intended for use by professional operators, such as technicians and physicians trained in molecular biological techniques and the operation of the EZ2 MDx system.

The product is intended for use by professional users, such as technicians and physicians trained in molecular biological techniques and the operation of the product.

INFORMATION 	Refer to the relevant User Manual for the EZ2 system for the appropriate intended use statements.
---	---

1.3.2 Requirements for working with the product

This section covers the general level of competence and training necessary for installation, use, daily maintenance, cleaning, preventive maintenance and servicing of the product. Appropriate training for all levels can be arranged with QIAGEN.

Task	Personnel	Knowledge
Delivery	No special requirements	No special requirements
Installation	Laboratory technicians or equivalent	Appropriately trained and experienced personnel familiar with the use of computers and automation in general. If the system is to be installed by a QIAGEN representative, that representative must be trained and certified.
Routine Use (running protocols)	Laboratory technicians or equivalent	Appropriately trained and experienced personnel, such as technicians or physicians, trained in molecular biology techniques.
Routine Maintenance	Laboratory technicians or equivalent	Appropriately trained and experienced personnel, such as technicians or physicians, trained in molecular biology techniques.
Servicing and Annual Maintenance	Certified QIAGEN representatives	Certified to service the system.

Table 1.3-1 Requirements for personnel

2 Hazards and Precautions

2.1 Introduction

Before using or servicing the EZ2, it is essential that you read this Service Manual as well as the current and appropriate User Manual carefully and pay particular attention to the safety information. The handbooks contain information and warnings that must be followed by the user and field personnel to ensure safe operation of the instrument and to maintain the instrument in a safe condition.

Possible hazards that could harm the user or result in damage to the instrument are clearly stated at the appropriate places throughout this manual.

The following safety conventions will be used throughout this manual.

WARNING 	The term WARNING is used to inform you about situation that could result in personal injury to you or other persons. Details about these circumstances are given in a box like this one. w0
CAUTION 	The term CAUTION is used to inform you about situations that could result in damage to the instrument or other equipment. Details about these circumstances are given in a box like this one. c0
INFORMATION 	The term INFORMATION is used to inform you about hints and tricks. Details about these circumstances are given in a box like this one.

The advice given in this handbook is intended to supplement, not supersede, the normal safety requirements prevailing in the user's country.

2.2 Safety

WARNING 	You must observe all safety requirements when installing and operating the product. Failure to install or use the system in a manner specified by QIAGEN may result in personal injury and / or equipment damage. w1
---	---

The mains plugs shall be easily accessible in case of the need of quickly disconnecting the equipment from the mains power.

The product shall be used in accordance with the instructions provided through dialogs appearing on the touchscreen, the manuals, instructions for use, within the limits set by the technical specifications.

Follow all generally accepted laboratory safety procedures and applicable laws and regulations. Use appropriate safety precautions when handling biological material.

2.3 Proper Use

WARNING 	RISK OF PERSONAL INJURY AND MATERIAL DAMAGE w2 Improper use of the EZ2 may cause personal injuries or damage to the system. The EZ2 must only be operated by qualified personnel who have been appropriately trained. Servicing of the EZ2 must only be performed by QIAGEN authorized representatives.
---	---

- Use only QIAGEN components, otherwise any right to make a claim under the guarantee may be invalidated.

In case of emergency, power OFF the EZ2 at the power switch located in front of the instrument and unplug the power cord from the power outlet.

CAUTION 	Damage to the instrument Avoid spilling water or chemicals onto the EZ2.
---	---

CAUTION 	Damage to the instrument Do not use spray bottles containing alcohol or disinfectant to clean surfaces of the EZ2 instrument. Spray bottles should be used only to clean items that have been removed from the worktable and if permitted by local laboratory operating practices.
--	---

WARNING 	Risk of fire or explosion When using ethanol or alcohol-based liquids on the EZ2, handle such liquids carefully and in accordance with the required safety regulations. If liquid has been spilled, wipe it off and leave the EZ2 hood open to allow flammable vapors to disperse.
---	---

CAUTION 	Damage to the instrument After wiping the worktable with paper towels, make sure that no residual pieces of paper towel remain. Pieces of paper towel remaining on the worktable could lead to a worktable collision.
---	--

WARNING 	Risk of Explosion The EZ2 is intended for use with reagents and substances supplied with QIAGEN kits as outlined in respective Information for use. Use of other reagents and substances may lead to fire or explosion.
---	--

If hazardous material is spilled on or inside the EZ2, the user is responsible for carrying out appropriate decontamination.

CAUTION 	Damage to the instrument Do not place items on top of the EZ2.	
CAUTION 	Damage to the instrument Ensure that the EZ2 is switched off before you manually move the mechanical components of the instrument.	
CAUTION 	Damage to the instrument Do not lean against the instrument or touchscreen.	

2.4 Electrical Safety

If operation of the instrument is interrupted in any way (for example, due to interruption of the power supply or a mechanical error), first switch off the EZ2 instrument, then disconnect the electrical cord from the power supply before attempting troubleshooting or service activity.

WARNING 	Disconnect the line power cords from the power outlet before servicing.	
WARNING 	ELECTRICAL HAZARD Disconnect the line power cords from the power outlet before servicing. Lethal voltages inside the instrument When the instrument is connected to line power, terminals may be live, and opening covers or removing parts is likely to expose live parts.	w5
WARNING 	ELECTRICAL HAZARD Any interruption of the protective conductor (earth/ground lead) inside or outside the instrument or disconnection of the protective conductor terminal is likely to make the instrument dangerous. Intentional interruption is prohibited.	w6
WARNING 	Damage to electronics Before powering ON the instrument, make sure that the correct supply voltage is used. Use of incorrect supply voltage may damage the electronics. To check the recommended supply voltage, refer to the specifications indicated in the type plate of the instrument.	

WARNING 	Risk of electric shock Do not open any panels on the EZ2. Risk of personal injury and material damage Only perform maintenance that is specifically described in this user manual, or service manual.
---	--

To ensure satisfactory and safe operation of the EZ2, follow the advice below:

- Make sure the line power cord is connected to a line power outlet that has a protective conductor (earth/ground).

INFORMATION 	The ground prong on the mains plugs must not be removed and the plugs should only be connected to grounded outlets as per local and national regulations. Use of power supplies that disable the grounding connections to the instrument is dangerous and prohibited.
---	---

- Keep mains plugs easily accessible in case the equipment needs to be disconnected quickly from the mains power.
- Use only power supplies and cords supplied with the system.
- Power supplies and cords should be inspected periodically and replaced if damaged or altered.
- Do not attempt to make any internal adjustments or replacements.
- Do not operate the instrument with any covers or parts removed.
- If water, liquids or reagents have spilled inside the instrument, switch off the instrument and disconnect it from the line power supply. Contact Global Product Support.

INFORMATION 	Note that the power switch does not disconnect the equipment completely from mains power.
---	---

- Servicing must be performed by trained QIAGEN certified personnel.
- If the instrument becomes electrically unsafe for use, make the instrument inoperative and secure it against unauthorized or unintentional operation. Contact Global Product Support.

The instrument is likely to be electrically unsafe when it:

- The EZ2 or the line power cord appear to be damaged.
- The EZ2 has been stored in unfavorable conditions for a prolonged period.
- The EZ2 has been subjected to severe transport stresses.
- Liquids have come into direct contact with electrical components of the EZ2.
- The power cord has been exchanged with a non-official power cord.
- Shows visible damage.
- The EZ2 has been stored in unfavorable conditions for a prolonged period.

WARNING 	Electric hazard Do not touch the EZ2 with wet hands.
---	--

WARNING 	Electric hazard Never install a fuse different from that specified in the service manual.	
---	---	--

2.5 Biological Safety

The EZ2 instruments are used in combination with biological material.

WARNING 	Always clean and decontaminate the instrument prior to service or use.	
WARNING 	Samples containing infectious agents Some samples used with the EZ2 instrument may contain infectious agents. Handle such samples with the greatest of care and in accordance with the required safety regulations. Always wear safety glasses, gloves, and a lab coat. The responsible body (for example, a laboratory manager) must take the necessary precautions to ensure that the surrounding workplace is safe, and that the instrument operators are suitably trained and not exposed to hazardous levels of infectious agents as defined in the applicable Material Safety Data Sheets (MSDSs) or the OSHA ¹ ,* ACGIH, [†] or COSHH [‡] documents. Venting for fumes and disposal of wastes must be in accordance with all national, state, and local health and safety regulations and laws.	

When handling biological material, use safe laboratory procedures as outlined in publications such as Biosafety in Microbiological and Biomedical Laboratories, HHS (HHS Publication Number) (<https://www.cdc.gov/labs/pdf/CDC-Biosafety-microbiologicalBiomedicalLaboratories-2009-P.pdf>).

* OSHA: Occupational Safety and Health Administration (United States of America).

† ACGIH: American Conference of Government Industrial Hygienists (United States of America).

‡ COSHH: Control of Substances Hazardous to Health (United Kingdom).

2.6 Chemical Safety

WARNING 	Hazardous chemicals w8 Some chemicals used with the EZ2 instrument may be hazardous or may become hazardous after completion of a purification. Always wear safety glasses, gloves, and a lab coat. The responsible body (e.g., laboratory manager) must take the necessary precautions to ensure that the surrounding workplace is safe, and that the operators are suitably trained and not exposed to hazardous levels of infectious agents as defined in the applicable Material Safety Data Sheets (MSDSs) or OSHA*, ACGIH†, or COSHH‡ documents. For more information, visit: www.qiagen.com/support/msds.aspx. Venting for fumes and disposal of wastes must be in accordance with all national, state, and local health and safety regulations and laws.
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* OSHA: Occupational Safety and Health Administration (United States of America).

† ACGIH: American Conference of Government Industrial Hygienists (United States of America).

‡ COSHH: Control of Substances Hazardous to Health (United Kingdom).

WARNING 	Toxic Fumes w9 Do not use bleach to clean or disinfect the EZ2 instrument. Bleach in contact with salts from the buffers can produce toxic fumes.
--	---

WARNING 	Toxic Fumes w10 Do not use bleach to disinfect used labware. Bleach in contact with salts from the buffers can produce toxic fumes.
---	---

Note: If you work with volatile solvents, toxic substances, etc., you must provide an efficient laboratory ventilation system to remove vapors that may be produced.

2.7 Waste disposal

Used consumables, such as reagent cartridges and disposable filter-tips, may contain hazardous chemicals or infectious agents from the purification process. Such waste must be collected and disposed of properly, according to local safety regulations.

For more information about how to dispose of the EZ2, see the user manual.

CAUTION 	Hazardous chemicals and infectious agents Waste may contain toxic or infectious material and must be disposed of properly. Refer to your local safety regulations for proper disposal procedures.
---	--

2.8 Mechanical Hazards

The hood of the EZ2 Instrument must remain closed during user operation of the instrument. The hood has a lock to prevent opening during runs.

CAUTION 	Moving parts A service switch allows the instrument to be operated with the hood open when the rear cover is removed. Use caution when working with the instrument and the covers are removed.	
---	---	--

The worktable of the EZ2 instrument moves during operation of the instrument. While loading the worktable, always stand clear of the instrument. Do not lean on the worktable when the robotic arm of the instrument is moving to reach loading position with its lid open. Wait until the robotic arm completes its movements before you start to load or unload.

WARNING 	Moving parts To avoid contact with moving parts during operation of the EZ2, the instrument must be operated with the lid closed. If there is a problem with the EZ2, great care should be taken if the instrument is operated with covers off during service.	w11
---	---	-----

WARNING 	Be careful when cleaning the piercing unit - the spikes are very sharp and can be contaminated.	w12
---	---	-----

WARNING 	Moving parts Avoid contact with moving parts during operation of the EZ2. Under no circumstances should hands be placed under the pipetting arm during movement. Do not attempt to remove any plasticware from the workdeck whilst the instrument is operating.	
---	---	--

2.9 Heat Hazard

The EZ2 worktable contains a heating system.

WARNING 	Hot surface The heating system can reach temperatures of up to 95°C (203°F). Avoid touching the heating system when it is hot.	w12
---	--	-----

CAUTION 	Risk of overheating To ensure proper ventilation, maintain a minimum clearance of 15 cm (5.9 in.) at the sides and rear of the EZ2. Slits and openings that ensure the ventilation of the EZ2 must not be covered.	c1
---	--	----

2.10 Radiation

The EZ2 instrument has a UV LED lamp. The wavelength of the UV light produced by the UV LED lamp is 270 to 285 nm. This wavelength corresponds to ultraviolet light type C, which can be used for decontamination procedures. A mechanical lock ensures that the hood must be closed for operation of the UV LED.

WARNING 	UV radiation Avoid looking directly into UV light. Do not expose your skin to UV light. The hood lock/sensor is disabled when the back cover is removed, enabling operation of the UV light even when the front hood is open.	
---	---	--

The EZ2 instrument has a 2D handheld barcode scanner to allow kit bar code and sample barcode scanning.

WARNING 	Risk of personal injury Hazard level 2 laser light: Do not stare into the light beam when using handheld barcode scanner.	
---	---	--

2.11 Maintenance safety

The user must follow the maintenance guideline outlined in the EZ2 Connect User Manual.

WARNING/ CAUTION 	RISK OF PERSONAL INJURY AND MATERIAL DAMAGE Only perform maintenance that is specifically described in the corresponding user manual or in this service manual.
--	---

CAUTION 	DAMAGE TO THE INSTRUMENT All printed circuit boards in the EZ2 have integrated circuits and can be severely damaged by static electricity. When working with the electronics, proper E.S.D grounding techniques must be followed at all times
---	---

CAUTION 	DAMAGE TO THE INSTRUMENT Do not use spray bottles containing alcohol or disinfectant to clean surfaces of the EZ2. Spray bottles should be used only to clean items that have been removed from the worktables.
---	---

2.12 Consumables

CAUTION 	Unsupported Consumables Do not connect or use any consumable, accessories, or external equipment other than those specified.	c2
---	--	----

3 General Information

3.1 Instrument Names

Instrument Type	Applications
EZ2 Connect LS	Life Science
EZ2 Connect FX	Forensic
EZ2 Connect MDx	Diagnostic

Table 3.1-1 EZ2 Connect Instruments

The EZ2 is available in different variants the EZ2, EZ2 FX and EZ2 MDx. The EZ2 is intended for general life sciences applications. The EZ2 FX is intended for forensic, human identification applications. The EZ2 MDx includes the regulated protocols for diagnostic applications. Use of the term EZ2 refers to all three variants, unless otherwise specified.

3.2 Technology - Principle

The EZ2 performs fully automated nucleic acid purification from up to 24 samples in molecular diagnostic and molecular biology applications using magnetic particle technology.

The EZ2 has pre-installed protocols that are used with QIAGEN kits to purify nucleic acids.

The instrument's touchscreen display allows the user to easily select protocols. The intuitive software/user interface guides the user through the run setup process, which includes selecting variable parameters.

Afterward, the operator loads labware and samples onto the EZ2 worktable, while following the instructions shown on the display. The order and contents of the labware are determined by each protocol. When the protocol run starts, the isolation of nucleic acids is carried out automatically in a modular workflow.

The workflow includes sample lysis, nucleic acid binding to magnetic particles, contaminant removal by washing, and, finally, elution from magnetic beads. The execution of individual steps varies and is optimized for each protocol.

The aspiration and dispensation of samples and reagents and the separation of magnetic particles are performed by the 24-channel pipettor head. If required by the protocol, the temperature of the liquids is controlled by the heating system.

The EZ2 Fx incorporates all the characteristics of the EZ2, as well as some additional features, which are particularly relevant for forensic workflows. These features include:

- An internal camera, which is used for load checks and bar code reading
- An external bar code reader, which is used for reading sample IDs and kit bar codes
- A sample recovery function for forensic protocols
- Extended user management
- Extended User Interface
- Additional reporting functions
- UV schedule function

The EZ2 MDx performs fully automated nucleic acid purification from up to 24 samples per single run in molecular diagnostic and molecular biology applications using magnetic particle technology. The EZ2 MDx is designed to automate EZ1 DSP Kits available from QIAGEN, processing of non-DSP kits is also supported.

The EZ2 MDx gives the option to start a protocol either in the IVD mode of the software (for validated IVD applications only) or in the Research mode of the software (for Molecular Biology Applications (MBA) only). The usage of IVD protocols is only possible and strictly restricted to the IVD mode of the software. This User Manual focuses on operating the EZ2 MDx in the IVD-software mode. For detailed instructions on how to operate the EZ2 MDx using the Research mode of the software (with MBA-protocols or any customized protocols), please refer to the User Manual of the EZ2 (available on the EZ2 product webpage under the Product Resources tab).

EZ2 MDx is pre-installed with various protocols for processing QIAGEN EZ1/EZ2 kits for purification of RNA, genomic DNA, and viral and bacterial nucleic acids. The user must first log in to the software in IVD mode using the touchscreen, then scans a kit barcode and loads labware, samples, and reagents onto the EZ2 MDx worktable. The user then closes the instrument hood and starts the protocol, which provides all necessary commands for sample lysis and purification. A fully automated load check helps to ensure correct loading of the worktable.

Offering an augmented user interface, users stay connected to their instrument through the built-in screen and also remotely with a computer or a mobile device (e.g., a tablet) and the QIASphere App using the associated QIASphere Connectivity Package.

The EZ2 MDX incorporates the following features:

- An internal camera, which is used for load checks and bar code reading
- An external bar code reader, which is used for reading sample IDs and kit bar codes
- Extended user management
- Extended User Interface
- Additional reporting functions
- UV schedule function

3.3 Definitions

EZ2: Overall System (refers to EZ2 Connect, EZ2 Connect FX, and EZ2 Connect MDX)

EZ2 Connect: Life Science (LS) system variant

EZ2 Connect FX: Forensics system variant

EZ2 FX: Short name for EZ2 Connect FX

EZ2 Connect MDX: MDX system variant

EZ2 MDX: Short name for EZ2 Connect MDX

GUI: Graphical User Interface

3.4 Product principle

The EZ2 instruments perform fully automated nucleic acid purification from up to 24 samples per single run using magnetic particle technology.

Addressing a wide range of sample types and applications for biomedical research, the EZ2 Connect comes with an elaborate kit portfolio. In-tip separation, on-board heating and reagent pipetting, as well as large volume capabilities enable an unprecedented level of benchtop automation.

The EZ2 Fx is tailored to Human ID and forensics customer requirements, ensuring performance, process safety and reliability.

The EZ2 MDx will address the challenges of medium throughput clinical diagnostics labs such as dealing with fluctuating numbers of sample, and a large variety of sample types and quality. A single-use cartridge per sample and easy push-button operation minimizes the risk of sampling errors.

The EZ2 MDx gives the option to start a protocol either in the IVD mode of the software (for validated IVD applications only) or in the Research mode of the software (for Molecular Biology Applications (MBA) only). The usage of IVD protocols is only possible and strictly restricted to the IVD mode of the software.

All EZ2 instruments are preinstalled with various protocols for processing QIAGEN EZ1&2 kits for purification of RNA, genomic DNA, and viral and bacterial nucleic acids, and RNA. The user must first log in to the software using the touchscreen, then scan a kit barcode and loads labware, samples, and reagents onto the EZ2 worktable. The user then closes the instrument hood and starts the protocol, which provides all necessary commands for sample lysis and purification. With the EZ2 Fx and EZ2 MDx, a fully automated load check helps to ensure correct loading of the worktable.

3.5 Instrument Overview

An introduction to the features of the instrument is provided here. For further information, refer to the relevant User Manual.

3.5.1 External features of the product



Picture 3.5-1 EZ2 Connect Fx

3.5.1.1 Touchscreen



Picture 3.5-1 EZ2 Touchscreen

The EZ2 has a 10.1-inch color touchscreen display with a resolution of 1280 x 800 pixels. The Graphical User Interface (GUI) is displayed on the touchscreen, allowing the user to completely operate the instrument.

3.5.1.2 Hood



Picture 3.5-1 EZ2 Hood in closed position

The EZ2 hood protects the interior of the instrument from external contamination during protocol runs. Additionally, the hood protects operators from UV irradiation during decontamination procedures.

The hood must be closed to enable a protocol run to start. The hood is locked at the beginning of a run, and remains locked throughout the duration of the run. This protects users from moving parts on the worktable. The hood can be manually opened to gain access to the worktable when no protocol is running.

3.5.1.3 Power Button



Picture 3.5-1 EZ2 Power Button

The power button is located on the front bottom right corner of the EZ2. Press the button to turn on the EZ2. After the button is pressed, a sound plays, the power button lights up, the startup screen appears on the touchscreen, and the instrument initializes.

To conserve energy, the EZ2 can be turned off when not in use. To turn off the EZ2, press the power button.

3.5.1.4 USB Ports



Picture 3.5-1 EZ2 has one USB port on the front next to the power button and two behind the touchscreen

The EZ2 has 3 USB ports. One is located on the front of the instrument next to the power button. Two are located on the upper back of the touchscreen display.

The USB ports allow you to connect a USB drive to the EZ2, in order to, for example, transfer report files.

In addition, the USB drive can be used to upload protocols, or update the software.

The bar code scanner is connected to the EZ2 using any one of the 3 available USB ports.

Important: Use only the USB flash drive provided by QIAGEN. Do not connect other USB flash drive devices to USB ports

Important: Do not remove the USB drive while downloading or transferring data or software to or from the instrument.

Important: Always turn off the EZ2 in order to plug or unplug the WiFi USB device. Plug-and-play of the Wi-Fi USB device while the instrument is turned on is not supported.

3.5.1.5 RJ-45 Ethernet port



Picture 3.5-1 Ethernet port (Labelled with 5)

The RJ-45 Ethernet port is located on the back of the instrument near the power connection. The Ethernet connection is 10/100 BASE T.

3.5.1.6 Power Cord Socket



Picture 3.5-1 Power cord socket (Labelled with 6)

The power cord socket is located at the back of the EZ2.

3.5.2 Internal features of the EZ2

3.5.2.1 Pipettor head



Picture 3.5-1 Pipettor head with 24 nozzles

The pipettor head is mounted above the worktable and moves in the Z-direction (meaning up and down) to reach the sample and reagent tubes on the worktable. The worktable itself moves in the Y-direction (meaning front to back) so that the pipettor head is above the appropriate position in the cartridge or tip rack during each action that is performed by the instrument.

The pipettor head contains 24 high-precision syringe pumps that are connected to tip adapters which can be attached to filter-tips. The syringe pumps operate simultaneously and can aspirate or dispense small volumes of liquid (20–1000 µl) through the attached filter-tips.

Another component of the pipettor head is the piercing unit that is located behind the tip adapters. The piercing unit is a row of 24 metal spikes that puncture the foil sealing the reagent cartridges. During operation, the piercing unit, controlled by the EZ2, opens all wells of the reagent cartridges in a dedicated order. The pipettor head then automatically picks up filter-tips from the tip rack and performs aspiration and dispensation operations at different locations on the worktable before ejecting the tips back into the tip rack at the end of the run.

3.5.2.2 Worktable



Picture 3.5-1 The EZ2 worktable

The EZ2 worktable contains two movable racks that hold all the labware required for a protocol run, and the heating system, which controls the temperature of samples during a run.

The cartridge rack consists of two separate parts. The left cartridge rack is used for cartridges in positions from 1 to 12. The right cartridge rack is used for cartridges in positions 13 to 24. On the worktable, the cartridge rack is located behind the tip rack. It holds up to 24 reagent cartridges.

Sealed reagent cartridges are prefilled and contain the reagents needed for a protocol run. Each cartridge consists of 10 sealed reagent wells and 2 empty heating positions. One heating position is a well and the other is a slot that can hold a tube.

The EZ2 tip rack consists of two separate parts. The left part of the tip rack is used for labware in positions from 1 to 12. The right part of the tip rack is used for labware in positions 13 to 24.

The tip racks are located at the front of the worktable. Each consists of four rows and 12 positions:

- Row A – which is the row that is closest to the cartridge rack, holds up to 24 sample tubes.
- Row B – usually holds up to 24 tip holders with filter tips, which are provided with the EZ1/EZ2 kits (see respective EZ1&2 kit handbook for more information).
- Row C – may contain a tube with manually filled components, such as carrier RNA or Ethanol (see respective EZ1&2 and EZ2 kit handbooks for more information).
- Row D – row closest to the user and holds up to 24 elution tubes.

Heating system



Picture 3.5-2 EZ2 heater unit

The heating system is located under the rear part of cartridge rack. If needed, it heats up the wells in positions 11 and 12.

Tray



Picture 3.5-3 EZ2 Tray

The tray is located underneath the tip and cartridge racks. Its role is to prevent contamination of the EZ2 that could be caused by liquids that are unintentionally spilled.

3.5.2.3 Magnet module



Picture 3.5-1 EZ2 magnet module

The EZ2 magnet module consists of magnets that are used to capture magnetic particles present in the liquid aspirated into the filter-tips.

3.5.2.4 Camera

The EZ2 has a built-in camera that is used for load checks and reading bar codes from cartridges.

Load checks are done before a protocol run starts. The camera checks if all the labware has been loaded in the correct positions. The results of the load check are shown on the screen. For more information on the load check.

The camera also reads the cartridge 2D bar codes. Information that is collected from the bar codes is included in run reports.

3.5.2.5 UV LED lamp

The EZ2 is equipped with a UV LED light for decontamination. During the maintenance decontamination procedure, the UV LED moves over the worktable.

Note: The hood must be closed before starting a maintenance procedure and is automatically locked during the procedure.

3.5.2.6 Internal light

The EZ2 has a built-in LED light. The internal light illuminates the worktable and informs about the current status of the run. There are two modes of the LED light:

Blinking light – indicates that operator activity is required (for example, when an error occurs).

Constant light – the default setting, used in all other situations.

3.5.3 Computer and Software

The EZ2 is operated through the integrated touchscreen.

3.5.4 Product Specification

Refer to the relevant user manual for detailed product specifications.

3.6 Tools

This section will describe the standard and special tools necessary for working with the EZ2.

3.6.1 Standard Tools

Standard tools are defined as tools that QIAGEN expects all trained engineers to have on hand during each service visit.

Standard Tools	Comment	Picture
Feeler Gauge Set	0.02mm - 1.0mm	
TORX Driver	T6, T9, T10, T20, T25, T30	
Angled TORX Key Set	T6, T9, T10, T20, T25, T30	
Phillips Screwdriver	PH#2	

Table 3.6-1 Tools

Standard Tools	Comment	Picture
Angled hex key	1.5 mm, 2 mm, 2.5 mm, 3 mm	
Flat Head Driver	small / medium	
Needle nose pliers	medium size	
Side cutter	standard	
Nut Driver (socket or handled)	5.5 mm	
Open end wrench	4 mm, 8 mm	
Digital multimeter (e.g. Fluke 116)	- Resolution: 0.1 mV - Accuracy: $\pm 1.0\%$ - Annual calibration according to manufacturer's prescription	
Thermometer incl. probe (e.g. Testo 926 with T needle probe)	- Resolution: 0.1°C - Accuracy: $\pm 0.5^\circ\text{C}$ - Annual calibration according to manufacturer's prescription	
Precision Scale	- Resolution 0.001 g - Accuracy ± 0.001 g - Annual calibration according to manufacturer's prescription	N/A

Table 3.6-1 Tools

4 Installation

System delivery and installationThe unpacking and installation of the product can be completed by the client. An installation service can be purchased where a QIAGEN authorized installer/specialist performs the installation of the EZ2 system.

4.0.1 EZ2 - Packing List

Parts			Quantity
Part #	Part	Picture	
9003210	EZ2		1
9003220	EZ2 Fx		
9003230	EZ2 MDx		
N/A	Quick Start Guide(s) for appropriate instrument: • HB-2924 MDx QSG EZ2 Connect • HB-2925 QSG EZ2 Connect		1
9027012	Cartridge Rack (Left)		1
	Cartridge Rack (Right)		1
9027009	Tip Rack (Left)		1
	Tip Rack (Right)		1

Table 4.0-1 EZ2 Packing List

Parts			Quantity
Part #	Part	Picture	
9027010	Flip Cap Rack (Left) <i>Only included with the Fx Instruments</i>		1 <i>Only included with the Fx</i>
	Flip Cap Rack (Right) <i>Only included with the Fx Instruments</i>		1 <i>Only included with the Fx</i>
9027102	Silicone grease		1
	Certificate of Manufacture EZ2		1
9026073	Handheld Barcode Reader <i>Only included with the MDx and FX Instruments</i>		1
9017850	USB Stick		1
9015126	Power Cord Set, International		1

Table 4.0-1 EZ2 Packing List

4.1 Installation of the EZ2

WARNING 	Risk of system damage If the EZ2 is transported in temperatures below 0° C (32° F), it is recommended to wait 24 hours before switching on the instrument.
--	---

4.1.1 Requirements

4.1.1.1 Site

The EZ2 is recommended be installed in a normal laboratory environment (18°C to 28°C and 15 - 75% relative humidity) for optimal performance. Refer to the appropriate user manual for full specifications.

The EZ2 must be located out of direct sunlight, away from heat sources, and away from sources of vibration and electrical interference. The site of installation should also be level, dry, clean and free of excessive drafts, excessive moisture, excessive dust, and not subject to large temperature fluctuations.

Approximately 110 cm (31.5 in.) clearance above the workbench is required to accommodate the EZ2 with the lids open. Allow at least 10 cm (4.9 in.) of free space behind and to the sides of the instrument for cooling and cabling.

The EZ2 must be placed within approximately 1.5 m (59 in.) of a properly grounded (earthed) AC power outlet. The power line to the product should be voltage regulated and surge protected.

INFORMATION 	It is recommended to plug the instrument directly into its own power outlet and not to share the power outlets with other lab equipment.
CAUTION 	Risk of overheating To ensure proper ventilation, maintain a minimum clearance of 10 cm (4.9 in.) at the sides and rear of the product. Slits and openings that ensure the ventilation of the EZ2 must not be covered. c2

4.1.1.2 Environment

- Temperature: 18°C to 30°C
- Humidity: 10% to 75% relative humidity
- Draft-free location not close to window
- Keep the instrument out of direct sunlight
- Altitude: Up to 2000 m (6500 ft.)

4.1.1.3 Power

The product operates at:

- 100–240 V AC, 50–60 Hz, 1000VA

Ensure that the voltage rating of the product is compatible with the AC voltage available at the installation site. Mains supply voltage fluctuations are not to exceed 10% of nominal supply voltages.

4.1.1.4 Grounding

To protect operating personnel, the product must be correctly grounded (earthed). The product is equipped with a 3-conductor AC power cord. To preserve this protection feature, do not operate the product from AC power outlet that has no ground (earth) connection.

4.1.2 Unpacking The EZ2

The EZ2 is delivered with all the necessary components for setting up and running the instrument. The package also contains a packing list and quick start guide. It is recommended to check the packing list to ensure all items are present and undamaged.



Picture 4.1-1 EZ2 Shipping Crate

1. Before unpacking the EZ2 move the package to the site of installation and check that the package does not have damage. If damage is noted, notify the local service manager and confirm installation before continuing.
2. Remove the straps securing the crate to the shipping pallet.
3. Cut the blue tape on the top of the box and open the flaps and reveal the accessories box.



Picture 4.1-2 Opened EZ2 shipping box and accessories box (external box already removed)

4. Remove the accessories box from within the foam and set aside. Check that all items were delivered and are not damaged.



5. To remove the exterior cardboard box, use the handles on the sides of the box to lift the box straight up and over the contents to remove the exterior cardboard box.



Picture 4.1-3 Handles on the sides of the box

6. Remove the two pieces of foam from atop the instrument.



Picture 4.1-4 Remove foam from atop the instrument

7. Remove the plastic covering from the instrument.



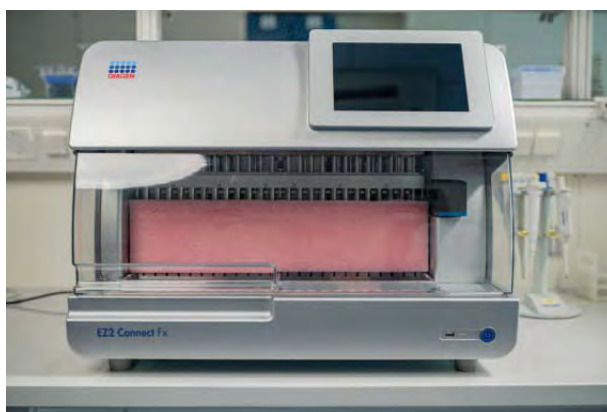
Picture 4.1-5 Plastic sheet removal

WARNING 	Risk of Injury Utilize at least two individuals when lifting the EZ2 instrument. When lifting, bend at the knees and keep your back straight.	
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8. Lift under both the left and right sides of the instrument and place the instrument into position on the bench.

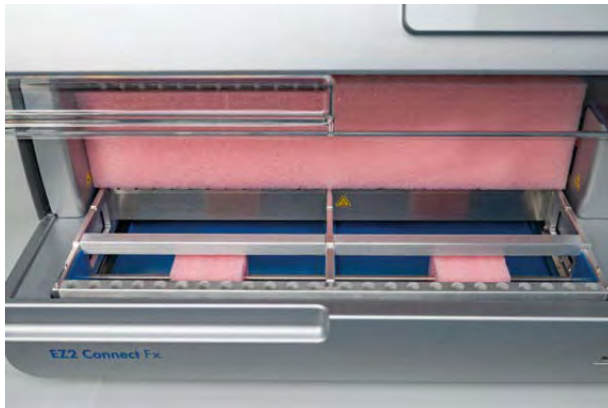
WARNING 	Risk of material damage Do not lift the EZ2 by holding the touchscreen or hood, as this may damage the instrument.	
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9. Lift open the clear cover hood to access the inside of the EZ2 instrument.



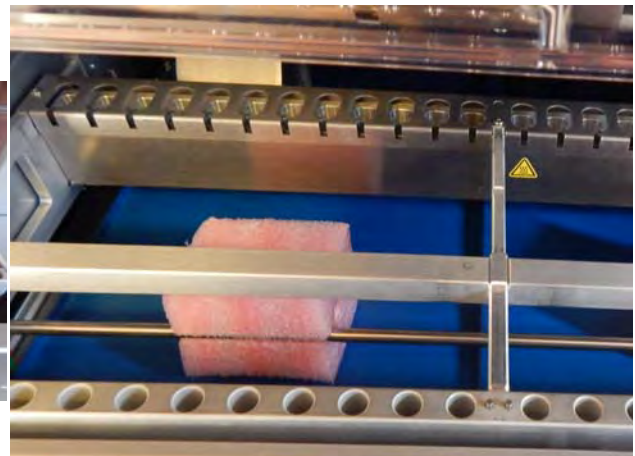
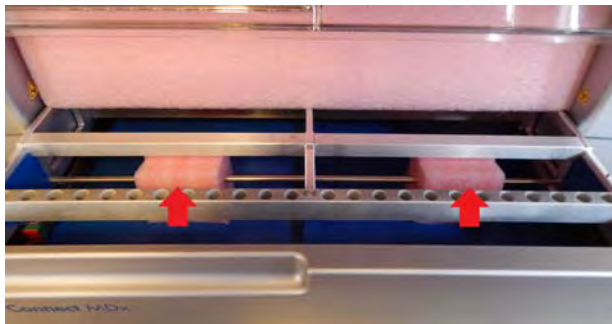
Picture 4.1-6 EZ2 with hood open

10. Remove the z- and p-axis transport lock foam by pulling the bottom of the large piece of foam toward the front of the instrument.



Picture 4.1-7 Foam Removal Z-P axis

11. Remove the small Y-foam blocks from the worktable by rotating until the split in the foam is toward the front and then pushing the bottom of the transport lock toward the rear direction. Remove the transport lock from the back.



Picture 4.1-8 Y Axis Foam Removal

12. Remove the silica packs from the unit.
13. Carefully remove the protective film from the touchscreen.
14. Insert the USB Wi-Fi adapter (if available) into the USB port above and behind the touchscreen.



Picture 4.1-9 Touchscreen USB ports, as viewed from the back of the instrument

15. Connect the handheld barcode reader (MDx and Fx versions only) to the front USB port or to a port behind the touchscreen.



Picture 4.1-10 Front USB Port

4.1.2.1 Power on the Instrument

1. Connect the appropriate power cord to the power inlet on the instrument and connect the power cord to the wall outlet.
2. Ensure the hood is closed.
3. Ensure that the EZ2 has equilibrated to ambient temperature before operating it.
4. Before powering on the instrument, the Barcode Reader and WiFi adapter (if available) must be installed. Confirm these are already installed.
5. Press the power button on the front of the instrument to power on the instrument.

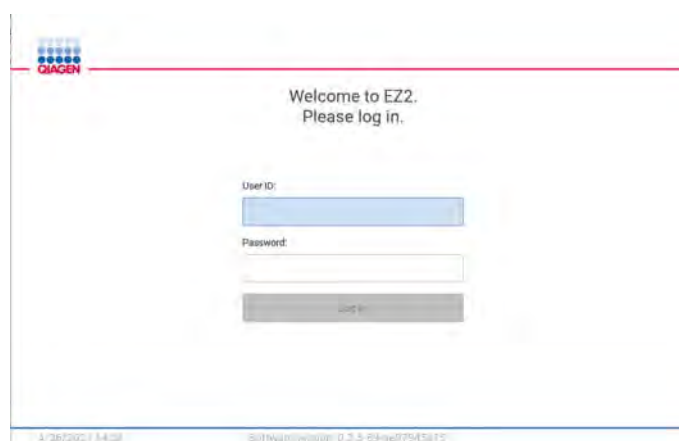


Picture 4.1-1 Location of the power button

6. Observe the instrument during the boot process: a sound will play, the power button will light up, the startup screen will appear on the touchscreen, and the instrument initializes.
7. The EZ2 must initialize and the touchscreen display the log on screen successfully.

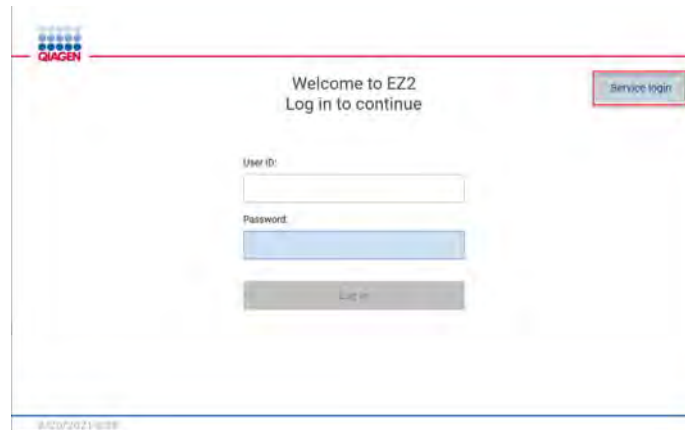
4.1.2.2 Software Configuration

The EZ2 is operated with the touchscreen, which requires a log in. In order to access the service functions, the use of a service key and unique password is required.



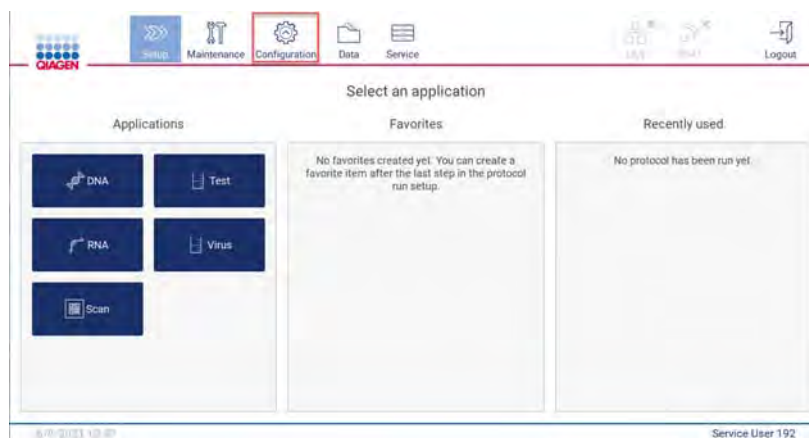
Picture 4.1-1 EZ2 GUI Log In Screen

1. Power down the instrument.
2. Insert the USB drive, which has the service.xml file in the root of the USB drive, into an available EZ2 USB port.
3. Press the power button to turn on the instrument. Allow the instrument to initialize and wait for the log on screen to be displayed.
4. Press on the Service button in the upper right corner and enter your specific EZ2 service password.
5. For log into just the User portion of the software, the default log in information is:
 - User ID: Admin
 - Password: Admin
6. Press on Login after entering the appropriate information.



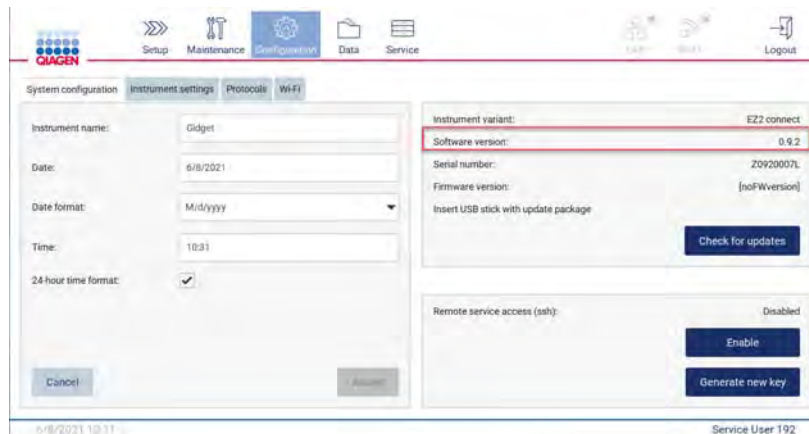
Picture 4.1-2 EZ2 Log In information

7. Click on the Configuration environment icon.



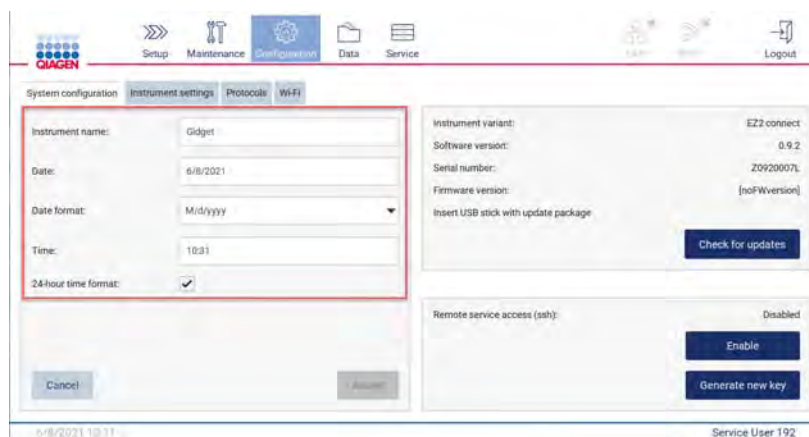
Picture 4.1-3 Configuration Environment Icon

8. Check the latest version of the User Software is installed. Update the software if not on the latest version.



Picture 4.1-4 Software Version

9. Set the system date, date format, and time by touching the areas on the screen and adjusting the settings to reflect the current data, time and clients preferred date format and hour format.



Picture 4.1-5 Instrument Name and Date

10. If necessary, change the device name.
11. After changing the settings, click Accept to save the settings.

4.1.2.3 Network Connections

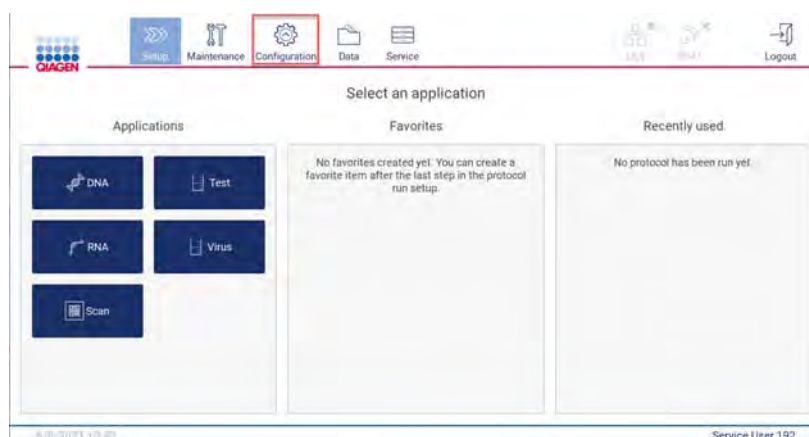
The EZ2 Connect series of instruments will have the option to be connected to a network for file transfer and instrument status monitoring via QIAsphere. This network connection can be made either with the Wireless Network (Wi-Fi) or with a wired network. This is an optional feature, so the following is only if the instrument will be connected to the network.

1. Connect the Ethernet cable to the designated network outlet in the lab, if applicable, or confirm the Wi-Fi adapter is inserted into one of the touchscreen USB ports. The instrument must be powered off to insert or remove the Wi-Fi adapter.

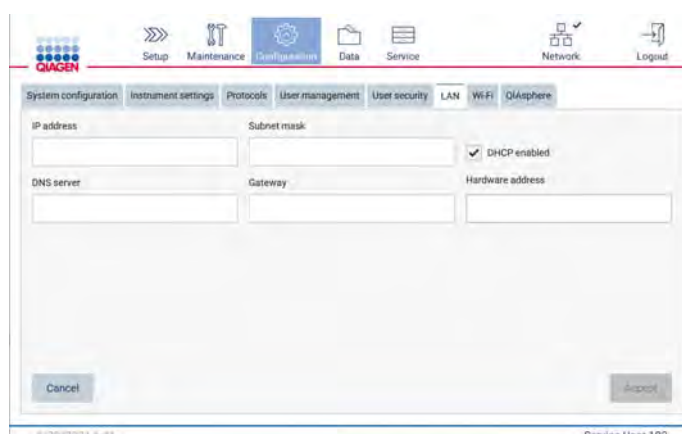


Picture 4.1-1 Touchscreen USB Ports

2. On the touchscreen software, press on the Configuration icon to enter the Configuration environment.



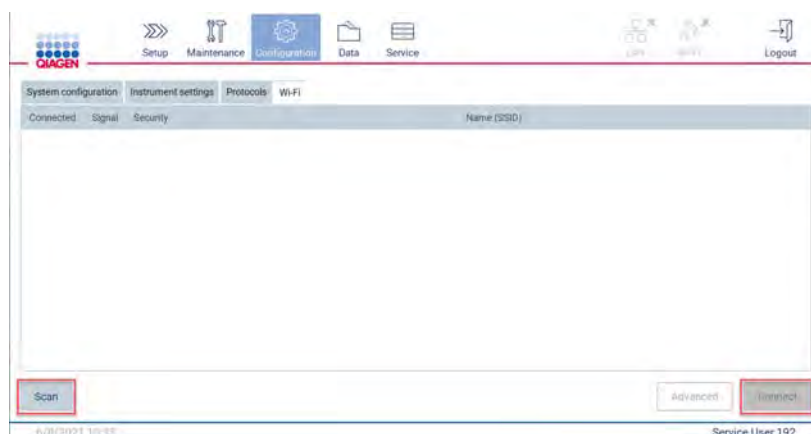
3. To configure the LAN connection:
 - 3.1. Press the LAN tab header to switch to configuration of the Local Area Network (LAN).



Picture 4.1-2 LAN Configuration tab

- 3.2. Enter the appropriate LAN information in the appropriate spaces.
4. To configure the Wi-Fi connection:
 - 4.1. Press the Wi-Fi tab header to switch to configuration of the Wi-Fi Network.
 - 4.2. Press the Scan button to scan for available Wi-Fi networks.

4.3. From the list of available WiFi networks, select the one to connect the EZ2 to and click on Connect.



4.4. Wi-Fi Configuration Screen

4.5. Enter the network password or other settings.

4.1.3 Camera Exposure Calibration

Before first use at the installation site, the camera exposure calibration must be performed on the EEZ2 Fx and EZ2 MDx systems. If the instrument is relocated or moved, the camera exposure calibration must be performed.

Perform the camera exposure calibration per [CP2 Camera Exposure Calibration](#).

4.2 EZ2 - Verification Testing

Before using the instrument, the operation of the instrument must be checked with the following tests.

4.2.1 LED Strip Test

Perform [VP1 - LED Strip Test](#).

4.2.2 Handheld Barcode Reader Test

Perform [VP9 - Barcode Reader Test](#).

4.2.3 Leakage Test

Perform [VP3 - Leakage Test](#).

4.2.4 Pipetting Test

Perform [VP4 - Pipetting Test](#).

4.2.5 Hood Lock Test

Perform [VP6 - Hood Lock Test](#).

4.2.6 Load Check Test

Perform [VP7 - Load Check Test](#).

4.2.7 Workflow System Test

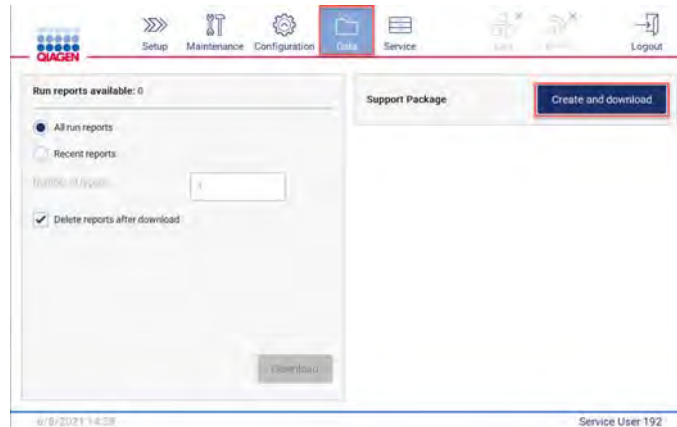
Perform [VP8 - Workflow System Test](#).

4.2.8 UV Test

Perform [VP2 - UV Light Test](#).

4.3 Installation Completion

1. Complete the Installation Checksheet (ICS-5000-0025), ensuring that all tasks have successfully been completed.
2. Download the Support Package from the instrument.
 - 2.1. Press Data to enter the Data environment.
 - 2.2. Press Create and Download next to Support Package to save the support package to the USB drive.



3. Upload the completed Installation Checksheet (ICS-5000-0025), the Excel Spreadsheet documenting the volumes from the workflow test, as well as the support package to Salesforce.
4. Mark any ISI notes that were implemented during the installation as complete in SFDC.
5. Close the work order or call according to the appropriate procedure.

5 EZ2 Modules

This chapter contains detailed information about the module in the EZ2. Each section will contain the following information:

- Functional Description/Principle of Operation
- Components

5.1 Design Parts - Covers

5.1.1 Principle of Operation

Covers protect and enhance the cosmetic appearance of the EZ2

5.1.2 Components - Design Parts (Covers)



5.2 Pipettor Module

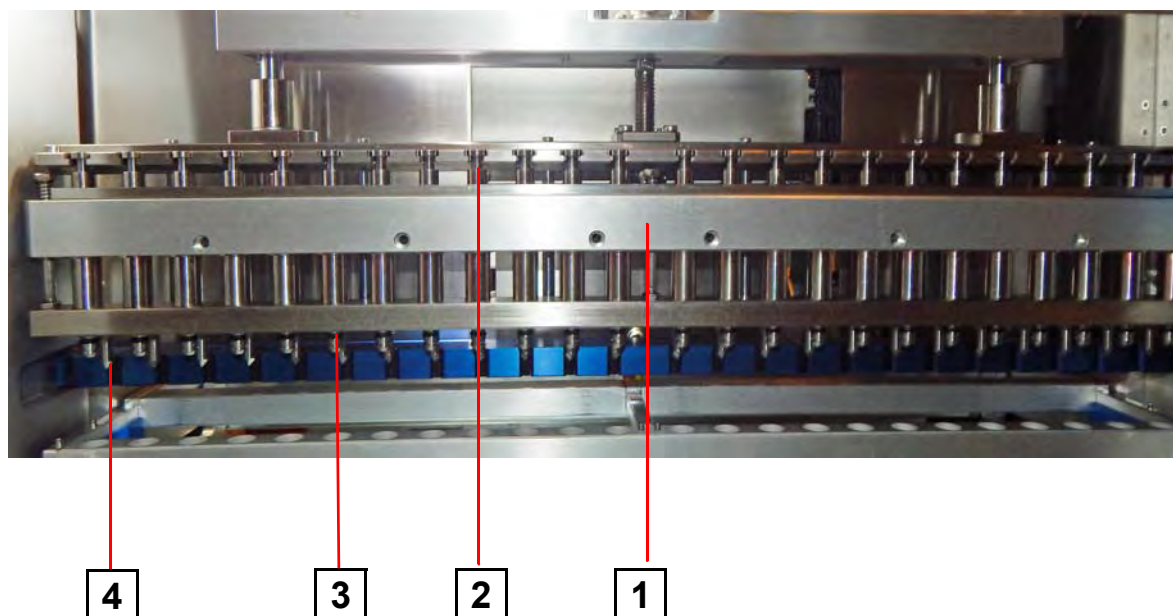
5.2.1 Principle of Operation

The pipettor module is fixed in the Y-direction and only moves up and down (z-direction). The pipettor module consists of 24 individual syringes that operate as a unit, meaning they move up and down simultaneously.

The pipettor module also contains the piercing unit, which contains 24 metal spikes used to puncture the foil that seals the reagent cartridges.

During operation, the piercing unit, controlled by the EZ2, opens all wells of the reagent cartridges in a dedicated order. The pipettor head then automatically picks up filter-tips from the tip rack and performs aspiration and dispensation operations at different locations on the worktable before ejecting the tips back into the tip rack at the end of the run.

5.3 Components - Pipettor Module



Picture 5.3-1 Pipettor Module

Pos	Quantity	Part Number	Name/Comment
1	1	9027187	Pipette Module, EZ2
2	24 (spare part is a set of 6)	9027084	Plunger Repair Set, EZ2 -contains Plunger/O-ring/Collar
3	24	9027085	D-ring, EZ2
4	24	9027185	Piercing Needle, EZ2

Table 5.3-1 Pipettor Module - Components

5.4 Heater Module

5.4.1 Principle of Operation

The heating system can heat wells in position 11 and 12 for each reagent cartridge.

5.4.2 Components - Heater Module

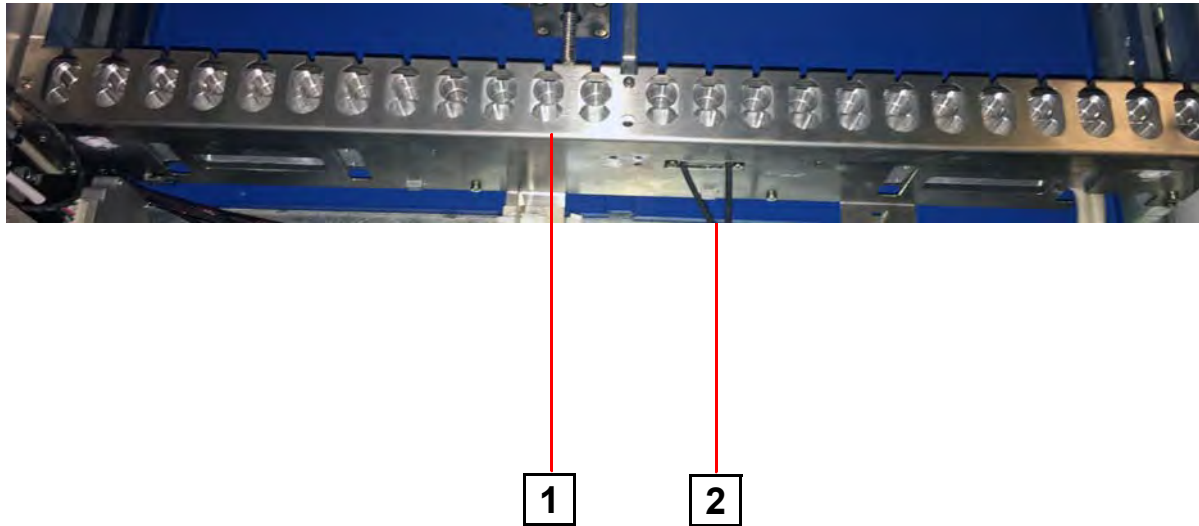


Table 5.4-1 Heater Module

Pos	Quantity	Part Number	Name/Comment
1	1	9027162	Heater Module, EZ2
2	1	9027166	Temperature Fuse, EZ2

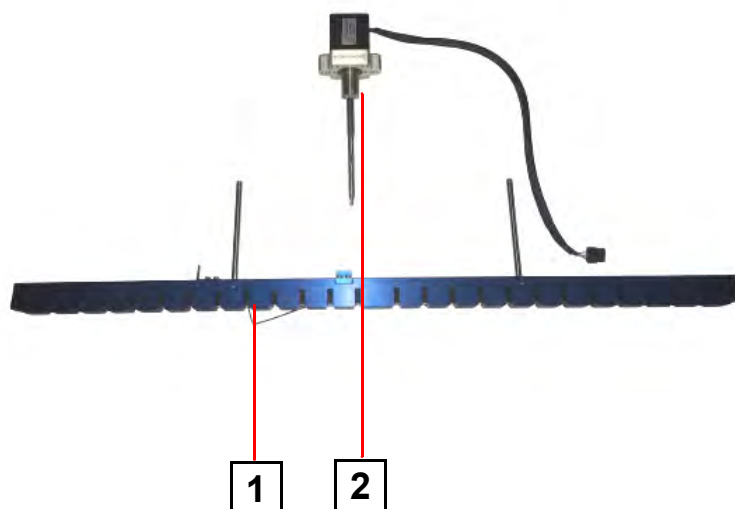
Table 5.4-2 Components- Heater Module

5.5 Magnet Module

5.5.1 Principle of Operation

The magnet module captures magnetic particles present in the liquid aspirated into the filter tips.

5.5.2 Components - Magnet Module



Picture 5.5-1 Magnet Module

Pos	Quantity	Part Number	Name/Comment
1	1	9027183	Magnet Assembly, EZ2
2	1	9027190	Magnet Motor, EZ2

Table 5.5-1 Magnet Module Components

5.6 UV Module

5.6.1 Principle of Operation

The UV module is a UV LED and moves in the X-direction. To decontaminate the entire workdeck, the workdeck also moves in the y-direction during decontamination.

5.6.2 Components - UV Module



Picture 5.6-1 UV Module

Pos	Quantity	Part Number	Name/Comment
1	1	9027152	UV LED Board PCB, EZ2

Table 5.6-1 UV Module Components

5.7 Camera Module

5.7.1 Principle of Operation

The camera is used for load checks and for reading cartridge bar codes.

5.7.2 Components - Camera Module



Picture 5.7-1 Camera Module

Pos	Quantity	Part Number	Name/Comment
1	1	9027158	Camera, EZ2

Table 5.7-1 Camera Components

5.8 HMI

5.8.1 Principle of Operation

The EZ2 HMI consists of the touchscreen and USB ports.

The EZ2 touchscreen allows the user to control the device via a GUI.

5.8.2 Components - HMI



Picture 5.8-1 HMI

Pos	Quantity	Part Number	Name/Comment
1	1	9027079	Module, LCD Display, EZ2

Table 5.8-1 HMI Components

6 Replacement Procedures (RP)

This chapter provides the removal and replacement procedures for the EZ2.

6.1 Protective Measures

In order to prevent possible exposure to electrical, mechanical or biological hazards, make sure to disconnect the instrument power cord and to use appropriate hand, eye, and body protection.

Material	Comment
Eye protection	Safety glasses
Hand protection	Gloves (rubber, nitrile, etc.)
Body protection	Lab coat

Table 6.1-1 Protective Material

WARNING 	Electrical Hazard Disconnect the line power cords from the power outlet before servicing. Lethal voltages inside the instrument When the instrument is connected to line power, terminals may be live, and opening covers or removing parts is likely to expose live parts.
WARNING 	Moving parts To avoid contact with moving parts during operation of the EZ2, the instrument must be operated with the door closed. If there is a problem with the EZ2, great care should be taken if the instrument is operated with door off.
WARNING 	Pinch and impact hazards The worktable and EZ2 components may move unexpectedly. Use extreme caution when placing your hands inside the instrument if the power is on.
WARNING 	Biological materials Handle biological material with the greatest of care and in accordance with the required safety regulations. Always wear safety glasses, gloves, and a lab coat. For more information, visit: www.qiagen.com/support/msds.aspx .

WARNING 	The following procedures are only to be performed by QIAGEN trained and certified individuals.
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6.2 Overview of Replacement Procedures

RP1 - Rear Cover

RP2 - Main Cover

RP3 - Left Stationary Cover

RP4 - Right Stationary Cover
RP5 - Internal Right Metal Cover
RP6 - Front Bottom Cover
RP7 - Air Filters
RP8 - Top Front Cover
RP9 - Guide Rail
RP10 - Magnet Motor
RP11 - Magnet Assembly
RP12 - Left Hood Solenoid
RP13 - Right Hood Solenoid
RP14 - Internal Right Plastic Cover
RP15 - Camera Enclosure
RP16 - Internal Left Cover
RP17 - Conston Springs
RP18 - Hood
RP19 - Hood Side Bracket Left/Right
RP20 - Service Switch
RP21 - Air Duct
RP22 - Fan/Bracket
RP23 - Thermocouple
RP24 - Hood Close Switch
RP25 - X Pulley
RP26 - X Timing Belt
RP27 - X Motor
RP28 - Z Motor
RP29 - P Motor
RP30 - Y Motor
RP31 - Y Timing Belt
RP32 - Y Idler Pulley
RP33 Power Button
RP34 - UV LED Board PCB
RP35 - Camera
RP36 - Camera LED Board PCB/Diffuser/Bracket
RP37 - Pipette Module
RP38 - Plunger/Pipetting Collar
RP39 - D-ring
RP40 - Piercing Needle
RP41 - Power Supply
RP42 - Motor Driver Board
RP43 - CPU Board
RP44 - Power Board
RP45 - Hood Guide Roller
RP46 - Foot
RP47 - LAN Port
RP48 - Ball Latches
RP49 - Speaker
RP50 - Worktable
RP51 - Heater Module
RP52 - Temperature Fuse
RP53 - Heater Fuse 20 Amp Power Board
RP54 - CPU Board 2.5 Amp Fuse
RP55 - Motor Board 0.16 Amp Fuse
RP56 - Motor Board 6.3 Amp Fuse
RP57 - UV 0.125 Amp Fuse
RP58 - CPU Coin Battery CR2450
RP59 - SD Micro Card
RP60 - AC Inlet
RP61 - 4 Amp 250 Volt Fuse

RP62 - E-Chain X Axis
 RP63 - E-Chain Y Axis
 RP64 - Z E-Chain
 RP65 - Cable From Motor Driver to Slot Sensor (P-Org)
 RP66 - Cable From Motor Driver-Sensor Z (Crash)
 RP67 - Cable From Motor Driver Board to Motor P-Axis
 RP68 - Cable Motor Driver Slot Sensor Z-EL
 RP69 - Cable Motor Driver Slot Sensor Z-Org
 RP70 - Cable Motor Driver Slot Sensor Z Crash
 RP71 - USB Long Cable to Front Bottom Cover
 RP72 - USB Short Cable to Top Front Cover
 RP73 - Cable From CPU Board to Camera
 RP74 - Camera Cable Adapter-LED Board
 RP75 - Cable From CPU Board to UV LED Board
 RP76 - Cable CPU Board-Lock Sensor Left
 RP77 - Cable Adapter Sensor ZEL
 RP78 - Cable Adapter Z Org Sensor
 RP79 - Cable CPU Board-Lock Sensor Right
 RP80 - Cable From Power Board to Power Supply
 RP81 - Cable Sensor Z Crash Adapter Sensor
 RP82 - Cable Adapter to P-Org Slot Sensor
 RP83 - Cable Lock Sensor Left Adapter
 RP84 - Cable Motor Driver-Slot Sensor M Org
 RP85 - Cable CPU Board-Display Backlight.
 RP86 - Cable CPU Board-Display LVDS Signal?
 RP87 - LED Strip
 RP88 - Cable CPU-Thermocouple Adapter
 RP89 - Cable Power Board-Heater Power Adapter
 RP90 - Cable Power Board-Temp Fuse Adapter
 RP91 - Z Motor Coupler
 RP92 - P Motor Coupler
 RP93 - FFC Cable
 RP94 - LCD Touchscreen

6.3 RP1 - Rear Cover

Part Required	Rear Cover
Tool(s) Required	T20
Calibration Required	NA
Testing Required	VP12 - System Initialization Test
Replacement Time	5 minutes
Calibration/Testing Time	2 Minutes
Total Repair Time	7 minutes

Table 6.3-1 RP1 Rear Cover

WARNING

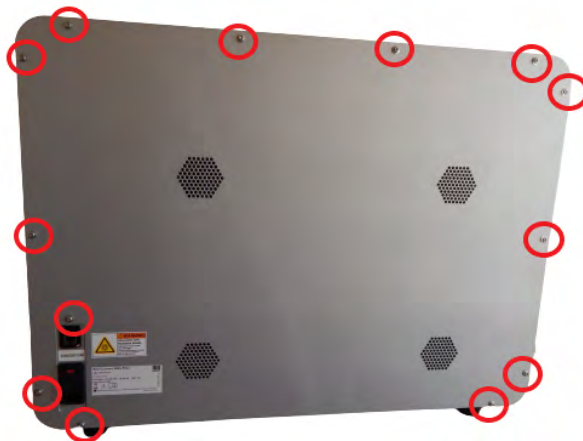
Removal of the back cover disengages the service switch. This allows the performance of runs without all of the covers in place, and with the hood up.

Ensure personal safety when working with the rear cover removed due to moving components.

The UV run may also be operated with the hood open. Caution should be exercised to limit exposure to the UV light.

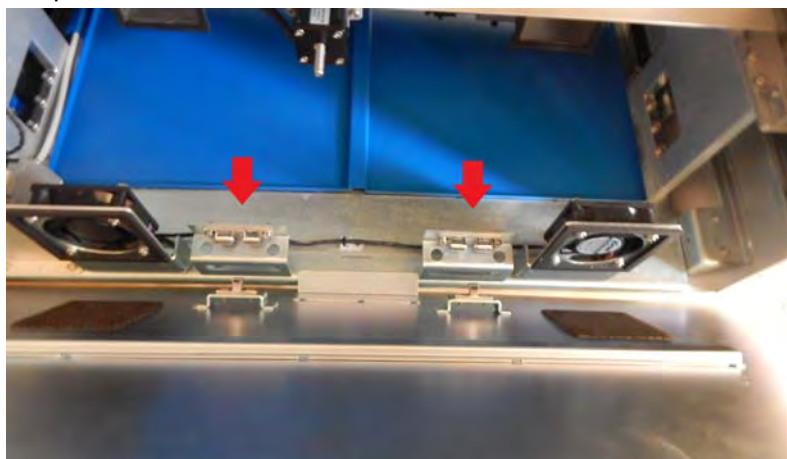
6.3.1 RP1 - Rear Cover - Removal Procedure

1. Turn off the power to the EZ2.
2. Remove the power cable and LAN cable.
3. Remove the 13 screws with a (T20) on the rear cover



Picture 6.3-1 Rear Cover

4. Lift the rear cover up and lean it back to remove.



Picture 6.3-2 Rear Cover Latches

6.3.2 RP1 - Rear Cover - Replacement Procedure

1. Slide the cover down to catch the ball latches.
2. Install all 13 (T10) screws back in position.
3. Plug in the power cord and LAN cable if available.
4. Perform [VP12 - System Initialization Test](#).

6.4 RP2 - Main Cover

Part Required	9027132 - Main Cover
Tool(s) Required	T20
0Calibration Required	NA
Testing Required	VP12 - System Initialization Test
Replacement Time	2 minutes
Calibration/Testing Time	2 minutes
Total Repair Time	4 minutes

Table 6.4-1 RP2 Main Cover

6.4.1 RP2 - Main Cover - Removal Procedure

1. Disconnect the power cable and LAN cable.
2. Remove rear cover per [RP1 - Rear Cover - Removal Procedure](#).
3. Push the main cover 3 inches to the rear of the instrument.



Picture 6.4-1 Main Cover Removal

4. Push the cover back to unhook the latches,



Picture 6.4-2 Main Cover Latches

5. Carefully pick up the cover to remove.

6.4.2 RP2 - Main Cover- Replacement Procedure

1. Carefully line the cover up on both sides of the instrument.
2. Slide the cover forward till it fastens in the brackets and meets the top front cover.
3. Put the rear cover back in position per [RP1 - Rear Cover - Replacement Procedure](#).
4. Perform [VP12 - System Initialization Test](#).

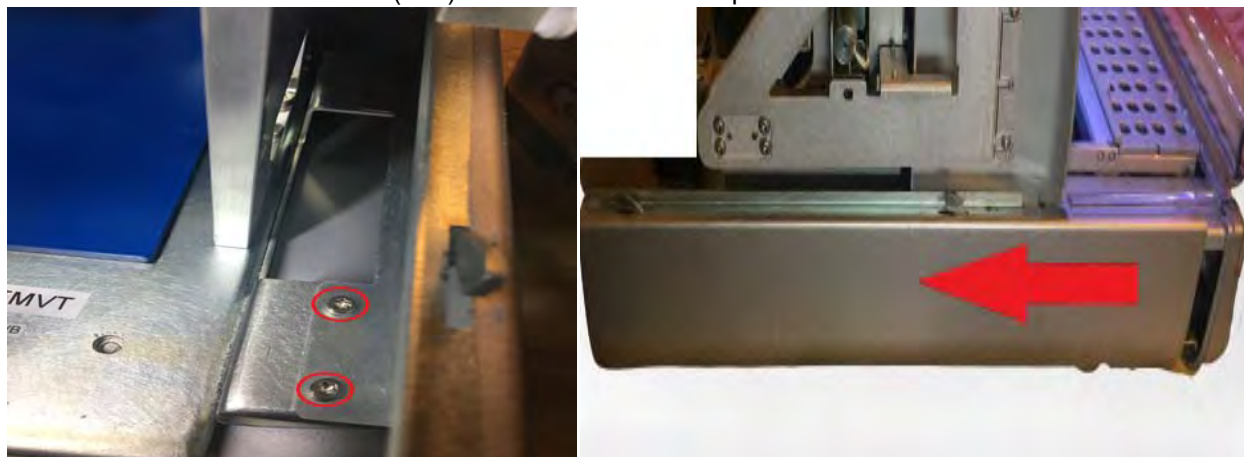
6.5 RP3 - Left Stationary Cover

Part Required	9027130 - Left Stationary Cover
Tool(s) Required	T10, T20
Calibration Required	NA
Testing Required	VP12 - System Initialization Test
Replacement Time	13 minutes
Calibration/Testing Time	2 minutes
Total Repair Time	15 minutes

Table 6.5-1 RP3 Left Stationary Cover

6.5.1 RP3 - Left Stationary Cover - Removal Procedure

1. Turn off the instrument.
2. Unplug the power cable and LAN cable.
3. Remove the rear cover per [RP1 - Rear Cover - Removal Procedure](#).
4. Remove the main cover per [RP2 - Main Cover - Removal Procedure](#).
5. Remove the 2 screws with a (T20) that holds the cover in place.



Picture 6.5-1 Left Stationary Cover

6. Slide the cover toward the back to remove.

6.5.2 RP3 - Left Stationary Cover - Replacement Procedure

1. Slide the cover forward to put back in position
2. Install both screws back in place with a (T20) that holds the cover in place.
3. Install the main cover per [RP2 - Main Cover- Replacement Procedure](#).
4. Install the rear cover per [RP1 - Rear Cover - Replacement Procedure](#).
5. Perform [VP12 - System Initialization Test](#).

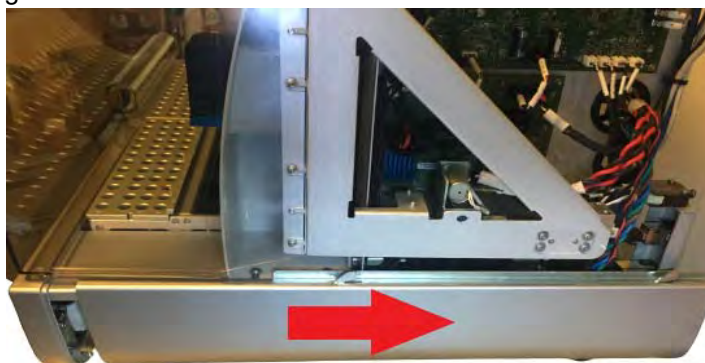
6.6 RP4 - Right Stationary Cover

Part Required	9027133 - Right Stationary Cover
Tool(s) Required	T20
Calibration Required	NA
Testing Required	VP12 - System Initialization Test
Replacement Time	13 minutes
Calibration/Testing Time	3 minutes
Total Repair Time	16 minutes

Table 6.6-1 RP4 Right Stationary Cover

6.6.1 RP4 - Right Stationary Cover - Removal Procedure

1. Turn off the instrument.
2. Unplug the power cable and LAN cable.
3. Remove the rear cover per [RP1 - Rear Cover - Removal Procedure](#).
4. Remove the main cover per [RP2 - Main Cover - Removal Procedure](#).
5. Remove the 2 screws (T20) at the back right side.



Picture 6.6-1 Right Side Stationary Cover

6. Slide the cover back 3 inches to remove.

6.6.2 RP4 - Right Stationary Cover - Replacement Procedure

1. Slide the cover forward into the bracket slot to put back in position.
2. Install the 2 screws back in place with a (T20) tool.
3. Install the main cover per [RP2 - Main Cover- Replacement Procedure](#).
4. Install the rear cover per [RP1 - Rear Cover - Replacement Procedure](#).
5. Perform [VP12 - System Initialization Test](#).

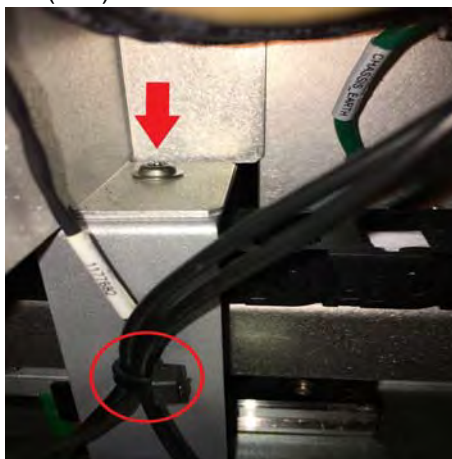
6.7 RP5 - Internal Right Metal Cover

Part Required	9027128 - Internal Right Cover
Tool(s) Required	T10, T20
Calibration Required	NA
Testing Required	VP12 - System Initialization Test
Replacement Time	30 minutes
Calibration/Testing Time	3 minutes
Total Repair Time	33 minutes

Table 6.7-1 RP5 Internal Right Cover

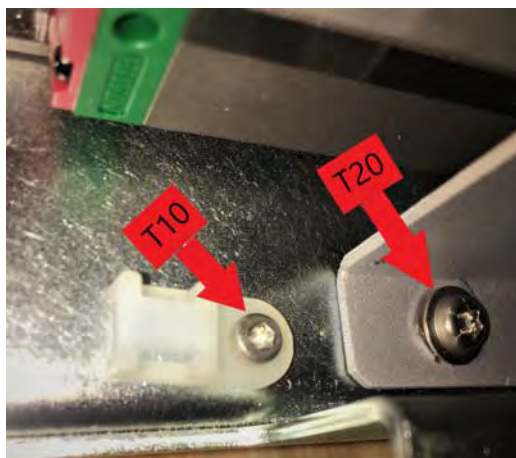
6.7.1 RP5 - Internal Right Metal Cover - Removal Procedure

1. Remove the front bottom cover per [RP6 - Front Bottom Cover - Removal Procedure](#).
2. Remove the right internal plastic cover per [RP14 - Internal Right Plastic Cover - Removal Procedure](#).
3. Cut the cable tie and remove the (T20) screw.



Picture 6.7-1 Right Cover Screw Top Side

4. Cut the cable ties over the screws and move the cables.
5. Remove the (T10) cable tie anchor screw.
6. Remove the (T20) bottom screw.



Picture 6.7-2 Right Cover Screw Bottom Side

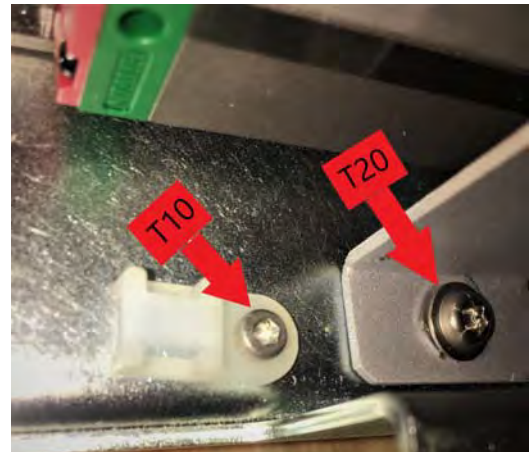
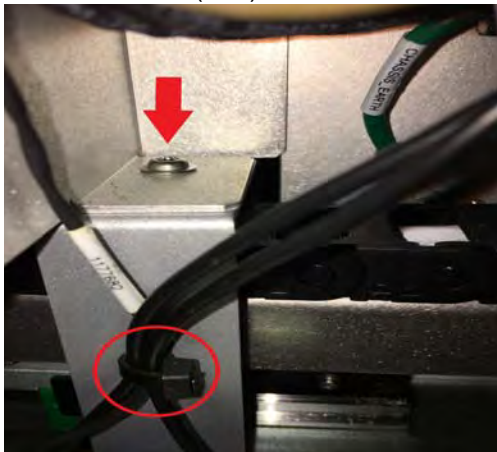
- Slide the cover towards the front of the instrument to remove.



Picture 6.7-3 Right Internal Cover Metal

6.7.2 RP5 - Internal Right Metal Cover- Replacement Procedure

- Slide the internal right cover in position.
- Align the cover so the 2 (T20) bracket screws can be replaced in the picture above.
- Install the 1 (T10) screw for the cable tie holder and replace the cable ties that were removed.



Picture 6.7-4 Internal Right Cover Screw Replacement

- Install the front bottom cover per [RP6 - Front Bottom Cover - Replacement Procedure](#).
- Replace the left stationary cover per [RP3 - Left Stationary Cover - Replacement Procedure](#).
- Replace the right stationary cover per [RP4 - Right Stationary Cover - Replacement Procedure](#).
- Replace the main cover per [RP2 - Main Cover- Replacement Procedure](#).
- Replace the rear cover per [RP1 - Rear Cover - Replacement Procedure](#).
- Perform [VP12 - System Initialization Test](#).

6.8 RP6 - Front Bottom Cover

Part Required	9027195 LS, 9027196 FX, 9027197 MDX, Front Bottom Cover
Tool(s) Required	T10, T20, T6
Calibration Required	NA
Testing Required	VP12 - System Initialization Test
Replacement Time	30 minutes
Calibration/Testing Time	3 minutes
Total Repair Time	33 minutes

Table 6.8-1 RP6 Front Bottom Cover

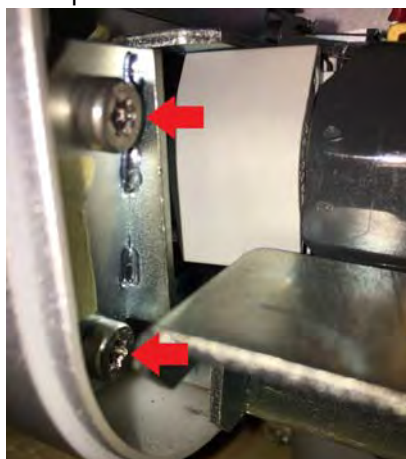
6.8.1 RP6 - Front Bottom Cover - Removal Procedure

1. Remove the rear cover per [RP1 - Rear Cover - Removal Procedure](#).
2. Remove the main cover per [RP2 - Main Cover - Removal Procedure](#).
3. Remove the stationary left cover per [RP3 - Left Stationary Cover - Removal Procedure](#).
4. Remove the right stationary cover per [RP4 - Right Stationary Cover - Removal Procedure](#).
5. Remove the left and right trays.
6. Remove all 10 (T20) screws on the back side of the front bottom cover.



Picture 6.8-1 Bottom Front Cover Screw Removal

7. Remove the 2 (T20) screws from the power button bracket.



Picture 6.8-2 Power Button Bracket Screws

8. Remove the 4 (T6) screws that hold the speaker on the cover.



Picture 6.8-3 Front Bottom Cover Speaker Removal

9. Cut any cable ties that anchor the cable for the speaker to the cover.

6.8.2 RP6 - Front Bottom Cover - Replacement Procedure

1. Replace the 4 (T6) screws that mount the speaker on the bottom front cover.
2. Replace the 10 (T20) screws that hold the front bottom cover in position.
3. Replace the 2 (T20) screws in the bracket that holds the power button.
4. Replace the 2 (T20) screws in the USB port bracket that holds it in position.
5. Replace the right stationary cover per [RP4 - Right Stationary Cover - Replacement Procedure](#).
6. Replace the left stationary cover per [RP3 - Left Stationary Cover - Replacement Procedure](#).
7. Replace the main cover per [RP2 - Main Cover- Replacement Procedure](#).
8. Replace the rear cover per [RP1 - Rear Cover - Replacement Procedure](#).
9. Perform [VP12 - System Initialization Test](#).

6.9 RP7 - Air Filters

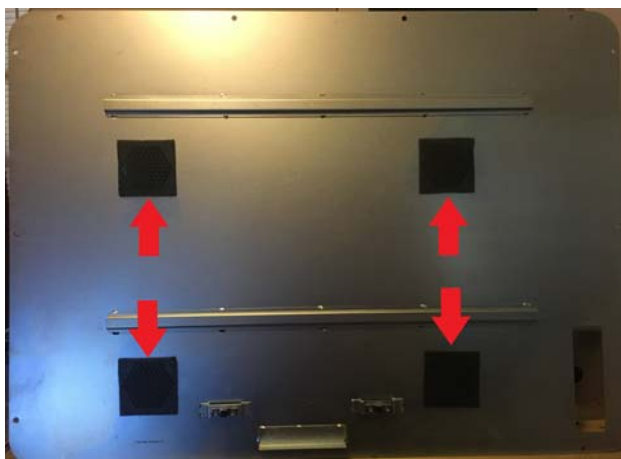
Part Required	9027154 - Air Filter
Tool(s) Required	T20
Calibration Required	NA
Testing Required	VP12 - System Initialization Test
Replacement Time	8 minutes
Calibration/Testing Time	2 minutes
Total Repair Time	10 minutes

Table 6.9-1 RP7 Air Filters

6.9.1 RP7 - Air Filters - Removal Procedure

1. Remove the rear cover per [RP1 - Rear Cover - Removal Procedure](#).

2. Remove the 4 air filters. Remove as much of the adhesive backing as possible, taking care to not scratch the panel.



Picture 6.9-1 Air Filters

6.9.2 RP7 - Air Filters - Replacement Procedure

1. Attach the 4 new air filters adhesive backing to the rear cover.
2. Replace the rear cover per [RP1 - Rear Cover - Replacement Procedure](#).
3. Perform [VP12 - System Initialization Test](#).

6.10 RP8 - Top Front Cover

Part Required	9027136 - Top Front Cover
Tool(s) Required	T30, T25, T20, T10
Calibration Required	NA
Testing Required	VP12 - System Initialization Test
Replacement Time	25 minutes
Calibration/Testing Time	2 Minutes
Total Repair Time	27 minutes

Table 6.10-1 RP8 Top Front Cover

6.10.1 RP8 - Top Front Cover - Removal Procedure

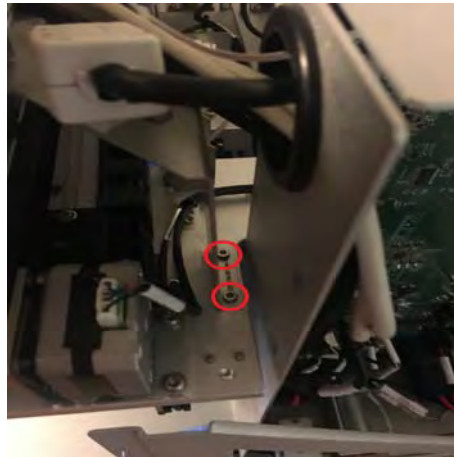
1. Power off the EZ2.
2. Remove the power cord and LAN cable.
3. Remove the rear cover per [RP1 - Rear Cover](#).
4. Remove the main cover per [RP2 - Main Cover](#).

5. Unplug the 5 cables in the picture below from the CPU board.



Picture 6.10-1 Cable Removal For Top Front Cover

6. Cut any cable ties so the cables can be pulled through the hole to the left.
7. Unhook both of the hood conston springs per [RP17 - Conston Springs - Removal Procedure](#).
8. Remove the 2 (T10) brace screws.



Picture 6.10-2 Top Front Cover Brace Screws

9. Cut any cable ties that hold the cables on the system.
10. Remove the 4 (T30) screws at the base of the top front cover.



Picture 6.10-3 Top Front Cover Screws

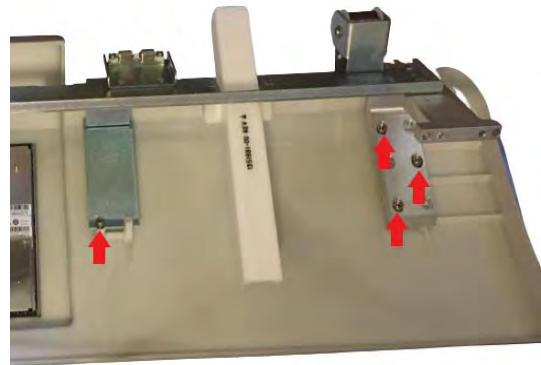


11. Lift upward on the top front cover to clear the pins and remove it from the unit.



Picture 6.10-4 Lift Front Top Cover

12. Remove the 1 (T20) grounding cable screw for the LCD screen on the side of the bracket.
 13. Cut any cable ties holding the cables on the brackets.
 14. Remove the 7 (T25) bracket screws.



Picture 6.10-5 Top Front Cover Bracket Screws

15. Remove the LCD screen per [RP94 - LCD Touchscreen - Removal Procedure](#).

6.10.2 RP8 - Top Front Cover - Replacement Procedure

1. Attach the LCD screen with adhesive that will come with the new top front cover per [RP94 - LCD Touchscreen - Replacement Procedure](#).
2. Insert any cable ties that hold the LCD cables in position.
3. Install the 1 (T20) grounding screw to the side of the left bracket.
4. Replace all 7 (T25) screws that holds the brackets on the top front cover.
5. Slide the 2 bottom brackets down over the pins while guiding the brace on the rear of the cover in position.
6. Replace the 2 (T10) screws on back brace of the cover.
7. Add cable ties fixing them to the correct position.
8. Install the 4 (T30) screws at the base of the brackets.
9. Connect the 5 cables back in position on the CPU board.
10. Replace the cable ties.
11. Reconnect the 2 conston springs to the hood.
12. Install the main cover.
13. Install the rear cover.
14. Perform [VP12 - System Initialization Test](#).

6.11 RP9 - Guide Rail

Part Required	9027129 - Guide Rail
Tool(s) Required	T30, T25, T20, T10
Calibration Required	NA
Testing Required	VP6 - Hood Lock Test
Replacement Time	25 minutes
Calibration/Testing Time	2 minutes
Total Repair Time	27 minutes

Table 6.11-1 RP9 Guide Rail

6.11.1 RP9 - Guide Rail - Removal Procedure

1. Remove the top front cover per [RP8 - Top Front Cover - Removal Procedure](#).
2. Remove the 2 (T20) screws holding the guide rail in position.



Picture 6.11-1 Guide Rail Screws

6.11.2 RP9 - Guide Rail - Replacement Procedure

1. Replace the 2 (T20) screws that fix the guide rail to the bracket.
2. Replace the top front cover per [RP8 - Top Front Cover - Replacement Procedure](#).
3. Perform [VP6 - Hood Lock Test](#).

6.12 RP10 - Magnet Motor

Part Required	9027190 - Magnet Motor
Tool(s) Required	T10,T20, 1.5 mm Hex Tool
Calibration Required	CP3 Pipette Z-axis Calibration , CP4 Worktable Calibration , CP6 Magnet Calibration
Testing Required	VP11 - M Axis Movement Test , VP8 - Workflow System Test
Replacement Time	10 minutes
Calibration/Testing Time	48 minutes
Total Repair Time	58 minutes

Table 6.12-1 RP10 Magnet Motor

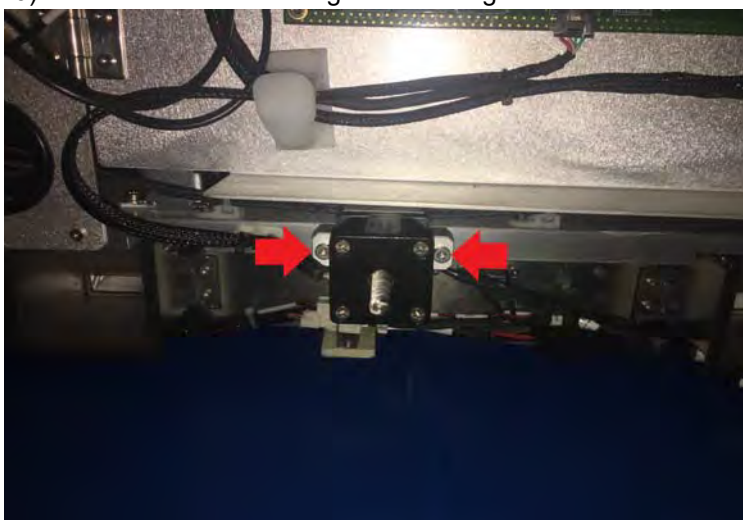
6.12.1 RP10 - Magnet Motor - Removal Procedure

1. Turn off the power and remove the power cable.
2. Remove the rear cover per [RP1 - Rear Cover - Removal Procedure](#).
3. Unplug the wiring harness for the magnet motor connected to the motor driver board (J25).



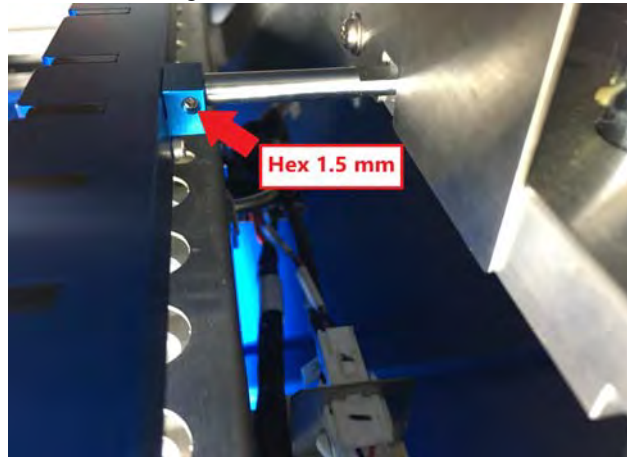
Picture 6.12-1 Magnet Motor J25 Connector

4. Remove the 2 (T20) screws on the left and right of the magnet motor.



Picture 6.12-2 M Motor Screws

5. Loosen the 2 set screws on the magnet motor shaft with a 1.5 mm hex tool.



Picture 6.12-3 Magnet Shaft 1.5 mm Screws

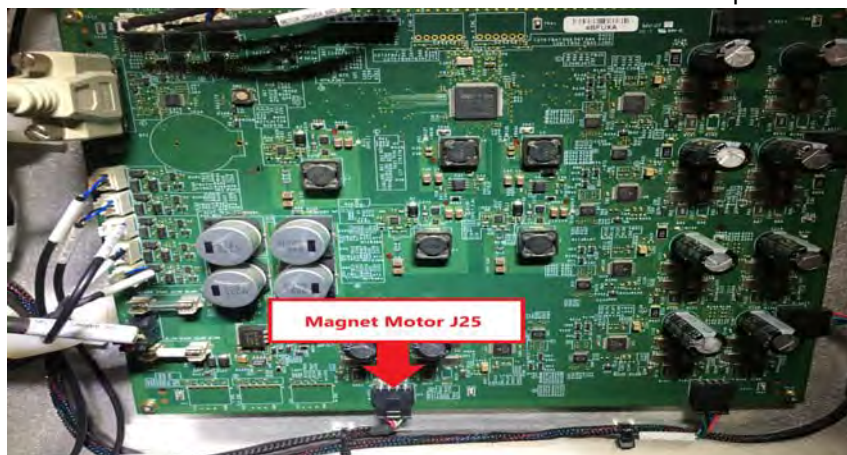
6. Pull the magnet motor straight out of the back of the instrument to remove.



Picture 6.12-4 Magnet Motor

6.12.2 RP10 - Magnet Motor - Replacement Procedure

1. Place the magnet motor back in position.
2. Install both (T20) screws that mounts the M motor back in position.
3. Tighten the 2 (1.5 mm) set screws that attaches the M motor shaft to the magnet bracket.
4. Plug the cable connector for the M motor back in the motor driver board in position J25.



Picture 6.12-5 M Motor Board Connection

5. Install the rear cover per [RP1 - Rear Cover - Replacement Procedure](#).

6. Perform [CP3 Pipette Z-axis Calibration](#), [CP4 Worktable Calibration](#), [CP6 Magnet Calibration](#), [VP11 - M Axis Movement Test](#), [VP8 - Workflow System Test](#).

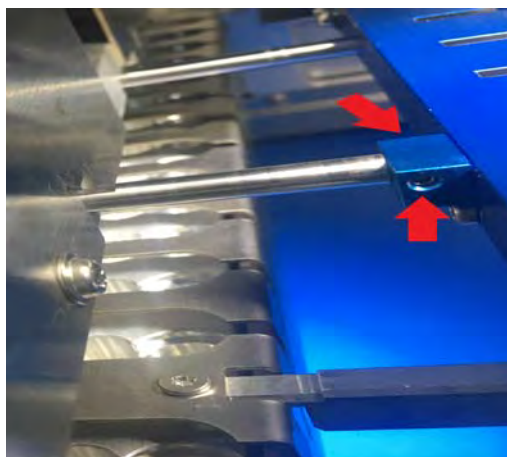
6.13 RP11 - Magnet Assembly

Part Required	9027183 - Magnet Assembly
Tool(s) Required	1.5 Hex Tool
Calibration Required	CP3 Pipette Z-axis Calibration , CP4 Worktable Calibration , CP6 Magnet Calibration
Testing Required	VP11 - M Axis Movement Test , VP8 - Workflow System Test
Replacement Time	10 minutes
Calibration/Testing Time	48 minutes
Total Repair Time	58 minutes

Table 6.13-1 RP11 Magnet Assembly

6.13.1 RP11 - Magnet Assembly - Removal Procedure

1. Turn off the instrument.
2. Loosen the 2 set screws that attach the magnet motor shaft to the magnet assembly with a 1.5 mm hex tool.



Picture 6.13-1 Magnet Module Set Screw

3. Pull the magnet assembly towards the front of the instrument to remove.



Picture 6.13-2 Magnet Assembly

6.13.2 RP11 - Magnet Assembly- Replacement Procedure

1. Turn off the instrument.
2. Slide the magnet assembly rails back in position.
3. Tighten the two set screws on both sides of the shaft.
4. Perform [CP3 Pipette Z-axis Calibration](#), [CP4 Worktable Calibration](#), [CP6 Magnet Calibration](#), [VP11 - M Axis Movement Test](#), [VP8 - Workflow System Test](#).

6.14 RP12 - Left Hood Solenoid

Part Required	9027087 - Solenoid
Tool(s) Required	T10, T20
Calibration Required	NA
Testing Required	VP6 - Hood Lock Test
Replacement Time	15 minutes
Calibration/Testing Time	2 minutes
Total Repair Time	17 minutes

Table 6.14-1 RP12 Left Hood Solenoid

6.14.1 RP12 - Left Hood Solenoid- Removal Procedure

1. Turn off the instrument.
2. Unplug the power cable and LAN cable.
3. Remove the rear cover per [RP1 - Rear Cover - Removal Procedure](#).
4. Remove the main cover per [RP2 - Main Cover - Removal Procedure](#).
5. Remove the 2 (T10) screws to open the panel.

*Picture 6.14-1 Back Panel*

6. Disconnect the solenoid connector.



Picture 6.14-2 Solenoid Connector

7. Remove the 2 (T20) solenoid bracket screws.



Picture 6.14-3 Left Solenoid Bracket Screws

8. Remove the solenoid.

6.14.2 RP12 - Left Hood Solenoid - Replacement Procedure

1. Return the solenoid to the original position.
2. Replace both of the (T20) screws in the bracket.
3. Plug in the solenoid connector.
4. Shut and replace the 2 (T10) screws in the back panel.
5. Place the main cover back in position per [RP2 - Main Cover- Replacement Procedure](#).
6. Put the rear cover back in position [RP1 - Rear Cover - Replacement Procedure](#).
7. Perform [VP6 - Hood Lock Test](#).

6.15 RP13 - Right Hood Solenoid

Part Required	9027087 - Solenoid
Tool(s) Required	T10, T20
Calibration Required	NA
Testing Required	VP6 - Hood Lock Test
Replacement Time	15 minutes
Calibration/Testing Time	2 minutes
Total Repair Time	17 minutes

Table 6.15-1 RP13 Right Hood Solenoid

6.15.1 RP13 - Right Hood Solenoid - Removal Procedure

1. Turn off the instrument.
2. Unplug the power cable and LAN cable.
3. Remove the rear cover per [RP1 - Rear Cover - Removal Procedure](#).
4. Remove the main cover per [RP2 - Main Cover - Removal Procedure](#).
5. Cut the small cable tie holding the connector and sensor wire together if needed.
6. Disconnect the solenoid cable connector.



Picture 6.15-1 Right Hood Solenoid

7. Remove the 2 (T20) screws holding the solenoid bracket in position.
8. Remove the solenoid.

6.15.2 RP13 - Right Hood Solenoid- Replacement Procedure

1. Return the solenoid to the original position.
2. Replace both of the (T20) screws in the bracket.
3. Plug in the solenoid connector.
4. Put a small cable tie on the connector wire and sensor wire.
5. Place the main cover back in position per [RP2 - Main Cover- Replacement Procedure](#).
6. Put the rear cover back in position per [RP1 - Rear Cover - Replacement Procedure](#).
7. Perform [VP6 - Hood Lock Test](#).

6.16 RP14 - Internal Right Plastic Cover

Part Required	9027143 (FX and MDX) 9027142 (LS)
Tool(s) Required	T10, T20
Calibration Required	NA
Testing Required	VP12 - System Initialization Test
Replacement Time	15 minutes
Calibration/Testing Time	2 minutes
Total Repair Time	17 minutes

Table 6.16-1 RP14 Internal Right Cover

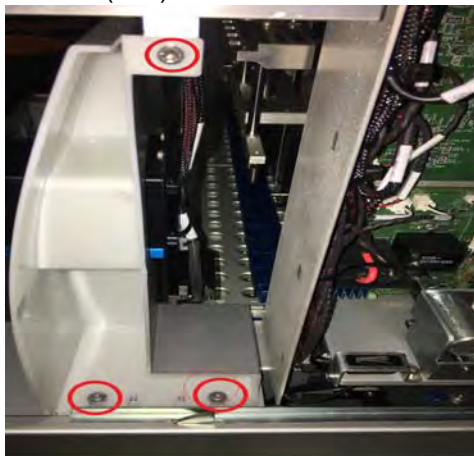
6.16.1 RP14 - Internal Right Plastic Cover - Removal Procedure

1. Turn off the power.
2. Remove the rear cover per [RP1 - Rear Cover - Removal Procedure](#).
3. Remove the main cover per [RP2 - Main Cover - Removal Procedure](#).
4. Carefully move the camera to the left of the instrument to allow space for removing the cover.



Picture 6.16-1 Internal Right Cover Removal Move Camera

5. Close the hood and remove the top (T20) screw from the cover.
6. Lift the hood and remove the 2 lower (T20) screws.



Picture 6.16-2 Internal Right Cover Screw removal

7. Lean the top of the cover toward the front of the instrument till the cover clears the top bracket.



Picture 6.16-3 Internal Right Cover Removal

8. Slide the bottom of the cover to the left to remove.

6.16.2 RP14 - Internal Right Plastic Cover - Replacement Procedure

1. Slide the top of the cover in position for the top bracket.
2. Align the bottom of the cover with the pins.
3. Install the 3 (T20) screws back in the cover.
4. Replace the main cover per [RP2 - Main Cover- Replacement Procedure](#).
5. Replace the rear cover per [RP1 - Rear Cover - Replacement Procedure](#).
6. Perform [VP12 - System Initialization Test](#).

6.17 RP15 - Camera Enclosure

Part Required	9027161 - Camera Enclosure
Tool(s) Required	T10
Calibration Required	NA
Testing Required	VP7 - Load Check Test
Replacement Time	2 minutes
Calibration/Testing Time	10 minutes
Total Repair Time	12 minutes

Table 6.17-1 RP15 Camera Enclosure

6.17.1 RP15 - Camera Enclosure - Removal Procedure

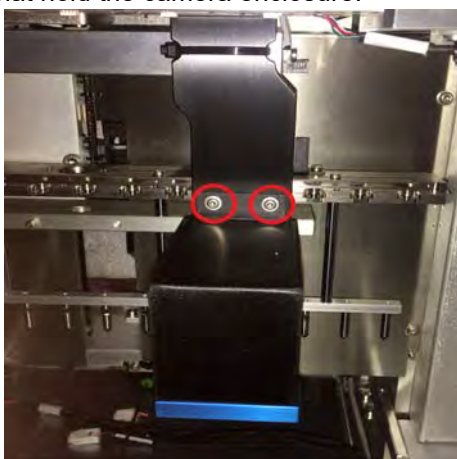
1. Turn off the power for the EZ2.

- Open the hood and carefully slide the camera to the left of the instrument.



Picture 6.17-1 Camera Position

- Remove the 2 (T10) screws that hold the camera enclosure.



Picture 6.17-2 Camera Enclosure Screws

- Remove the camera enclosure.

6.17.2 RP15 - Camera Enclosure- Replacement Procedure

- Open the hood.
- Place the camera enclosure back in position over the camera.
- Put the 2 (T10) screws back in position.
- Perform [VP7 - Load Check Test](#).

6.18 RP16 - Internal Left Cover

Part Required	9027141 - Internal Left Cover
Tool(s) Required	T10, T20
Calibration Required	NA
Testing Required	VP12 - System Initialization Test
Replacement Time	15 minutes
Calibration/Testing Time	2 minutes
Total Repair Time	17 minutes

Table 6.18-1 RP16 Internal Left Cover

6.18.1 RP16 - Internal Left Cover - Removal Procedure

1. Turn off the power.
2. Remove the power cord and LAN cable.
3. Remove the rear cover per [RP1 - Rear Cover - Removal Procedure](#).
4. Remove the main cover per [RP2 - Main Cover - Removal Procedure](#).
5. Lift up the hood and remove the 2 (T20) screws on the bottom of the cover.
6. Close the hood and remove the 1 (T20) screw on the top of the cover.



Picture 6.18-1 Left Internal Cover Screws

7. Pick up the cover and move it toward the front of the instrument.

6.18.2 RP16 - Internal Left Cover- Replacement Procedure

1. Put the cover back in position.
2. Install the 2 (T20) screws at the bottom.
3. Install the 1 (T20) screw at the top.
4. Put the main cover back in position per [RP2 - Main Cover- Replacement Procedure](#).
5. Return the rear cover per [RP1 - Rear Cover - Replacement Procedure](#).
6. Perform [VP12 - System Initialization Test](#).

6.19 RP17 - Conston Springs

Part Required	9027071 - Conston Springs
Tool(s) Required	T10, T20
Calibration Required	NA
Testing Required	VP6 - Hood Lock Test
Replacement Time	18 minutes
Calibration/Testing Time	2 minutes
Total Repair Time	20 minutes

Table 6.19-1 RP17 Conston Springs

6.19.1 RP17 - Conston Springs - Removal Procedure

1. Turn off the power.
2. Remove the rear cover per [RP1 - Rear Cover - Removal Procedure](#).
3. Remove the main cover per [RP2 - Main Cover - Removal Procedure](#).

4. Remove the 2 (T20) screws on the left and right side that attach the hood to the conston spring.



Picture 6.19-1 Conston Spring Hood Screws

5. Remove both the left and right (T20) screws.



Picture 6.19-2 Conston Spring Screws Left and Right

6. Remove the conston spring shaft.



Picture 6.19-3 Conston Spring Shaft

6.19.2 RP17 - Conston Spring - Replacement Procedure

1. Slide the conston spring shaft in the middle of the spring housing.
2. Place the Conston spring in the bracket.
3. Install the (T20) left and right screw.
4. Attach the spring to the hood with the (T20) screw.
5. Replace the main cover per [RP2 - Main Cover- Replacement Procedure](#).
6. Replace the rear cover per [RP1 - Rear Cover - Replacement Procedure](#).

7. Perform [VP6 - Hood Lock Test](#).

6.20 RP18 - Hood

Part Required	9027140 - Hood
Tool(s) Required	T30, T25, T20, T10
Calibration Required	NA
Testing Required	VP6 - Hood Lock Test
Replacement Time	30 minutes
Calibration/Testing Time	2 minutes
Total Repair Time	32 minutes

Table 6.20-1 RP18 Hood

6.20.1 RP18 - Hood - Removal Procedure

1. Turn off the power.
2. Remove the power cord and LAN cable.
3. Remove the rear cover per [RP1 - Rear Cover - Removal Procedure](#).
4. Remove the main cover per [RP2 - Main Cover - Removal Procedure](#).
5. Remove the right door spring (T20) screw.
6. Remove the left door spring (T20) screw.



Picture 6.20-1 Hood Spring Removal

7. Remove the 4 (T10) screws on the right side of the hood.



Picture 6.20-2 Hood Side Screws

8. Remove the 4 (T20) screws on the left side of the hood.



Picture 6.20-3 Hood Screws

9. Remove the top front cover per [RP8 - Top Front Cover - Removal Procedure](#).
10. Slide the hood out through the front of the instrument.

6.20.2 RP18 - Hood - Replacement Procedure

1. Slide the hood in position.
2. Align the left and right sides of the hood with the hinge brackets.
3. Put the 4 (T10) screws in each side of the hood attaching it to the bracket.
4. Install the top front cover per [RP8 - Top Front Cover - Replacement Procedure](#).
5. Attach both (T20) screws back in the hood spring.
6. Perform [VP6 - Hood Lock Test](#).

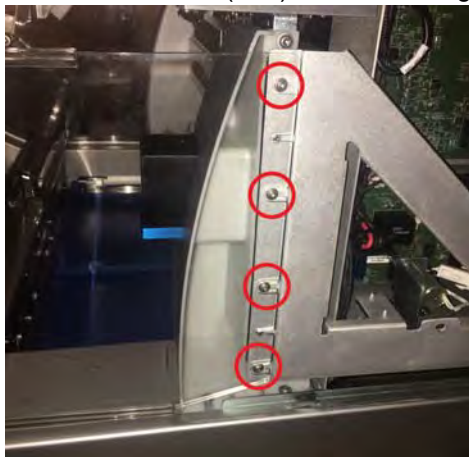
6.21 RP19 - Hood Side Bracket Left/Right

Part Required	9027219 Left, 9027220 Right - Hood Side Bracket
Tool(s) Required	T10, T20
Calibration Required	NA
Testing Required	VP6 - Hood Lock Test
Replacement Time	20 minutes
Calibration/Testing Time	2 minutes
Total Repair Time	22 minutes

Table 6.21-1 RP19 Hood Side Bracket

6.21.1 RP19 - Hood Side Bracket - Removal Procedure

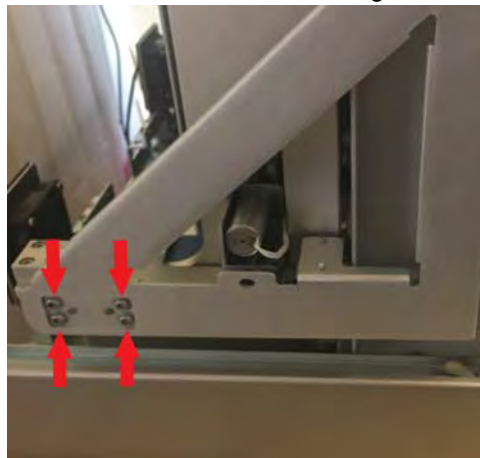
1. Turn off the power to the EZ2.
2. Remove the rear cover per [RP1 - Rear Cover - Removal Procedure](#).
3. Remove the main cover per [RP2 - Main Cover - Removal Procedure](#).
4. Remove the 4 (T10) screws holding the hood to the bracket.



Picture 6.21-1 Hood Screw Removal



5. Remove the 4 (T20) screws that hold the bracket on the hinge.



Picture 6.21-2 Hood Bracket Screws

6.21.2 RP19 - Hood Side Bracket - Replacement Procedure

1. Install the 4 (T20) screws that hold the bracket to the hinge.
2. Replace the 4 (T10) screws that attach the bracket to the hood.
3. Install the main cover per [RP2 - Main Cover- Replacement Procedure](#).
4. Install the rear cover per [RP1 - Rear Cover - Replacement Procedure](#).
5. Perform [VP6 - Hood Lock Test](#).

6.22 RP20 - Service Switch

Part Required	9027148 - Service Switch
Tool(s) Required	T10, T20
Calibration Required	NA
Testing Required	VP13 - Service Switch
Replacement Time	13 minutes
Calibration/Testing Time	3 minutes
Total Repair Time	16 minutes

Table 6.22-1 RP20 Service Switch

6.22.1 RP20 - Service Switch - Removal Procedure

1. Turn off the EZ2 power.
2. Unplug the power cord and LAN cable.
3. Remove the rear cover per [RP1 - Rear Cover - Removal Procedure](#).
4. Remove the 2 (T10) screws mounting the switch.
5. Unplug the 2 wire connectors on the back of the switch.



Picture 6.22-1 Service Switch

6. Remove the service switch.

6.22.2 RP20 - Service Switch - Replacement Procedure

1. Plug the 2 wire connectors in on the back of the switch.
2. Replace the 2 (T10) screws that hold the switch in place.
3. Install the rear cover per [RP1 - Rear Cover - Replacement Procedure](#).
4. Plug in the power cable and LAN cable.
5. Perform [VP13 - Service Switch](#).

6.23 RP21 - Air Duct

Part Required	9027153 - Air Duct
Tool(s) Required	T10, T20
Calibration Required	TBD
Testing Required	VP14 - Fan Test
Replacement Time	10 minutes
Calibration/Testing Time	2 minutes
Total Repair Time	12 minutes

Table 6.23-1 RP21 Air Duct

6.23.1 RP21 - Air Duct - Removal Procedure

1. Turn off the power.
2. Remove the rear cover per [RP1 - Rear Cover - Removal Procedure](#).
3. Remove the 4 (T10) screws.



Picture 6.23-1 Air Duct Removal

4. Pull the air duct to remove.

6.23.2 RP21 - Air Duct- Replacement Procedure

1. Install the 4 (T10) screws in the airduct.
2. Attach the rear cover per [RP1 - Rear Cover - Replacement Procedure](#).
3. Perform [VP14 - Fan Test](#).

6.24 RP22 - Fan/Bracket

Part Required	9027138 Right, 9027138 Left - Fan With Bracket
Tool(s) Required	T10, T20
Calibration Required	NA
Testing Required	VP14 - Fan Test
Replacement Time	13 minutes
Calibration/Testing Time	2 minutes
Total Repair Time	15 minutes

Table 6.24-1 RP22 Fan with Bracket

6.24.1 RP22 - Fan/Bracket - Removal Procedure

1. Turn off the power.
2. Remove the rear cover per [RP1 - Rear Cover - Removal Procedure](#).
3. Remove the 2 bottom bracket (T20) screws.
4. Cut the cable ties.



Picture 6.24-1 Fan removal

5. Remove the main cover per [RP2 - Main Cover - Removal Procedure](#).
6. Unplug the fan connector from the CPU board.



Picture 6.24-2 Fan connectors Left and Right

7. Remove the bracket with the fan.

6.24.2 RP22 - Fan/Bracket - Replacement Procedure

1. Install the 2 (T20) screws in the bottom of the bracket.
2. Plug in the connector for the fan.
3. Use a cable ties to fasten the connector wiring in place.
4. Reassemble the main cover per [RP2 - Main Cover- Replacement Procedure](#).
5. Reassemble the rear cover per [RP1 - Rear Cover - Replacement Procedure](#).
6. Plug in the power cord and LAN cable.
7. Perform [VP14 - Fan Test](#).

6.25 RP23 - Thermocouple

Part Required	9027167 - Thermocouple
Tool(s) Required	T10, T20
Calibration Required	TBD
Testing Required	VP5 - Temperature Test
Replacement Time	12 minutes
Calibration/Testing Time	20 minutes
Total Repair Time	32 minutes

Table 6.25-1 RP23 Thermocouple

6.25.1 RP23 - Thermocouple - Removal Procedure

1. Turn off the power.
2. Remove the power cord and LAN cable.
3. Remove the rear cover per [RP1 - Rear Cover - Removal Procedure](#).
4. Remove the (T10) screw holding the thermocouple in position.
5. Cut the cable tie securing the harness to the bracket.



Picture 6.25-1 Thermocouple Removal

6. Unplug the harness connector.
7. Remove the thermocouple.

6.25.2 RP23 - Thermocouple- Replacement Procedure

1. Install the (T10) screw holding the thermocouple in position.
2. Plug the wiring harness connector in position.
3. Replace the cable tie that secures the wiring harness to the bracket.

4. Install the rear cover per [RP1 - Rear Cover - Replacement Procedure](#).
5. Plug in the power cord and LAN cable.
6. Perform [VP5 - Temperature Test](#).

6.26 RP24 - Hood Close Switch

Part Required	9027088 - Hood Close Switch
Tool(s) Required	T10, T6, T20
Calibration Required	NA
Testing Required	VP6 - Hood Lock Test
Replacement Time	10 minutes
Calibration/Testing Time	2 minutes
Total Repair Time	12 minutes

Table 6.26-1 RP24 Hood Close Switch

6.26.1 RP24 - Hood Close Switch - Removal Procedure

1. Turn off the power.
2. Remove the power cord and LAN cable.
3. Remove the rear cover per [RP1 - Rear Cover - Removal Procedure](#)
4. Remove the main cover per [RP2 - Main Cover - Removal Procedure](#)
5. Raise the hood.
6. Remove the 2 (T6) screws that hold the switch in position.
7. Unplug the 2 cable connectors at the bottom of the switch.



Picture 6.26-1 Door Switch

8. Remove the door switch.

6.26.2 RP24 - Hood Close Switch- Replacement Procedure

1. Insert the 2 cable connectors on the door switch. The white wire goes on the left and the black wire goes on the middle connector.
2. Install the 2 (T6) screws that hold the door switch in place.
3. Install the main cover per [RP2 - Main Cover- Replacement Procedure](#).
4. Install the rear cover per [RP1 - Rear Cover - Replacement Procedure](#).
5. Plug in the power cord and LAN cable.
6. Perform [VP6 - Hood Lock Test](#).

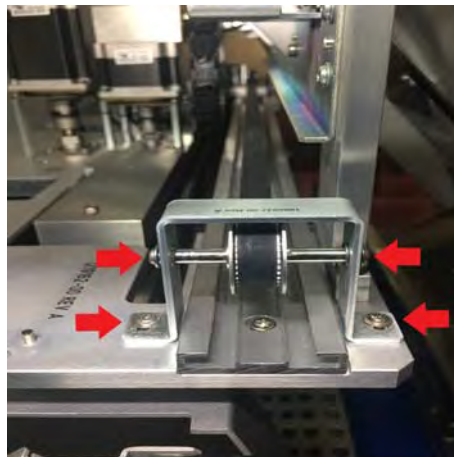
6.27 RP25 - X Pulley

Part Required	9027213 - X Pulley
Tool(s) Required	T10, T20
Calibration Required	CP7 X-Belt Tension , CP1 Camera Calibration , CP2 Camera Exposure Calibration
Testing Required	VP7 - Load Check Test , VP2 - UV Light Test
Replacement Time	17 minutes
Calibration/Testing Time	26 minutes
Total Repair Time	43 minutes

Table 6.27-1 RP25 X Pulley

6.27.1 RP25 - X Pulley - Removal Procedure

1. Turn off the power.
2. Remove the power cord and LAN cable.
3. Remove the rear cover per [RP1 - Rear Cover - Removal Procedure](#).
4. Remove the main cover per [RP2 - Main Cover - Removal Procedure](#).
5. Remove the 2 (T10) screws that hold the pulley in place.
6. Remove the 2 (T10) screws that are on both sides of the idler shaft.



Picture 6.27-1 X Pulley

6.27.2 RP25 - X Pulley- Replacement Procedure

1. Put the X belt on the pulley and put the (T10) on each side of the shaft.
2. Install the 2 (T10) screws on the base of the X pulley leave the screws loose.
3. Perform [CP7 X-Belt Tension](#).
4. Install the main cover per [RP2 - Main Cover- Replacement Procedure](#).
5. Install the rear cover per [RP1 - Rear Cover - Replacement Procedure](#).
6. Plug in the power cord and LAN cable.
7. Perform [CP1 Camera Calibration](#), [CP2 Camera Exposure Calibration](#), [VP7 - Load Check Test](#), [VP2 - UV Light Test](#).

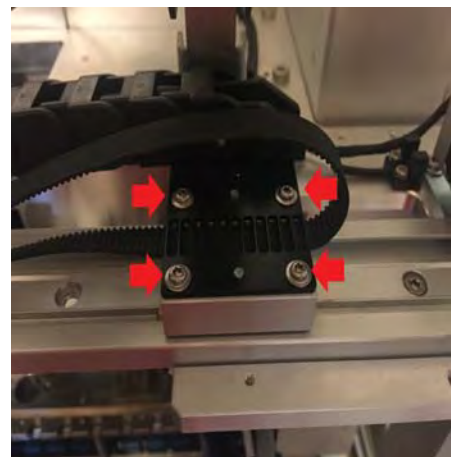
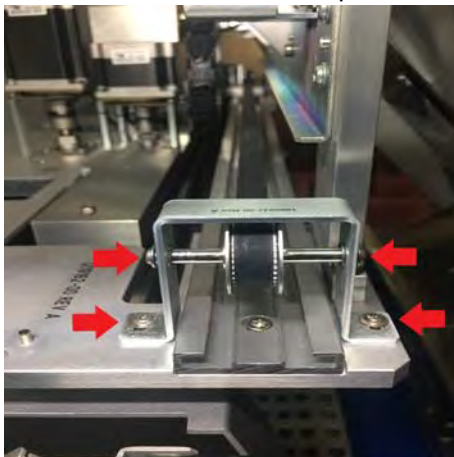
6.28 RP26 - X Timing Belt

Part Required	9027168 - X Timing Belt
Tool(s) Required	TBD
Calibration Required	CP7 X-Belt Tension , CP1 Camera Calibration , CP2 Camera Exposure Calibration
Testing Required	VP7 - Load Check Test , VP2 - UV Light Test
Replacement Time	22 minutes
Calibration/Testing Time	26 minutes
Total Repair Time	48 minutes

Table 6.28-1 RP26 X Timing Belt

6.28.1 RP26 - X Timing Belt - Removal Procedure

1. Turn off the power.
2. Remove the rear cover per [RP1 - Rear Cover - Removal Procedure](#).
3. Remove the main cover per [RP2 - Main Cover - Removal Procedure](#).



Picture 6.28-1 X Belt Replacement

4. Remove the 2 (T10) screws that hold the pulley in place.
5. Remove the 2 (T10) screws that are on both sides of the idler shaft.
6. Slide the belt off the X motor shaft.
7. Remove the 4 (T10) screws on the X rail bracket plate that holds the belt for the X movement.
8. Pull up on the plate and remove the belt.

6.28.2 RP26 - X Timing Belt - Replacement Procedure

1. Install 4 (T10) screws that holds the belt on the X rail bracket.
2. Place the X belt over the X motor shaft.
3. Put the X belt on the pulley and put the (T10) on each side of the shaft.
4. Install the 2 (T10) screws on the base of the X pulley leave the screws loose.
5. Perform [CP7 X-Belt Tension](#).
6. Install the main cover per [RP2 - Main Cover- Replacement Procedure](#).
7. Install the rear cover per [RP1 - Rear Cover - Replacement Procedure](#).
8. Plug in the power cord and LAN cable.
9. Perform [CP1 Camera Calibration](#), [CP2 Camera Exposure Calibration](#), [VP7 - Load Check Test](#), [VP2 - UV Light Test](#).

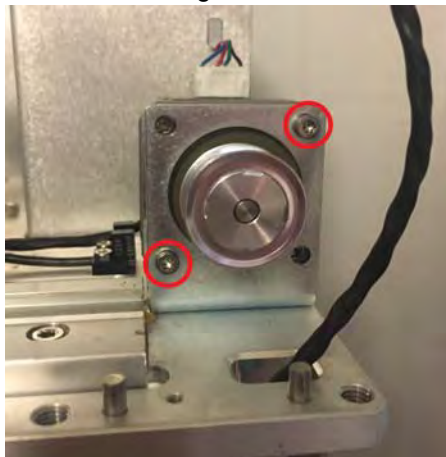
6.29 RP27 - X Motor

Part Required	9027149 - X Motor
Tool(s) Required	T10, T15, T20
Calibration Required	CP7 X-Belt Tension, CP1 Camera Calibration, CP2 Camera Exposure Calibration
Testing Required	VP7 - Load Check Test, VP2 - UV Light Test
Replacement Time	20 minutes
Calibration/Testing Time	26 minutes
Total Repair Time	46 minutes

Table 6.29-1 RP27 X Motor

6.29.1 RP27 - X Motor - Removal Procedure

1. Turn off the power.
2. Remove the power cord and LAN cable.
3. Remove the rear cover per [RP1 - Rear Cover](#)
4. Remove the main cover [RP2 - Main Cover](#).
5. Remove the X belt per [RP26 - X Timing Belt](#).
6. Remove the 2 (T10) from the X motor mounting bracket.



Picture 6.29-1 X Motor

7. Remove the 2 (T10) from the damper and separate it from the motor.



Picture 6.29-2 Damper Removal

6.29.2 RP27 - X Motor- Replacement Procedure

1. Install the damper on the motor with the 2 (T10) screws.
2. Install the 2 (T10) screws through the X motor bracket mounting the motor in position.
3. Place the belt over the X motor shaft.
4. Perform [CP7 X-Belt Tension](#).
5. Install the main cover per [RP2 - Main Cover- Replacement Procedure](#).
6. Install the rear cover per [RP1 - Rear Cover - Replacement Procedure](#).
7. Perform [CP1 Camera Calibration](#), [CP2 Camera Exposure Calibration](#), [VP7 - Load Check Test](#), [VP2 - UV Light Test](#).

6.30 RP28 - Z Motor

Part Required	9027192 - Z Motor
Tool(s) Required	T10, T20, 2.5 mm Hex Tool
Calibration Required	CP3 Pipette Z-axis Calibration
Testing Required	VP10 - Z Axis Movement Test , VP8 - Workflow System Test
Replacement Time	18 minutes
Calibration/Testing Time	38 minutes
Total Repair Time	56 minutes

Table 6.30-1 RP28 Z Motor

6.30.1 RP28 - Z Motor - Removal Procedure

1. Turn off the power no the EZ2 and remove the power cord.
2. Remove the rear cover per [RP1 - Rear Cover - Removal Procedure](#).
3. Remover the main cover per [RP2 - Main Cover - Removal Procedure](#).
4. Remove the 2 (T20) screws on the base of the Z motor.



Picture 6.30-1 Z Motor

5. Pull the Z motor straight up to remove.

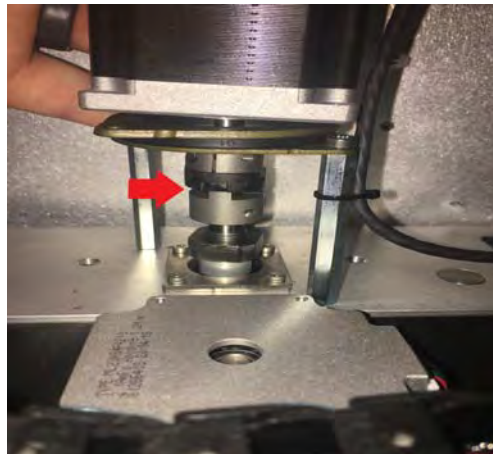
6. Remove the Z motor coupler with a 2.5 mm hex tool.



Picture 6.30-2 Z Motor Coupler

6.30.2 RP28 - Z Motor - Replacement Procedure

1. Install the coupler on the Z motor shaft with a 2.5 mm hex tool.
2. Ensure that the Z motor coupler joint fits together correctly before installing the screws that hold the Z motor in position.



Picture 6.30-3 Z Motor Coupler Joint

3. The cable connector position should be on the facing right side.
4. Install the 2 (T20) screws that mount the Z motor in place.
5. Plug in the cable connector into the motor.
6. Install the main cover per [RP2 - Main Cover- Replacement Procedure](#).
7. Install the rear cover per [RP1 - Rear Cover - Replacement Procedure](#).
8. Perform [CP3 Pipette Z-axis Calibration](#), [VP10 - Z Axis Movement Test](#), [VP8 - Workflow System Test](#).

6.31 RP29 - P Motor

Part Required	9027191 - P Motor
Tool(s) Required	T10, T20, 2.5 mm Hex Tool
Calibration Required	CP5 Pipette P-Axis Calibration ,
Testing Required	VP3 - Leakage Test , VP4 - Pipetting Test , VP8 - Workflow System Test
Replacement Time	17 minutes
Calibration/Testing Time	42 minutes
Total Repair Time	59 minutes

Table 6.31-1 RP29 P Motor

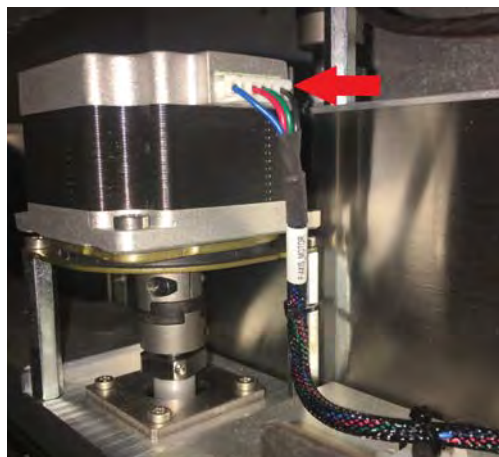
6.31.1 RP29 - P Motor - Removal Procedure

1. Turn off the EZ2 power.
2. Remove the rear cover per [RP1 - Rear Cover - Removal Procedure](#).
3. Remove the main cover per [RP2 - Main Cover - Removal Procedure](#).
4. Remove the 2 (T20) screws on the base of the P motor.



Picture 6.31-1 P motor screws

5. Unplug the P motor cable connector.



Picture 6.31-2 P Motor Cable Connector

6. Pull the P motor straight up to remove.

7. Remove the coupler with a 2.5 mm hex tool.



Picture 6.31-3 P Motor Coupler

6.31.2 RP29 - P Motor - Replacement Procedure

1. Install the coupler with a 2.5 mm hex tool.
2. Ensure that the P motor coupler joint fits together correctly before installing the screws that hold the P motor in position.



Picture 6.31-4 P Motor Coupler joining

3. The cable connector position should be on the facing right side.
4. Install the 2 (T20) screws that mount the P motor in place.
5. Plug in the cable connector into the motor.
6. Install the main cover per [RP2 - Main Cover- Replacement Procedure](#).
7. Install the rear cover per [RP1 - Rear Cover - Replacement Procedure](#).
8. Perform [CP5 Pipette P-Axis Calibration](#), [VP3 - Leakage Test](#), [VP4 - Pipetting Test](#), [VP8 - Workflow System Test](#).

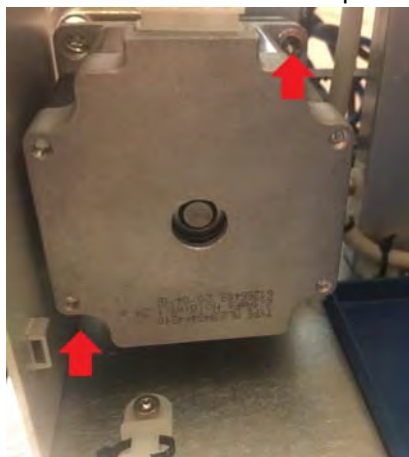
6.32 RP30 - Y Motor

Part Required	9027191 - Y Motor
Tool(s) Required	T10, T20, 2 mm Hex Tool
Calibration Required	CP8 Y-Belt Tension , CP3 Pipette Z-axis Calibration , CP4 Worktable Calibration
Testing Required	VP8 - Workflow System Test
Replacement Time	25
Calibration/Testing Time	50 minutes
Total Repair Time	1 hours 15 minutes

Table 6.32-1 RP30 Y Motor

6.32.1 RP30 - Y Motor - Removal Procedure

1. Turn off the EZ2 power.
2. Remove the rear cover per [RP1 - Rear Cover - Removal Procedure](#).
3. Remove the main cover per [RP2 - Main Cover - Removal Procedure](#).
4. Remove the right stationary cover per [RP4 - Right Stationary Cover - Removal Procedure](#).
5. Unplug the wire connector from the top of the motor.
6. Remove the 2 (T20) mounting screws that hold the motor in position.



Picture 6.32-1 Y Motor Mounting Screws

7. After the Y motor mounting screws are removed, this will allow enough movement to slide the Y belt off the pulley.



Picture 6.32-2 Y Belt Removal

8. Once the belt is removed, the 2(2 mm) set screws can be loosened.

*Picture 6.32-3 Y Pulley Set Screws*

9. Remove the pulley from the Y motor shaft and remove the motor.

6.32.2 RP30 - Y Motor - Replacement Procedure

1. Place the Y motor in position with the shaft through the motor bracket.
2. Align the flat section of the Y motor shaft with a set screw on the pulley.
3. Tighten the (2 mm) set screw with a hex tool.

*Picture 6.32-4 Y Motor Shaft Pulley & Shaft Alignment*

4. Place the Y belt on the Y motor pulley.
5. Pull the Y motor to tighten the tension on the Y belt.
6. Install the 2 (T25) Y motor mounting screws.
7. Plug the wire connector in the top of the motor.
8. Perform [CP8 Y-Belt Tension](#).
9. Replace the main cover per [RP2 - Main Cover- Replacement Procedure](#).
10. Replace the rear cover per [RP1 - Rear Cover - Replacement Procedure](#)
11. Perform [CP3 Pipette Z-axis Calibration](#), [CP4 Worktable Calibration](#), [VP8 - Workflow System Test](#).

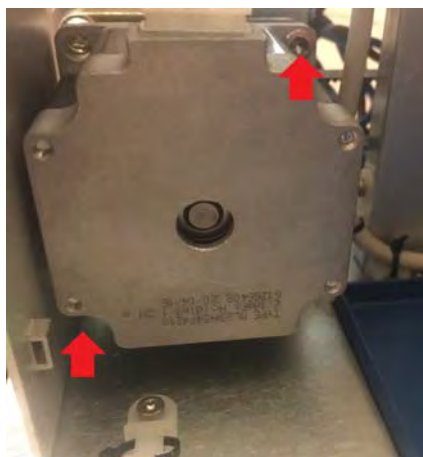
6.33 RP31 - Y Timing Belt

Part Required	9027193 - Y Timing Belt
Tool(s) Required	T10, T20, T25, 2 mm Hex Tool
Calibration Required	CP8 Y-Belt Tension , CP3 Pipette Z-axis Calibration , CP4 Worktable Calibration
Testing Required	VP8 - Workflow System Test
Replacement Time	45 minutes
Calibration/Testing Time	50 minutes
Total Repair Time	1 hour 35 minutes

Table 6.33-1 RP31 Y Timing Belt

6.33.1 RP31 - Y Timing Belt - Removal Procedure

1. Remove the rear cover per [RP1 - Rear Cover - Removal Procedure](#).
2. Remove the main cover per [RP2 - Main Cover - Removal Procedure](#).
3. Remove the 2 (T25) screws from the Y motor.
4. Remove the Y drive belt from the Y motor pulley.



Picture 6.33-1 Y Belt Motor Pulley Removal



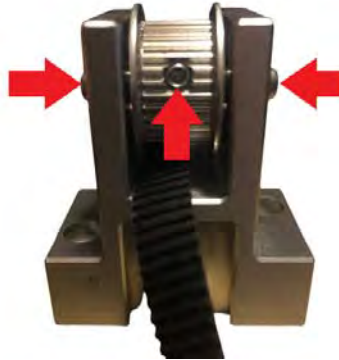
5. Remove the 2 (T20) screws from the base of the Y idler pulley.



Picture 6.33-2 Y Idler Pulley Removal

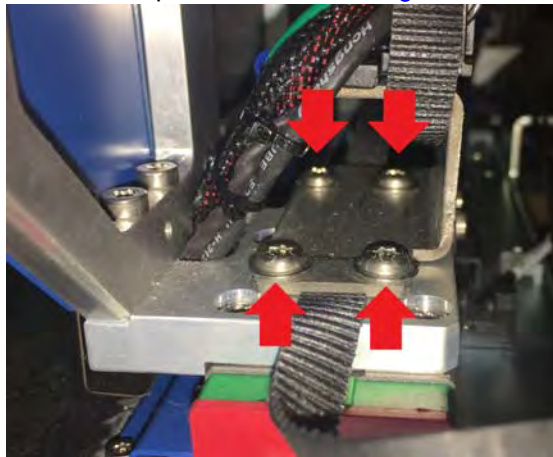
6. Loosen the 2 (2 mm) Screws on the pulley.

7. Remove the 2 (T10) screws that hold the pulley shaft in place.



Picture 6.33-3 Y Idler Pulley Set Screws

8. Remove the left stationary cover per [RP3 - Left Stationary Cover - Removal Procedure](#).
9. Remove the left internal cover per [RP14 - Internal Right Plastic Cover - Removal Procedure](#).
10. Remove the bottom front cover per [RP6 - Front Bottom Cover - Removal Procedure](#).
11. Remove the internal right metal cover per [RP5 - Internal Right Metal Cover - Removal Procedure](#).



Picture 6.33-4 Y Belt Bracket Screws

12. Remove the 4 (T20) screws that connect the Y belt to the Y bearing rail.
13. Remove the Y belt.

6.33.2 RP31 - Y Timing Belt - Replacement Procedure

1. Install the 4 (T20) screws that hold the Y belt on the Y bearing rail.
2. Place the Y belt on the Y idler pulley per [RP32 - Y Idler Pulley - Replacement Procedure](#).
3. Place the Y belt on the Y motor pulley per [RP30 - Y Motor - Replacement Procedure](#).
4. Perform [CP8 Y-Belt Tension](#).
5. Install the right internal metal cover per [RP5 - Internal Right Metal Cover- Replacement Procedure](#).
6. Install the front bottom cover per [RP6 - Front Bottom Cover - Replacement Procedure](#).
7. Perform [CP3 Pipette Z-axis Calibration](#), [CP4 Worktable Calibration](#), and [VP8 - Workflow System Test](#).

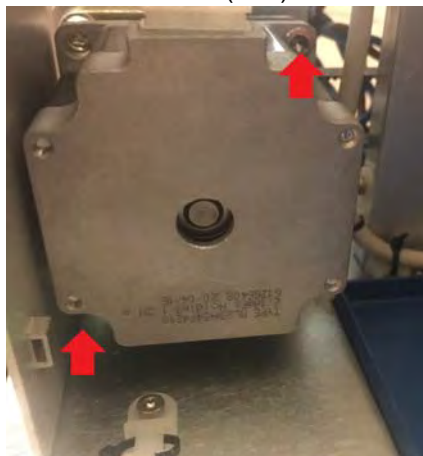
6.34 RP32 - Y Idler Pulley

Part Required	9027184 - Y Idler Pulley
Tool(s) Required	T10, T20, T25, 2 mm Hex Tool
Calibration Required	CP8 Y-Belt Tension , CP3 Pipette Z-axis Calibration , CP4 Worktable Calibration
Testing Required	VP8 - Workflow System Test
Replacement Time	52 minutes
Calibration/Testing Time	50 minutes
Total Repair Time	1 hour 42 minutes

Table 6.34-1 RP32 Y Idler Pulley

6.34.1 RP32 - Y Idler Pulley - Removal Procedure

1. Remove the rear cover per [RP1 - Rear Cover - Removal Procedure](#).
2. Remove the main cover per [RP2 - Main Cover - Removal Procedure](#).
3. Remove the power button for easier access to the Y idler pulley per [RP33 - Power Button - Removal Procedure](#).
4. Remove the 2 (T25) screws from the Y motor.



Picture 6.34-1 Y Belt Motor Pulley Removal

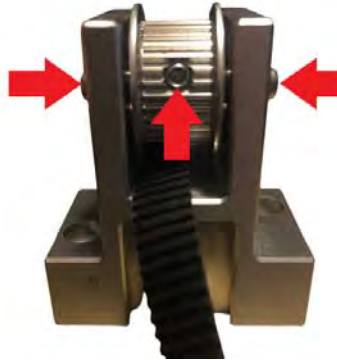


5. Remove the Y drive belt from the Y motor pulley.



Picture 6.34-2 Y Idler Pulley Removal

- Remove the 2 (T20) screws from the base of the Y idler pulley.



Picture 6.34-3 Y Idler Pulley Set Screws

- Loosen the 2 (2 mm) Screws on the pulley.
- Remove the 2 (T10) screws that hold the pulley shaft in place.
- Remove the shaft from the Idler pulley.

6.34.2 RP32 - Y Idler Pulley - Replacement Procedure

- Place the Y Belt on the Y idler shaft
- Insert the 2 (T10) screws on both sides of the idler shaft.
- Tighten the 2 (2 mm) set screws.
- Install the 2 (T20) screws that secure the Y idler pulley to the base of the instrument.
- Perform [CP8 Y-Belt Tension](#).
- Replace the power button per [RP33 - Power Button - Replacement Procedure](#).
- Replace the main cover per [RP2 - Main Cover- Replacement Procedure](#)
- Install the rear cover per [RP1 - Rear Cover - Replacement Procedure](#).
- Perform [CP3 Pipette Z-axis Calibration](#), [CP4 Worktable Calibration](#), [VP8 - Workflow System Test](#).

6.35 RP33 Power Button

Part Required	9027086 - Power Button
Tool(s) Required	T10, T20, T25, Phillips
Calibration Required	NA
Testing Required	VP12 - System Initialization Test
Replacement Time	33minutes
Calibration/Testing Time	2 minutes
Total Repair Time	35 minutes

Table 6.35-1 RP33 Power Button

6.35.1 RP33 - Power Button - Removal Procedure

- Remove the rear cover per [RP1 - Rear Cover - Removal Procedure](#).
- Remove the main cover per [RP2 - Main Cover - Removal Procedure](#).
- Remove the right stationary cover per [RP4 - Right Stationary Cover - Removal Procedure](#).

4. Remove the 2 (T25) screws from the power button bracket.



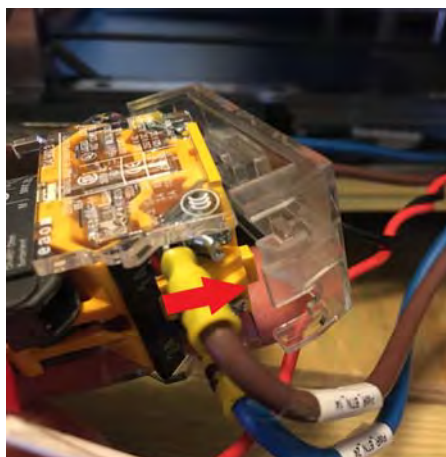
Picture 6.35-1 Power Button Bracket Screws

5. Separate the power button in 2 pieces by prying up on the top black tab and prying down on the bottom tab.
6. Remove the bracket holding the front half of the button.



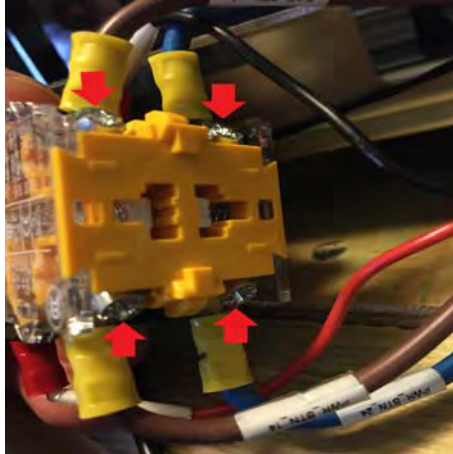
Picture 6.35-2 Power Button Wire removal

7. Pry the clear plastic tab outward and remove the wire connection cover.



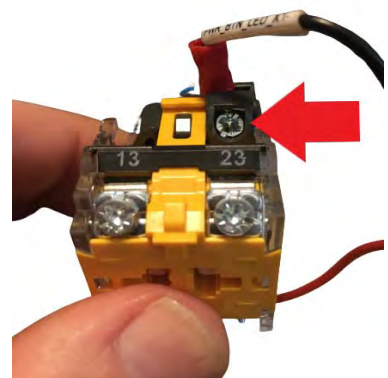
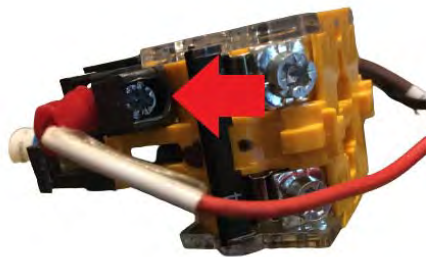
Picture 6.35-3 Power Button Connection Cover

8. Loosen the 4 screws with a Phillips screwdriver to remove the wires.



Picture 6.35-4 Power button connectors

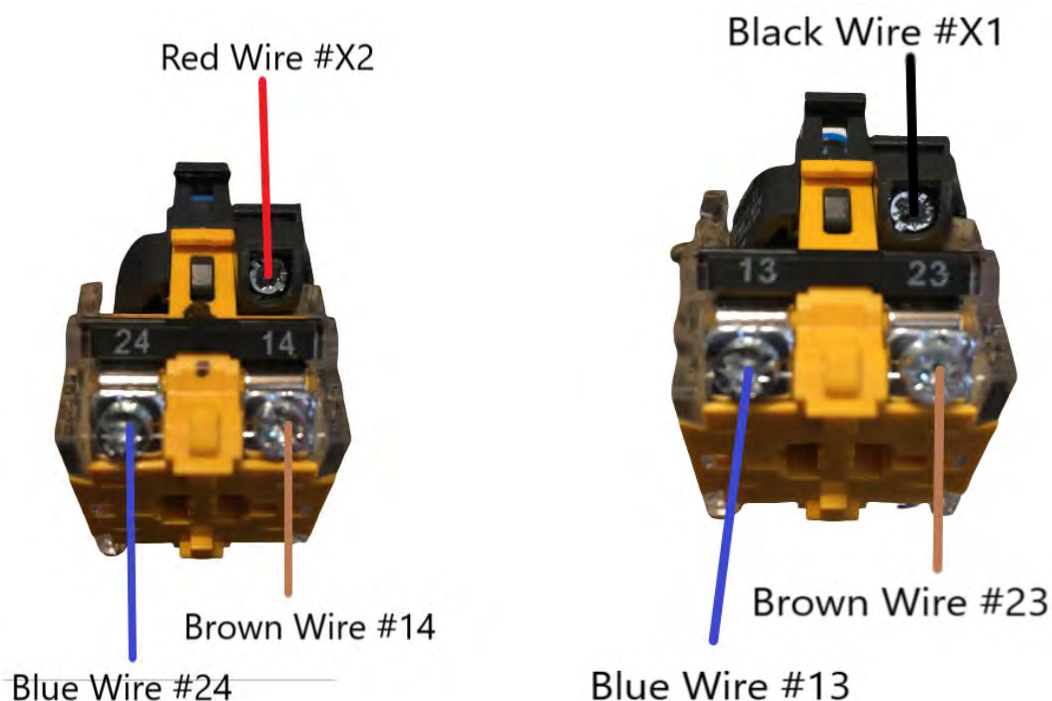
9. Use a Phillips screwdriver to disconnect the red and black wire from the power button.



Picture 6.35-5 Power Button Red and Black Wire

6.35.2 RP33 - Power Button - Replacement Procedure

1. Connect the wiring as shown.



Picture 6.35-6 Power Button Wiring

2. The red wiring connector goes on the top side of the button.
3. Install the front half of the button with the 2 (T25) screws.
4. Place the clear plastic wiring cover over the connectors.
5. Snap the wire connection portion of the button back in position with the front portion.
6. Replace the stationary right cover per [RP4 - Right Stationary Cover - Replacement Procedure](#).
7. Replace the main cover per [RP2 - Main Cover- Replacement Procedure](#).
8. Replace the rear cover per [RP1 - Rear Cover - Replacement Procedure](#).
9. Perform [VP12 - System Initialization Test](#).

6.36 RP34 - UV LED Board PCB

Part Required	9027152 - UV LED Board PCB
Tool(s) Required	T10
Calibration Required	NA
Testing Required	VP2 - UV Light Test
Replacement Time	5 minutes
Calibration/Testing Time	2 minutes
Total Repair Time	7 minutes

Table 6.36-1 RP34 UVC LED Board PCB

6.36.1 RP34 - UV LED Board PCB - Removal Procedure

1. Power off the instrument.
2. Remove the power cord.
3. Slide the camera module to the left for access to the screws.
4. Remove the 2 (T10) screws from the UV LED bracket.



Picture 6.36-1 UV LED Board screws

5. Unplug the UV LED board connector.



Picture 6.36-2 UV LED Bracket

6. Remove the mounting bracket with UV LED board PCB.



Picture 6.36-3 UV LED module

6.36.2 RP34 - UV LED Board PCB - Replacement Procedure

1. Install the bracket with the UV LED board on the bottom of the camera arm.
2. Install the 2 (T10) screws that hold the UV LED bracket in position.
3. Plug in the cable connector to the UV LED board.
4. Perform [VP2 - UV Light Test](#).

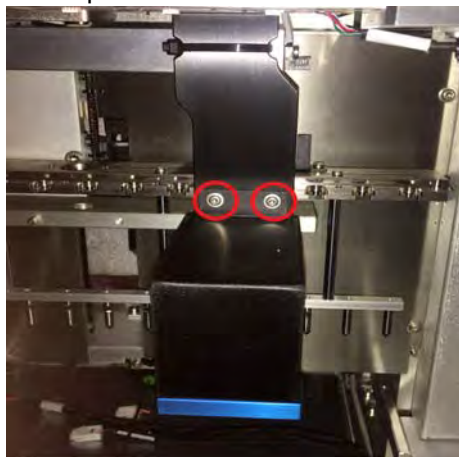
6.37 RP35 - Camera

Part Required	Camera
Tool(s) Required	T10, T6
Calibration Required	CP1 Camera Calibration , CP2 Camera Exposure Calibration
Testing Required	VP7 - Load Check Test
Replacement Time	7 minutes
Calibration/Testing Time	14 minutes
Total Repair Time	21 minutes

Table 6.37-1 RP35 Camera

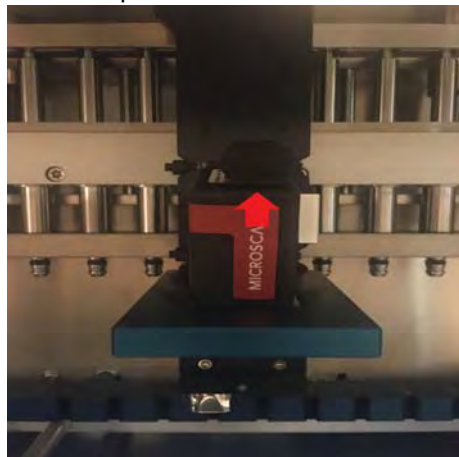
6.37.1 RP35 - Camera - Removal Procedure

1. Power off the EZ2.
2. Remove the power cord.
3. Manually move the camera arm to the center of the instrument.
4. Remove the 2 (T10) screws at the top of the camera enclosure and remove the enclosure.



Picture 6.37-1 Camera Enclosure Removal

5. Pull upward on the camera cable connector to unplug the cable.



Picture 6.37-2 Unplug Camera Cable

6. Hold the camera while removing the 2 (T6) camera bracket screws that holds the camera in position.



Picture 6.37-3 Camera Bracket Screws

6.37.2 RP35 - Camera- Replacement Procedure

1. Insert the camera in position and insert the 2 (T6) screws that hold the camera to the bracket.
2. Plug the cable in to the camera.
3. Place the camera enclosure over the camera and install the 2 (T10) screws.
4. Perform [CP1 Camera Calibration](#), [CP2 Camera Exposure Calibration](#), [VP7 - Load Check Test](#).

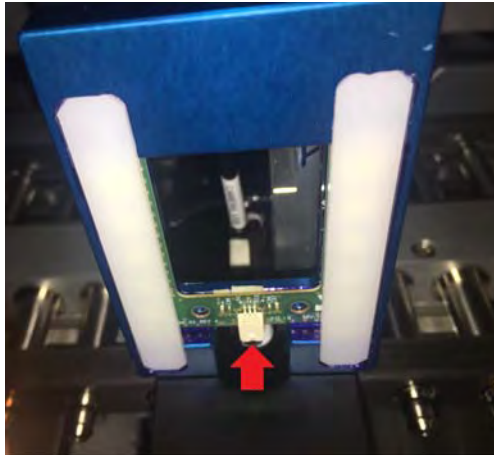
6.38 RP36 - Camera LED Board PCB/Diffuser/Bracket

Part Required	9027145 - Camera LED Board PCB/Diffuser/Bracket
Tool(s) Required	T10,T20
Calibration Required	CP1 Camera Calibration , CP2 Camera Exposure Calibration
Testing Required	VP7 - Load Check Test
Replacement Time	8 minutes
Calibration/Testing Time	14 minutes
Total Repair Time	22 minutes

Table 6.38-1 RP36 Camera LED Board PCB/Diffuser/Bracket

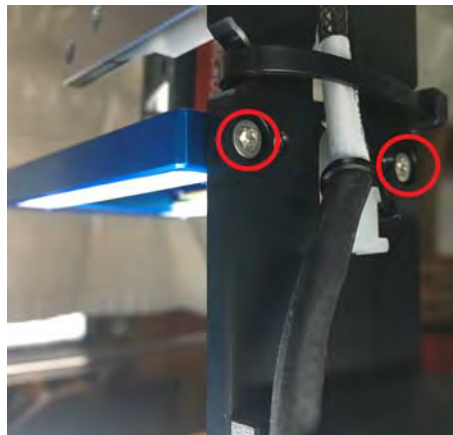
6.38.1 RP36 - Camera LED Board PCB/Diffuser/Bracket- Removal Procedure

1. Turn off the power and remove the power cord.
2. Move the camera to the left.
3. Remove the internal right cover to get easier access to the back side of the bracket.
4. Remove the camera enclosure cover.
5. Open the hood and unplug the small connector on the camera LED board.



Picture 6.38-1 Camera LED Board, Diffuser and Bracket

6. Move the camera module X bracket to the left to have better access for screw removal.
7. Remove the 2 (T10) screws to remove the camera LED bracket.



Picture 6.38-2 Camera LED Bracket Screws

6.38.2 RP36 - Camera LED Board PCB/Diffuser/Bracket- Replacement Procedure

1. Install the 2 (T10) screws that attach the camera LED bracket.
2. Plug the small connector into the camera LED board.
3. Put the camera enclosure cover back in position and the 2 (T10) screws that hold it in place.
4. Install the internal right cover and the 3 (T20) screws that hold the cover in position.
5. Perform [CP1 Camera Calibration](#), [CP2 Camera Exposure Calibration](#), [VP7 - Load Check Test](#).

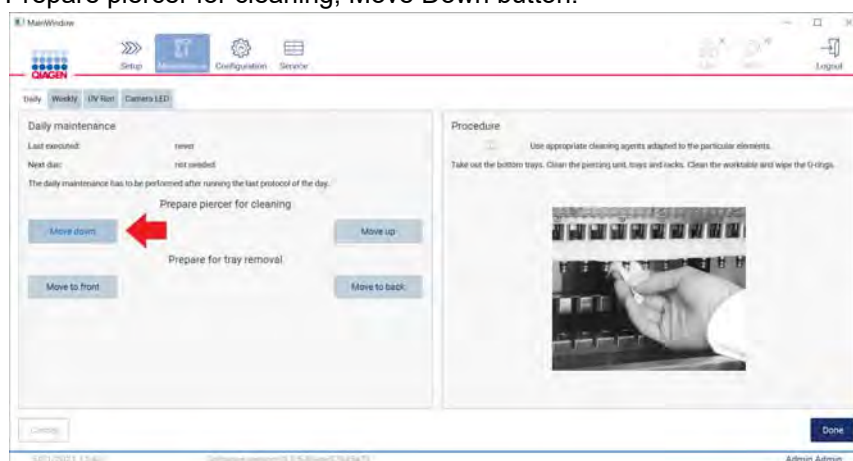
6.39 RP37 - Pipette Module

Part Required	9027187 - Pipette Module
Tool(s) Required	T10, T20
Calibration Required	CP3 Pipette Z-axis Calibration , CP4 Worktable Calibration , CP5 Pipette P-Axis Calibration
Testing Required	VP3 - Leakage Test , VP4 - Pipetting Test , VP8 - Workflow System Test
Replacement Time	20 minutes
Calibration/Testing Time	52 minutes
Total Repair Time	1 hour 2 minutes

Table 6.39-1 RP37 Pipette Module

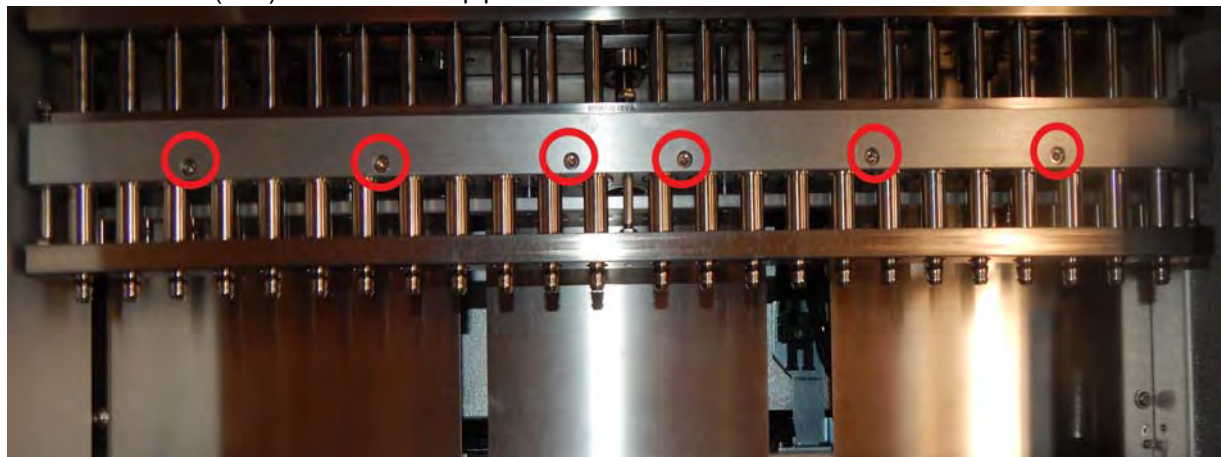
6.39.1 RP37 - Pipette Module - Removal Procedure

1. Power on the EZ2.
2. Log in to the software.
3. Select the Maintenance tab.
4. Click on the Prepare piercer for cleaning, Move Down button.



Picture 6.39-1 Pipette Module Removal

5. Do not press the Done button, turn off the EZ2 power. Doing this will give you easier access to some of the screws for removal.
6. Remove the 6 (T20) that mount the pipette module.



Picture 6.39-2 Pipette Module Screws

7. Remove the 13 (T10) screws on the plunger holding bracket.



Picture 6.39-3 Pipette Top Screws

8. After removing all the screws and the top bracket pull the pipette module toward the front of the instrument to remove.



Picture 6.39-4 Pipette Unit Removal

6.39.2 RP37 - Pipette Module- Replacement Procedure

1. Insert the pipette module in the mounting position.
2. Install the 6 (T20) screws that run through the front of the pipette module.
3. Put the plunger holding bracket in position.



Picture 6.39-5 Plunger Holding Bracket

4. Install the 13 (T10) screws in the top of the plunger holding bracket.

5. Perform [CP3 Pipette Z-axis Calibration](#), [CP4 Worktable Calibration](#), [CP5 Pipette P-Axis Calibration](#), [VP3 - Leakage Test](#), [VP4 - Pipetting Test](#), [VP8 - Workflow System Test](#).

6.40 RP38 - Plunger/Pipetting Collar

Part Required	9027084 - Plunger Repair Set of 6 With O-rings and Collars, 9027073 -Silicon Grease
Tool(s) Required	T10, T20, Forceps
Calibration Required	CP3 Pipette Z-axis Calibration , CP4 Worktable Calibration , CP5 Pipette P-Axis Calibration
Testing Required	VP3 - Leakage Test , VP4 - Pipetting Test , VP8 - Workflow System Test .
Replacement Time	23 minutes
Calibration/Testing Time	52 minutes
Total Repair Time	1 hour 5 minutes

Table 6.40-1 RP38 Plunger/Collar/O-ring

6.40.1 RP38 - Plunger/Collar/O-ring - Removal Procedure

1. Power off the EZ2 and remove the power cord.
2. Open the hood.
3. Remove the pipette module per [RP37 - Pipette Module - Removal Procedure](#).
4. Remove the 12 (T10) plunger holder plate screws.
5. Remove the O-ring holder plate.



Picture 6.40-1 O-ring Holder Plate

6. Pull the plunger up to remove it.



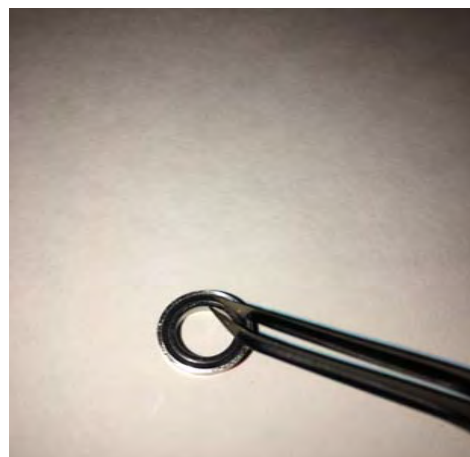
Picture 6.40-2 Plunger Removal



7. Keep all the plungers in order of the position they were removed in.
8. If the collar is on the plunger, slide the collar off.
9. Use a forceps tool to remove the top O-ring and collar.



Picture 6.40-3 Collar and O-rings



10. Remove the bottom O-ring with the forceps tool.

6.40.2 RP38 - Plunger/Collar/O-ring - Replacement Procedure



Picture 6.40-4 Silicon Grease Part Number 9027073

1. Lubricate the O-ring with silicon grease and place the bottom in the pipette module.
2. Put the collar on top of the bottom O-ring.

3. Lubricate the O-ring with silicon grease and place the top O-ring on the collar and slide the plunger down in the pipette module plunger position.
4. When installing the plungers and collars, keep them in the same positions they were removed in.
5. Install the pipette module in position per [RP37 - Pipette Module- Replacement Procedure](#).
6. Perform [CP3 Pipette Z-axis Calibration](#), [CP4 Worktable Calibration](#), [CP5 Pipette P-Axis Calibration](#), [VP3 - Leakage Test](#), [VP4 - Pipetting Test](#), [VP8 - Workflow System Test](#).

6.41 RP39 - D-ring

Part Required	9027085 - D-ring set of 24, 9027073 - Silicon Grease
Tool(s) Required	T10, T20, Forceps
Calibration Required	NA
Testing Required	VP3 - Leakage Test
Replacement Time	25 minutes
Calibration/Testing Time	7 minutes
Total Repair Time	32 minutes

Table 6.41-1 RP39 D-ring

6.41.1 RP39 - D-ring - Removal Procedure

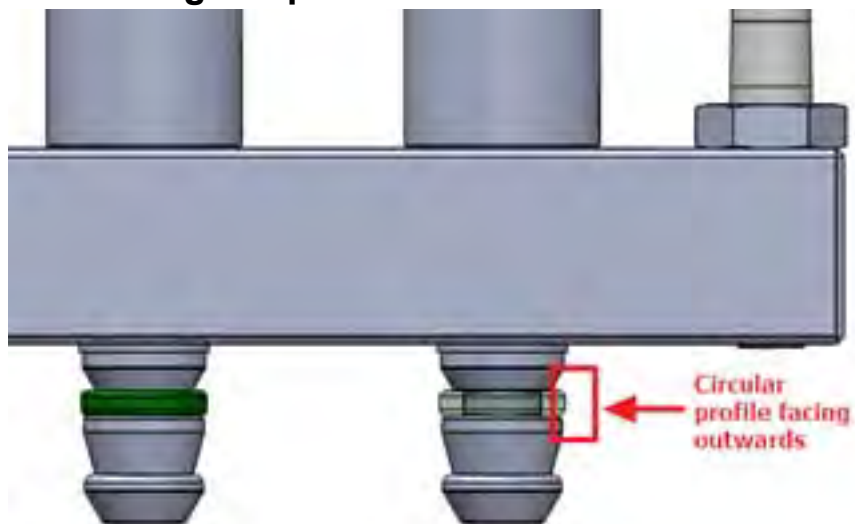
1. Turn off the power to the EZ2.
2. Remove the power cord.
3. Open the hood.



Picture 6.41-1 D-Ring Removal

4. Remove the D-ring with a small tweezer or similar tool.

6.41.2 RP39 - D-ring - Replacement Procedure



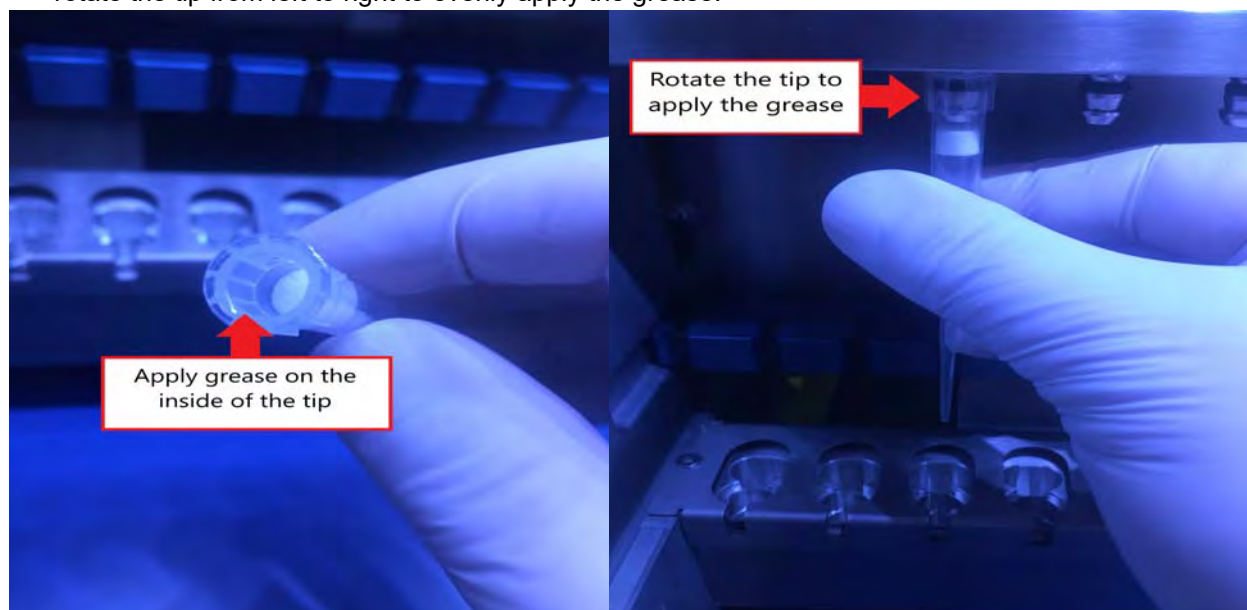
Picture 6.41-2 D-ring Orientation

1. Install the D-ring by placing it over the pipette head and rolling it till in the circular profile is facing the outward position.



Picture 6.41-3 Silicon Grease Part Number 9027073

2. Apply grease on a pipette tip, inside above the filter and push the tip up over the D-ring to grease and rotate the tip from left to right to evenly apply the grease.



Picture 6.41-4 Greasing D-rings

3. Perform [VP3 - Leakage Test](#).

6.42 RP40 - Piercing Needle

Part Required	9027185 - Piercing Needle
Tool(s) Required	4 mm Wrench
Calibration Required	NA
Testing Required	VP8 - Workflow System Test
Replacement Time	24 minutes
Calibration/Testing Time	30 minutes
Total Repair Time	54 minutes

Table 6.42-1 RP40 Piercing Needle

6.42.1 RP40 - Piercing Needle - Removal Procedure

WARNING 	The piercing needles are sharp. Use caution when working near the needles. We recommend double gloves for this procedure.
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1. Turn off the power.
2. Open the hood.
3. Remove the pipette module per [RP37 - Pipette Module - Removal Procedure](#).

CAUTION 	When removing a piercing needle, ensure you place your wrench on the flat surfaces that are milled into the part for that purpose. c0
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4. Use a 4 mm wrench to turn and remove the piercing needle.



Picture 6.42-1 Piercing Needle Removal

6.42.2 RP40 - Piercing Needle - Replacement Procedure

1. Open the hood.
2. Use a 4mm wrench to screw the piercing needle in position.
3. Replace the pipette module per [RP37 - Pipette Module- Replacement Procedure](#).
4. Perform [VP8 - Workflow System Test](#).

6.43 RP41 - Power Supply

Part Required	9027147 - Power Supply
Tool(s) Required	T10, T9, T20, Phillips
Calibration Required	NA
Testing Required	VP12 - System Initialization Test
Replacement Time	22 minutes
Calibration/Testing Time	2 minutes
Total Repair Time	24 minutes

Table 6.43-1 RP41 Power Supply

6.43.1 RP41 - Power Supply - Removal Procedure

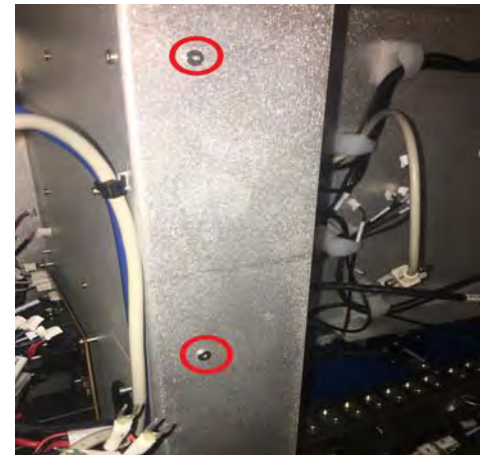
1. Turn off the EZ2 and remove the power cord and LAN cable.
2. Remove the rear cover per [RP1 - Rear Cover - Removal Procedure](#).
3. Remove the main cover per [RP2 - Main Cover - Removal Procedure](#).
4. Disconnect the wires from the CPU PCB board.
5. Remove the CPU PCB board [RP43 - CPU Board - Removal Procedure](#).
6. Remove the plastic covers over the wire connectors.
7. Unhook the 5 wires connected to the power supply with a phillips screwdriver.



Picture 6.43-1 Power Supply Wire Removal



8. Remove the 6 (T9) screws that mount the power supply.



Picture 6.43-2 Power Supply Screw Removal

6.43.2 RP41 - Power Supply- Replacement Procedure

1. Insert the 6 (T9) mounting screws that mount the power supply in position.
2. reconnect the 5 wires to the power supply with a Phillips.
3. Place the small plastic covers over the wire connectors.
4. Install the CPU PCB board per [RP43 - CPU Board - Replacement Procedure](#).
5. Put the main cover back in position per [RP2 - Main Cover- Replacement Procedure](#).
6. Put the rear cover back in position per [RP1 - Rear Cover - Replacement Procedure](#).
7. Plug in the power cable and LAN cable.
8. Perform [VP12 - System Initialization Test](#).

6.44 RP42 - Motor Driver Board

Part Required	9027082 - Motor Driver Board
Tool(s) Required	T10, T20
Calibration Required	NA
Testing Required	VP8 - Workflow System Test
Replacement Time	14 minutes
Calibration/Testing Time	30 minutes
Total Repair Time	44 minutes

Table 6.44-1 RP42 Motor Driver Board

6.44.1 RP42 - Motor Driver Board - Removal Procedure

1. Turn off the EZ2 and remove the power cord and LAN cable.
2. Remove the rear cover per [RP1 - Rear Cover - Removal Procedure](#).
3. Unplug all cable connections on the motor driver board.

4. Remove the 4 (T10) screws on the corners of the motor driver board.

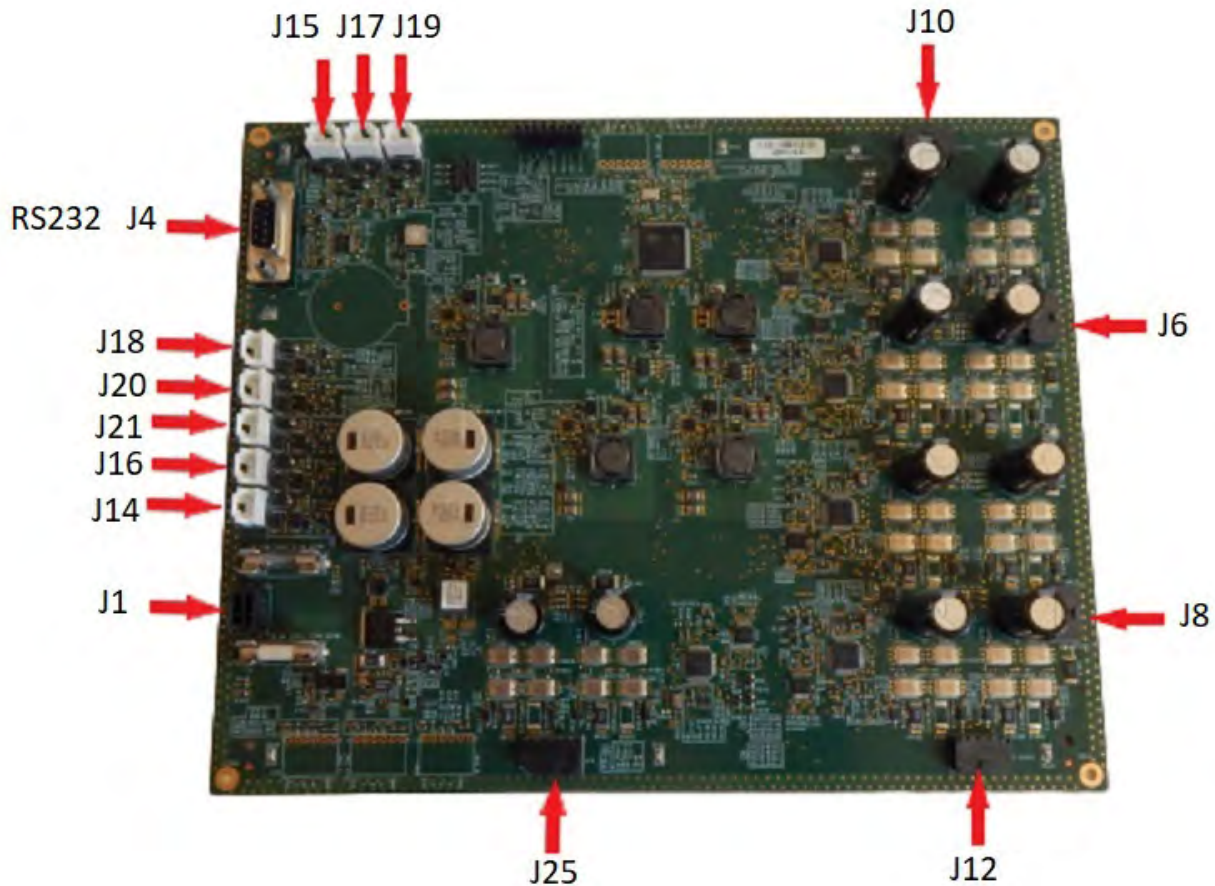


Picture 6.44-1 Motor Driver Board Screws

6.44.2 RP42 - Motor Driver Board - Replacement Procedure

1. Install the 4 (T10) screws on the corners of the motor driver board.
- 1.

2. Plug in all cable connections in the correct positions.



Picture 6.44-2 .Motor Driver PCB Connections

Connection	Name	Description
J1	Power Input	Input supply & I2C data line
J3	MCU JTAG	Programming port
J4	DB9 CPU	UART line
J6	Motor Z Axis	Motor connection
J8	Motor P Axis	Motor connection
J10	Motor X Axis	Motor connection
J12	Motor Y Axis	Motor connection
J25	Motor M Axis	Motor connection
J15	Slot Sensor Z Crash	Slot sensor
J17	Slot Sensor P Org	Slot sensor
J19	Slot Sensor X Org	Slot sensor
J18	Slot Sensor Y Org	Slot sensor
J20	Slot Sensor Y EL	Slot sensor
J21	Slot Sensor M Org	Slot sensor
J14	Slot Sensor Z Org	Slot sensor
J16	Slot Sensor Z EL	Slot sensor

Table 6.44-2 Motor Driver Board Connection Functions

3. Install the main cover per [RP2 - Main Cover- Replacement Procedure](#)
4. Replace the rear cover per [RP1 - Rear Cover - Replacement Procedure](#).
5. Perform [VP8 - Workflow System Test](#).

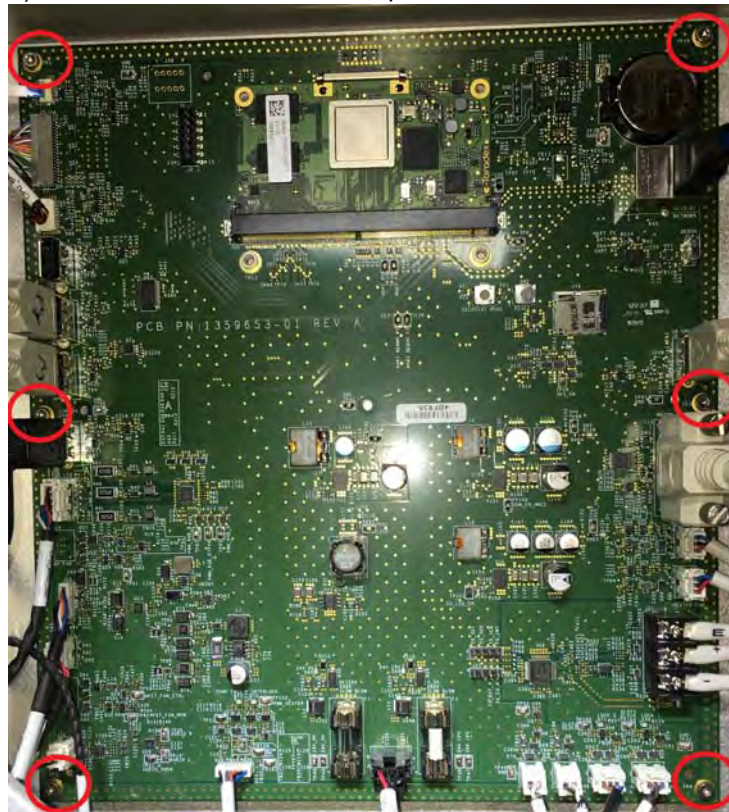
6.45 RP43 - CPU Board

Part Required	9027081 - CPU Board
Tool(s) Required	T10, T20, Phillips
Calibration Required	CP1 Camera Calibration , CP2 Camera Exposure Calibration , CP3 Pipette Z-axis Calibration , CP4 Worktable Calibration , CP5 Pipette P-Axis Calibration , CP6 Magnet Calibration , CP9 Set Time/Date
Testing Required	VP8 - Workflow System Test
Replacement Time	18 minutes
Calibration/Testing Time	56 minutes
Total Repair Time	1 hour 14 minutes

Table 6.45-1 RP43 CPU PCB Board

6.45.1 RP43 - CPU Board - Removal Procedure

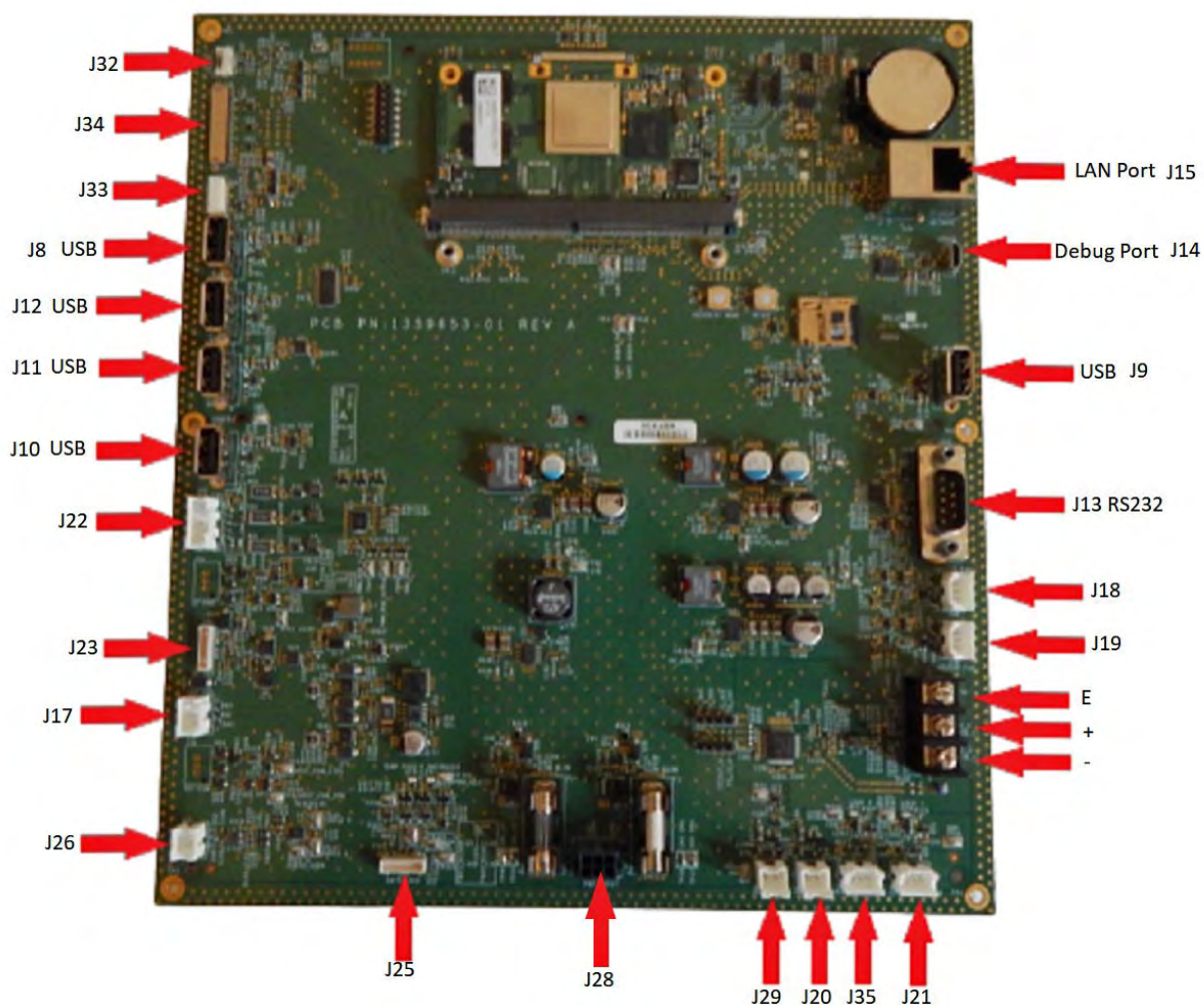
1. Turn off the EZ2 and remove the power cord and LAN cable.
2. Remove the rear cover per [RP1 - Rear Cover - Removal Procedure](#).
3. Remove the main cover per [RP2 - Main Cover - Removal Procedure](#).
4. Disconnect all cables from the CPU board.
5. Remove the 6 (T10) screws that hold the board in place.



Picture 6.45-1 CPU Board Screws

6.45.2 RP43 - CPU Board - Replacement Procedure

1. Install the 6 (T10) screws in the CPU board.
2. Plug all cable connectors back in the CPU board.



Picture 6.45-2 CPU Cable connections

3. Install the main cover per [RP2 - Main Cover- Replacement Procedure](#).
4. Install the rear cover per [RP1 - Rear Cover - Replacement Procedure](#).
5. Perform [CP1 Camera Calibration](#), [CP2 Camera Exposure Calibration](#), [CP3 Pipette Z-axis Calibration](#), [CP4 Worktable Calibration](#), [CP5 Pipette P-Axis Calibration](#), [CP6 Magnet Calibration](#), [CP9 Set Time/ Date](#), [VP8 - Workflow System Test](#).

6.46 RP44 - Power Board

Part Required	9027083 - Power Board
Tool(s) Required	T10, T20
Calibration Required	NA
Testing Required	VP8 - Workflow System Test
Replacement Time	15 minutes
Calibration/Testing Time	30 minutes
Total Repair Time	45 minutes

Table 6.46-1 RP44 Power Board

6.46.1 RP44 - Power Board - Removal Procedure

1. Turn off the power to the EZ2 and remove the power cord and LAN cable.
2. Remove the rear cover per [RP1 - Rear Cover - Removal Procedure](#).
3. Remove the main cover per [RP2 - Main Cover - Removal Procedure](#).
4. Unplug the connectors on the power board.
5. Remove the 4 (T10) screws on the corners of the power board.

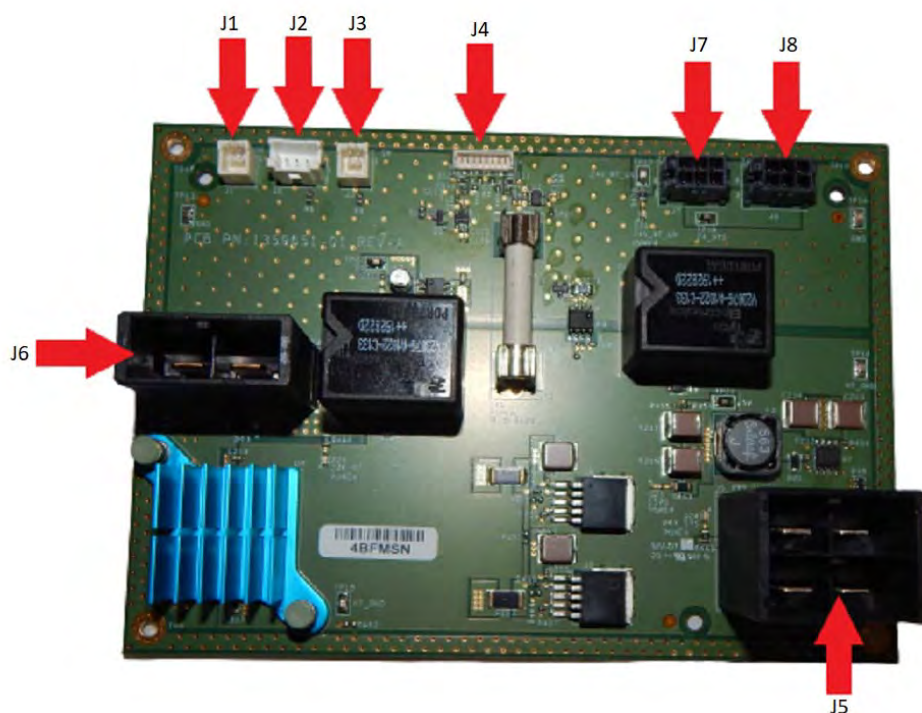


Picture 6.46-1 Power Board Screws

6.46.2 RP44 - Power Board - Replacement Procedure

1. Replace the 4 screws that hold the power board in position.

2. Plug in all the connectors on the power board.



Picture 6.46-2 Power Board Connectors

3. Install the main cover per [RP2 - Main Cover- Replacement Procedure](#).
4. Replace the rear cover per [RP1 - Rear Cover - Replacement Procedure](#).
5. Perform [VP12 - System Initialization Test](#).

6.47 RP45 - Hood Guide Roller

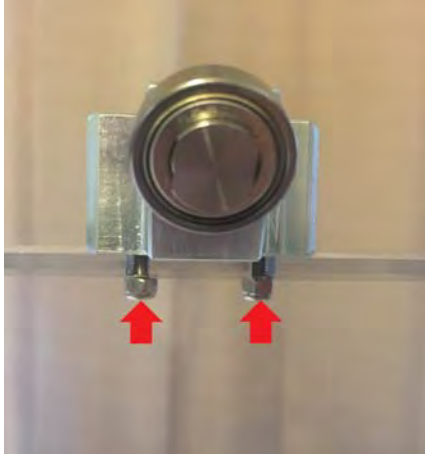
Part Required	9027211 - Hood Guide Roller
Tool(s) Required	5.5 mm Driver, T10, T20, T30
Calibration Required	NA
Testing Required	VP6 - Hood Lock Test
Replacement Time	23 minutes
Calibration/Testing Time	2 minutes
Total Repair Time	25 minutes

Table 6.47-1 RP45 Hood Guide Roller

6.47.1 RP45 - Hood Guide Roller - Removal Procedure

1. Remove the rear cover per [RP1 - Rear Cover - Removal Procedure](#).
2. Remove the main cover per [RP2 - Main Cover - Removal Procedure](#).
3. Open the hood.
4. Use a 5.5 mm driver to hold the nuts.

- Loosen the (T10) screws from the bottom nuts.



Picture 6.47-1 Hood Roller Removal



- Slide the guide roller off the hood.

6.47.2 RP45 - Hood Guide Roller - Replacement Procedure

- Place the (10) screws through the top of the hood guide roller.
- Screw the 5.5 mm nuts on the bottom of the hood guide roller but keep the nuts loose.
- Slide the hood guide roller in the open groves of the hood.
- Tighten the screws until the hood guide roller is firmly in position.
- Replace the main cover per [RP2 - Main Cover- Replacement Procedure](#).
- Replace the rear cover per [RP1 - Rear Cover - Replacement Procedure](#).
- Perform [VP6 - Hood Lock Test](#).

6.48 RP46 - Foot

Part Required	9027188 - Foot Black, 9027189 Foot Gray
Tool(s) Required	T25
Calibration Required	NA
Testing Required	VP12 - System Initialization Test
Replacement Time	5 minutes
Calibration/Testing Time	2 minutes
Total Repair Time	7 minutes

Table 6.48-1 RP46 Foot

6.48.1 RP46 - Foot - Removal Procedure

- It is best to use two people when replacing a foot for safety.
- Slide the EZ2 corner of the foot you would like to replace off the table for access to the foot.

3. Remove the (T25) screw in the center of the foot to remove the foot.



Picture 6.48-1 Foot Removal

6.48.2 RP46 - Foot - Replacement Procedure

1. Use the (T25) screw through the center foot and screw it to the base of the instrument.
2. Perform [VP12 - System Initialization Test](#).

6.49 RP47 - LAN Port

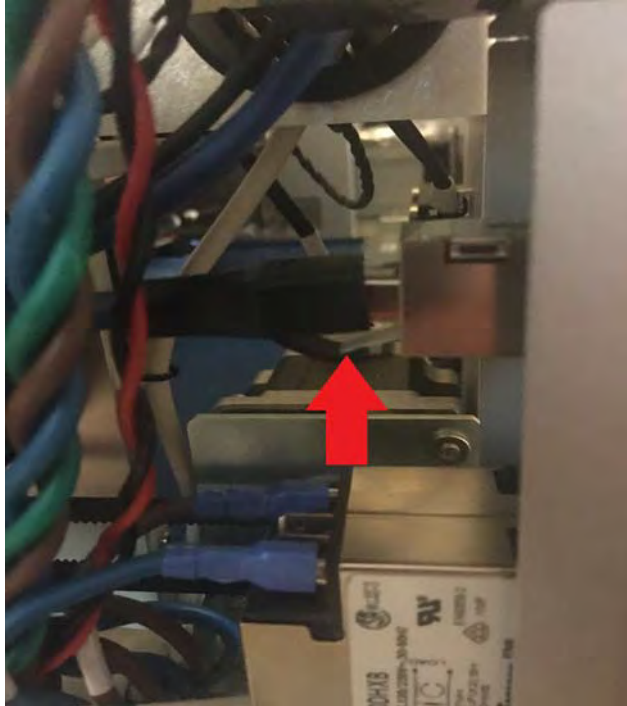
Part Required	9027137 - LAN Port
Tool(s) Required	T10, T20
Calibration Required	NA
Testing Required	VP12 - System Initialization Test
Replacement Time	12 minutes
Calibration/Testing Time	2 minutes
Total Repair Time	14 minutes

Table 6.49-1 RP46 LAN Port

6.49.1 RP47 - LAN Port - Removal Procedure

1. Remove the rear cover per [RP1 - Rear Cover - Removal Procedure](#).
2. Remove the main cover per [RP2 - Main Cover - Removal Procedure](#).
3. Unplug the LAN cable from the back of the port.

4. Remove the 2 (T10) screws to remove the port.



Picture 6.49-1 LAN Port Removal

6.49.2 RP47 - LAN Port - Replacement Procedure

1. Replace the 2 (T10) screws in the front of the port.
2. Plug in the LAN cable to the back of the port.
3. Replace the main cover per [RP2 - Main Cover- Replacement Procedure](#).
4. Replace the rear cover per [RP1 - Rear Cover - Replacement Procedure](#).
5. Perform [VP12 - System Initialization Test](#).

6.50 RP48 - Ball Latches

Part Required	Ball Latches
Tool(s) Required	T10, T20
Calibration Required	NA
Testing Required	VP12 - System Initialization Test
Replacement Time	6 minutes
Calibration/Testing Time	2 minutes
Total Repair Time	8 minutes

Table 6.50-1 RP48 Ball Latches

6.50.1 RP48 - Ball Latches - Removal Procedure

1. Remove the rear cover per [RP1 - Rear Cover - Removal Procedure](#).

2. Remove both of the (T20) screws to remove the ball latch.



Picture 6.50-1 Ball Latches

6.50.2 RP48 - Ball Latches - Replacement Procedure

1. Install the 2 (T20) screws that mount the ball latch in position.
2. Install the rear cover per [RP1 - Rear Cover - Replacement Procedure](#).
3. Perform [VP12 - System Initialization Test](#).

6.51 RP49 - Speaker

Part Required	9027218 - Speaker
Tool(s) Required	T10, T6, T20, T25
Calibration Required	NA
Testing Required	VP15 - Speaker Test
Replacement Time	28 minutes
Calibration/Testing Time	2 minutes
Total Repair Time	30 minutes

Table 6.51-1 RP49 Speaker

6.51.1 RP49 - Speaker - Removal Procedure

1. Remove the rear cover per [RP1 - Rear Cover - Removal Procedure](#).
2. Remove the main cover per [RP2 - Main Cover - Removal Procedure](#).
3. Remove the stationary left cover per [RP3 - Left Stationary Cover - Removal Procedure](#).
4. Remove the right stationary cover per [RP4 - Right Stationary Cover - Removal Procedure](#).
5. Remove the left and right trays per [RP95 - Left and Right Tray - Removal Procedure](#).

6. Remove all 8 (T20) screws on the back side of the front bottom cover.



Picture 6.51-1 Bottom Front Cover Screw Removal

7. Remove the 2 (T20) screws from the power button bracket.



Picture 6.51-2 Power Button Bracket Screws

8. Remove the 4 (T6) screws that hold the speaker on the cover.



Picture 6.51-3 Front Bottom Cover Speaker Removal

9. Cut any cable ties that anchor the cable for the speaker to the cover.
10. Unplug the speaker cable from position J26 on the CPU board.

6.51.2 RP49 - Speaker - Replacement Procedure

1. Replace the 4 (T6) screws that mount the speaker on the bottom front cover.
2. Plug in the cable connector in J26 position on the CPU board.
3. Install any cable ties that had been cut when the speaker was removed.
4. Replace the 10 (T20) screws that hold the front bottom cover in position.
5. Replace the 2 (T20) screws in the bracket that holds the power button.
6. Replace the 2 (T20) screws in the USB port bracket that holds it in position.
7. Replace the right stationary cover per [RP4 - Right Stationary Cover - Replacement Procedure](#).
8. Replace the left stationary cover per [RP3 - Left Stationary Cover - Replacement Procedure](#).
9. Replace the main cover per [RP2 - Main Cover- Replacement Procedure](#).
10. Replace the rear cover per [RP1 - Rear Cover - Replacement Procedure](#).
11. Perform [VP15 - Speaker Test](#).

6.52 RP50 - Worktable

Part Required	9027169 - Worktable
Tool(s) Required	T10, T20
Calibration Required	CP3 Pipette Z-axis Calibration , CP4 Worktable Calibration , CP6 Magnet Calibration
Testing Required	VP8 - Workflow System Test
Replacement Time	32 minutes
Calibration/Testing Time	45 minutes
Total Repair Time	1 hour 17 minutes

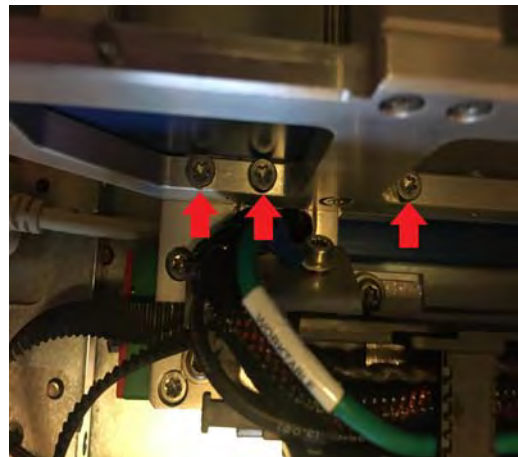
Table 6.52-1 RP50 Worktable

6.52.1 RP50 - Worktable - Removal Procedure..

1. Remove the rear cover per [RP1 - Rear Cover - Removal Procedure](#).
2. Remove the main cover per [RP2 - Main Cover - Removal Procedure](#).
3. Remove the heater module per [RP51 - Heater Module - Removal Procedure](#)
4. Remove the bottom front cover per [RP6 - Front Bottom Cover - Removal Procedure](#).
5. Remove the right internal metal cover per [RP5 - Internal Right Metal Cover - Removal Procedure](#)
6. Remove the 2 (T20) screws on the right side of the worktable.
7. Remove the 3 (T20) screws on the left side of the worktable.



Picture 6.52-1 Worktable Screw Removal



8. Remove the worktable.

6.52.2 RP50 - Worktable - Replacement Procedure

1. Align the screw positions with the left and right Y rails.
2. Install the 2 (T20) screws on the right side of the worktable.
3. Install the 3 (T20) screws on the left side of the worktable.
4. Install the heater unit per?
5. Replace the front bottom cover per [RP6 - Front Bottom Cover - Replacement Procedure](#).
6. Replace the right internal metal cover per [RP5 - Internal Right Metal Cover- Replacement Procedure](#).
7. Perform [CP3 Pipette Z-axis Calibration](#), [CP4 Worktable Calibration](#), [CP6 Magnet Calibration](#), [VP8 - Workflow System Test](#).

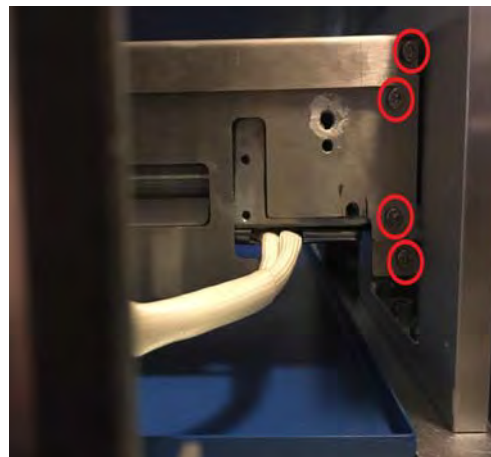
6.53 RP51 - Heater Module

Part Required	9027162 - Heater Module
Tool(s) Required	T6, T10, T20, T25
Calibration Required	NA
Testing Required	VP5 - Temperature Test
Replacement Time	22 minutes
Calibration/Testing Time	23 minutes
Total Repair Time	45 minutes

Table 6.53-1 RP51 Heater Module

6.53.1 RP51 - Heater Module - Removal Procedure

1. Remove the rear cover per [RP1 - Rear Cover - Removal Procedure](#).
2. Remove both the left and right air ducts per [RP21 - Air Duct - Removal Procedure](#).
3. Remove the 4 (T10) screws on the left and right side of the heater.



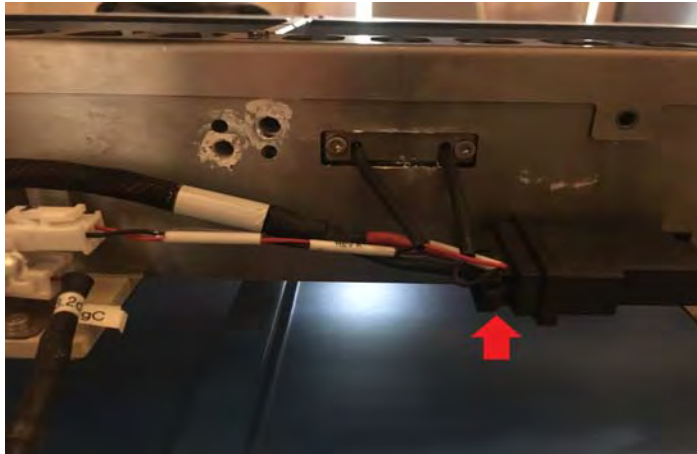
Picture 6.53-1 Heater Left & Right Screws

4. Remove the 3 (T10) screws in the middle of the heater.



Picture 6.53-2 Heater Middle Screws

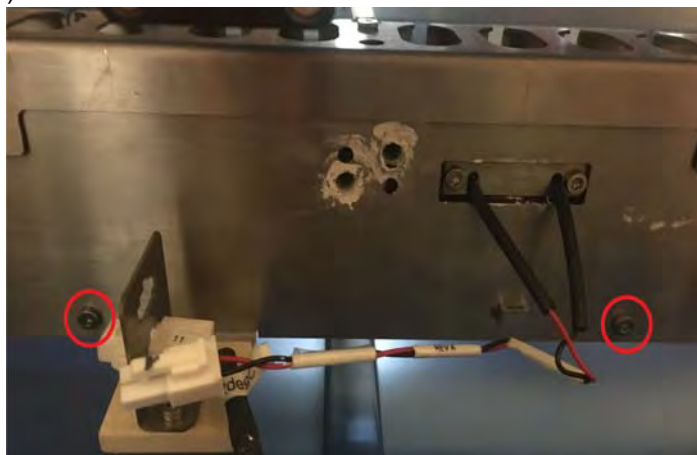
5. Remove the temperature fuse per [RP52 - Temperature Fuse - Removal Procedure](#)
6. Unplug the heater power connector, labeled as "Heater Power Adapter".



Picture 6.53-3 Heater Power Connector

Picture 6.53-4 Center Bottom Screws

7. Remove the 2 (T10) screws.



Picture 6.53-5 Top Screws

8. Remove the 3 top (T6) screws and remove the stainless steel cover on top of the heater.



9. After the top cover is removed, pull the heater towards the back and remove it.
10. Remove the 3 (T25) screws on the bottom of the heater.



Picture 6.53-6 3 Bottom Screws

6.53.2 RP51 - Heater Module - Replacement Procedure

1. Insert the 3 (T25) screws in the bottom of the heater.
2. Slide the heater back in position with the connector on the right.
3. Install top stainless steel cover with the 3 (T6) screws in top of the cover.
4. Install the 12 (T10) screws on the back of the heater.
5. Install the thermocouple per [RP23 - Thermocouple- Replacement Procedure](#).
6. Install the temperature fuse per [RP52 - Temperature Fuse - Replacement Procedure](#).
7. Put the rear cover back in position per [RP1 - Rear Cover - Replacement Procedure](#).
8. Perform [VP5 - Temperature Test](#).

6.54 RP52 - Temperature Fuse

Part Required	9027166 - Temperature Fuse
Tool(s) Required	T20, T10
Calibration Required	NA
Testing Required	VP5 - Temperature Test
Replacement Time	6 minutes
Calibration/Testing Time	23 minutes
Total Repair Time	29 minutes

Table 6.54-1 RP52 Heater Fuse

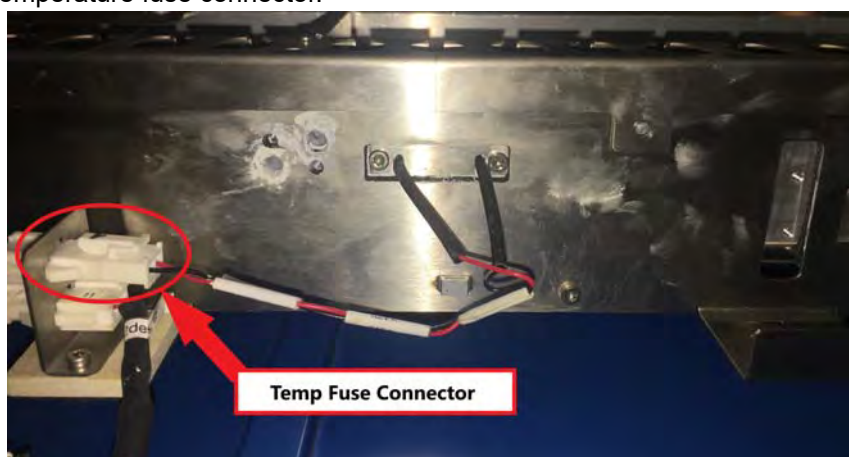
6.54.1 RP52 - Temperature Fuse - Removal Procedure

1. Remove the rear cover per [RP1 - Rear Cover - Removal Procedure](#).
2. Remove the 2 (T10) screws that hold the temperature fuse in place.



Picture 6.54-1 Temperature Fuse

3. Unplug the temperature fuse connector.



Picture 6.54-2 Temperature Fuse Connector

6.54.2 RP52 - Temperature Fuse - Replacement Procedure

1. Install the 2 (T10) screws that hold the temperature fuse in position.

2. Plug in the connector, marked on the cable as "temp fuse adapter".
3. Put the rear cover back in position per [RP1 - Rear Cover - Replacement Procedure](#).
4. Perform [VP5 - Temperature Test](#).

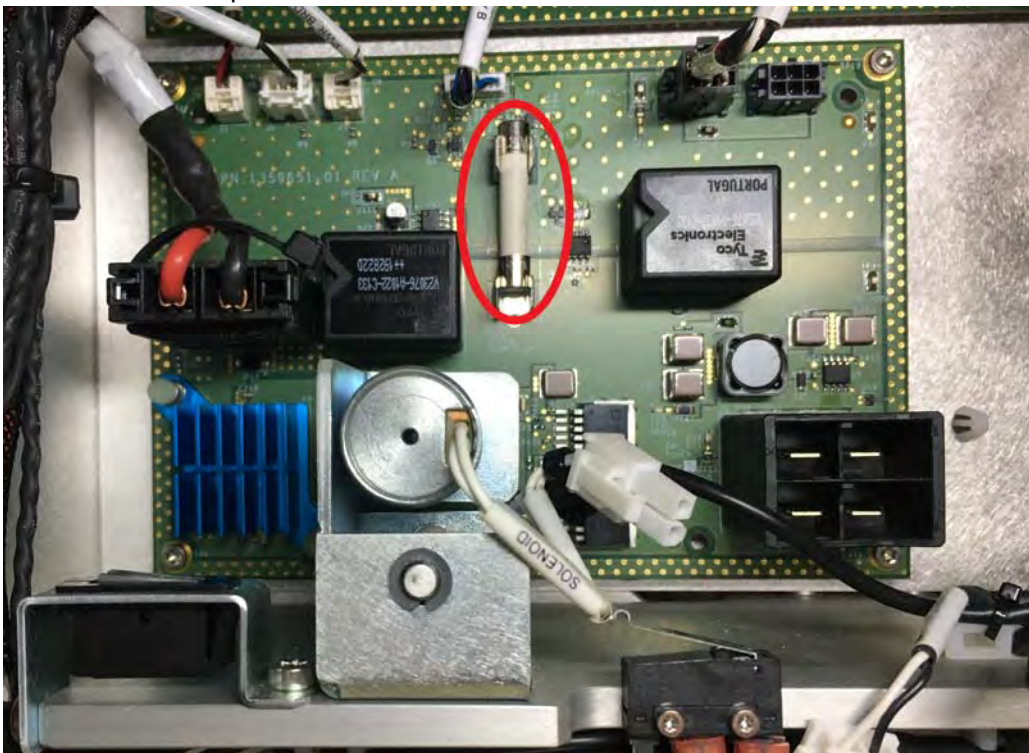
6.55 RP53 - Heater Fuse 20 Amp Power Board

Part Required	9027091 - Heater Fuse 20 Amp
Tool(s) Required	T20
Calibration Required	NA
Testing Required	VP5 - Temperature Test
Replacement Time	8 minutes
Calibration/Testing Time	23 minutes
Total Repair Time	31 minutes

Table 6.55-1 RP53 Heater Fuse 20 Amp

6.55.1 RP53 - Heater Fuse 20 Amp Power Board- Removal Procedure

1. Remove the rear cover per [RP1 - Rear Cover - Removal Procedure](#).
2. Remove the main cover per [RP2 - Main Cover - Removal Procedure](#).



Picture 6.55-1 Heater 20 Amp Fuse

3. Remove the 20 Amp heater fuse from the power board.

6.55.2 RP53 - Heater Fuse 20 Amp Power Board - Replacement Procedure

1. Install the 20 Amp heater fuse on the power board.
2. Replace the main cover per [RP2 - Main Cover- Replacement Procedure](#).

3. Replace the rear cover per [RP1 - Rear Cover - Replacement Procedure](#).
4. Perform [VP5 - Temperature Test](#).

6.56 RP54 - CPU Board 2.5 Amp Fuse

Part Required	9027089 - CPU Board 2.5 Amp Fuse
Tool(s) Required	T20
Calibration Required	TBD
Testing Required	VP8 - Workflow System Test
Replacement Time	8 minutes
Calibration/Testing Time	30 minutes
Total Repair Time	38 minutes

Table 6.56-1 RP54 CPU Board 2.5 Amp Fuse

6.56.1 RP54 - CPU Board 2.5 Amp Fuse - Removal Procedure

1. Remove the rear cover per [RP1 - Rear Cover - Removal Procedure](#).
2. Remove the main cover per [RP2 - Main Cover - Removal Procedure](#).

Picture 6.56-1

3. Remove the 2.5 Amp fuse from the CPU board.



Picture 6.56-2 CPU Board 2.5 Amp Fuse

6.56.2 RP54 - CPU Board 2.5 Amp Fuse - Replacement Procedure

1. Replace the 2.5 Amp fuse on the CPU board.
2. Install the main cover per [RP2 - Main Cover- Replacement Procedure](#).
3. Install the rear cover per [RP1 - Rear Cover - Replacement Procedure](#).
4. Perform [VP8 - Workflow System Test](#).

6.57 RP55 - Motor Board 0.16 Amp Fuse

Part Required	9027078 - Motor Board 0.16 Amp Fuse
Tool(s) Required	T20
Calibration Required	NA
Testing Required	VP12 - System Initialization Test
Replacement Time	6 minutes
Calibration/Testing Time	2 minutes
Total Repair Time	8 minutes

Table 6.57-1 RP55 Motor Board 0.16 Amp Fuse

6.57.1 RP55 - Motor Board 0.16 Amp Fuse- Removal Procedure

1. Remove the rear cover per [RP1 - Rear Cover - Removal Procedure](#).
2. Remove the 0.16 Amp fuse from the motor board.



Picture 6.57-1 0.16 Amp Fuse

6.57.2 RP55 - Motor Board 0.16 Amp Fuse - Replacement Procedure

1. Replace the 0.16 Amp fuse on the motor drive board.
2. Replace the rear cover per [RP1 - Rear Cover - Replacement Procedure](#).
3. Perform [VP12 - System Initialization Test](#).

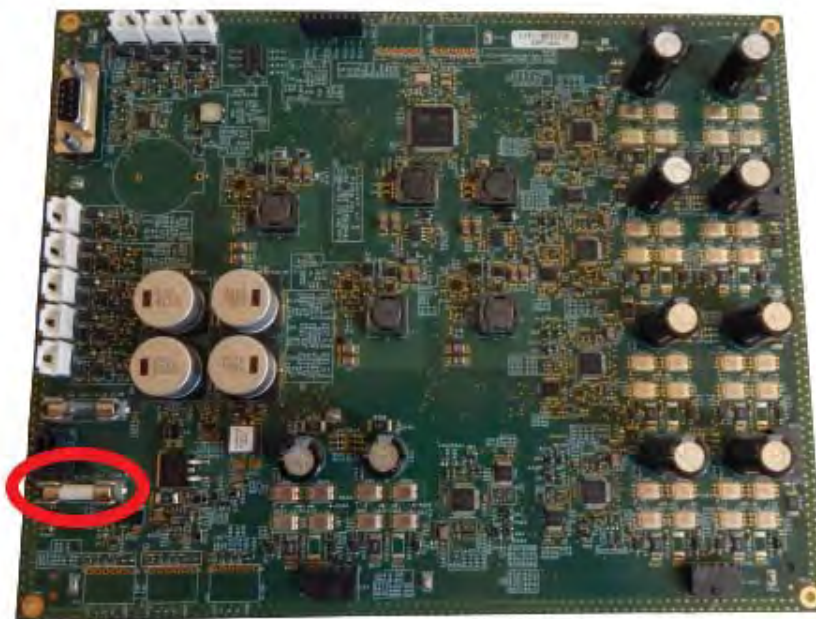
6.58 RP56 - Motor Board 6.3 Amp Fuse

Part Required	9027080 - Motor Board 6.3 Amp Fuse
Tool(s) Required	T20
Calibration Required	NA
Testing Required	VP12 - System Initialization Test
Replacement Time	6 minutes
Calibration/Testing Time	2 minutes
Total Repair Time	8 minutes

Table 6.58-1 RP56 Motor Board 6.3 Amp Fuse

6.58.1 RP56 - Motor Board 6.3 Amp Fuse - Removal Procedure

1. Remove the rear cover per [RP1 - Rear Cover - Removal Procedure](#).
2. Remove the 6.3 Amp fuse from the motor driver board.



Picture 6.58-1 6.3 Amp Fuse

6.58.2 RP56 - Motor Board 6.3 Amp Fuse - Replacement Procedure

1. Install the 6.3 Amp fuse on the motor driver board.
2. Install the rear cover per [RP1 - Rear Cover - Replacement Procedure](#).
3. Perform [VP12 - System Initialization Test](#).

6.59 RP57 - UV 0.125 Amp Fuse

Part Required	9027090 - UV 0.125 Amp Fuse
Tool(s) Required	T20
Calibration Required	NA
Testing Required	VP2 - UV Light Test
Replacement Time	8 minutes
Calibration/Testing Time	2 minutes
Total Repair Time	10 minutes

Table 6.59-1 RP57 UV LED 0.125 Amp Fuse on CPU Board

6.59.1 RP57 - UV 0.125 Amp Fuse - Removal Procedure

1. Remove the rear cover per [RP1 - Rear Cover - Removal Procedure](#).
2. Remove the main cover per [RP2 - Main Cover - Removal Procedure](#).
3. Remove the UV 0.125 Amp fuse from the CPU board.



Picture 6.59-1 UV 0.125 Amp Fuse

6.59.2 RP57 - UV 0.125 Amp Fuse - Replacement Procedure

1. Replace the UV 0.125 Amp fuse on the CPU board.
2. Install the main cover per [RP2 - Main Cover- Replacement Procedure](#).
3. Install the rear cover per [RP1 - Rear Cover - Removal Procedure](#).
4. Perform [VP2 - UV Light Test](#).

6.60 RP58 - CPU Coin Battery CR2450

Part Required	NA-Acquire locally. This is a standard CR2450 coin type battery.
Tool(s) Required	T20
Calibration Required	CP9 Set Time/Date
Testing Required	VP12 - System Initialization Test
Replacement Time	8 minutes
Calibration/Testing Time	4 minutes
Total Repair Time	12 minutes

Table 6.60-1 RP58 CPU Coin Battery CR2450

6.60.1 RP58 - CPU Coin Battery CR2450 - Removal Procedure

1. Remove the rear cover per [RP1 - Rear Cover - Removal Procedure](#).
2. Remove the main cover per [RP2 - Main Cover - Removal Procedure](#).
3. Remove the battery from the CPU board.



Picture 6.60-1

6.60.2 RP58 - CPU Coin Battery CR2450 - Replacement Procedure

1. Replace the coin battery with #CR2450 type battery on the CPU board.
2. Install the main cover per [RP2 - Main Cover- Replacement Procedure](#).
3. Install the rear cover per [RP1 - Rear Cover - Removal Procedure](#).
4. Perform [CP9 Set Time/Date](#), [VP12 - System Initialization Test](#).

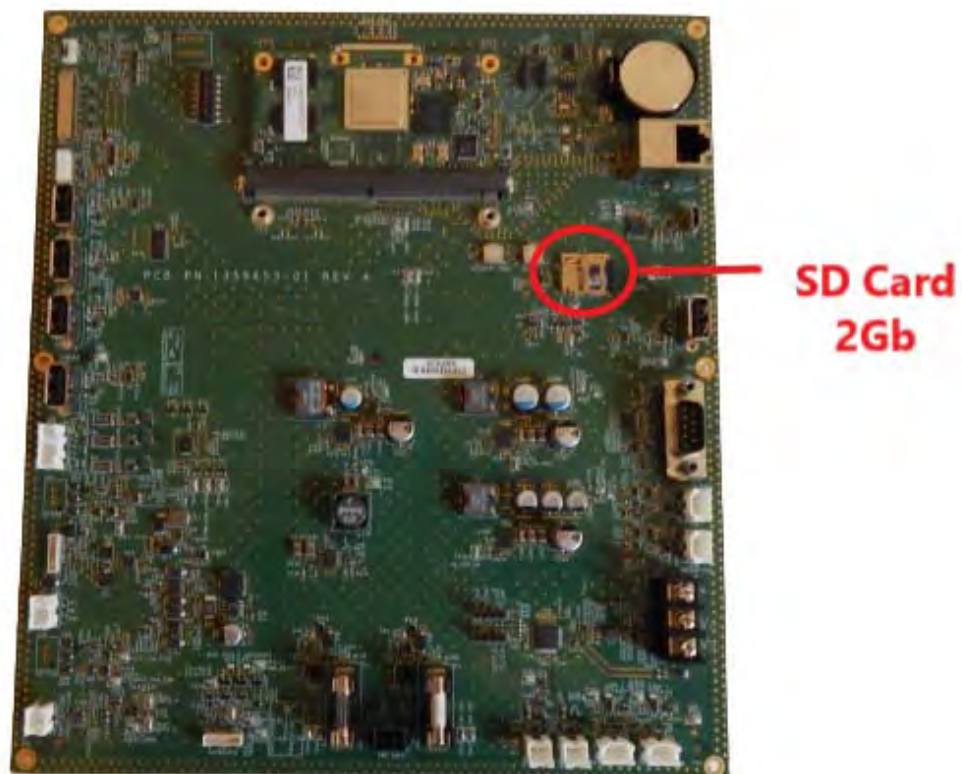
6.61 RP59 - SD Micro Card

Part Required	9027144 - SD Micro Card
Tool(s) Required	T20
Calibration Required	NA
Testing Required	VP12 - System Initialization Test
Replacement Time	8 minutes
Calibration/Testing Time	2 minutes
Total Repair Time	10 minutes

Table 6.61-1 RP59 SD Micro Card 2Gb

6.61.1 RP59 - SD Micro Card - Removal Procedure

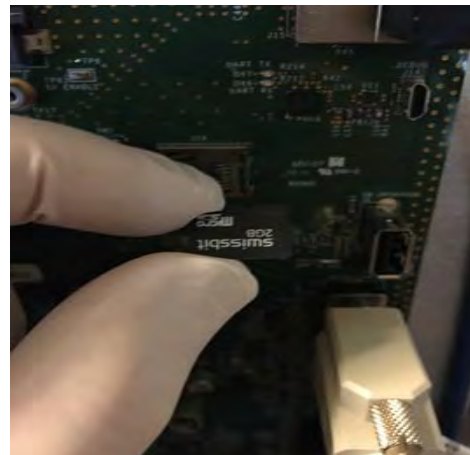
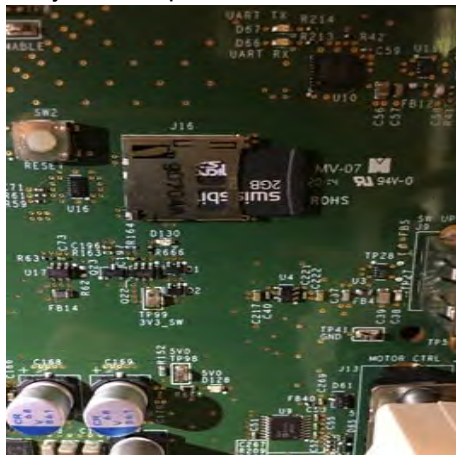
1. Remove the rear cover per [RP1 - Rear Cover - Removal Procedure](#).
2. Remove the main cover per [RP2 - Main Cover - Removal Procedure](#).
3. Carefully pull the SD micro card out of the slot to remove it.



Picture 6.61-1 CPU SD Micro Card

6.61.2 RP59 - SD Micro Card - Replacement Procedure

1. Check the orientation of the SD micro card when inserting it into the slot, the wording should be facing you and upside down.



Picture 6.61-2 SD Micro Card Replacement.

2. Install the main cover per [RP2 - Main Cover- Replacement Procedure](#).
3. Install the rear cover per [RP1 - Rear Cover - Replacement Procedure](#).
4. Perform [VP12 - System Initialization Test](#).

6.62 RP60 - AC Inlet

Part Required	9027127 - AC Inlet
Tool(s) Required	T20
Calibration Required	NA
Testing Required	VP12 - System Initialization Test
Replacement Time	10 minutes
Calibration/Testing Time	2 minutes
Total Repair Time	12 minutes

Table 6.62-1 RP60 AC Inlet

6.62.1 RP60 - AC Inlet - Removal Procedure

1. Remove the rear cover per [RP1 - Rear Cover - Removal Procedure](#).
2. Remove the main cover per [RP2 - Main Cover - Removal Procedure](#).

3. Press the 2 black tabs down on the top of the AC inlet while pulling outward on the inlet.



Picture 6.62-1 AC Inlet

4. Press the 2 black tabs up on the bottom of the AC inlet with a flat screwdriver while pulling outward on the inlet.
5. Remove the 3 wires connected to the back of the AC inlet and remove the AC inlet.

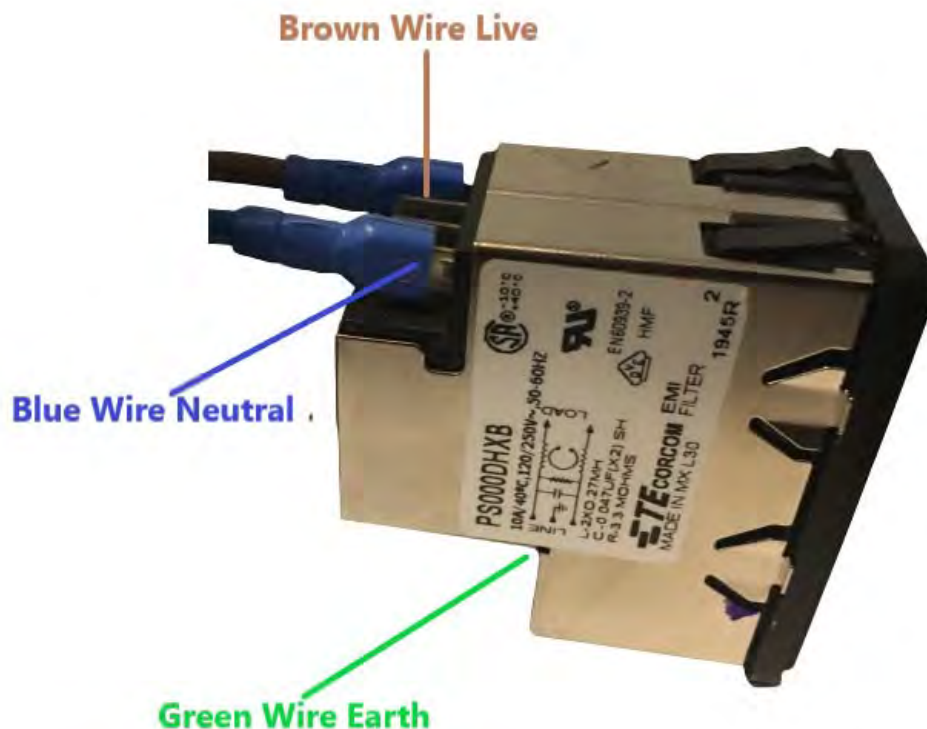


Picture 6.62-2 AC Inlet Wires

6.62.2 RP60 - AC Inlet - Replacement Procedure

1. Push the AC inlet through the hole till it locks in place.

2. Connect the wiring to the AC inlet.



Picture 6.62-3 AC Inlet Wiring

3. Replace the main cover per [RP2 - Main Cover- Replacement Procedure](#).
4. Replace the rear cover per [RP1 - Rear Cover - Replacement Procedure](#).
5. Perform [VP12 - System Initialization Test](#).

6.63 RP61 - 4 Amp 250 Volt Fuse

Part Required	9027072 - AC Inlet 4 Amp 250 Volt Fuse
Tool(s) Required	Flat Head Screwdriver
Calibration Required	NA
Testing Required	VP12 - System Initialization Test
Replacement Time	2 minutes
Calibration/Testing Time	2 minutes
Total Repair Time	4 minutes

Table 6.63-1 RP61 4 Amp 250 Volt Fuse

6.63.1 RP61 - 4 Amp 250 Volt Fuse - Removal Procedure

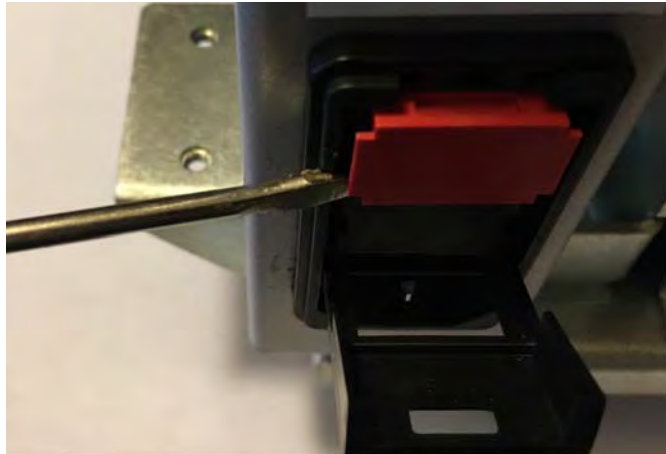
1. Remove the power cord.

2. Open the top of the AC inlet.



Picture 6.63-1 Open Fuse Holder

3. Use a small flat head screwdriver and pry the red fuse holder out of the AC inlet.
4. Remove the 4 Amp fuse from the fuse holder.



Picture 6.63-2 Fuse Holder Removal

5. The fuse holder will have 2 fuses, one on each side of the fuse holder.



Picture 6.63-3 Fuse Holder

6.63.2 RP61 - 4 Amp 250 Volt Fuse - Replacement Procedure

1. Insert the 4 Amp fuse in the fuse holder.
2. Place the fuse holder back inside the AC inlet.
3. Shut the black portion of the AC inlet that locks the fuse holder in position.
4. Perform [VP12 - System Initialization Test](#).

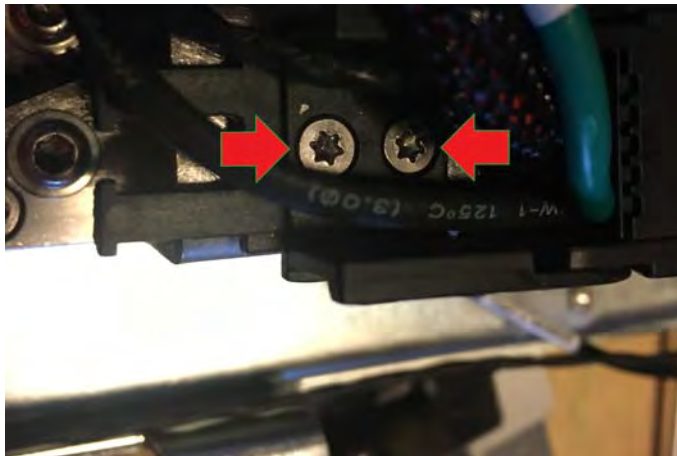
6.64 RP62 - E-Chain X Axis

Part Required	9027160 - E-Chain X Axis
Tool(s) Required	T10, T20
Calibration Required	NA
Testing Required	VP8 - Workflow System Test
Replacement Time	32 minutes
Calibration/Testing Time	30 minutes
Total Repair Time	1 hour 2 minutes

Table 6.64-1 RP62 E-Chain X Axis

6.64.1 RP62 - E-Chain X Axis - Removal Procedure

1. Remove the rear cover per [RP1 - Rear Cover - Removal Procedure](#).
2. Remove the main cover per [RP2 - Main Cover - Removal Procedure](#).
3. Remove the 2 (T10) screws on both ends of the E-chain.

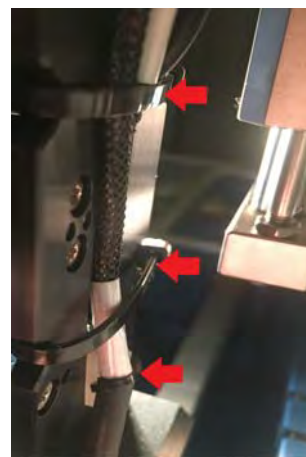


Picture 6.64-1 E-Chain X Screws

4. Cut the 4 cable ties on the camera arm.



Picture 6.64-2 Cable Ties



5. Unplug the LED board and UV board.



Picture 6.64-3 Unplug Boards

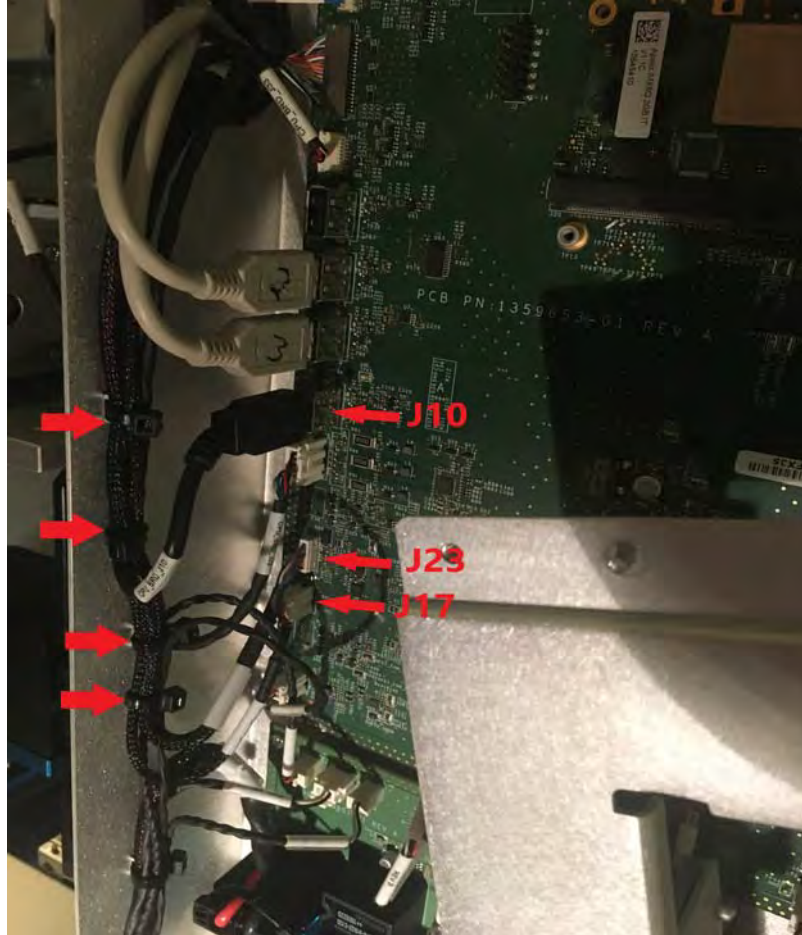


6. Unplug the camera cable.



Picture 6.64-4 Unplug Camera

7. Cut the cable ties and unplug J10, J23 and J17.



Picture 6.64-5 Unplug E Chain Cables

8. Remove the X E-Chain and lay the E-chain flat then remove all 3 cables running through the E-chain.

6.64.2 RP62 - E-Chain X Axis - Replacement Procedure

1. Lay the E-chain flat.
2. Feed the 3 cables J10, J23 and J17 through the center of the E-chain, one cable at a time.
3. Attach the E-chain to the brackets using the 2 (T10) screws.
4. Plug in the camera, UV, and LED board.
5. Plug the J10, J23 and J17 cables in the CPU board.
6. Replace the main cover per [RP2 - Main Cover- Replacement Procedure](#).
7. Replace the rear cover per [RP1 - Rear Cover - Replacement Procedure](#).
8. Perform [VP8 - Workflow System Test](#).

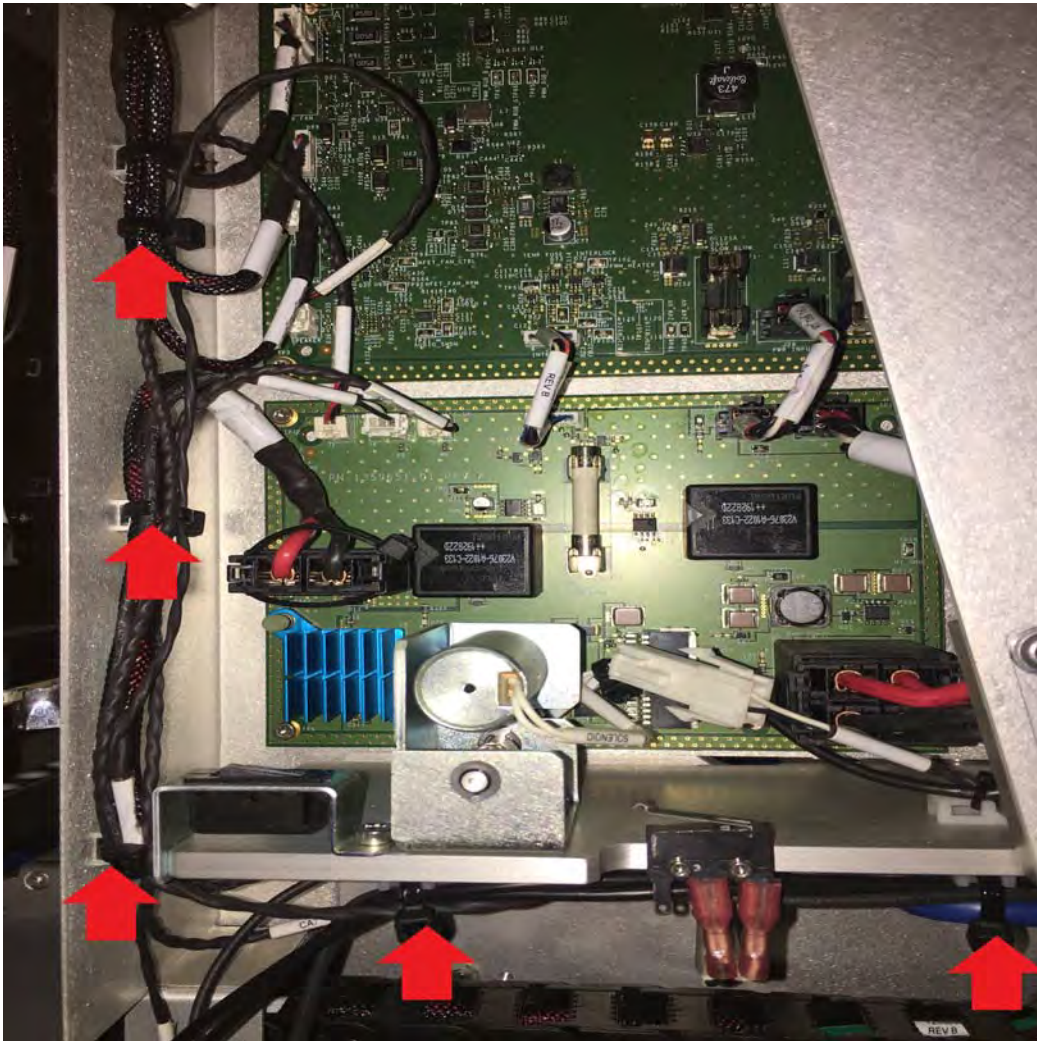
6.65 RP63 - E-Chain Y Axis

Part Required	9027180 E-Chain Y Axis
Tool(s) Required	T10, T20
Calibration Required	NA
Testing Required	VP8 - Workflow System Test
Replacement Time	50 minutes
Calibration/Testing Time	30 minutes
Total Repair Time	1 hour 20 minutes

Table 6.65-1 RP63 E-Chain Y Axis

6.65.1 RP63 - E-Chain Y Axis - Removal Procedure

1. Remove the rear cover per [RP1 - Rear Cover - Removal Procedure](#).
2. Remove the main cover per [RP2 - Main Cover - Removal Procedure](#).
3. Remove the hood per [RP18 - Hood - Removal Procedure](#).
4. Remove the internal right metal cover per [RP5 - Internal Right Metal Cover - Removal Procedure](#).
5. Cut the 5 cable ties shown in the picture.



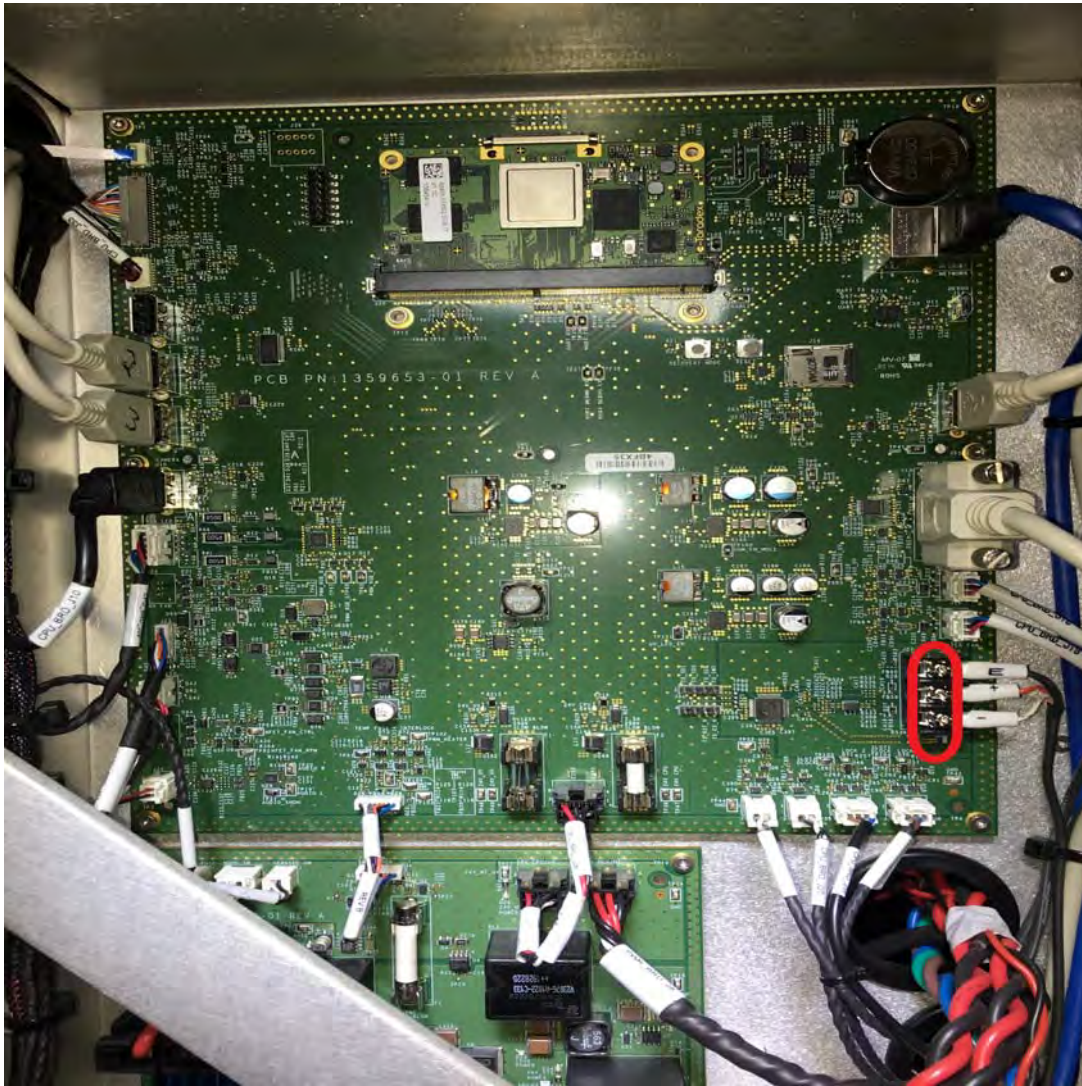
Picture 6.65-1 Cable Ties

6. Unplug the 2 connectors J6 and J1 from the power board.



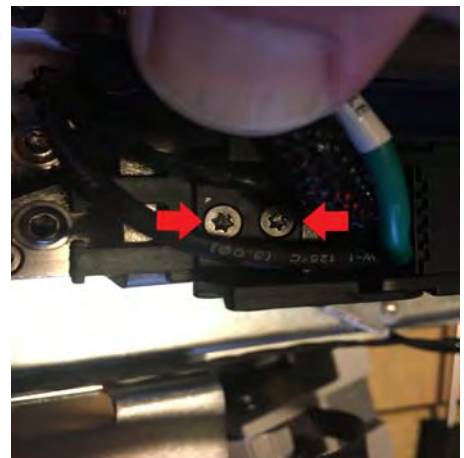
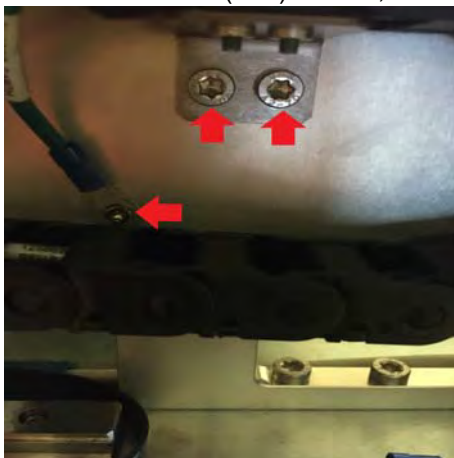
Picture 6.65-2 E Chain Cable Connectors Power Board

7. Loosen the 3 Phillips screws that hold the E, + and - cables in position on the CPU board and remove them.



Picture 6.65-3 E Chain CPU Connections

8. Remove the 2 (T20) bracket screws and the 1 (T10) ground screw.
9. Remove the 4 (T10) screws, each end of the E-chain will have 2 screws.



Picture 6.65-4 Y E-Chain Screws

10. Lay the E-chain flat and pull the cables through it.

6.65.2 RP63 - E-Chain Y Axis- Replacement Procedure

1. Lay the E-chain flat and run the cables, J1, J6, E+- and ground one at time.
2. Once the cables are in position, install the 2 (T10) screws on each end of the E-chain.
3. Plug the J1, J6 connectors on the power board.
4. Install the screw back in the ground cable.
5. Install the E,+,- cables on the CPU board.
6. Replace the 2 (T20) screws in the E-chain bracket.
7. Install the right internal metal cover per [RP5 - Internal Right Metal Cover- Replacement Procedure](#).
8. Perform [VP8 - Workflow System Test](#).

6.66 RP64 - Z E-Chain

Part Required	9027181 - Z E-Chain
Tool(s) Required	T10, T20
Calibration Required	NA
Testing Required	VP8 - Workflow System Test
Replacement Time	33 minutes
Calibration/Testing Time	30 minutes
Total Repair Time	1 hour 3 minutes

Table 6.66-1 RP64 Z E-Chain

6.66.1 RP64 - Z E-Chain - Removal Procedure

1. Remove the rear cover per [RP1 - Rear Cover - Removal Procedure](#).
2. Remove the main cover per [RP2 - Main Cover - Removal Procedure](#).
3. Remove the pipette module per [RP37 - Pipette Module - Removal Procedure](#).
4. Open the Z access door.
5. Cut the cable tie at the bottom side of the Z E-chain and remove the 2 (T10) screws.



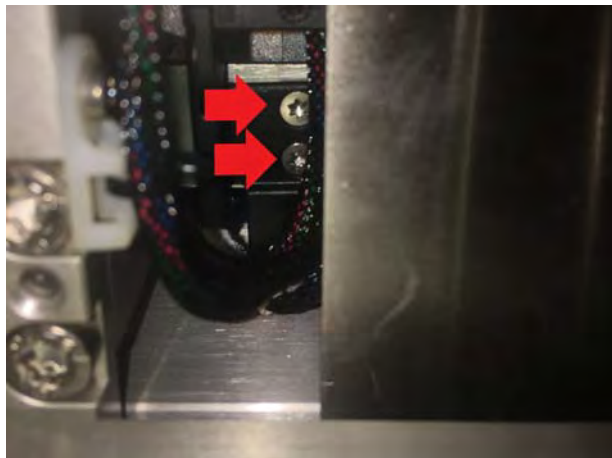
Picture 6.66-1 Z E-Chain Bottom Side

6. Open the Z access door and cut the cable tie on the bottom side of the E-chain.



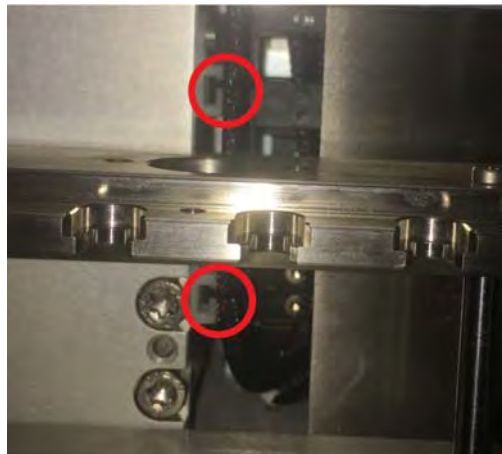
Picture 6.66-2 Cable Tie

7. Remove the 2 (T10) screws from the E-chain bracket.



Picture 6.66-3 E-Chain Screws

8. Cut the 2 small cable ties.



Picture 6.66-4 Small Cable Ties

9. Unplug the 3 connectors J15, J17 and J8.



Picture 6.66-5 Cable locations

10. Cut all the cable ties and lay the E-chain flat and pull the 3 cables out of the E-chain.

6.66.2 RP64 - Z E-Chain - Replacement Procedure

1. Lay the Z E-chain flat and pull the J15, J17 and J8 cables through one at a time.
2. Put the Z E-chain in position and install the 2 (T10) screws in the top bracket and the bottom bracket.
3. Plug in the 3 cables J15, J17 and J8 into the motor driver board.
4. Install the cable ties on the access door and on the top and bottom of the E-chain.
5. Install the 2 (T10) screws in the Z access door.
6. Replace the main cover per [RP2 - Main Cover- Replacement Procedure](#).
7. Replace the rear cover per [RP1 - Rear Cover - Replacement Procedure](#).
8. Perform [VP8 - Workflow System Test](#).

6.67 RP65 - Cable From Motor Driver to Slot Sensor (P-Org)

Part Required	9027172 - Cable From Motor Driver to Slot Sensor P-Org
Tool(s) Required	T10, T20
Calibration Required	CP5 Pipette P-Axis Calibration
Testing Required	VP3 - Leakage Test
Replacement Time	7 minutes
Calibration/Testing Time	12 minutes
Total Repair Time	19 minutes

Table 6.67-1 RP65 Cable from Motor Driver to Slot Sensor (P-Org)

6.67.1 RP65 - Cable from Motor Driver to Slot Sensor (P-Org)- Removal Procedure

1. Remove the rear cover per [RP1 - Rear Cover - Removal Procedure](#).
2. Remove the Z E-Chain per [RP64 - Z E-Chain - Removal Procedure](#).
3. Unplug the J17 (P-Org) sensor cable from the motor board and cut any cable ties needed to remove it.



Picture 6.67-1 P-Org Cable J17

4. Disconnect the white connector and cut the small cable tie.



Picture 6.67-2 P-Org Sensor Side

5. Remove the cable from the Z E-chain.

6.67.2 RP65 - Cable From Motor Driver-Sensor P Org - Replacement Procedure

1. Run the cable through the Z E-chain.
2. Install the Z E-chain per [RP64 - Z E-Chain - Replacement Procedure](#)
3. Plug in the white connector to the P-org sensor and put the small cable tie in place.
4. Plug the cable connector J17 into the motor board.
5. Install the main cover per [RP2 - Main Cover- Replacement Procedure](#).

6. Install the rear cover per [RP1 - Rear Cover - Replacement Procedure](#).
7. Perform [CP5 Pipette P-Axis Calibration](#), [VP3 - Leakage Test](#).

6.68 RP66 - Cable From Motor Driver-Sensor Z (Crash)

Part Required	9027175 - Cable From Motor Driver Sensor Z Crash
Tool(s) Required	T10, T20
Calibration Required	NA
Testing Required	VP12 - System Initialization Test
Replacement Time	8 minutes
Calibration/Testing Time	2 minutes
Total Repair Time	10 minutes

Table 6.68-1 RP66 Cable from Motor Driver-Sensor Z (Crash)

6.68.1 RP66 - Cable from Motor Driver-Sensor Z (Crash)- Removal Procedure

1. Remove the rear cover per [RP1 - Rear Cover - Removal Procedure](#).
2. Remove the Z E-Chain per [RP64 - Z E-Chain - Removal Procedure](#).
3. Disconnect J15 from the motor driver board and cut the cable ties.



Picture 6.68-1 J15 Z Crash Cable

4. Disconnect the white connector from the Z crash sensor.



Picture 6.68-2 Z Crash Connector

5. Pull the J15 Z crash cable through the Z E-chain.

6.68.2 RP66 - Cable from Motor Driver-Sensor Z (Crash) - Replacement Procedure

1. Run the J15 Z crash cable through the Z E-chain.
2. Install the Z E-chain per [RP64 - Z E-Chain - Replacement Procedure](#).
3. Plug in the white connector to the Z crash sensor and put the small cable tie in place.
4. Plug the cable connector J15 into the motor board.
5. Install the main cover per [RP2 - Main Cover- Replacement Procedure](#).
6. Install the rear cover per [RP1 - Rear Cover - Replacement Procedure](#).
7. Perform [VP12 - System Initialization Test](#).

6.69 RP67 - Cable From Motor Driver Board to Motor P-Axis

Part Required	9027171 Cable From Motor Driver Board to Motor P-Axis
Tool(s) Required	T10, T20
Calibration Required	NA
Testing Required	VP12 - System Initialization Test
Replacement Time	33 minutes
Calibration/Testing Time	2 minutes
Total Repair Time	35 minutes

Table 6.69-1 RP67 Cable from Motor Driver Board to Motor P-Axis

6.69.1 RP67 - Cable from Motor Driver Board to Motor P-Axis - Removal Procedure

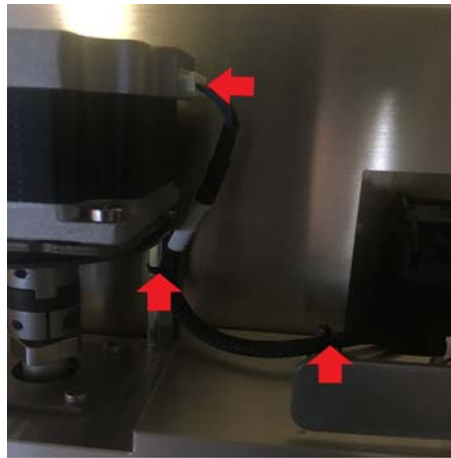
1. Remove the rear cover per [RP1 - Rear Cover - Removal Procedure](#).
2. Remove the Z E-Chain per [RP64 - Z E-Chain - Removal Procedure](#).

3. Unplug J8 from the motor driver board and cut the 2 cable ties holding the cable on the door.



Picture 6.69-1 J8 P-Axis Connector

4. Unplug the cable from the P motor and cut the 2 small cable ties.



Picture 6.69-2 P Axis Motor Cable

5. Pull the cable out of the Z E-chain.

6.69.2 RP67 - Cable from Motor Driver Board to Motor P-Axis - Replacement Procedure

1. Run the J8 P-axis cable through the Z E-chain.
2. Install the Z E-chain per [RP64 - Z E-Chain - Replacement Procedure](#).
3. Plug in the connector to the P motor and put the 2 small cable ties in place.
4. Plug the cable connector J8 into the motor board.
5. Install the main cover per [RP2 - Main Cover- Replacement Procedure](#).
6. Install the rear cover per [RP1 - Rear Cover - Replacement Procedure](#).
7. Perform [VP12 - System Initialization Test](#).

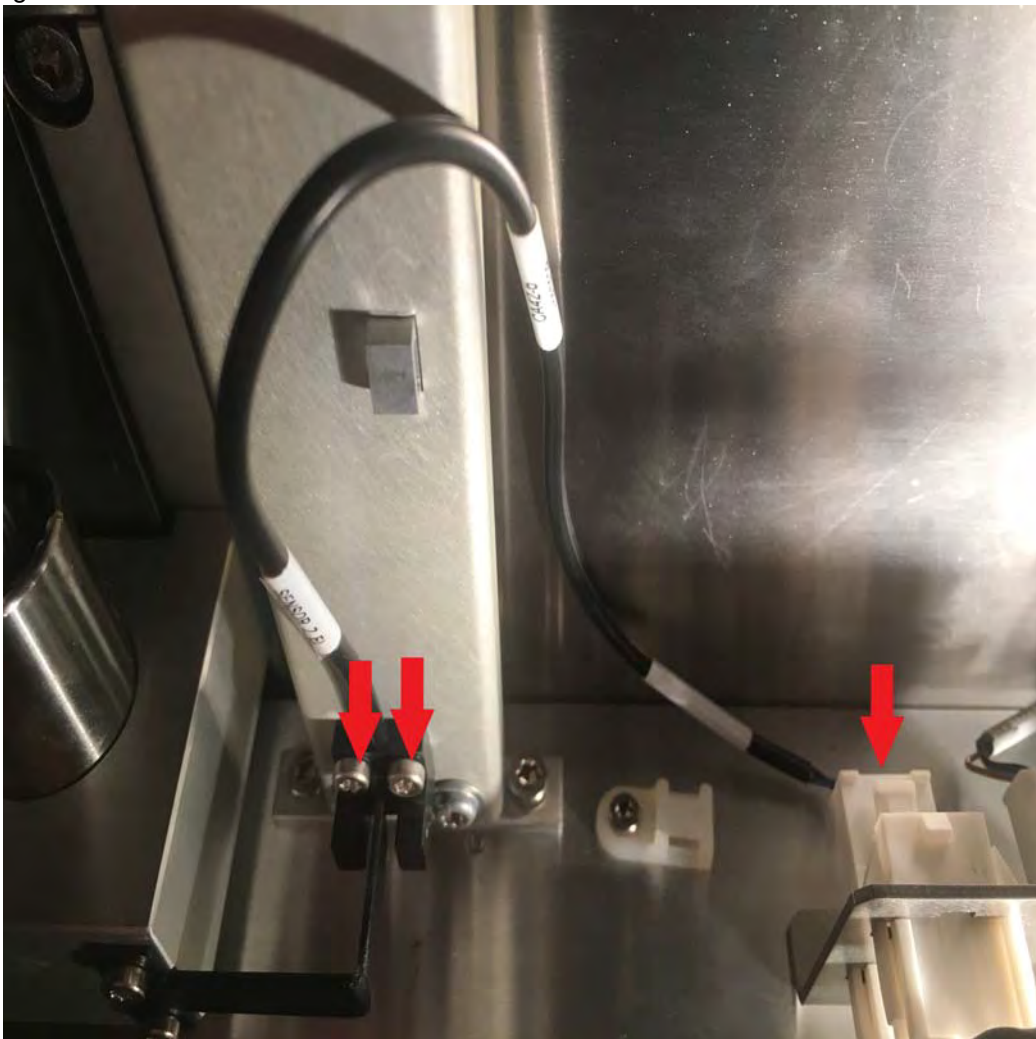
6.70 RP68 - Cable Motor Driver Slot Sensor Z-EL

Part Required	9027204 - Cable Driver Slot Sensor Z-EL
Tool(s) Required	T10, T20
Calibration Required	CP3 Pipette Z-axis Calibration
Testing Required	VP10 - Z Axis Movement Test
Replacement Time	7 minutes
Calibration/Testing Time	10 minutes
Total Repair Time	17, minutes

Table 6.70-1 RP68 Cable Motor Driver Slot Sensor Z-EL

6.70.1 RP68 - Cable Motor Driver Slot Sensor Z-EL - Removal Procedure

1. Remove the rear cover per [RP1 - Rear Cover - Removal Procedure](#).
2. Open the Z access door.
3. Remove the 2 (T10) screws from the slot sensor.
4. Unplug the cable.



Picture 6.70-1 Cable Motor Driver Slot Sensor Z-EL

6.70.2 RP68 - Cable Motor Driver Slot Sensor Z-EL - Replacement Procedure

1. Plug the slot sensor cable in.
2. Install the 2 (T10) screws in the sensor.
3. Close the Z access door and replace the 2 (T10) screws.
4. Install the rear cover per [RP1 - Rear Cover - Replacement Procedure](#).
5. Perform [CP3 Pipette Z-axis Calibration](#), [VP10 - Z Axis Movement Test](#).

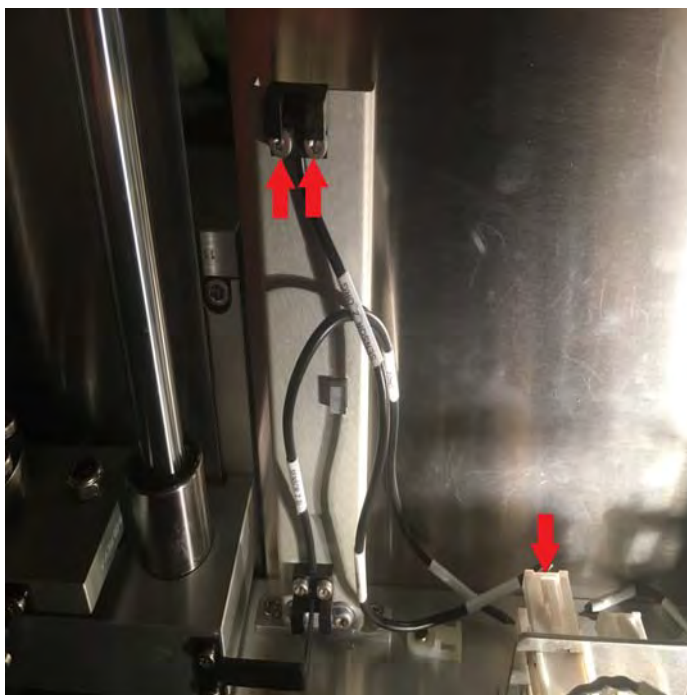
6.71 RP69 - Cable Motor Driver Slot Sensor Z-Org

Part Required	9027205 - Cable Motor Driver Slot Sensor Z-Org
Tool(s) Required	T10, T20
Calibration Required	CP3 Pipette Z-axis Calibration
Testing Required	VP10 - Z Axis Movement Test
Replacement Time	7 minutes
Calibration/Testing Time	10 minutes
Total Repair Time	17 minutes

Table 6.71-1 RP69 Cable Motor Driver Slot Sensor Z-Org

6.71.1 RP69 - Cable Motor Driver Slot Sensor Z-Org - Removal Procedure

1. Remove the rear cover per [RP1 - Rear Cover - Removal Procedure](#).
2. Open the Z access door.
3. Remove the 2 (T10) screws from the slot sensor.
4. Unplug the cable.



Picture 6.71-1 Cable Motor Driver Slot Sensor Z-Org

6.71.2 RP69 - Cable Motor Driver Slot Sensor Z-Org - Replacement Procedure

1. Plug the slot sensor cable in.
2. Install the 2 (T10) screws in the sensor.
3. Close the Z access door and replace the 2 (T10) screws.
4. Install the rear cover per [RP1 - Rear Cover - Replacement Procedure](#).
5. Perform [CP3 Pipette Z-axis Calibration](#), [VP10 - Z Axis Movement Test](#).

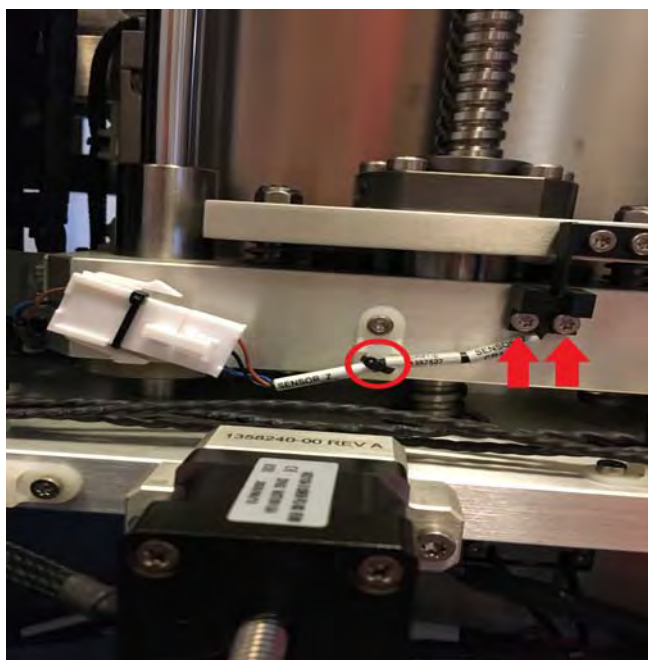
6.72 RP70 - Cable Motor Driver Slot Sensor Z Crash

Part Required	9027175 - Cable Motor Driver Slot Sensor Z Crash
Tool(s) Required	T10, T20
Calibration Required	CP3 Pipette Z-axis Calibration
Testing Required	VP10 - Z Axis Movement Test
Replacement Time	7 minutes
Calibration/Testing Time	10 minutes
Total Repair Time	17 minutes

Table 6.72-1 RP70 Cable Motor Driver Slot Sensor Z Crash

6.72.1 RP70 - Cable Motor Driver Slot Sensor Z Crash - Removal Procedure

1. Remove the rear cover per [RP1 - Rear Cover - Removal Procedure](#).
2. Open the Z access door.
3. Remove the 2 (T10) screws from the slot sensor.
4. Cut the small cable tie.
5. Unplug the cable.



Picture 6.72-1 Cable Motor Driver Slot Sensor Z-EL

6.72.2 RP70 - Cable Motor Driver Slot Sensor Z Crash - Replacement Procedure

1. Plug the slot sensor cable in.
2. Install the 2 (T10) screws in the sensor.
3. Replace the small cable tie.
4. Close the Z access door and replace the 2 (T10) screws.
5. Install the rear cover per [RP1 - Rear Cover - Replacement Procedure](#).
6. Perform [CP3 Pipette Z-axis Calibration](#), [VP10 - Z Axis Movement Test](#).

6.73 RP71 - USB Long Cable to Front Bottom Cover

Part Required	9027150 - USB Long Cable to Front Bottom Cover
Tool(s) Required	T10, T20
Calibration Required	NA
Testing Required	VP16 - USB Port Test
Replacement Time	8 minutes
Calibration/Testing Time	2 minutes
Total Repair Time	10 minutes

Table 6.73-1 RP71 USB Long Cable to Front Bottom Cover

6.73.1 RP71 - USB Long Cable to Front Bottom Cover - Removal Procedure

1. Remove the rear cover per [RP1 - Rear Cover - Removal Procedure](#).
2. Remove the main cover per [RP2 - Main Cover - Removal Procedure](#).
3. Remove the bottom front cover [RP6 - Front Bottom Cover - Replacement Procedure](#).
4. Remove the 2 (T10) screws that mount the screws to the front cover.



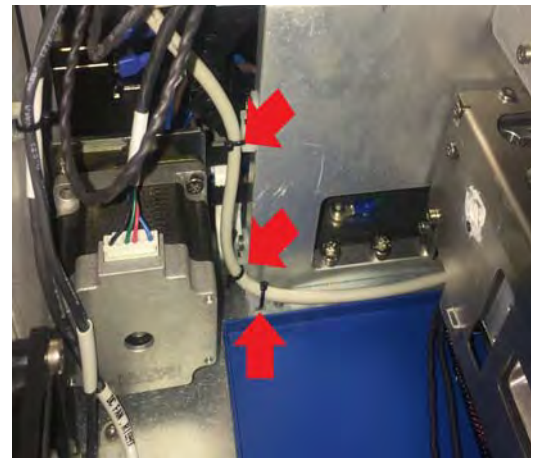
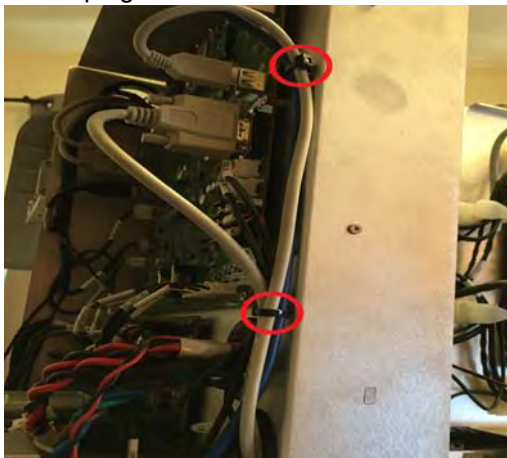
Picture 6.73-1 USB Mounting Screws

5. Remove the 2 (T10) USB cover screws on the base of the instrument.



Picture 6.73-2 Base USB Cable Cover

6. Unplug the USB cable from the CPU board and cut the cable ties.



Picture 6.73-3 Cut Cable Ties

7. Pull the USB cable through the front to remove it.

6.73.2 RP71 - USB Long Cable to Front Bottom Cover - Replacement Procedure

1. Route the USB cable through the front mounting hole.
2. Plug the USB cable in to the CPU board.
3. Install the 2 (T10) screws mounting the cable to the front bottom cover.
4. Install the 2 (T10) screws in the base cable cover.
5. Replace the 5 cable ties around the USB cable.
6. Install the front bottom cover per [RP6 - Front Bottom Cover - Replacement Procedure](#).
7. Install the main cover per [RP2 - Main Cover- Replacement Procedure](#).
8. Install the rear cover per [RP1 - Rear Cover - Replacement Procedure](#).
9. Perform [VP16 - USB Port Test](#).

6.74 RP72 - USB Short Cable to Top Front Cover

Part Required	9027151 - USB Short Cable to Top Front Cover
Tool(s) Required	T10, T20, T30
Calibration Required	NA
Testing Required	VP16 - USB Port Test
Replacement Time	23 minutes
Calibration/Testing Time	2 minutes
Total Repair Time	25 minutes

Table 6.74-1 RP72 USB Short Cable to Top Front Cover

6.74.1 RP72 - USB Short Cable to Top Front Cover - Removal Procedure

1. Remove the rear cover per [RP1 - Rear Cover - Removal Procedure](#).
2. Remove the main cover per [RP2 - Main Cover - Removal Procedure](#).
3. Remove the top front cover per [RP8 - Top Front Cover - Removal Procedure](#).
4. Remove the 2 (T10) screws holding the end of the USB cable and remove the cable.



Picture 6.74-1 USB Short Cable Screws

6.74.2 RP72 - USB Short Cable to Top Front Cover - Replacement Procedure

1. Install the 2 (T10) screws that attach the USB cable to the front top cover.
2. Replace the front top cover per [RP8 - Top Front Cover - Replacement Procedure](#).
3. Replace the main cover per [RP2 - Main Cover- Replacement Procedure](#).
4. Replace the rear cover per [RP1 - Rear Cover - Replacement Procedure](#).
5. Preform [VP16 - USB Port Test](#).

6.75 RP73 - Cable From CPU Board to Camera

Part Required	9027156 - Cable From CPU Board to Camera
Tool(s) Required	T10, T20
Calibration Required	NA
Testing Required	VP7 - Load Check Test
Replacement Time	33 minutes
Calibration/Testing Time	10 minutes
Total Repair Time	43 minutes

Table 6.75-1 RP73 Cable From CPU Board to Camera

6.75.1 RP73 - Cable From CPU Board to Camera - Removal Procedure

1. Remove the rear cover per [RP1 - Rear Cover - Removal Procedure](#).
2. Remove the main cover per [RP2 - Main Cover - Removal Procedure](#).
3. Remove the camera enclosure per [RP15 - Camera Enclosure - Removal Procedure](#).
4. Remove the X E-chain per [RP62 - E-Chain X Axis - Removal Procedure](#).
5. Unplug the cable from the camera and pull it through the X E-chain.



Picture 6.75-1 Unplug Camera Cable



6. Unplug the camera cable from the CPU board and cut the cable ties holding it in place.



Picture 6.75-2 Camera Cable to CPU

6.75.2 RP73 - Cable From CPU Board to Camera - Replacement Procedure

1. Insert the camera cable through the X E-chain.
2. Install the X E-chain per [RP62 - E-Chain X Axis - Replacement Procedure](#).
3. Plug the cable in to the camera.
4. Plug the cable in to the CPU board and install cable ties.
5. Replace the camera enclosure per [RP15 - Camera Enclosure- Replacement Procedure](#).
6. Replace the main cover per [RP2 - Main Cover- Replacement Procedure](#).
7. Replace the rear cover per [RP1 - Rear Cover - Replacement Procedure](#).
8. Perform [VP7 - Load Check Test](#).

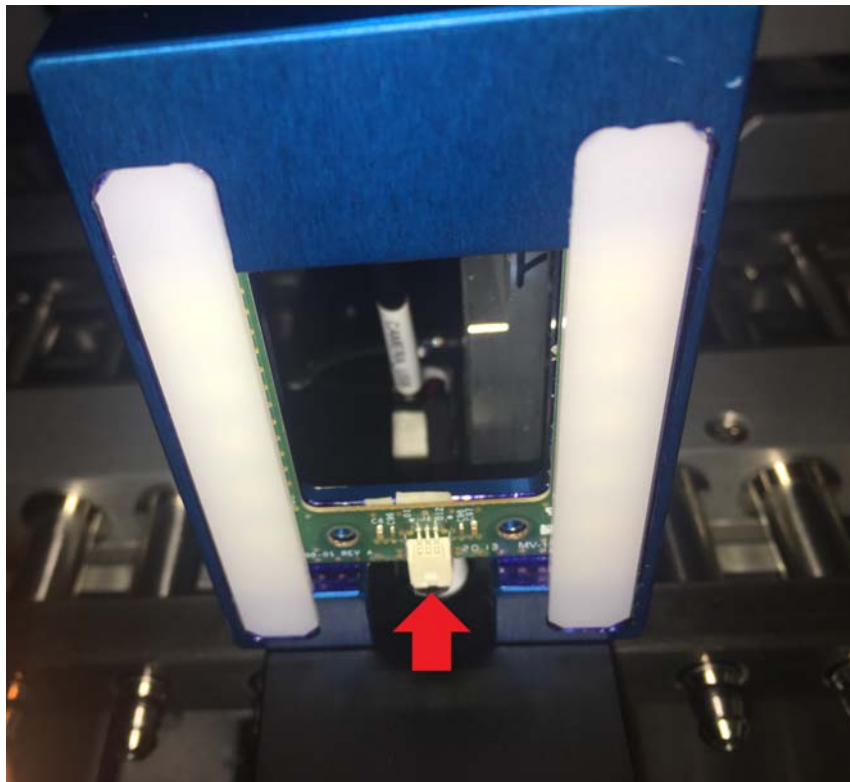
6.76 RP74 - Camera Cable Adapter-LED Board

Part Required	9027155 - Camera Cable Adapter-LED Board
Tool(s) Required	T10, T20
Calibration Required	NA
Testing Required	VP7 - Load Check Test
Replacement Time	33 minutes
Calibration/Testing Time	10 minutes
Total Repair Time	43 minutes

Table 6.76-1 RP74 Camera Cable Adapter-LED Board

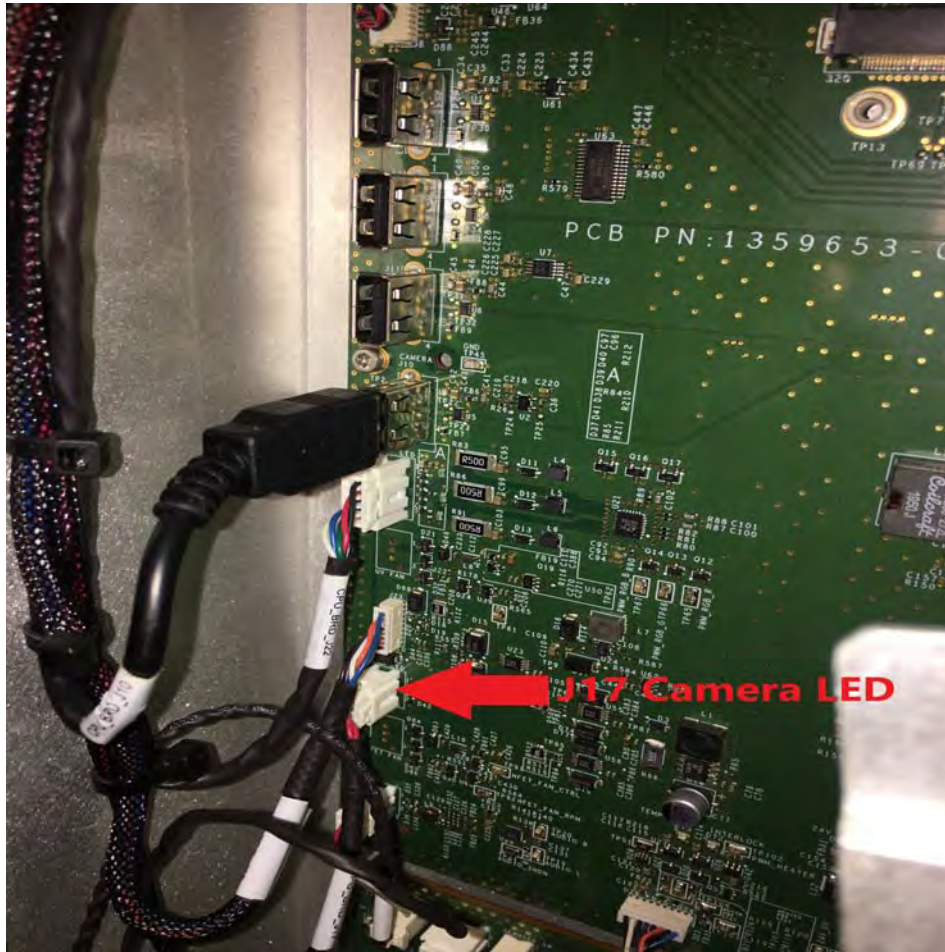
6.76.1 RP74 - Camera Cable Adapter-LED - Removal Procedure

1. Remove the rear cover per [RP1 - Rear Cover - Removal Procedure](#).
2. Remove the main cover per [RP2 - Main Cover - Removal Procedure](#).
3. Remove the X E-chain per [RP62 - E-Chain X Axis - Removal Procedure](#).
4. Unplug the LED cable connector from the board.



Picture 6.76-1 LED Cable Connector

5. Unplug the J17 camera LED connector and cut the cable ties.



Picture 6.76-2 Camera LED Cable J17

6. Pull the camera LED cable through the X E-chain.

6.76.2 RP74 - Camera Cable LED Adapter-LED - Replacement Procedure

1. Insert the camera cable through the X E-chain.
2. Install the X E-chain per [RP62 - E-Chain X Axis - Replacement Procedure](#).
3. Plug the cable in to the camera LED board.
4. Plug the cable in to J17 on the CPU board and install cable ties.
5. Replace the main cover per [RP2 - Main Cover- Replacement Procedure](#).
6. Replace the rear cover per [RP1 - Rear Cover - Replacement Procedure](#).
7. Perform [VP7 - Load Check Test](#).

6.77 RP75 - Cable From CPU Board to UV LED Board

Part Required	9027157 - Cable From CPU Board to UV LED Board
Tool(s) Required	T10, T20
Calibration Required	TBD
Testing Required	VP2 - UV Light Test
Replacement Time	33 minutes
Calibration/Testing Time	2 minutes
Total Repair Time	35 minutes

Table 6.77-1 RP75 Cable From CPU Board to UV LED Board

6.77.1 RP75 - Cable From CPU Board to UV LED Board - Removal Procedure

1. Remove the rear cover per [RP1 - Rear Cover - Removal Procedure](#).
2. Remove the main cover per [RP2 - Main Cover - Removal Procedure](#).
3. Remove the X E-chain per [RP62 - E-Chain X Axis - Removal Procedure](#).
4. Unplug the UV LED cable.



Picture 6.77-1 Unplug UV LED Cable

5. Unplug the UV LED cable J23 from the CPU board and remove any cable ties.



Picture 6.77-2 UV LED Cable to CPU Board J23

6. Pull the cable through the X E-chain to remove it.

6.77.2 RP75 - Cable From CPU Board to UV LED Board - Replacement Procedure

1. Insert the UV LED cable in the X axis E-chain per [RP62 - E-Chain X Axis - Replacement Procedure](#).
2. Plug the cable in to the UV LED board.
3. Plug the cable in to the CPU board J23 connector and replace the cable ties.
4. Install the main cover per [RP2 - Main Cover- Replacement Procedure](#).
5. Install the rear cover per [RP1 - Rear Cover - Replacement Procedure](#).
6. Perform [VP2 - UV Light Test](#).

6.78 RP76 - Cable CPU Board-Lock Sensor Left

Part Required	9027200 - Cable CPU Board-Lock Sensor Left
Tool(s) Required	T10, T20
Calibration Required	NA
Testing Required	VP6 - Hood Lock Test
Replacement Time	12 minutes
Calibration/Testing Time	2 minutes
Total Repair Time	14 minutes

Table 6.78-1 RP76 Cable CPU Board Lock Sensor Left

6.78.1 RP76 - Cable CPU Board Lock Sensor Left - Removal Procedure

1. Remove the rear cover per [RP1 - Rear Cover - Removal Procedure](#).
2. Remove the main cover per [RP2 - Main Cover - Removal Procedure](#).
3. Open the rear Z access panel.
4. Unplug the white connector.
5. Press in on both sides of the connector and put it from the bracket.



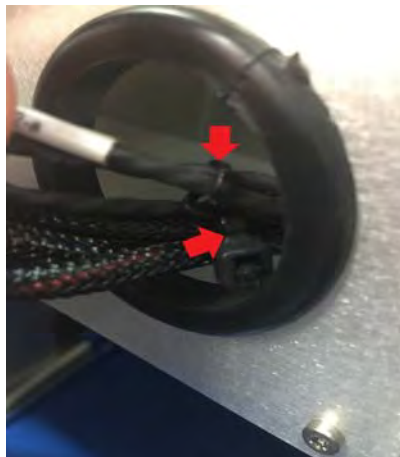
Picture 6.78-1 CPU Cable Lock Sensor Left

6. Cut and remove the 3 cable ties.



Picture 6.78-2 Cable Ties

7. Cut and remove the 2 cable ties to the left of the access door.



Picture 6.78-3 Left Side Cable Ties

8. Pull the cable through both holes in the back panels.



Picture 6.78-4 Cable Removal

9. Cut the small cable tie.
10. Unplug the J21 cable connector and remove the cable.



Picture 6.78-5 J21 CPU Board Left Hood Lock Sensor

6.78.2 RP76 - Cable CPU Board Lock Sensor Left - Replacement Procedure

1. Plug the cable in position J21 on the CPU board.
2. Route the larger end of the cable back through the 2 holes in the panel of the instrument.

3. Insert the cable connector back in the bracket and plug in the male side of the connector.



Picture 6.78-6 Cable Connector

4. Install the 6 cable ties that were removed.
5. Close the Z access door and replace the 2 (T10) screws in the door.
6. Install the main cover per [RP2 - Main Cover- Replacement Procedure](#).
7. Install the rear cover per [RP1 - Rear Cover - Replacement Procedure](#).
8. Perform [VP6 - Hood Lock Test](#).

6.79 RP77 - Cable Adapter Sensor ZEL

Part Required	9027209 - Cable Adapter Sensor ZEL
Tool(s) Required	T10, T20
Calibration Required	NA
Testing Required	VP12 - System Initialization Test
Replacement Time	12 minutes
Calibration/Testing Time	2 minutes
Total Repair Time	14 minutes

Table 6.79-1 RP77 Cable Adapter Sensor ZEL

6.79.1 RP77 - Cable Adapter Sensor ZEL - Removal Procedure

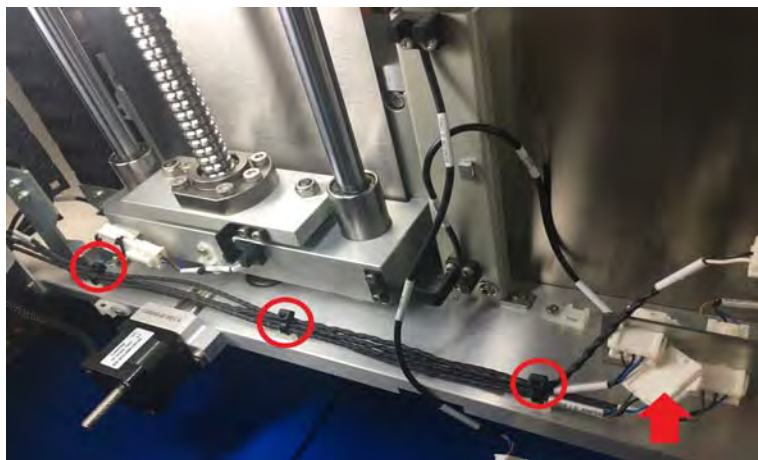
1. Remove the rear cover per [RP1 - Rear Cover - Removal Procedure](#).
2. Remove the main cover per [RP2 - Main Cover - Removal Procedure](#).
3. Open the rear Z access panel.
4. Unplug the white bottom connector.

5. Press in on both sides of the connector and put it from the bracket.



Picture 6.79-1 Cable Adapter Sensor ZEL

6. Cut and remove the 3 cable ties.



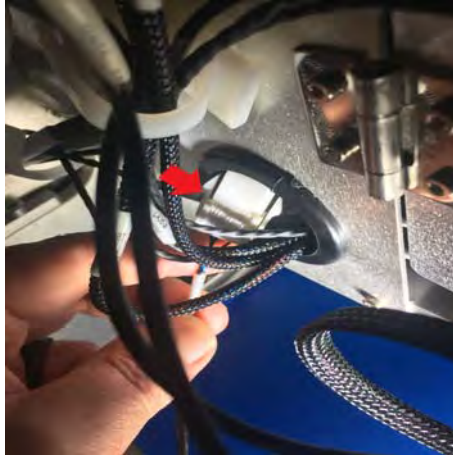
Picture 6.79-2 Cable Ties

7. Cut and remove the 2 cable ties to the left of the access door.



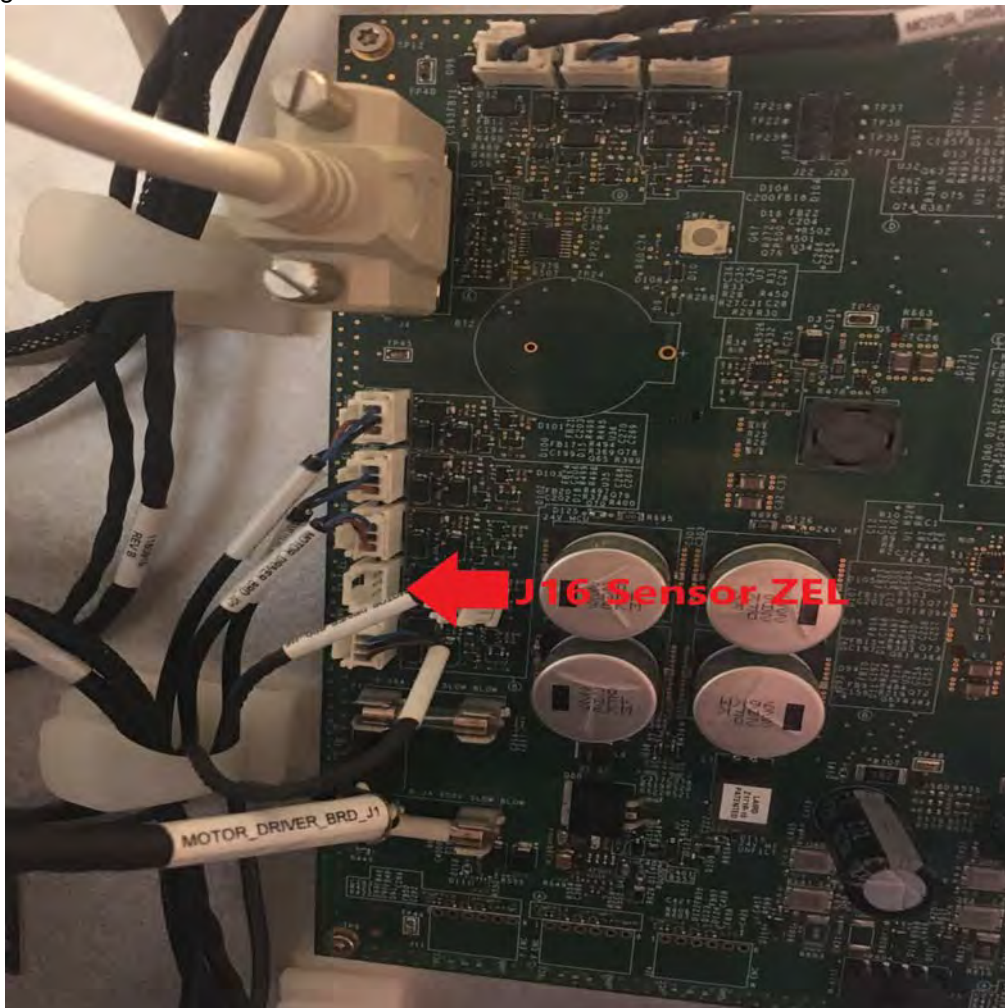
Picture 6.79-3 Left Side Cable Ties

8. Pull the cable through the hole in the back panel.



Picture 6.79-4 Cable Removal

9. Unplug the J16 cable connector on the motor board and remove the cable.



Picture 6.79-5 J16 Connector on Motor Board

6.79.2 RP77 - Cable Adapter Sensor ZEL - Replacement Procedure

1. Plug the cable in position J16 on the motor board.
2. Route the larger end of the cable back through the hole in the panel of the instrument.

3. Insert the cable connector back in the bracket and plug in the male side of the connector.



Picture 6.79-6 Cable Connector

4. Install the 5 cable ties that were removed.
5. Close the Z access door and replace the 2 (T10) screws in the door.
6. Install the main cover per [RP2 - Main Cover- Replacement Procedure](#).
7. Install the rear cover per [RP1 - Rear Cover - Replacement Procedure](#).
8. Perform [VP12 - System Initialization Test](#).

6.80 RP78 - Cable Adapter Z Org Sensor

Part Required	9027210 Cable Adapter Z Org Sensor
Tool(s) Required	T10, T20
Calibration Required	NA
Testing Required	VP12 - System Initialization Test
Replacement Time	12 minutes
Calibration/Testing Time	2 minutes
Total Repair Time	14 minutes

Table 6.80-1 RP78 Cable Sensor Z Org Adapter

6.80.1 RP78 - Cable Adapter Z Org Sensor - Removal Procedure

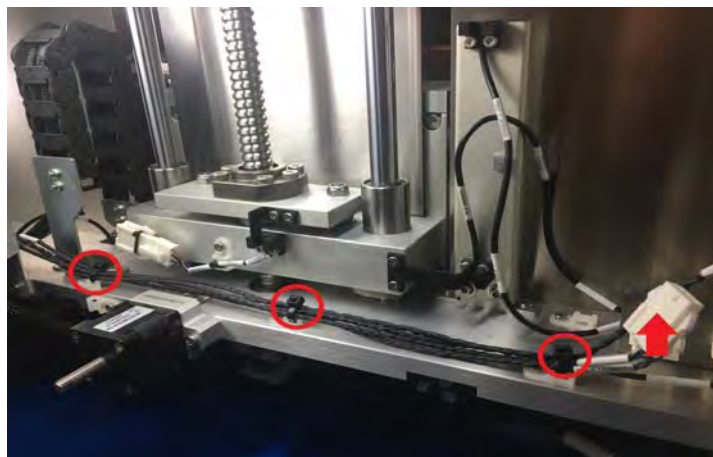
1. Remove the rear cover per [RP1 - Rear Cover - Removal Procedure](#).
2. Remove the main cover per [RP2 - Main Cover - Removal Procedure](#).
3. Open the rear Z access panel.
4. Unplug the white connector.

5. Press in on both sides of the connector and put it from the bracket.



Picture 6.80-1 Z Org Sensor Adapter Cable

6. Cut and remove the 3 cable ties.



Picture 6.80-2 Cable Ties

7. Cut and remove the 2 cable ties to the left of the access door.



Picture 6.80-3 Left Side Cable Ties

8. Pull the cable through the hole in the back panel.



Picture 6.80-4 Cable Removal

9. Unplug the J14 cable connector from the motor board and remove the cable.



Picture 6.80-5 J14 Motor Board Z Org Sensor Adapter Cable

6.80.2 RP78 - Cable Adapter Z Org Sensor - Replacement Procedure

1. Plug the cable in position J14 on the motor board.

2. Route the larger end of the cable back through the hole in the panel of the instrument.



Picture 6.80-6 Cable Connector

3. Insert the cable connector back in the bracket and plug in the male side of the connector.
4. Install the 5 cable ties that were removed.
5. Close the Z access door and replace the 2 (T10) screws in the door.
6. Install the main cover per [RP2 - Main Cover- Replacement Procedure](#).
7. Install the rear cover per [RP1 - Rear Cover - Replacement Procedure](#).
8. Perform [VP12 - System Initialization Test](#).

6.81 RP79 - Cable CPU Board-Lock Sensor Right

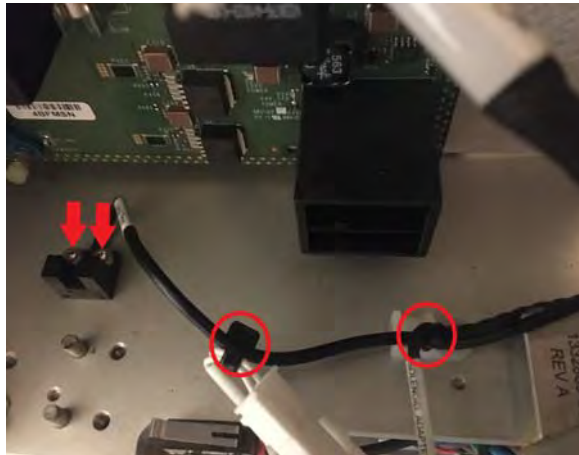
Part Required	9027201 - Cable CPU Board-Lock Sensor Right
Tool(s) Required	T10, T20
Calibration Required	NA
Testing Required	VP6 - Hood Lock Test
Replacement Time	9 minutes
Calibration/Testing Time	2 minutes
Total Repair Time	11 minutes

Table 6.81-1 RP79 Cable CPU Board-Lock Sensor Right

6.81.1 RP79 - Cable CPU Board-Lock Sensor Right - Removal Procedure

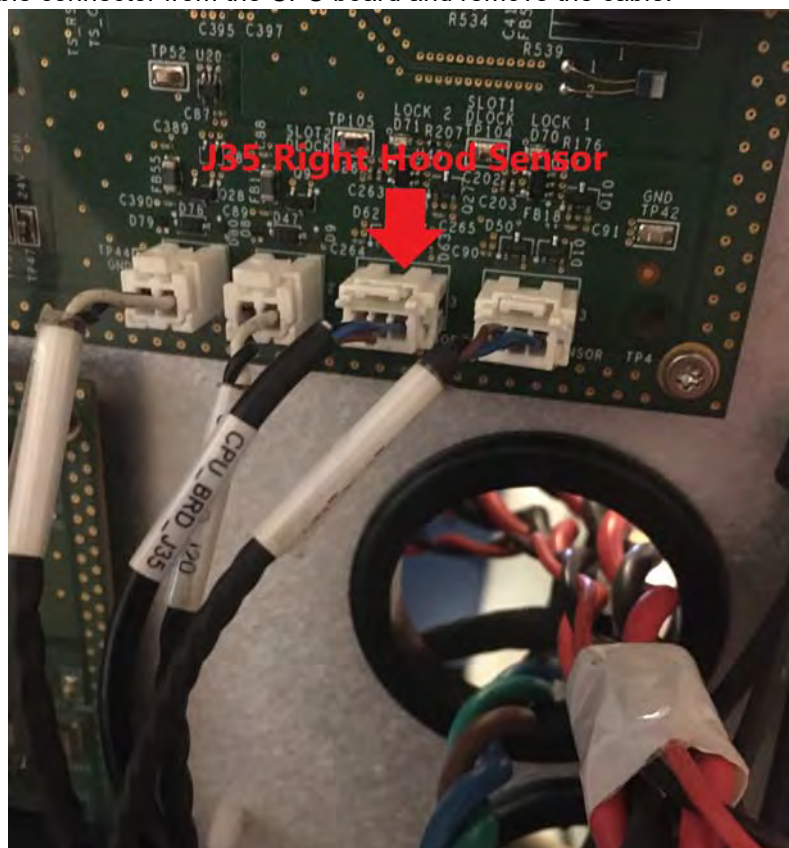
1. Remove the rear cover per [RP1 - Rear Cover - Removal Procedure](#).
2. Remove the main cover per [RP2 - Main Cover - Removal Procedure](#).
3. Remove the right door lock solenoid per [RP13 - Right Hood Solenoid - Removal Procedure](#).

4. Remove the 2 (T10) screws from the slot sensor and cut the 2 cable ties.



Picture 6.81-1 Right Hood Sensor Removal

5. Unplug J35 cable connector from the CPU board and remove the cable.



Picture 6.81-2 J35 Connector

6.81.2 RP79 - Cable CPU Board-Lock Sensor - Replacement Procedure

1. Insert the cable connector in position J35 on the CPU board.
2. Replace the 2 (T10) screws and replace the 2 small cable ties.
3. Install the right hood lock solenoid per [RP13 - Right Hood Solenoid- Replacement Procedure](#).
4. Perform [VP6 - Hood Lock Test](#).

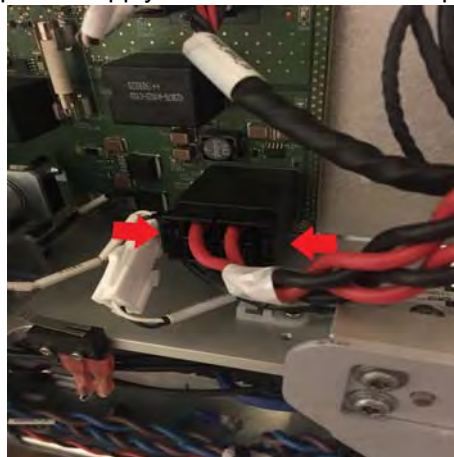
6.82 RP80 - Cable From Power Board to Power Supply

Part Required	9027206 - Cable From Power Board to Power Supply
Tool(s) Required	T10, T20
Calibration Required	NA
Testing Required	VP5 - Temperature Test
Replacement Time	10 minutes
Calibration/Testing Time	23minutes
Total Repair Time	33 minutes

Table 6.82-1 RP80 Cable From Power Board to Power Supply

6.82.1 RP80 - Cable From Power Board to Power Supply - Removal Procedure

1. Remove the rear cover per [RP1 - Rear Cover - Removal Procedure](#).
2. Remove the main cover per [RP1 - Rear Cover - Removal Procedure](#).
3. Press in on both sides of the power supply cable connector and unplug it from the power board.



Picture 6.82-1 Power Cable Connector J8

4. Slide the small clear plastic cover off the screw connectors.
5. Remove the positive and negative cables from the power supply with a Phillips screwdriver.



Picture 6.82-2 Positive & Negative Power Supply Cables

6.82.2 RP80 - Cable From Power Board to Power Supply - Replacement Procedure

1. Attach the positive cable on the right side of the power supply.
2. Attach the negative cable on the left side of the power supply.
3. Pull the connector through the hole and plug the connector in position J8.
4. Replace the main cover per [RP2 - Main Cover- Replacement Procedure](#).
5. Replace the rear cover per [RP1 - Rear Cover - Replacement Procedure](#).
6. Perform [VP5 - Temperature Test](#).

6.83 RP81 - Cable Sensor Z Crash Adapter Sensor

Part Required	9027208 - Cable Sensor Z Crash Adapter Sensor
Tool(s) Required	T10, T20
Calibration Required	CP3 Pipette Z-axis Calibration
Testing Required	VP10 - Z Axis Movement Test
Replacement Time	32 minutes
Calibration/Testing Time	10 minutes
Total Repair Time	42 minutes

Table 6.83-1 RP81 Cable sensor Z Crash Adapter Sensor

6.83.1 RP81 - Cable sensor Z Crash Adapter Sensor - Removal Procedure

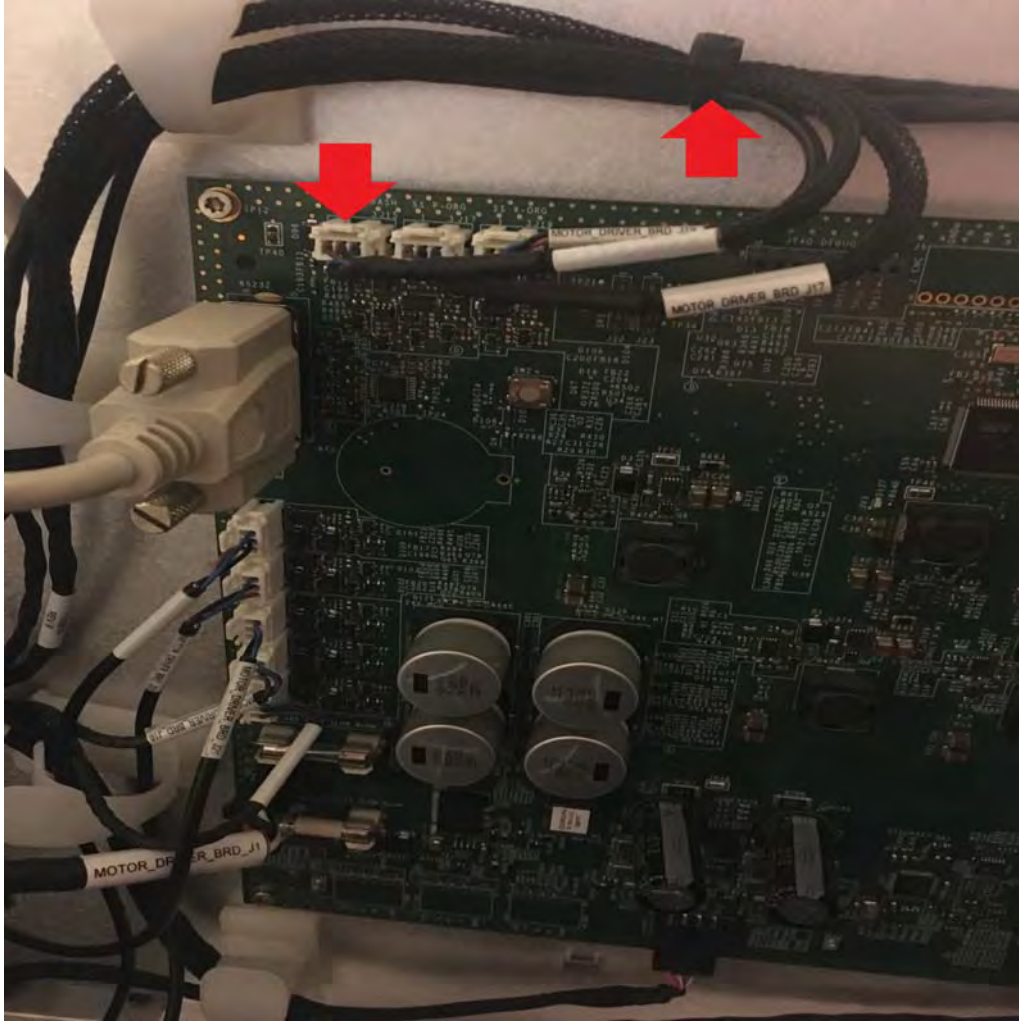
1. Remove the rear cover per [RP1 - Rear Cover - Removal Procedure](#).
2. Remove the main cover per [RP2 - Main Cover - Removal Procedure](#).
3. Unplug the cable adapter and cut the small cable tie.



Picture 6.83-1 Z Crash Cable Adapter Connection

4. Remove the Z E-chain per [RP64 - Z E-Chain - Removal Procedure](#).

5. Unplug connector J15 from the motor board and remove the cable tie.



Picture 6.83-2 Z Crash Adapter Cable to Motor Board

6. Pull the cable through the Z E-chain and remove it.

6.83.2 RP81 - Cable sensor Z Crash Adapter Sensor - Replacement Procedure

1. Run the cable through the Z E-chain.
2. Replace the Z E-chain per [RP64 - Z E-Chain - Replacement Procedure](#).
3. Plug the cable in to J15 on the motor board.
4. Plug the connector in to the Z crash cable slot sensor connector and add the cable tie.
5. Perform [CP3 Pipette Z-axis Calibration](#), [VP10 - Z Axis Movement Test](#).

6.84 RP82 - Cable Adapter to P-Org Slot Sensor

Part Required	9027207 - Cable Adapter to P-Org Slot Sensor
Tool(s) Required	T10, T20
Calibration Required	NA
Testing Required	VP12 - System Initialization Test
Replacement Time	4 minutes
Calibration/Testing Time	2 minutes
Total Repair Time	6 minutes

Table 6.84-1 RP82 Cable Adapter to P-Org Slot Sensor

6.84.1 RP82 - Cable Adapter to P-Org Slot Sensor - Removal Procedure

1. Power off the instrument.
2. Open the hood.
3. Unplug the connector and remove the cable tie.
4. Remove the 2 (T10) screws holding the slot sensor in position.



Picture 6.84-1 P-Org Adapter Connector

6.84.2 RP82 - Cable Adapter to P-Org Slot Sensor - Replacement Procedure

1. Connect the cable connector.
2. Install the 2 (T10) screws that hold the sensor in position.
3. Replace the cable tie.
4. Power on the instrument and preform [VP12 - System Initialization Test](#).

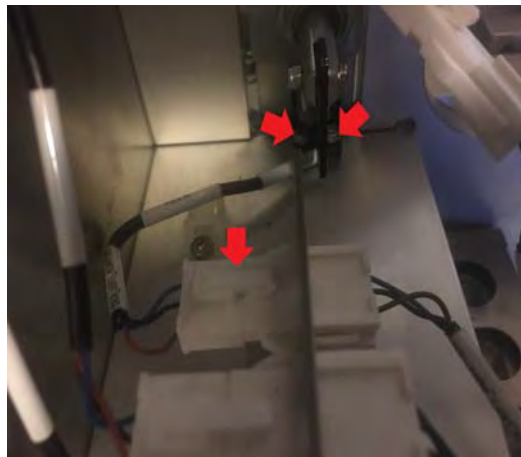
6.85 RP83 - Cable Lock Sensor Left Adapter

Part Required	9027202 - Cable Lock Sensor Left Adapter
Tool(s) Required	T10, T20
Calibration Required	NA
Testing Required	VP6 - Hood Lock Test
Replacement Time	5 minutes
Calibration/Testing Time	2 minutes
Total Repair Time	7 minutes

Table 6.85-1 RP83 Cable Lock Sensor Left Adapter

6.85.1 RP83 - Cable Lock Sensor Left Adapter - Removal Procedure

1. Remove the rear cover per [RP1 - Rear Cover - Removal Procedure](#).
2. Open the rear access panel by removing the 2 (T10) screws.
3. Unplug the cable from the connector.
4. Remove the 2 (T10) screws on both sides of the sensor and remove the cable.



Picture 6.85-1 Left Lock Sensor Cable Adapter

6.85.2 RP83 - Cable Lock Sensor Left Adapter - Replacement Procedure

1. Plug the connector in.
2. Replace the 2 (T10) screws that hold the sensor in position.
3. Close the rear access panel and replace the 2 (T10) screws.
4. Replace the rear cover per [RP1 - Rear Cover - Replacement Procedure](#).
5. Perform [VP6 - Hood Lock Test](#).

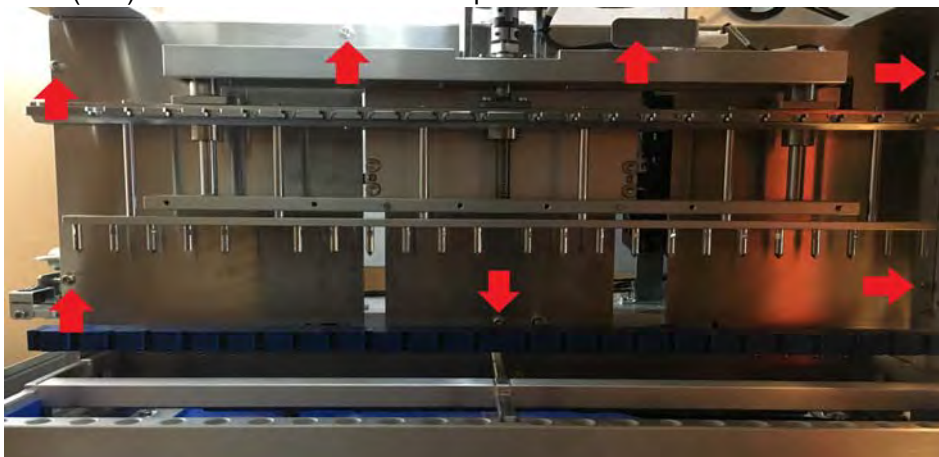
6.86 RP84 - Cable Motor Driver-Slot Sensor M Org

Part Required	9027203 - Cable Motor Driver-Slot Sensor M Org
Tool(s) Required	T10, T20, T30
Calibration Required	CP1 Camera Calibration, CP2 Camera Exposure Calibration, CP6 Magnet Calibration,
Testing Required	VP11 - M Axis Movement Test, VP7 - Load Check Test
Replacement Time	50 minutes
Calibration/Testing Time	31 minutes
Total Repair Time	1 hour 21 minutes

Table 6.86-1 RP84 Cable Motor Driver-Slot Sensor M Org

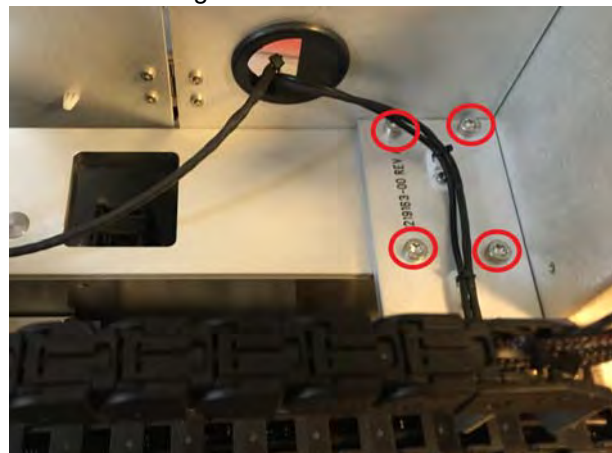
6.86.1 RP84 - Cable Motor Driver-Slot Sensor M Org - Removal Procedure

1. Remove the rear cover per [RP1 - Rear Cover - Removal Procedure](#).
2. Remove the main cover per [RP2 - Main Cover - Removal Procedure](#).
3. Remove the top front cover per [RP8 - Top Front Cover - Removal Procedure](#).
4. Remove the magnet assembly [RP11 - Magnet Assembly - Removal Procedure](#).
5. Remove the 7 (T20) screws that attach the back plate.



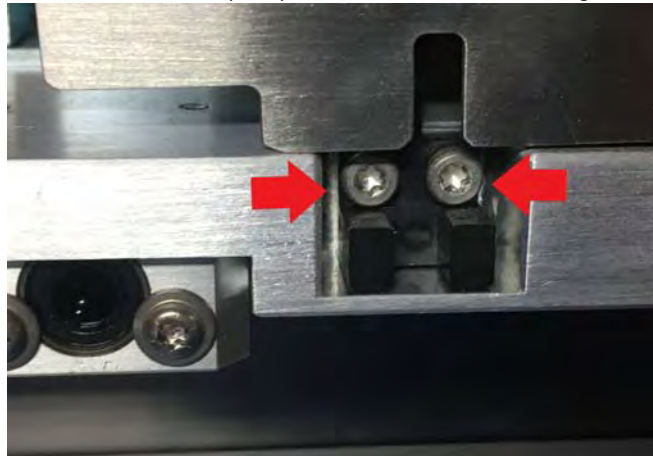
Picture 6.86-1 Back Plate Screws

6. Remove the 8 (T30) top mounting bracket screws on the left and right.



Picture 6.86-2 Mounting Bracket Screw Removal

7. Lift up on the brackets to make space to move the back plate.
8. With the back plate lifted, remove the 2 (T10) screws that hold the magnet sensor in position.

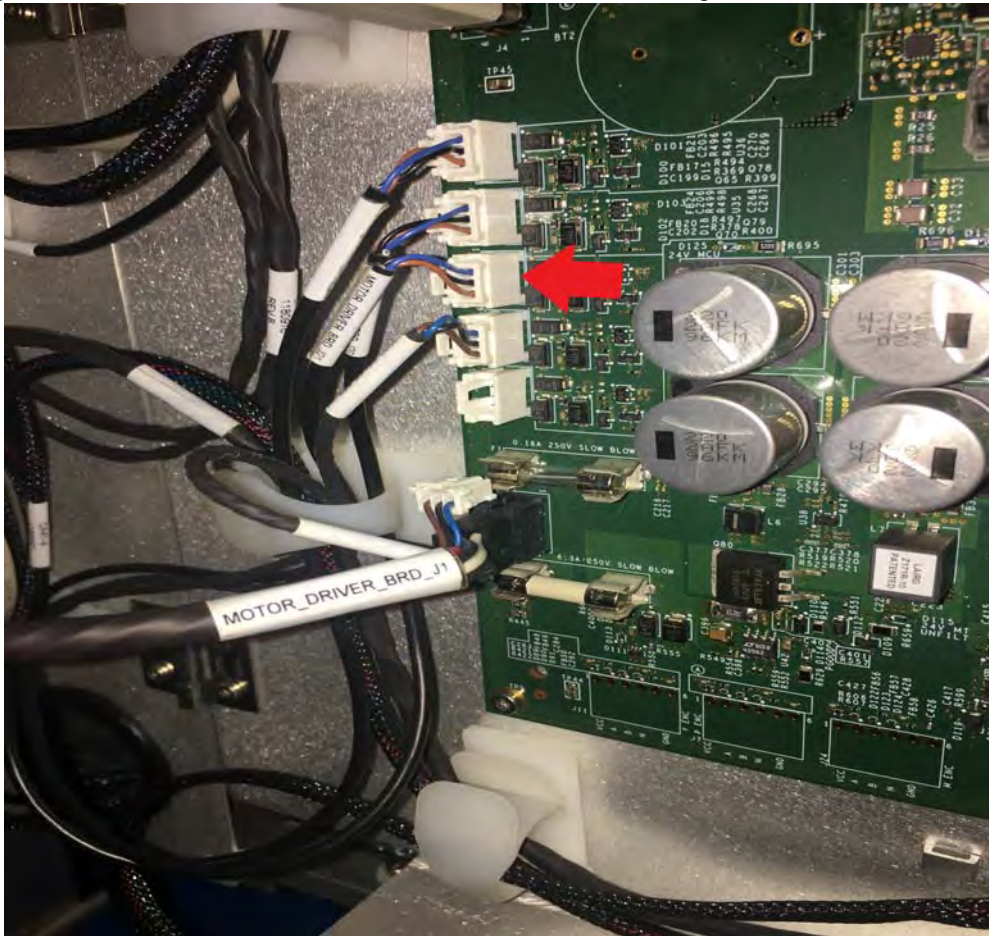


Picture 6.86-3 M Sensor Cable Slot Screws

9. Remove the 2 (T10) screws from the back panel and open it.
10. Cut the 2 cable ties that come through the hole on back left.



11. Unplug the J21 connector and remove the sensor and cable through the hole on the back left.



Picture 6.86-4 J21

6.86.2 RP84 - Cable Motor Driver-Slot Sensor M Org - Replacement Procedure

1. Feed the sensor end of the cable through the hole on the left of the motor board.
2. Insert the sensor in position and install the 2 (T10) screws.
3. Plug the connector of the cable in the J21 position on the motor board.
4. Insert the 7 (T20) screws in the back plate.
5. Replace the magnet assembly per [RP11 - Magnet Assembly- Replacement Procedure](#).
6. Replace the 8 (T30) screws in the top two bracket sides.
7. Replace the top front cover per [RP8 - Top Front Cover - Replacement Procedure](#).
8. Replace the main cover per [RP2 - Main Cover- Replacement Procedure](#).
9. Replace the rear cover per [RP1 - Rear Cover - Replacement Procedure](#).
10. Perform [CP1 Camera Calibration](#), [CP2 Camera Exposure Calibration](#), [CP6 Magnet Calibration](#), [VP11 - M Axis Movement Test](#), [VP7 - Load Check Test](#).

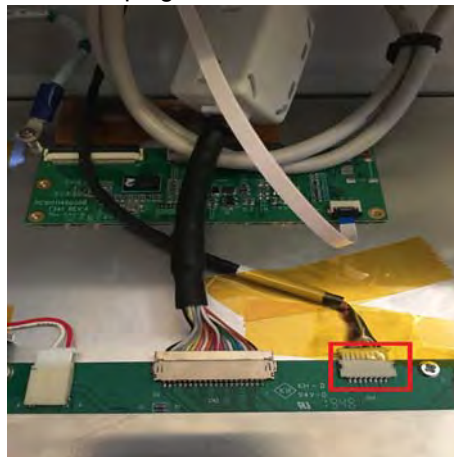
6.87 RP85 - Cable CPU Board-Display Backlight.

Part Required	9027198 - Cable CPU Board-Display Backlight
Tool(s) Required	T10, T20, T30
Calibration Required	NA
Testing Required	VP12 - System Initialization Test
Replacement Time	27 minutes
Calibration/Testing Time	3 minutes
Total Repair Time	30 minutes

Table 6.87-1 RP85 Cable CPU Board-Display Backlight

6.87.1 RP85 - Cable CPU Board-Display Backlight - Removal Procedure

1. Remove the rear cover per [RP1 - Rear Cover - Removal Procedure](#).
2. Remove the main cover per [RP2 - Main Cover - Removal Procedure](#).
3. Remove the top front cover per [RP8 - Top Front Cover - Removal Procedure](#).
4. Remove the tape from the cable and unplug it from the back of the LCD touchscreen.



Picture 6.87-1 Cable CPU Board-Display Backlight

6.87.2 RP85 - Cable CPU Board-Display Backlight - Replacement Procedure

1. Plug the cable in to the back of the LCD touchscreen and tape it down in position.
2. Replace the top front cover per [RP8 - Top Front Cover - Replacement Procedure](#).
3. Replace the main cover per [RP2 - Main Cover- Replacement Procedure](#).
4. Replace the rear cover per [RP1 - Rear Cover - Replacement Procedure](#).
5. Perform [VP12 - System Initialization Test](#).

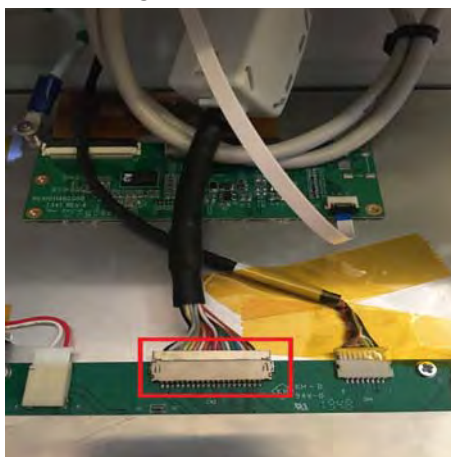
6.88 RP86 - Cable CPU Board-Display LVDS Signal

Part Required	9027199 - Cable CPU Board-Display LVDS Signal
Tool(s) Required	T10, T20, T30
Calibration Required	NA
Testing Required	VP12 - System Initialization Test
Replacement Time	27 minutes
Calibration/Testing Time	3 minutes
Total Repair Time	30 minutes

Table 6.88-1 RP86 Cable CPU Board-Display LVDS Signal

6.88.1 RP86 - Cable CPU Board-Display LVDS Signal - Removal Procedure

1. Remove the rear cover per [RP1 - Rear Cover - Removal Procedure](#).
2. Remove the main cover per [RP2 - Main Cover - Removal Procedure](#).
3. Remove the top front cover per [RP8 - Top Front Cover - Removal Procedure](#).
4. Remove the cable from the back of the LCD.



Picture 6.88-1 Cable CPU Board-Display LVDS Signal

6.88.2 RP86 - Cable CPU Board-Display LVDS Signal - Replacement Procedure

1. Plug the cable in to the connector at the back of the LCD screen.
2. Replace the top front cover per [RP8 - Top Front Cover - Replacement Procedure](#).
3. Replace the main cover per [RP2 - Main Cover- Replacement Procedure](#).
4. Replace the rear cover per [RP1 - Rear Cover - Replacement Procedure](#).
5. Perform [VP12 - System Initialization Test](#).

6.89 RP87 - LED Strip

Part Required	9027214 - LED Strip
Tool(s) Required	T20
Calibration Required	NA
Testing Required	VP1 - LED Strip Test
Replacement Time	18 minutes
Calibration/Testing Time	2 minutes
Total Repair Time	20 minutes

Table 6.89-1 RP87 LED Strip

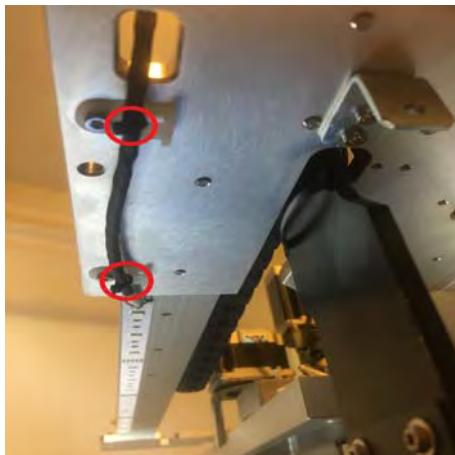
6.89.1 RP87 - LED Strip - Removal Procedure

1. Remove the rear cover per [RP1 - Rear Cover - Removal Procedure](#).
2. Remove the main cover per [RP2 - Main Cover - Removal Procedure](#).
3. The LED Strip has a adhesive on the backing and will need to be scraped to remove the LED strip.

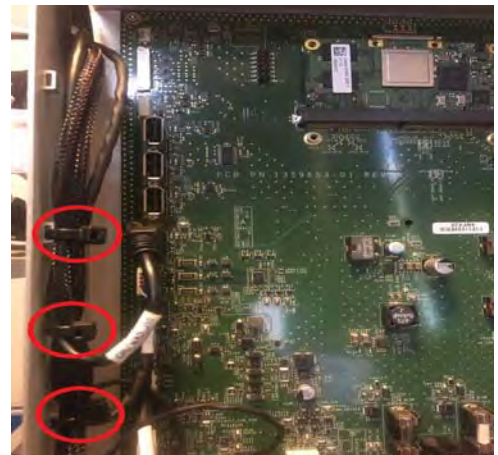


Picture 6.89-1 LED Strip

4. Remove the cable ties that hold the LED cable in position.



Picture 6.89-2 LED Cable Ties

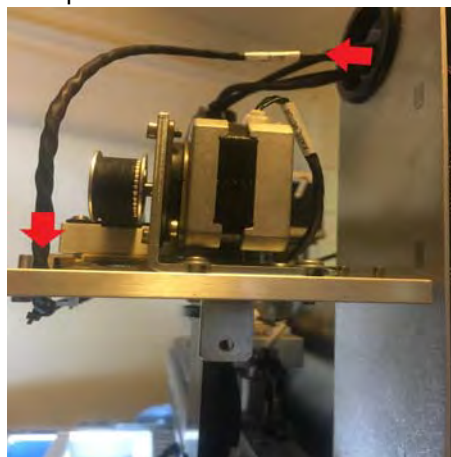


5. Unplug the J22 connector from the CPU board.



Picture 6.89-3 LED J22 Connector

6. Pull the LED cable through the top hole in the CPU board cover and through the bracket hole of the LED and remove the cable and LED strip.



Picture 6.89-4 LED Cable Removal

6.89.2 RP87 - LED Strip - Replacement Procedure

1. Place the new LED strip in position under the X rail bracket.
2. Feed the cable through the hole at the end of the X rail bracket and through the hole on the top left of the CPU board housing.
3. Plug the cable connector in position J22 of the CPU board.
4. Replace all cable ties that have been removed.
5. Replace the main cover per [RP2 - Main Cover- Replacement Procedure](#).
6. Replace the rear cover per [RP1 - Rear Cover - Replacement Procedure](#).

7. Perform [VP1 - LED Strip Test](#).

6.90 RP88 - Cable CPU-Thermocouple Adapter

Part Required	9027170 - Cable CPU-Thermocouple Adapter
Tool(s) Required	T10, T20
Calibration Required	NA
Testing Required	VP5 - Temperature Test
Replacement Time	40 minutes
Calibration/Testing Time	20 minutes
Total Repair Time	1 hour

Table 6.90-1 RP88 Cable CPU-Thermocouple Adapter

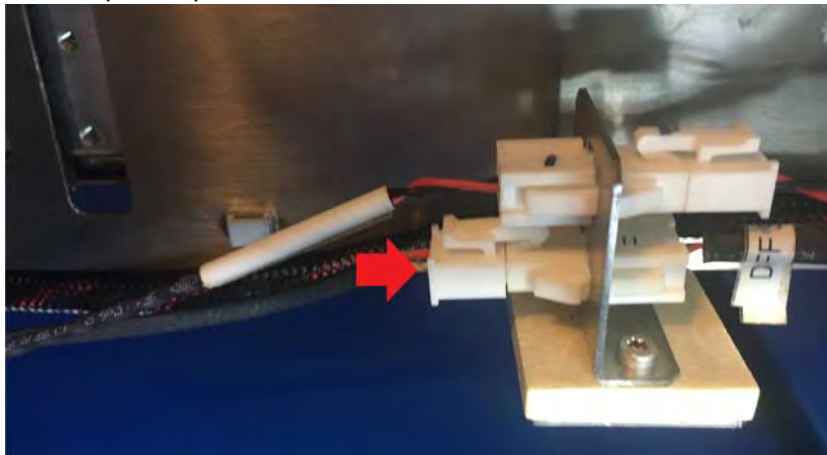
6.90.1 RP88 - Cable CPU-Thermocouple Adapter - Removal Procedure

1. Remove the rear cover per [RP1 - Rear Cover - Removal Procedure](#).
2. Remove the main cover per [RP2 - Main Cover - Removal Procedure](#).
3. Remove the Y E-Chain per [RP63 - E-Chain Y Axis - Removal Procedure](#).
4. Remove the 3 (T10) screws on the right side of the worktable to remove the plate for access to the heater cables.



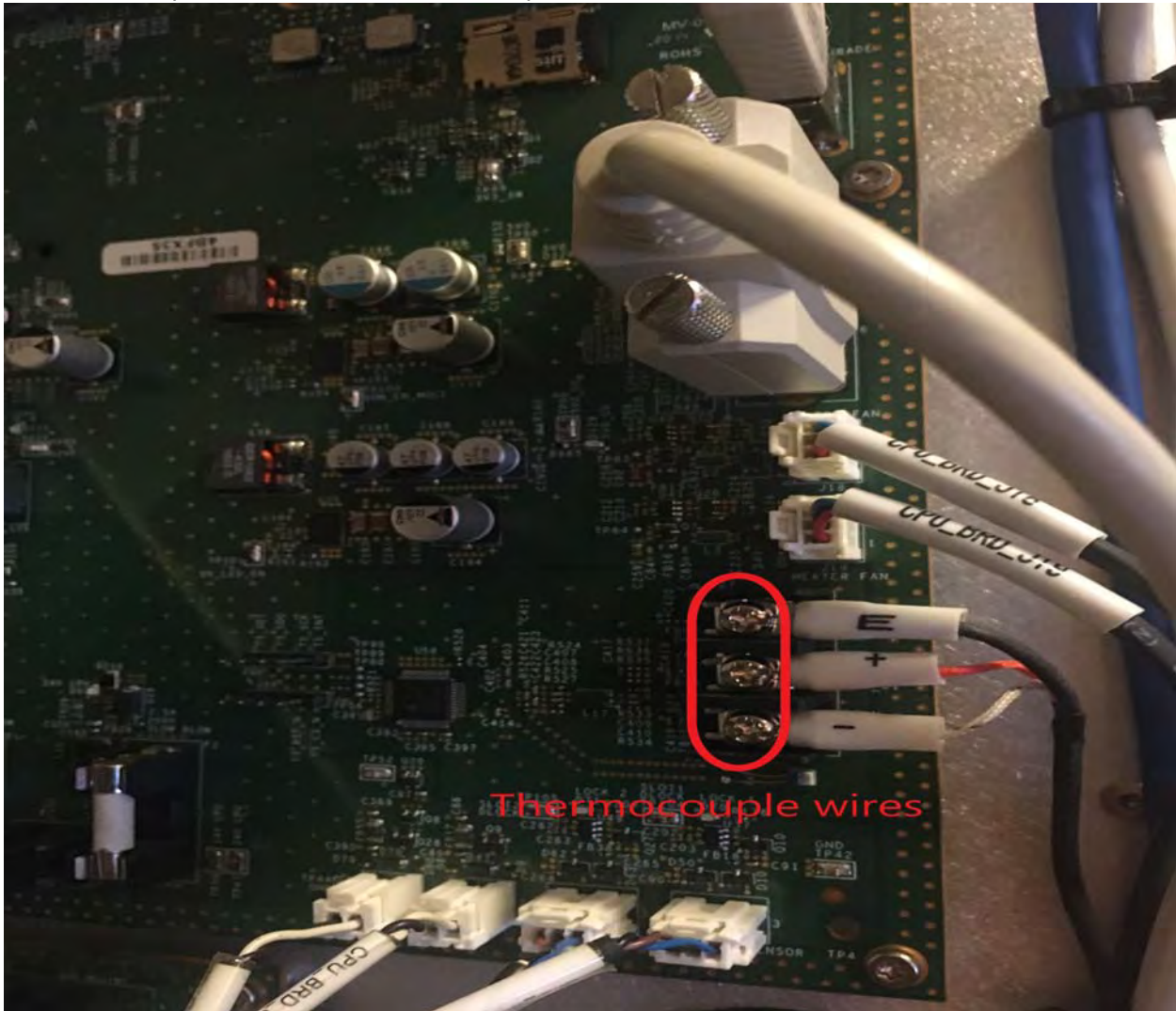
Picture 6.90-1 Worktable screws

5. Unplug the thermocouple adapter from the connector.



Picture 6.90-2 Thermocouple Adapter Cable

6. Use a Phillips and remove the 3 thermocouple connector wires from the CPU board.



Picture 6.90-3 Thermocouple CPU Connectors

7. Remove the thermocouple wire from the Y axis E-chain.

6.90.2 RP88 - Cable CPU-Thermocouple Adapter - Replacement Procedure

1. Pull the thermocouple adapter wire through the Y axis E-chain.
2. Connect the large white connector to the back of the heater unit thermocouple wire.
3. Replace the Y axis E-chain per [RP63 - E-Chain Y Axis- Replacement Procedure](#).
4. Use a Phillips to connect the 3 wire connectors to the CPU board.
5. Replace the 3 (T10) screws in the heater cable cover on the inside right of the worktable.
6. Replace the main cover per [RP2 - Main Cover- Replacement Procedure](#).
7. Replace the rear cover per [RP1 - Rear Cover - Replacement Procedure](#).
8. Perform [VP5 - Temperature Test](#).

6.91 RP89 - Cable Power Board-Heater Power Adapter

Part Required	9027176 - Cable Power Board-Heater Power Adapter
Tool(s) Required	T10, T20
Calibration Required	NA
Testing Required	VP5 - Temperature Test
Replacement Time	40 minutes
Calibration/Testing Time	20 minutes
Total Repair Time	1 hour

Table 6.91-1 RP89 Cable Power Board-Heater Power Adapter

6.91.1 RP89 - Cable Power Board-Heater Power Adapter - Removal Procedure

1. Remove the rear cover per [RP1 - Rear Cover - Removal Procedure](#).
2. Remove the main cover per [RP2 - Main Cover - Removal Procedure](#).
3. Remove the Y E-Chain per [RP63 - E-Chain Y Axis - Removal Procedure](#).
4. Remove the 3 (T10) screws on the right side of the worktable to remove the plate for access to the heater cables.



Picture 6.91-1 Heater Cables Side Plate

5. Unplug the heater power connector on the back side of the heater module.



Picture 6.91-2 Heater Power Connector Back Side

6. Unplug J6 and work cable through the Y axis E-chain and remove the cable.



Picture 6.91-3 J6 Power Board Connector for Heater

6.91.2 RP89 - Cable Power Board-Heater Power - Replacement Procedure

1. Run the cable through the Y axis E-chain.
2. Install the Y axis E-chain per [RP63 - E-Chain Y Axis- Replacement Procedure](#).
3. Plug in the power connector on the back of the heater module.
4. Plug in the power connector on the power board in position J6.
5. Install the heater cable cover panel on the inside of the worktable.
6. Replace any cable ties that have been removed.
7. Replace the main cover per [RP2 - Main Cover- Replacement Procedure](#).
8. Install the rear cover per [RP1 - Rear Cover - Replacement Procedure](#).
9. Perform [VP5 - Temperature Test](#).

6.92 RP90 - Cable Power Board-Temp Fuse Adapter

Part Required	9027177 - Cable Power Board -Temp Fuse Adapter
Tool(s) Required	T10, T20
Calibration Required	NA
Testing Required	VP5 - Temperature Test
Replacement Time	40 minutes
Calibration/Testing Time	20 minutes
Total Repair Time	1 hour

Table 6.92-1 RP90 Cable Power Board-Temp Fuse Adapter

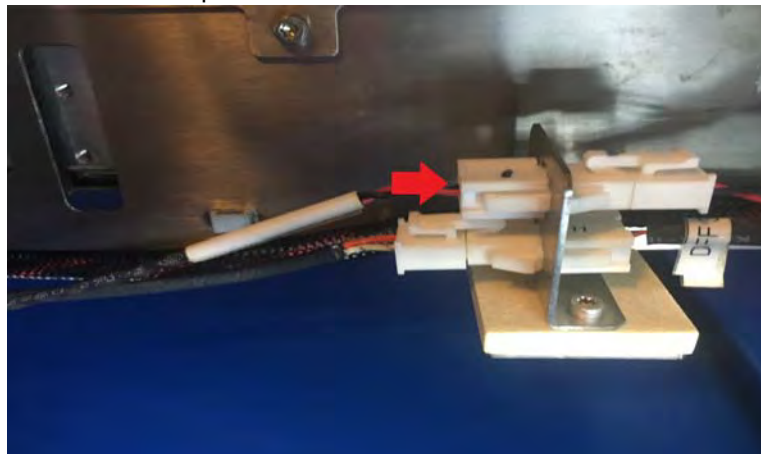
6.92.1 RP90 - Cable Power Board-Temp Fuse Adapter - Removal Procedure

1. Remove the rear cover per [RP1 - Rear Cover - Removal Procedure](#).
2. Remove the main cover per [RP2 - Main Cover - Removal Procedure](#).
3. Remove the Y E-Chain per [RP63 - E-Chain Y Axis - Removal Procedure](#).
4. Remove the 3 (T10) screws on the right side of the worktable to remove the plate for access to the heater cables.



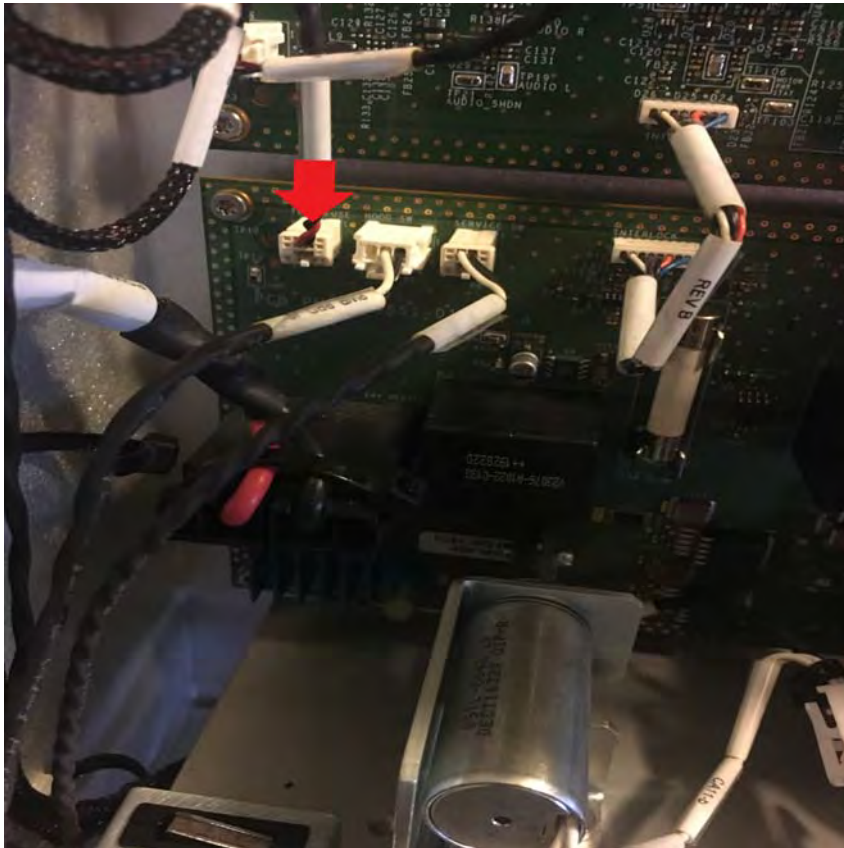
Picture 6.92-1 Heater Cables Side Plate

5. Unplug the temperature fuse adapter cable connector at the back of the heater unit.



Picture 6.92-2 Temp Fuse Adapter Cable Connector

6. Unplug the J1 connector on the Power board.



Picture 6.92-3 J1 Temperature Fuse Connector on CPU Board

7. Pull the temperature fuse adapter cable through the Y axis E-chain and remove it.

6.92.2 RP90 - Cable Power Board-Temp Fuse Adapter - Replacement Procedure

1. Run the cable through the Y axis E-chain.
2. Install the Y axis E-chain per [RP63 - E-Chain Y Axis- Replacement Procedure](#).
3. Plug in the fuse connector on the back of the heater module.
4. Plug in the fuse connector on the Power board in position J1.
5. Install the heater cable cover panel on the inside right of the worktable.
6. Replace any cable ties that have been removed.
7. Replace the main cover per [RP2 - Main Cover- Replacement Procedure](#).
8. Install the rear cover per [RP1 - Rear Cover - Replacement Procedure](#).
9. Perform [VP5 - Temperature Test](#).

6.93 RP91 - Z Motor Coupler

Part Required	9027179 - Z Motor Coupler
Tool(s) Required	T10, T20, 2.5 mm Hex Tool
Calibration Required	CP3 Pipette Z-axis Calibration
Testing Required	VP10 - Z Axis Movement Test , VP8 - Workflow System Test
Replacement Time	18 minutes
Calibration/Testing Time	38 minutes
Total Repair Time	56 minutes

Table 6.93-1 RP91 Z Motor Coupler

6.93.1 RP91 - Z Motor Coupler - Removal Procedure

1. Turn off the power no the EZ2 and remove the power cord.
2. Remove the rear cover per [RP1 - Rear Cover - Removal Procedure](#).
3. Remover the main cover per [RP2 - Main Cover - Removal Procedure](#).
4. Remove the 2 (T20) screws on the base of the Z motor.
5. Unplug the motor connector.
6. Pull the Z motor straight up to remove.



Picture 6.93-1 Z Motor

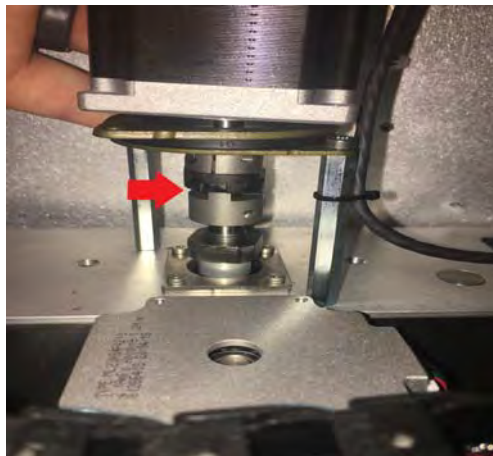
7. Remove the Z motor coupler upper and lower portion with a 2.5 mm hex tool.



Picture 6.93-2 Z Motor Coupler

6.93.2 RP91 - Z Motor Coupler- Replacement Procedure

1. Install the coupler upper and lower portion on the Z motor shaft with a 2.5 mm hex tool.
2. Ensure that the Z motor coupler joint fits together correctly before installing the screws that hold the Z motor in position.



Picture 6.93-3 Z Motor Coupler Joint

3. The cable connector position should be on the facing right side.
4. Install the 2 (T20) screws that mount the Z motor in place.
5. Plug in the cable connector into the motor.
6. Install the main cover per [RP2 - Main Cover- Replacement Procedure](#).
7. Install the rear cover per [RP1 - Rear Cover - Replacement Procedure](#).
8. Perform [CP3 Pipette Z-axis Calibration](#), [VP10 - Z Axis Movement Test](#), [VP8 - Workflow System Test](#).

6.94 RP92 - P Motor Coupler

Part Required	9027178 - P Motor Coupler
Tool(s) Required	T10, T20, 2.5 mm Hex Tool
Calibration Required	CP5 Pipette P-Axis Calibration ,
Testing Required	VP3 - Leakage Test , VP4 - Pipetting Test , VP8 - Workflow System Test
Replacement Time	17 minutes
Calibration/Testing Time	42 minutes
Total Repair Time	59 minutes

Table 6.94-1 RP92 P Motor

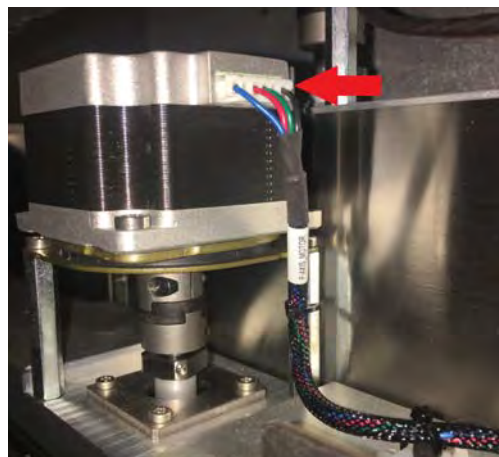
6.94.1 RP92 - P Motor Coupler - Removal Procedure

1. Turn off the EZ2 power.
2. Remove the rear cover per [RP1 - Rear Cover - Removal Procedure](#).
3. Remove the main cover per [RP2 - Main Cover - Removal Procedure](#).
4. Remove the 2 (T20) screws on the base of the P motor.



Picture 6.94-1 P motor screws

5. Unplug the P motor cable connector.



Picture 6.94-2 P Motor Cable Connector

6. Pull the P motor straight up to remove it.

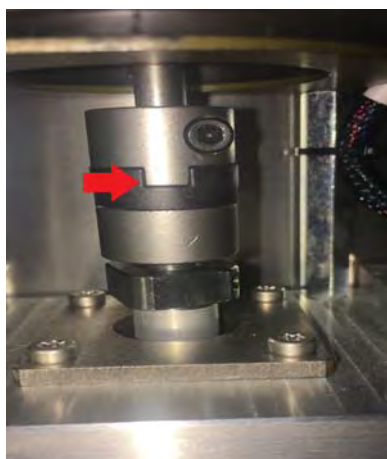
7. Remove the upper and lower portion of the coupler with a 2.5 mm hex tool.



Picture 6.94-3 P Motor Coupler

6.94.2 RP92 - P Motor Coupler - Replacement Procedure

1. Install the upper and lower portion of the coupler with a 2.5 mm hex tool.
2. Ensure that the P motor coupler joint fits together correctly before installing the screws that hold the P motor in position.



Picture 6.94-4 P Motor Coupler joining

3. The cable connector position should be on the facing right side.
4. Install the 2 (T20) screws that mount the P motor in place.
5. Plug in the cable connector into the motor.
6. Install the main cover per [RP2 - Main Cover- Replacement Procedure](#).
7. Install the rear cover per [RP1 - Rear Cover - Replacement Procedure](#).
8. Perform [CP5 Pipette P-Axis Calibration](#), [VP3 - Leakage Test](#), [VP4 - Pipetting Test](#), [VP8 - Workflow System Test](#).

6.95 RP93 - FFC Cable

Part Required	9027199 - Cable CPU Board-Display LVDS Signal
Tool(s) Required	T10, T20, T30
Calibration Required	NA
Testing Required	VP12 - System Initialization Test
Replacement Time	27 minutes
Calibration/Testing Time	3 minutes
Total Repair Time	30 minutes

Table 6.95-1 RP93 FFC Cable

6.95.1 RP93 - FFC Cable - Removal Procedure

1. Remove the rear cover per [RP1 - Rear Cover - Removal Procedure](#).
2. Remove the main cover per [RP2 - Main Cover - Removal Procedure](#).
3. Remove the top front cover per [RP8 - Top Front Cover - Removal Procedure](#).
4. Remove the cable from the back of the LCD.



Picture 6.95-1 FFC Cable

6.95.2 RP93 - FFC Cable - Replacement Procedure

1. Plug the cable in to the connector at the back of the LCD screen.
2. Replace the top front cover per [RP8 - Top Front Cover - Replacement Procedure](#).
3. Replace the main cover per [RP2 - Main Cover- Replacement Procedure](#).
4. Replace the rear cover per [RP1 - Rear Cover - Replacement Procedure](#).
5. Perform [VP12 - System Initialization Test](#).

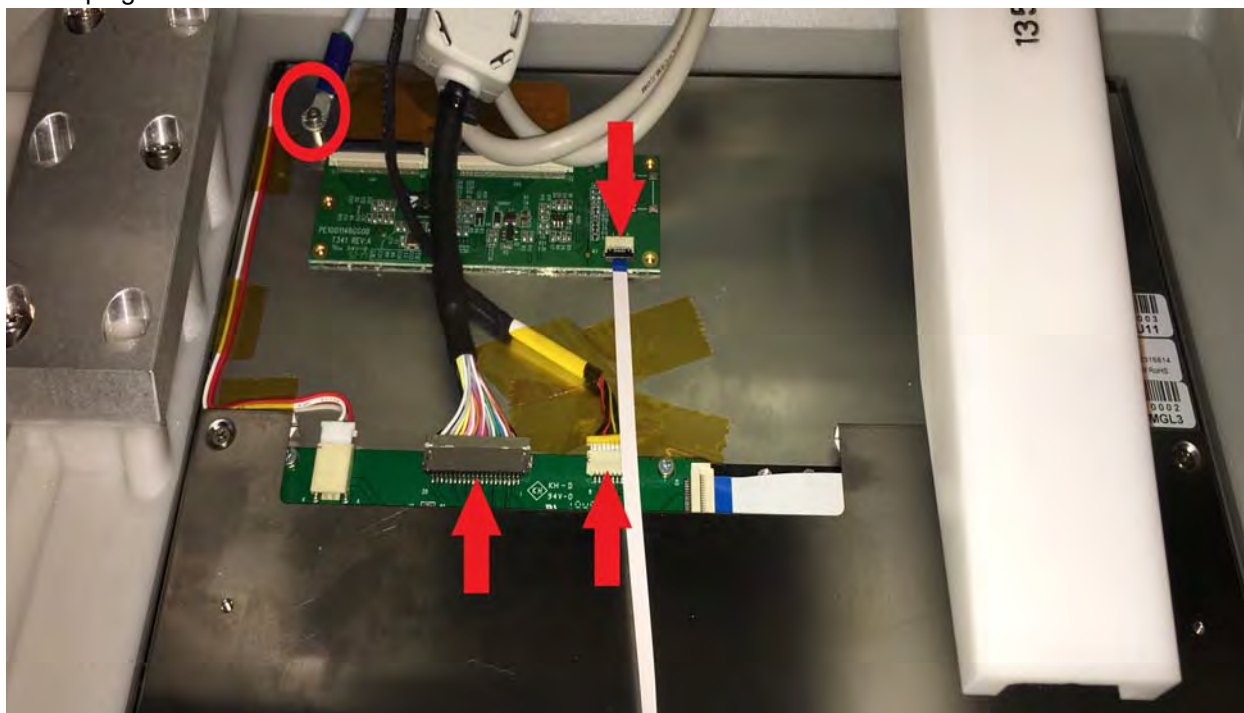
6.96 RP94 - LCD Touchscreen

Part Required	9027079- LCD Touchscreen
Tool(s) Required	T30, T25, T20, T10
Calibration Required	NA
Testing Required	VP12 - System Initialization Test
Replacement Time	25 minutes
Calibration/Testing Time	2 Minutes
Total Repair Time	27 minutes

Table 6.96-1 RP94 LCD Touchscreen

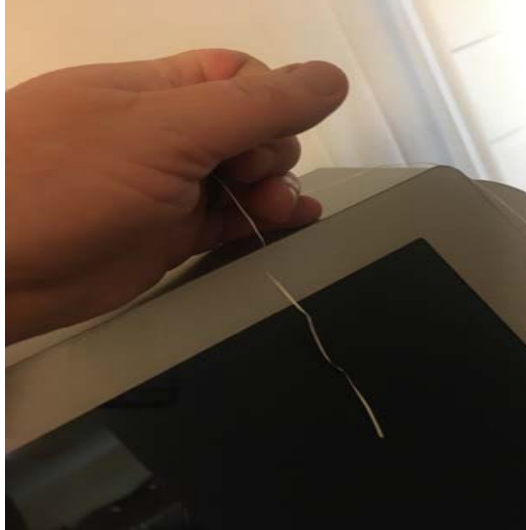
6.96.1 RP94 - LCD Touchscreen - Removal Procedure

1. Remove the rear cover per [RP1 - Rear Cover - Removal Procedure](#).
2. Remove the main cover per [RP2 - Main Cover - Removal Procedure](#).
3. Remove the top front cover per [RP8 - Top Front Cover - Removal Procedure](#).
4. Remove the (T10) screw that attaches the ground cable to the back of the touchscreen.
5. Unplug the 3 cables from the back of the touchscreen.



Picture 6.96-1 Touchscreen Cables

6. Use a thin piece plastic or dental floss to pry the screen out of the front top cover by each of the corners.



Picture 6.96-2 Touchscreen Removal With Floss

7. Be careful not to damage the cover when you pry the touchscreen out of the top front cover.
8. Pull the touchscreen from the front to remove it.

6.96.2 RP94 - LCD Touchscreen - Replacement Procedure

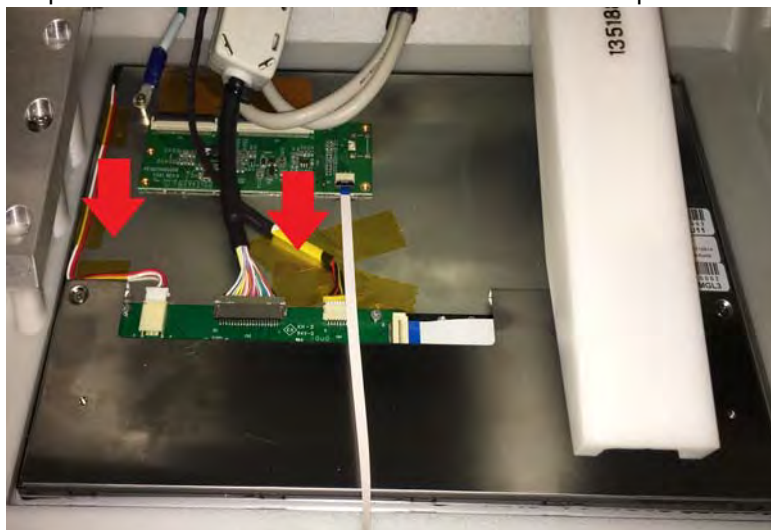
1. Take the new LCD Panel and peel off the backing adhesive before align it to the Top Front Cover Plastic.



Picture 6.96-3 Touchscreen Alignment

2. Align the 4 corners of the touchscreen before pushing the screen in place.
3. Attach the (T20) ground screw on the back side of the touchscreen.
4. Plug in the 3 cables and attach the ground wire with the (T10) screw.

5. Apply the Kapton tape to hold the cables on the back of the screen in position.



Picture 6.96-4 Kapton Tape

6. Replace the top front cover per [RP8 - Top Front Cover - Replacement Procedure](#).
7. Replace the main cover per [RP2 - Main Cover- Replacement Procedure](#).
8. Replace the rear cover per [RP1 - Rear Cover - Replacement Procedure](#).

6.97 RP95 - Left and Right Trays

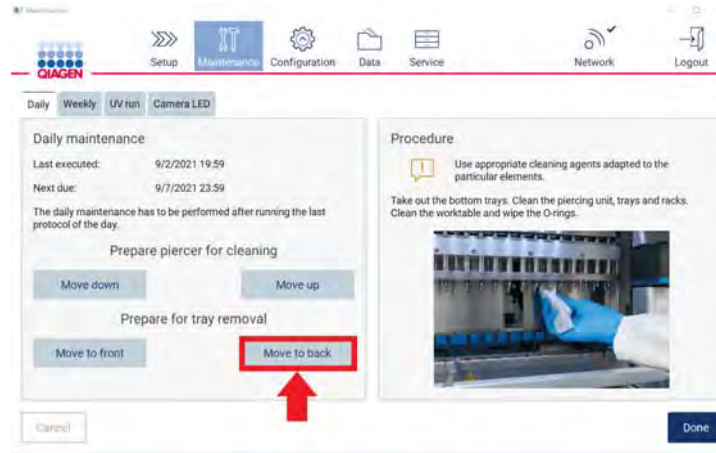
Part Required	9027131 - Left Tray, 9027134 - Right Tray
Tool(s) Required	None
Calibration Required	None
Testing Required	TBD
Replacement Time	1 minute
Calibration/Testing Time	None
Total Repair Time	1 minute

Table 6.97-1 RP95 Left & Right Trays

6.97.1 RP95 - Left and Right Tray - Removal Procedure

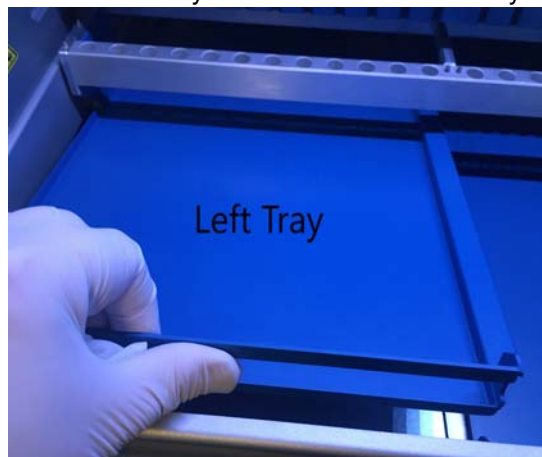
1. Turn on the EZ2 Connect.
2. Select the Maintenance environment and press daily.

3. Press the **“Move to back”** button.



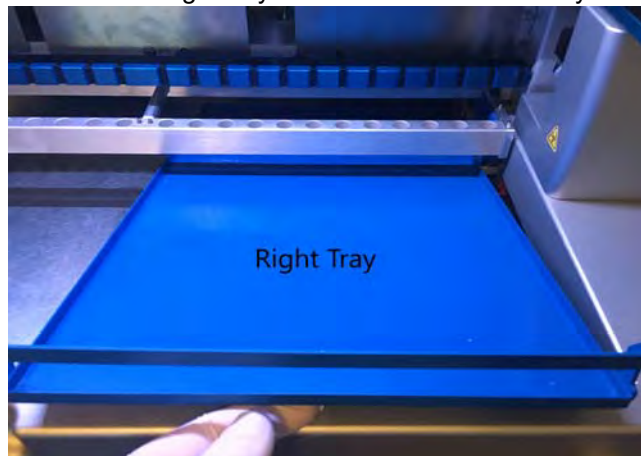
Picture 6.97-1 Move to back

4. Grab the handle on the front of the left tray and lift the front of the tray and pull out the tray.



Picture 6.97-2 Left Tray

5. Grab the handle on the front of the right tray and lift the front of the tray and pull out the tray.



Picture 6.97-3 Right Tray

6.97.2 RP95 - Left and Right Tray - Replacement Procedure

1. Insert the right tray first.
2. Insert the left tray.
3. Press the **“Done”** button on the maintenance screen.

7 Calibration Procedures (CP)

Calibration procedures are used to adjust/calibrate modules or system parameters.

7.1 Adjustment/Calibration Procedures

7.1.1 CP1 Camera Calibration

Purpose	This calibrates the camera x-axis position.
Tool(s) Required	None
Calibration/Testing Time	5 minutes

Table 7.1-1 CP1 Camera Calibration

1. Turn on the power to the EZ2.
2. After the instrument initializes select "Service" and log in as service.



Picture 7.1-1 Camera Calibration

3. Select "Service".
4. Select "Calibration".
5. Select "Camera Calibration".
6. Insert the empty tip rack containing position 24 on the right side of the workspace.
7. Clear any items from the rack so it is empty.
8. Remove any other racks from the workspace.
9. Close the hood.
10. Press the start button to start the calibration.
11. Once completed press the "OK" button.

7.1.2 CP2 Camera Exposure Calibration

Purpose	This calibration optimizes the camera exposure for the actual light environment on the tip rack inside the instrument.
Tool(s) Required	None
Calibration/Testing Time	5 minutes

Table 7.1-2 CP2 Camera Exposure Calibration

1. Turn on the power to the EZ2.

- After the instrument initializes click on the service tab and log in as service.



Picture 7.1-2 Camera Exposure Calibration

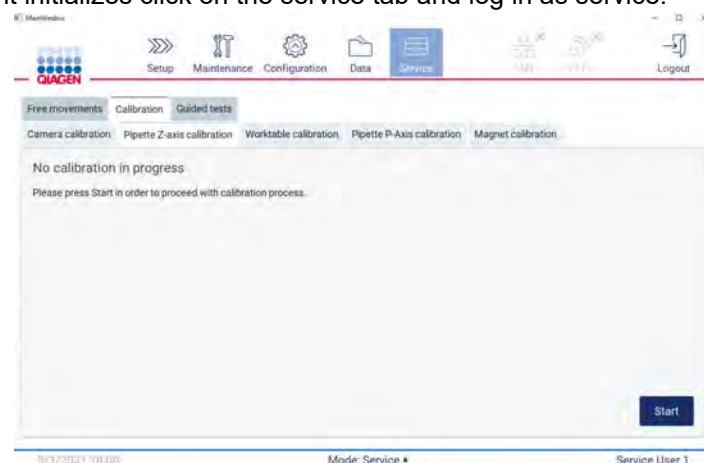
- Select "Maintenance".
- Select "Camera LED"
- Load the left and right tip racks empty, with no other racks loaded.
- Close the hood.
- Select the "Start" button.
- Once the calibration is completed select "Finished".

7.1.3 CP3 Pipette Z-axis Calibration

Purpose	This calibrates the Z pipette positions.
Tool(s) Required	T20, 0.02 mm Feeler Gauge
Calibration/Testing Time	6 minutes

Table 7.1-3 CP3 Pipette Z-axis Calibration

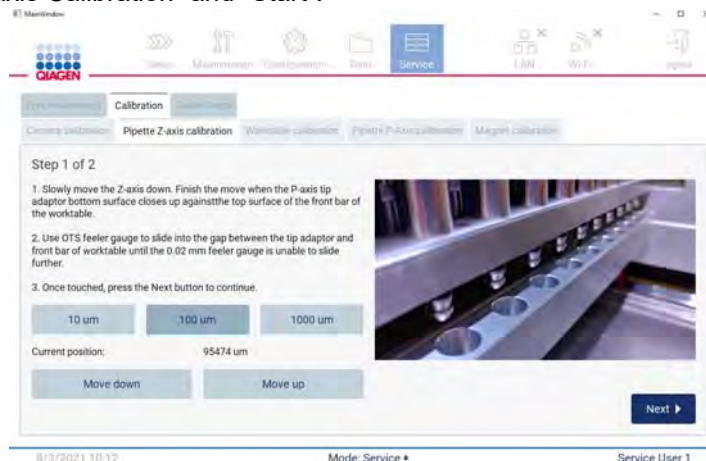
- Remove the rear cover per [RP1 - Rear Cover - Removal Procedure](#).
- Turn on the power to the EZ2.
- After the instrument initializes click on the service tab and log in as service.



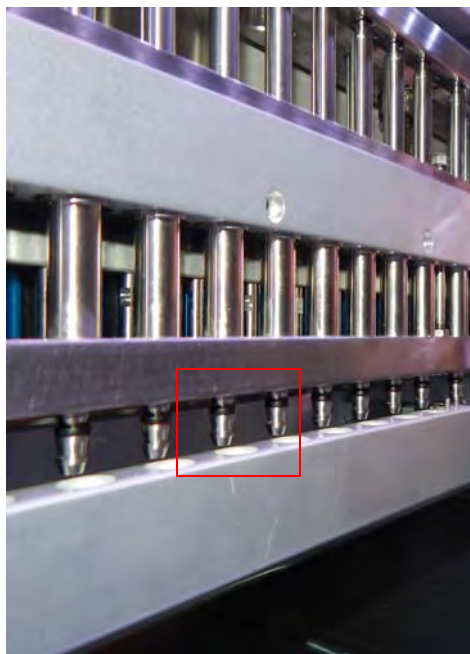
Picture 7.1-3 Start Pipette Z Axis Calibration

- Remove all racks from the system.
- Select "Service".
- Select "Calibration".

7. Select "Pipette Z-axis Calibration" and "Start".

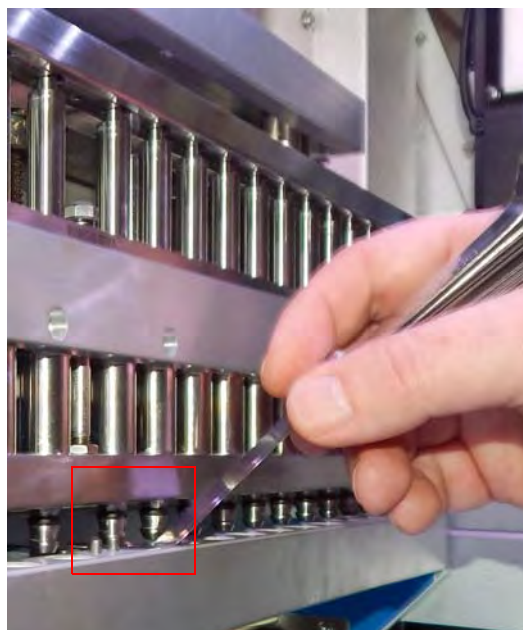
*Picture 7.1-4 Pipette Z Movement*

8. Slowly move the Z-axis down so the P-axis tip adapter bottom is close to the top surface of the front bar of the worktable.

*Picture 7.1-5 Pipette Z-Axis Calibration*

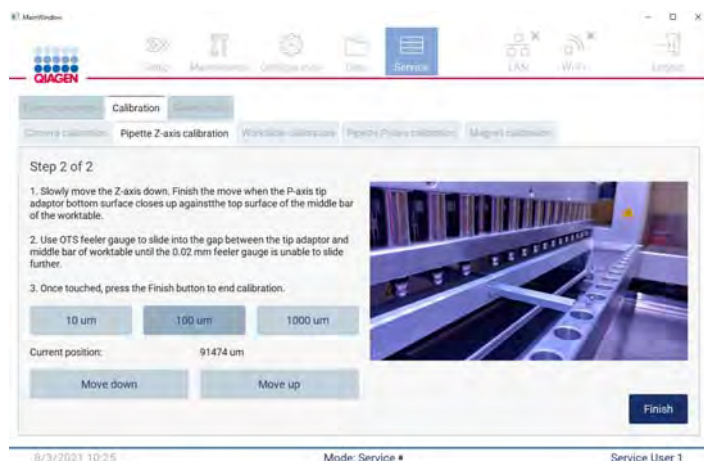
9. Place a 0.02 mm feeler gauge between the tip adapter and the front bar of the worktable.

10. Bring the axis down until the feeler gauge is unable to slide between the positions, check all the positions.



Picture 7.1-6 Feeler Gauge

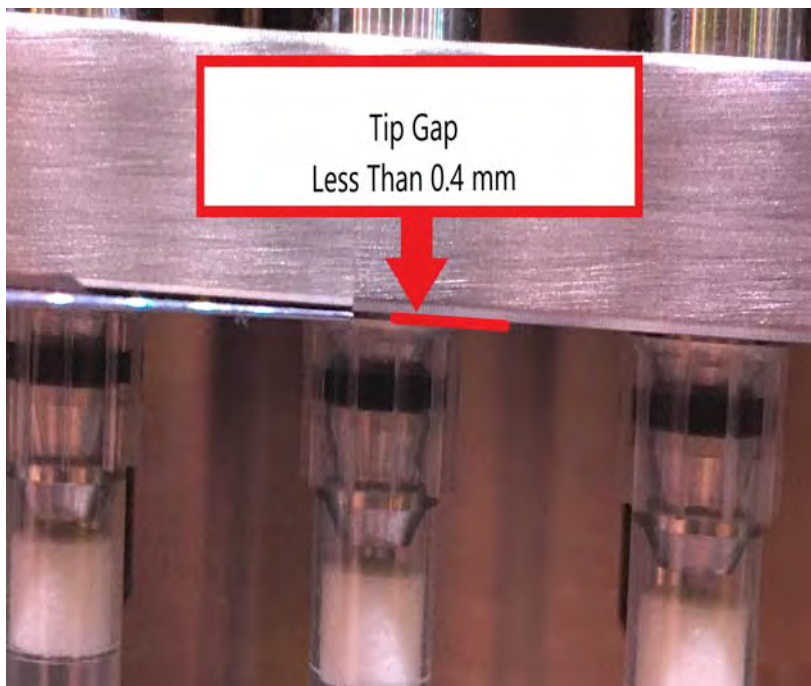
11. Select "Next".



Picture 7.1-7 Pipette Z-Axis Middle Bar

12. Slowly move the Z-axis down so the P-axis tip adapter bottom close to the top surface of the middle bar of the worktable.
13. Use the 0.02 mm feeler gauge between the tip adapter and the middle bar of the worktable.
14. Bring the axis down until the feeler gauge is unable to slide between the positions.
15. Check all positions with the 0.02 feeler gauge, if a position is too tight, move up 10 UM and check again.
16. Select the "Finish" button and select "Yes" to save the calibration.
17. Next check the tip gap from the ejector bar to the top of the tip with a feeler gauge on all 24 positions.
18. To pick up the tips and check the gap select the **Free Movements** tab.

19. Go to the software chapter in the service manual [Worktable and Pipette](#) to pick up and release tips in the free movements environment.



Picture 7.1-8 Tip Gap Less Than 0.4 mm

20. To adjust the tip gap, open the Z access panel.

21. Remove the lock nut on the whichever side the gap is over 0.4 mm with a 10 mm wrench.



Picture 7.1-9 Remove Lock Nut

22. Use a 4 mm hex tool to turn the screw till the gap is smaller than 0.4 mm.



Picture 7.1-10 Tip Gap Screw Adjustment

23. Turn the adjustment screw clockwise to decrease the tip gap.

24. Once the gap is less than 0.4 mm apply Loctite 680 to the screw and put the lock nut back in place.



Picture 7.1-11 Loctite 680

25. Recalibrate the pipette Z axis if the tip gap adjustment has been made.

26. Install the 2 (10) screws in the Z access panel.

27. Replace the rear cover per [RP1 - Rear Cover - Removal Procedure](#).

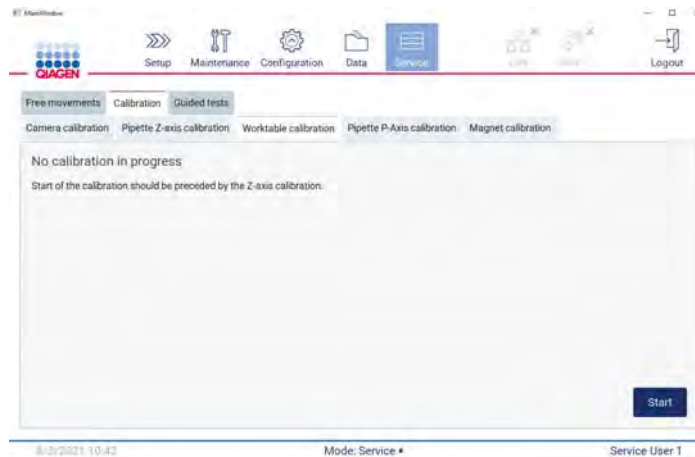
7.1.4 CP4 Worktable Calibration

Purpose	This calibration sets the reference point for the y-axis.
Tool(s) Required	T20, 0.02 mm Feeler gauge
Calibration/Testing Time	11 minutes

Table 7.1-4 Worktable Calibration

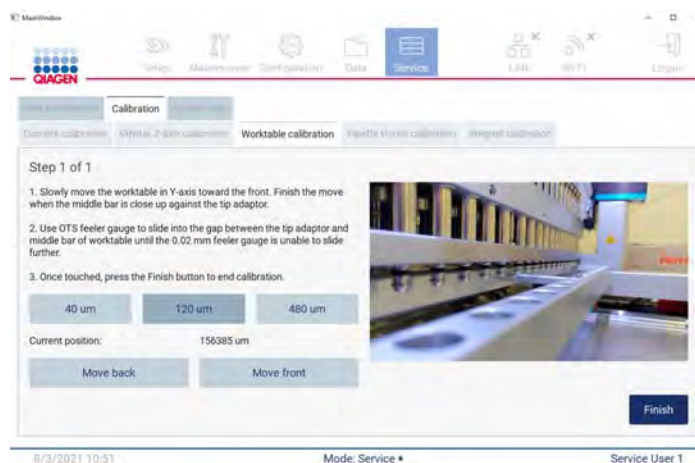
1. Remove the rear cover per [RP1 - Rear Cover - Removal Procedure](#).
2. Turn on the power to the EZ2.

3. Perform CP3 Pipette Z-axis Calibration.



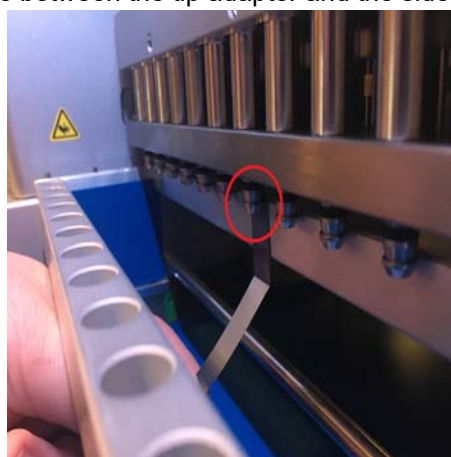
Picture 7.1-12 Worktable Calibration Page

4. Remove all racks from the system.
5. Select **"Service"**.
6. Select **"Worktable Calibration"**.
7. Select **"Start"**.



Picture 7.1-13 Worktable Movement Page

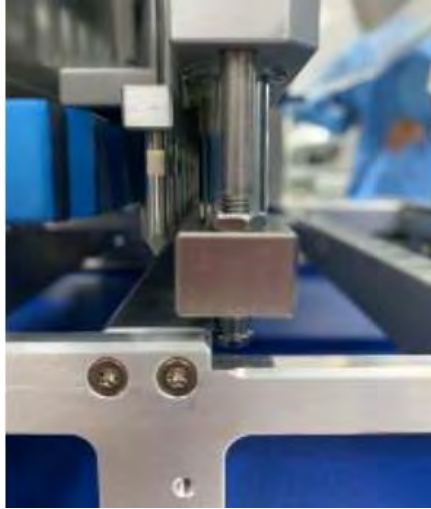
8. Slowly move the Y-axis worktable toward the front so middle bar is close against the tip adapter.
9. Use the 0.02 mm feeler gauge between the tip adapter and the side of the middle bar of the worktable.



Picture 7.1-14 Feeler Gauge Position

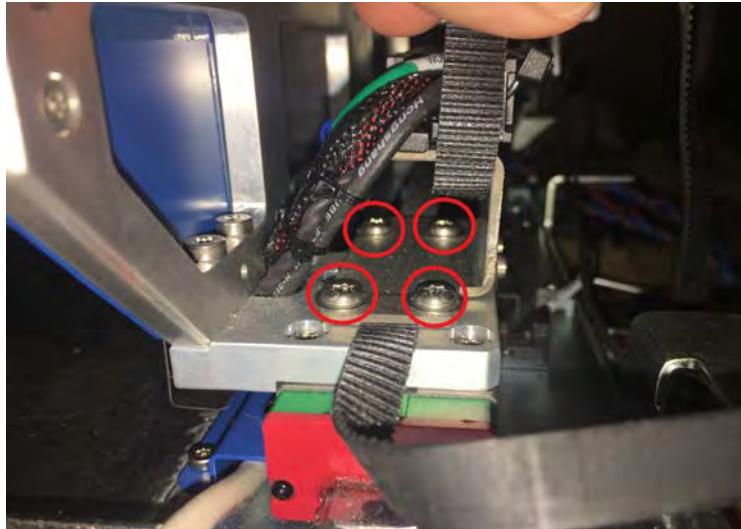
10. Move the Y-axis until the feeler gauge is unable to slide between the tip adapter.
11. Calibrate with the tightest position with the 0.02 mm feeler gauge.
12. Check all the remaining positions gaps are less than 0.4 mm with a feeler gauge.
13. If the gaps are less than the 0.4 mm on all the tip adapters, select the **"Finish"** button and **"Yes"** to input the new calibration.

14. If the gaps are over 0.4 mm turn off the instrument.
15. Remove the right internal metal cover per [RP5 - Internal Right Metal Cover - Removal Procedure](#).
16. Manually push the worktable to the back and bring down the Z-Axis until the tip adapter and worktable middle bar touching with each other.



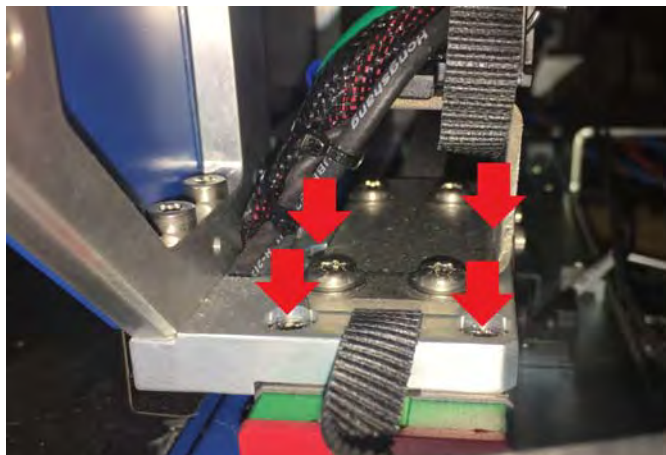
Picture 7.1-15 Adapters Against Middle Bar

17. Remove the 4 screws on the Y right bearing carrier.



Picture 7.1-16 Remove Screws

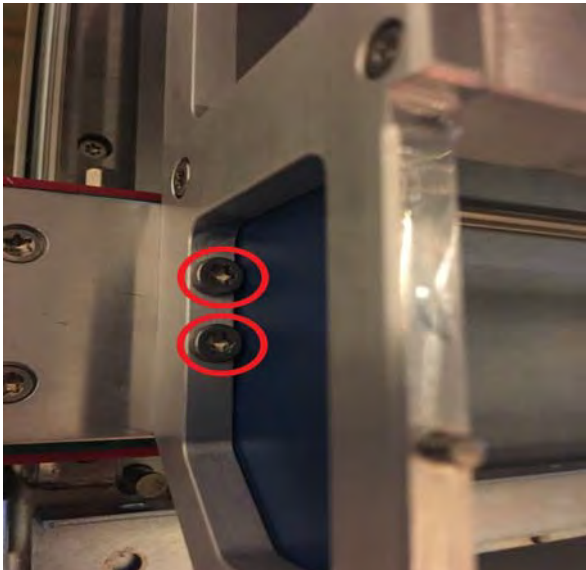
18. Remove the 4 (T20) screws.



Picture 7.1-17 Loosen 4 Screws

19. Loosen the 4 (T20) screws on the Y bearing carrier.

20. Loosen the 2 (T20) screws on the left base of the worktable.



Picture 7.1-18 Right Worktable Screws

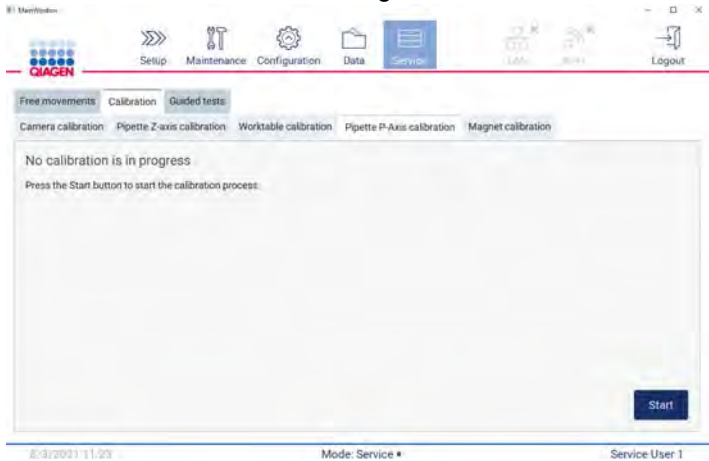
21. Align the tip adapters with the middle bar of the worktable so the back of the 24 adapters touches the middle bar all the way across.
22. Tighten the 2 (T20) screws on the left side of the worktable and tighten the 4 screws on the right side of the worktable.
23. Install the 4 screws that hold the Y E-chain in position.
24. Repeat steps 1-13 again by calibrating the worktable after the alignments have been made.
25. Replace the internal right metal cover per [RP5 - Internal Right Metal Cover- Replacement Procedure](#).

7.1.5 CP5 Pipette P-Axis Calibration

Purpose	This calibration sets the tip release position.
Tool(s) Required	T20, Feeler Gauge Set 1 mm - 0.02 mm, 8 mm wrench
Calibration/Testing Time	5 minutes but variable if adjustment is required

Table 7.1-5 CP5 Pipette P-Axis Calibration

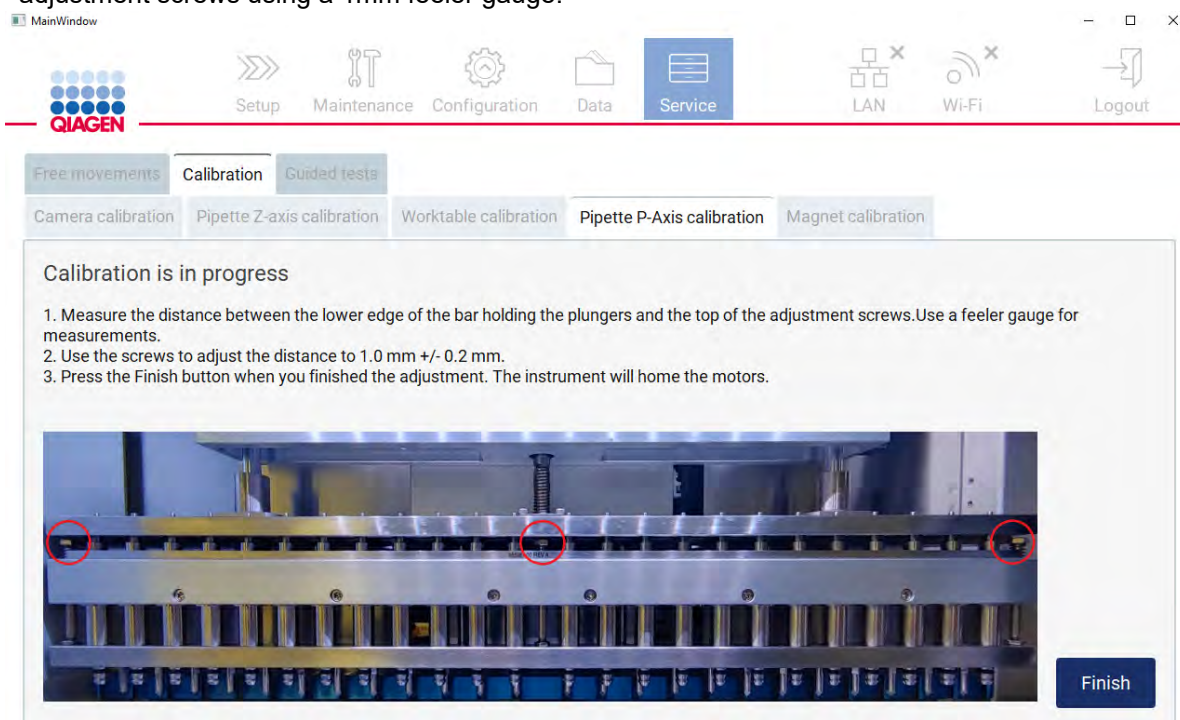
1. Remove the rear cover per [RP1 - Rear Cover - Removal Procedure](#).
2. Turn on the power to the EZ2.
3. After the instrument initializes select “Service” to log in.



Picture 7.1-19 Pipette P-Axis Calibration Page

4. Select “Service”.

5. Select the "Pipette P-Axis Calibration" and select "Start".
6. Measure the distance between the bottom of the holding bar of the plungers and the top of the adjustment screws using a 1mm feeler gauge.

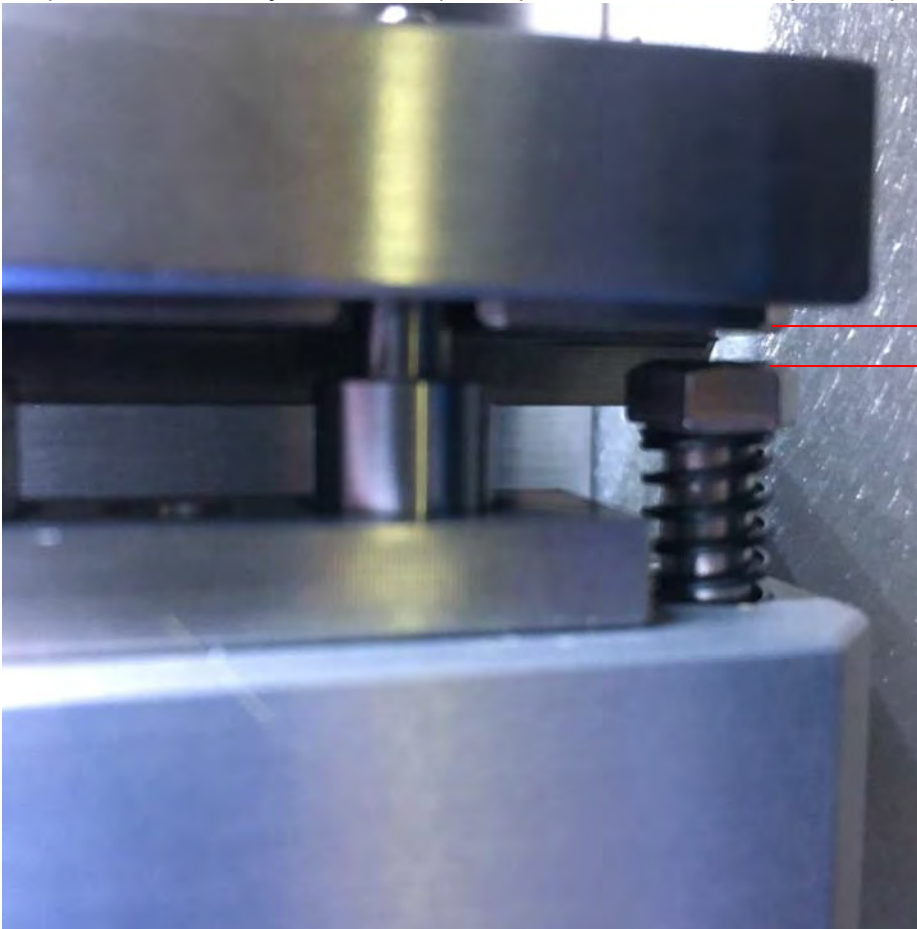


8/3/2021 11:14
Picture 7.1-20 Pipette P axis Calibration

Mode: Service ▾

Service User 1

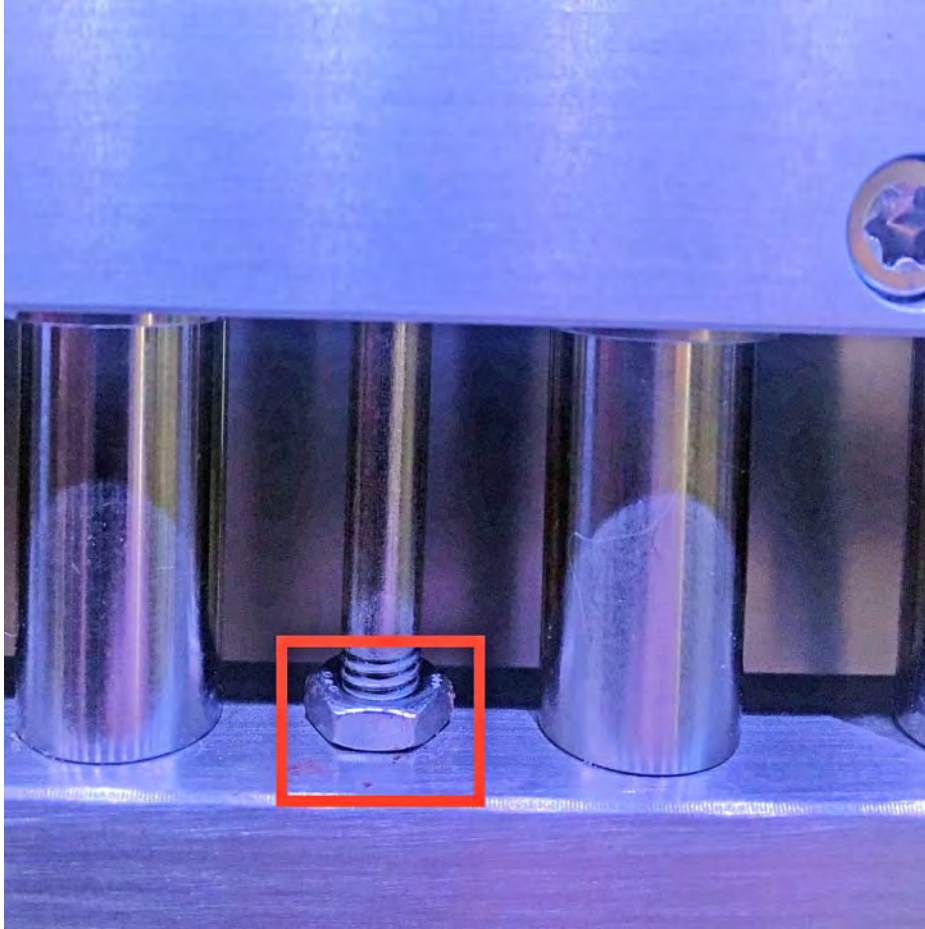
7. The distance must be set to 1.0 mm $\pm$ 0.2 mm. If the distance is correct, proceed to step 11 of this procedure. If an adjustment is required, proceed to the next step in this procedure.



Measure this gap.

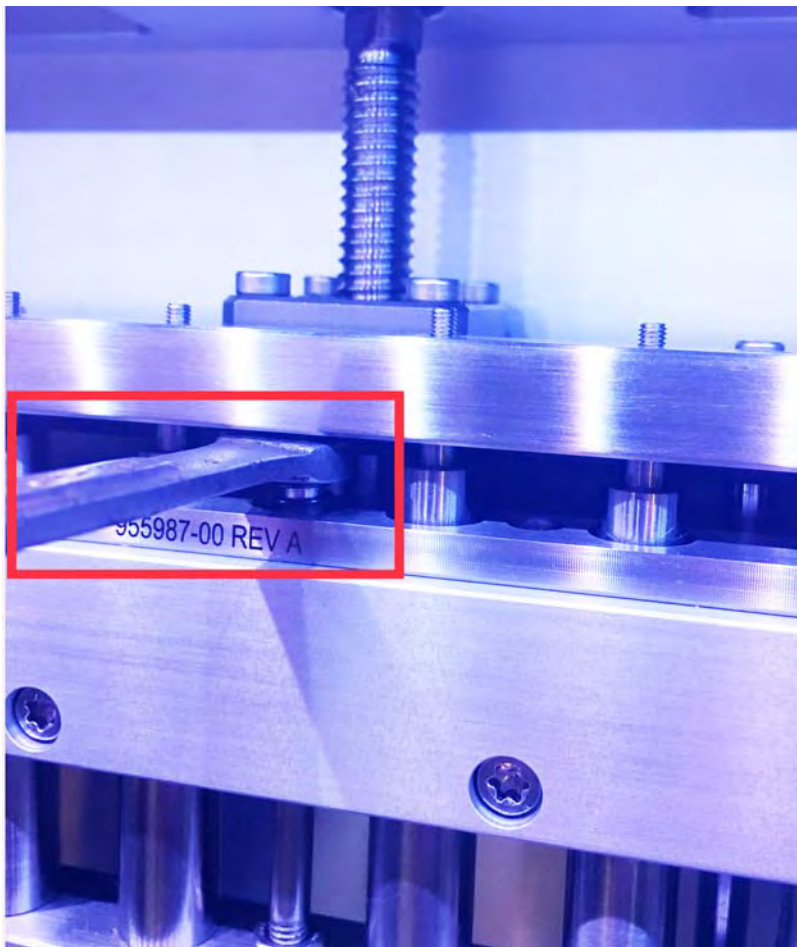
Picture 7.1-21 Gap Measurement

8. To adjust the distance, loosen the nut at the bottom of the screw with a 8 mm wrench.



Picture 7.1-22 P Axis Nut

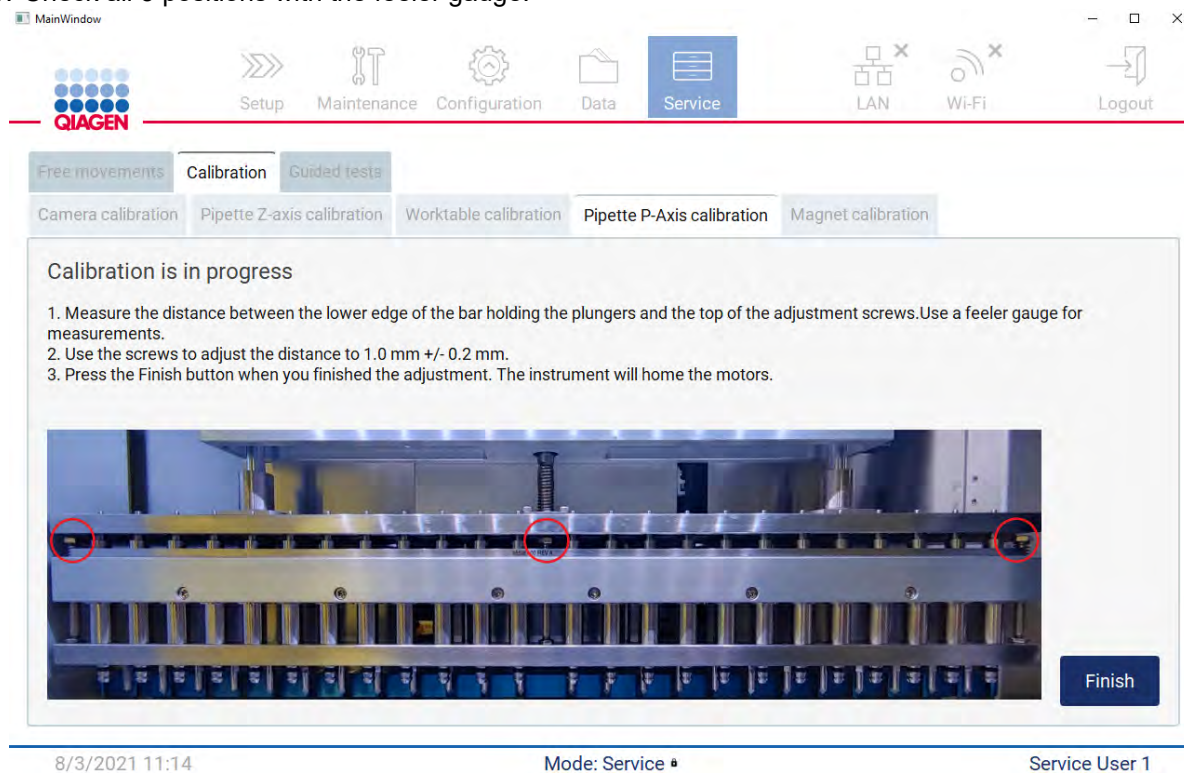
9. Use an 8 mm wrench at the top of the screw to turn it up or down, as required to obtain the correct distance.



Picture 7.1-23 Adjustment

10. After the adjustment has been made tighten the nut at the bottom of the screw with the 8 mm wrench.

11. Check all 3 positions with the feeler gauge.



Picture 7.1-24 Measured position

12. Select the "Finished" button when you have completed your adjustments.

13. Replace the rear cover per [RP1 - Rear Cover - Removal Procedure](#).

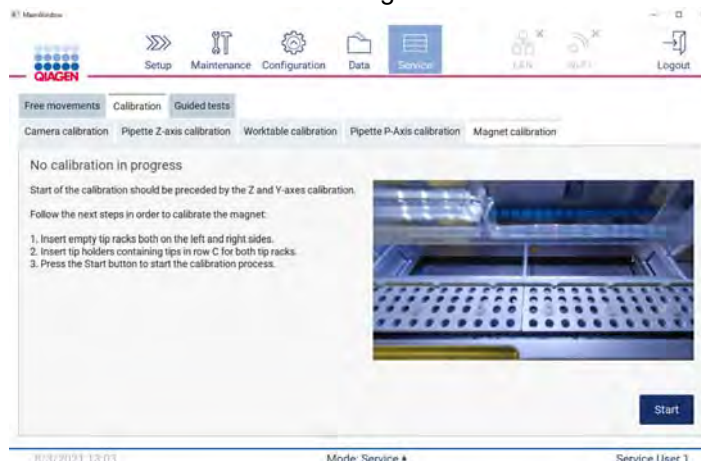
7.1.6 CP6 Magnet Calibration

Purpose	This calibration will adjust for the correct distance from the tip to the magnet for the separation of beads in the protocol.
Tool/Consumables Required	T10, 7 tips and tip holders
Calibration/Testing Time	6 Minutes

Table 7.1-6 CP6 Magnet Calibration

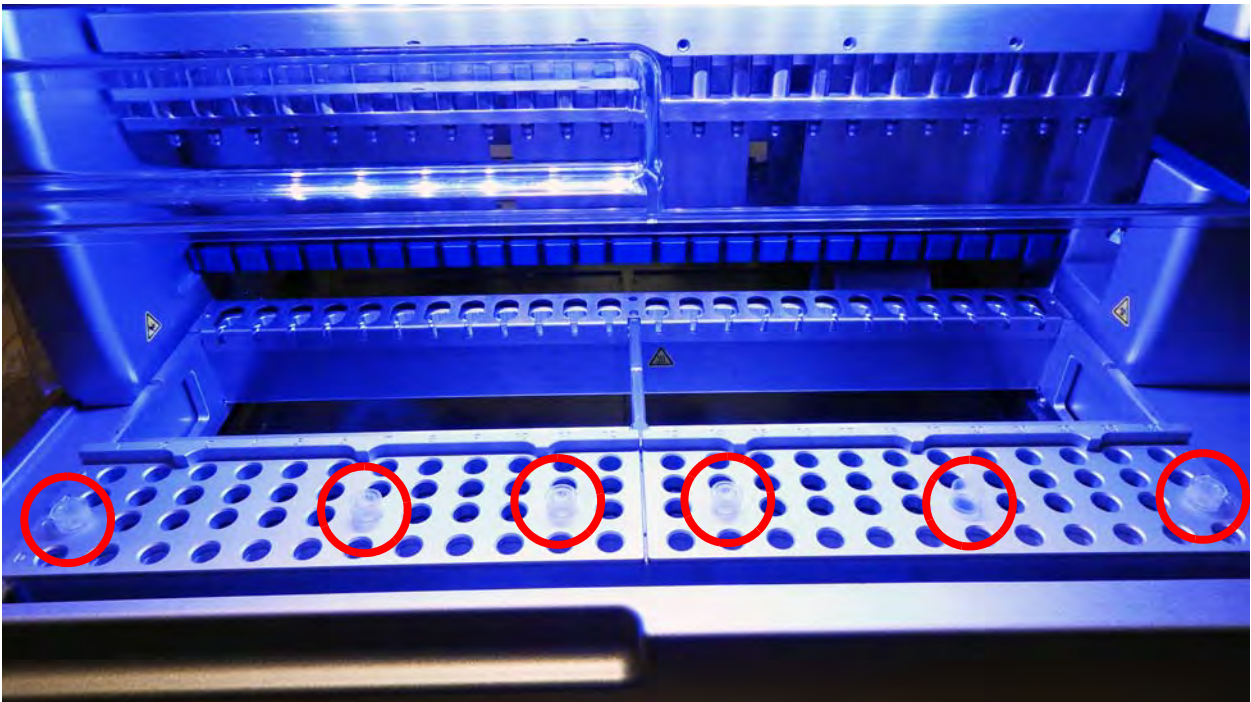
1. Remove the rear cover per [RP1 - Rear Cover - Removal Procedure](#).
2. Turn on the power to the EZ2.

3. After the instrument initializes select “Service” to log in.



Picture 7.1-25 Magnet Calibration Screen

4. Select “Service”.
5. Select “Calibration”.
6. Select “Magnet Calibration”.
7. Insert both the left and right tip racks and place tips in 1,4,8,12,16,20, and 24 of row C on both racks.
8. Select “Start” to continue the calibration.



Picture 7.1-26 Magnet Calibration Tips

9. Slowly move the M-axis magnet towards the tips until the magnet unit just touches the tips, but does not push or deflect the tips.



Picture 7.1-27 Tips Touching the Magnet

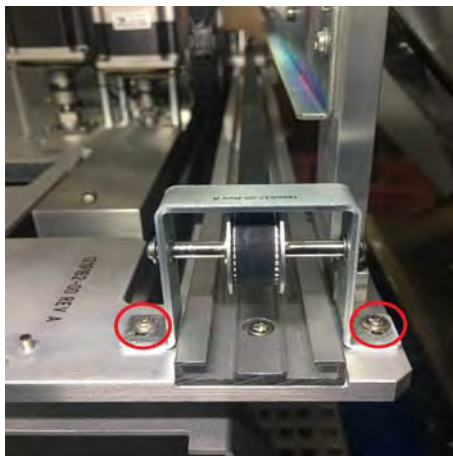
10. Select “Finish” and “Yes” to complete.
11. Replace the rear cover per [RP1 - Rear Cover - Replacement Procedure](#).

7.1.7 CP7 X-Belt Tension

Purpose	This calibration will adjust for the correct tension of the X belt for the X axis to work correctly.
Tool(s) Required	T10, T20
Calibration/Testing Time	15 Minutes

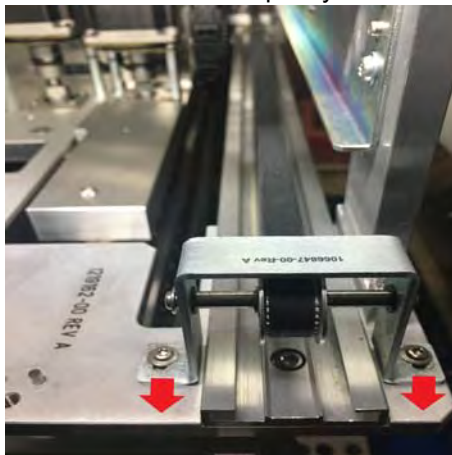
Table 7.1-7 CP7 X-Belt Tension

1. Remove the rear cover per [RP1 - Rear Cover - Removal Procedure](#).
2. Remove the main cover per [RP2 - Main Cover - Removal Procedure](#).



Picture 7.1-28 X Belt Pulley Bracket Screws

- Loosen the 2 (T10) screws at the base of the X belt pulley bracket.



Picture 7.1-29 Setting X Belt Tension

- Pull the X belt pulley bracket to the left of the instrument until all of the slack is removed from the screw holes in the base of the bracket.
- Tighten the 2 (T10) screws at the base of the bracket to set the tension.
- Calibrate the camera per [CP1 Camera Calibration](#).
- Calibrate the camera exposure per [CP2 Camera Exposure Calibration](#).

7.1.8 CP8 Y-Belt Tension

Purpose	This calibration will adjust for the correct tension of the Y belt for the Y axis to work correctly.
Tool(s) Required	T20, 0.02 mm Feeler Gauge
Calibration/Testing Time	20 Minutes

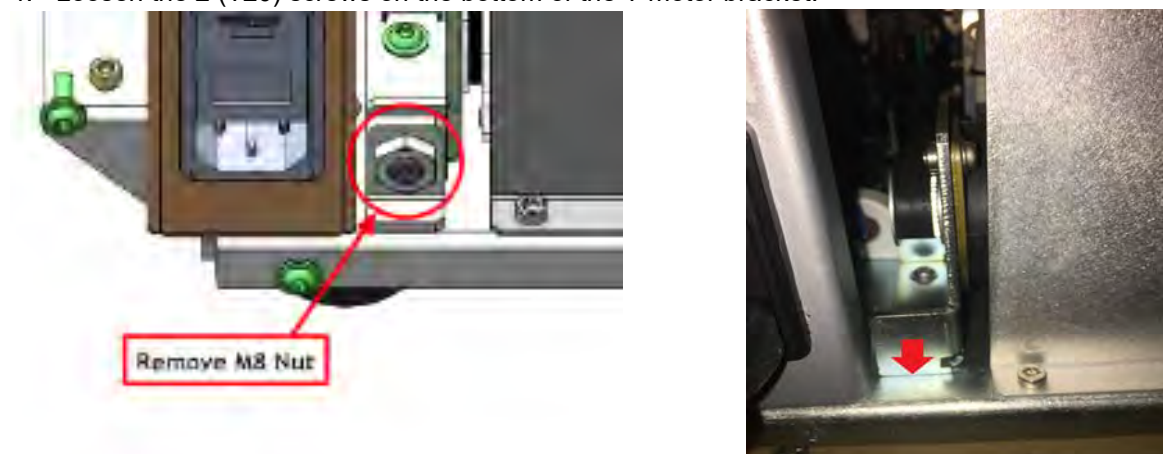
Table 7.1-8 CP8 Y-Belt Tension

- Remove the rear cover per [RP1 - Rear Cover - Removal Procedure](#).
- Remove the main cover per [RP2 - Main Cover - Removal Procedure](#).
- Remove the bottom right stationary cover per [RP4 - Right Stationary Cover - Removal Procedure](#).



Picture 7.1-30 Y Motor Bracket Screws

- Loosen the 2 (T20) screws on the bottom of the Y motor bracket.



Picture 7.1-31 Y Belt Tension

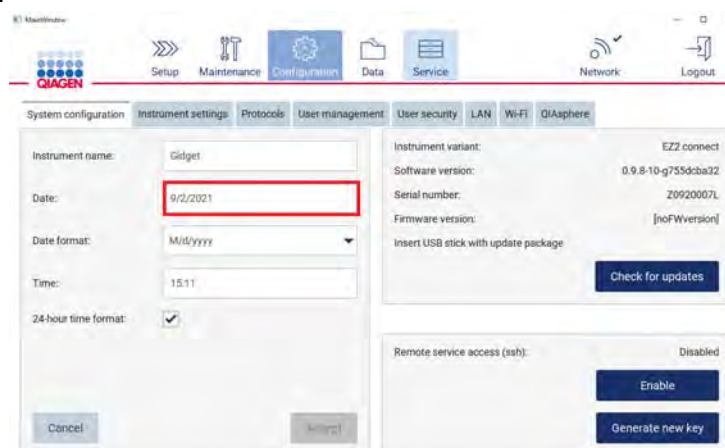
- Pull the end of the bracket towards the back of the instrument till all the slack is removed from the screws and tighten the 2 (T20) screws.
- Replace the right stationary cover per [RP4 - Right Stationary Cover - Replacement Procedure](#).
- Replace the main cover per [RP2 - Main Cover- Replacement Procedure](#).
- Replace the rear cover per [RP1 - Rear Cover - Replacement Procedure](#).
- Calibrate the worktable per [CP4 Worktable Calibration](#).

7.1.9 CP9 Set Time/Date

Purpose	This calibration will set the correct date and time in the software.
Tool(s) Required	None
Calibration/Testing Time	1 Minutes

Table 7.1-9 CP9 Set Time/Date

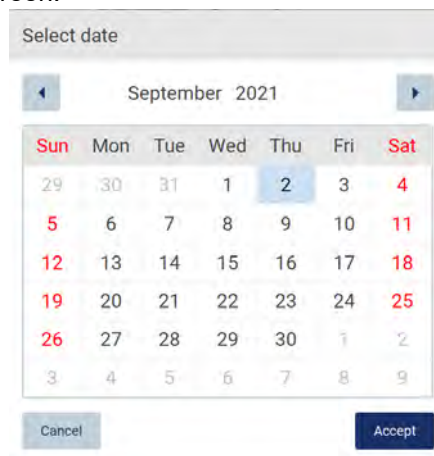
- Turn on the EZ2 Connect power button.
- Log in as service.



Picture 7.1-32 Date Setting

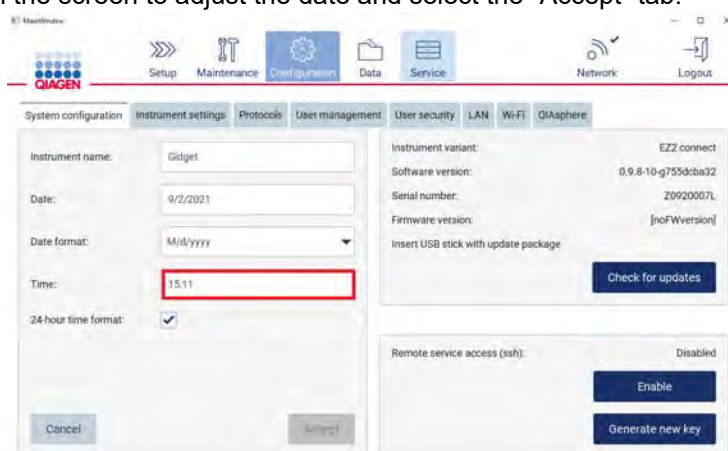
- Select the Configuration environment.
- Select the "System Configuration" tab.

5. Touch the date area on the screen.



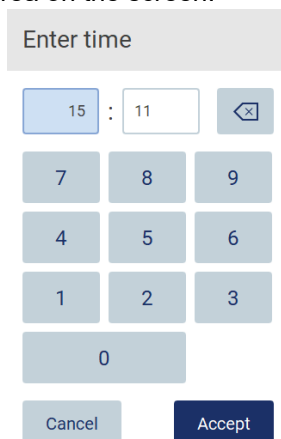
Picture 7.1-33 Date Change

6. Touch the date on the screen to adjust the date and select the “Accept” tab.



Picture 7.1-34 Time Setting

7. To change the time, touch the time area on the screen.



Picture 7.1-35 Change Time

8. Use the touch screen to change the time and select “Accept”.
 9. Press “Accept” again.
 10. The time and date should now be corrected.

8 Verification Procedures (VP)

Verification Procedures (VP) are tests or procedures that are performed to confirm that the system is operating to specification. [Verification Procedures \(VP\)](#)

8.1 Overview of Verification Procedures

VP #	Name	Purpose
VP1	LED Strip Test	This is to verify the function of the LED lighting of the worktable area.
VP2	UV Light Test	This is to verify the function of the UV lighting for decontamination of the work area.
VP3	Leakage Test	This test is to verify the tips are on correctly and that the O-rings and plungers in the pipette module are sealed and working correctly.
VP4	Pipetting Test	This test is to verify that all 24 pipette channels are pipetting the correct volume of fluid.
VP5	Temperature Test	This test is to verify that the heater module is reaching the correct temperature when heating is needed for protocols.
VP6	Hood Lock Test	This test is to verify that the hood is locked in the down position and can not open when the instrument is operating for the safety of the operator.
VP7	Load Check Test	This test verify the camera function to check when a protocol is ran with all testing materials are in the correct position.
VP8	Full System Test	This test will verify that the protocol is transferring fluids correctly and all calibrations were correctly implemented.
VP9	Barcode Reader Test	This test is to verify that the barcode ID is correctly read and translated to the software.
VP10	Z Axis Movement Test	This test is to verify for step loss after parts have been replaced or calibrations have been completed.
VP11	M Axis Movement Test	This test is to verify for step loss after parts have been replaced or calibrations have been completed.
VP12	System Initialization	This verification test will check the powering up of the instrument and that no errors have occurred.
VP13	Service Switch	This test is to check that the service switch is only active when the rear cover has been removed.
VP14	Fan Test	This is to check that the fans are functional.
VP15	Speaker Test	This test is to check for sound when the instrument alarms.
VP16	USB Test	This is a verification that run reports and software updates can be loaded on a USB stick and transferred to and from the instrument.
VP17	X Axis Movement Test	This is to verify the movement of the camera and UV modules are moving correctly across the X axis.
VP18	P Axis Movement Test	This is to verify that the P motor position is staying in the correct position after movements are made.
VP19	Y Axis Movement Test	This is to verify that the Y motor position of the worktable is staying in the correct position after movements are made.

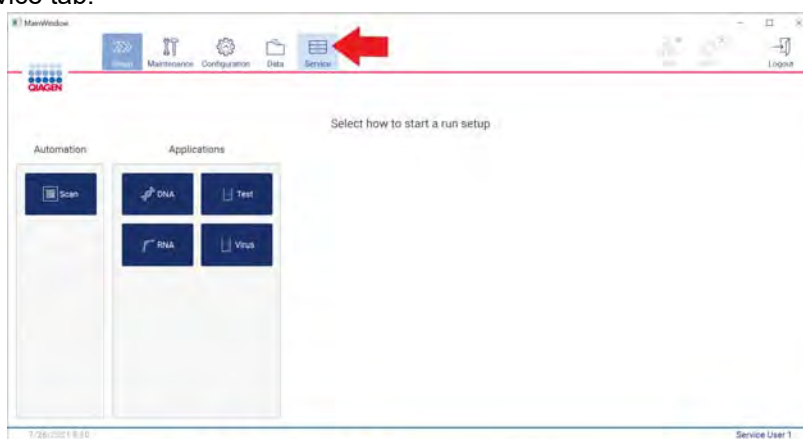
Table 8.1-1 Verification Procedure Overview

8.2 VP1 - LED Strip Test

Purpose	This is to verify the function of the LED lighting of the worktable area.
Tool(s) Required	NA
Verification Time	1 minutes

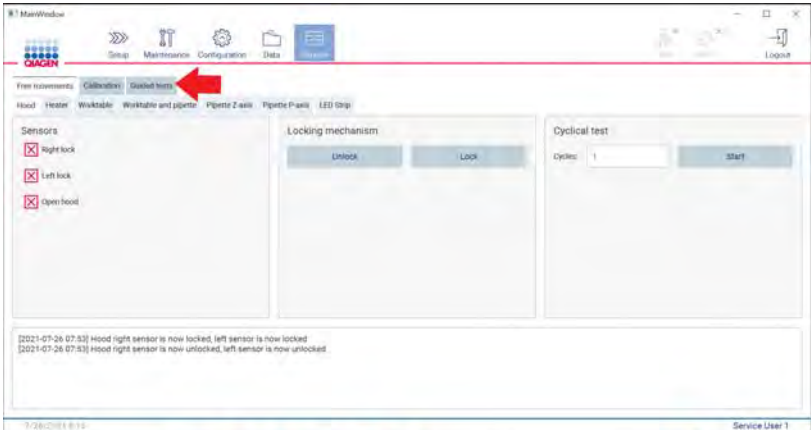
Table 8.2-1 VP1 LED Strip Test

1. Log in to the EZ2 software as a service user.
2. Select the service tab.



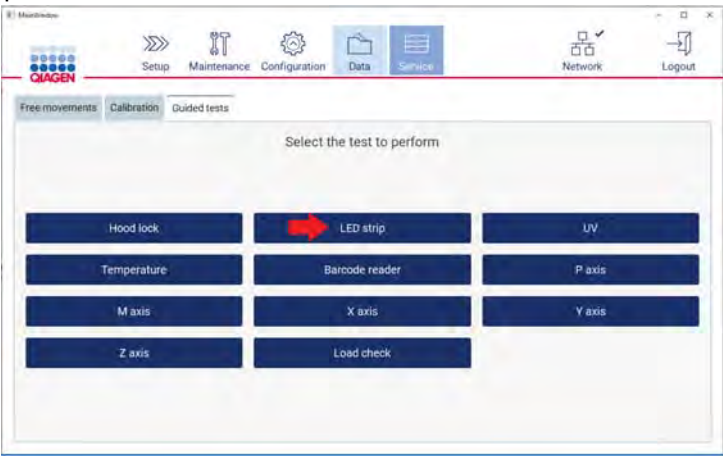
Picture 8.2-1 Service Tab

3. Select the guided test tab.



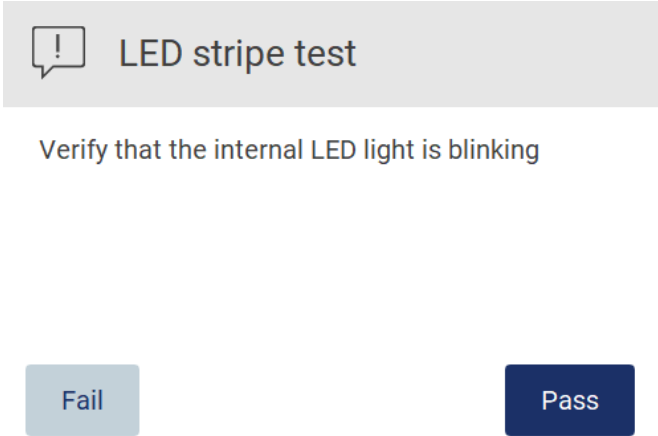
Picture 8.2-2 Guided test tab

4. Select the LED strip tab.



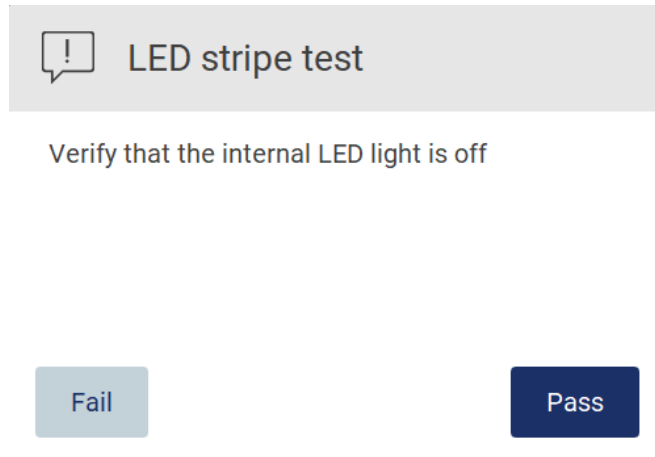
Picture 8.2-3 LED strip test

5. Check that the LED is blinking and select the “Pass” tab.



Picture 8.2-4 LED blinking

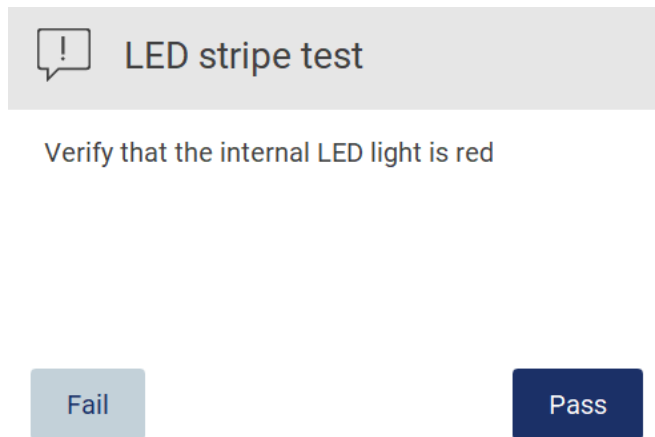
6. Check that the LED is off and select the “Pass” tab.



The screenshot shows a software interface for an "LED stripe test". At the top, there is a header bar with a speech bubble icon containing an exclamation mark and the text "LED stripe test". Below this, the instruction "Verify that the internal LED light is off" is displayed. At the bottom of the interface, there are two buttons: a light blue "Fail" button on the left and a dark blue "Pass" button on the right.

Picture 8.2-5 LED off

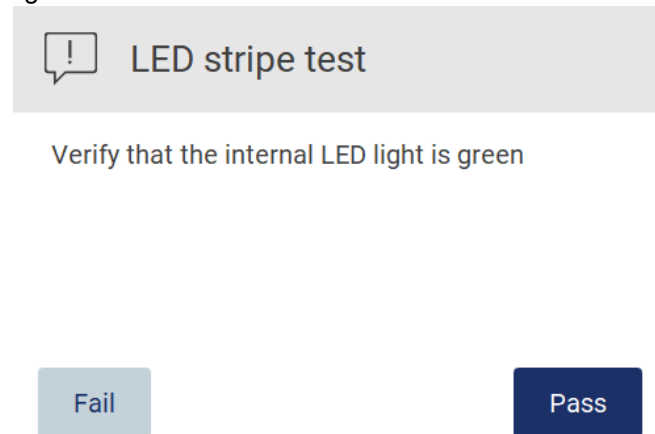
7. Check that the LED is red and select the “Pass” tab.



The screenshot shows the same "LED stripe test" interface. The header bar and instruction "Verify that the internal LED light is red" are present. The "Fail" button is light blue, and the "Pass" button is dark blue.

Picture 8.2-6 Red LED

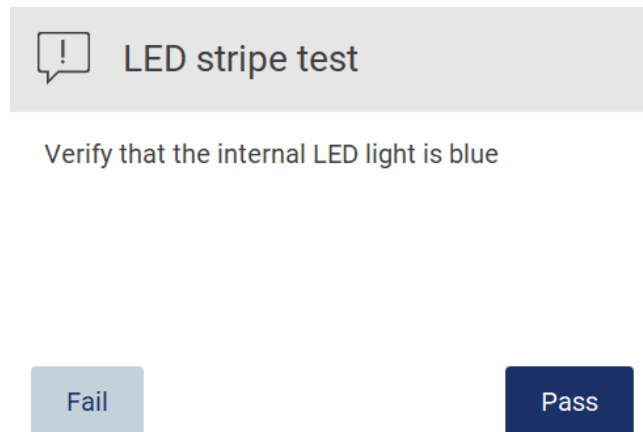
8. Check that the LED is green and select the “Pass” tab.



The screenshot shows the same "LED stripe test" interface. The header bar and instruction "Verify that the internal LED light is green" are present. The "Fail" button is light blue, and the "Pass" button is dark blue.

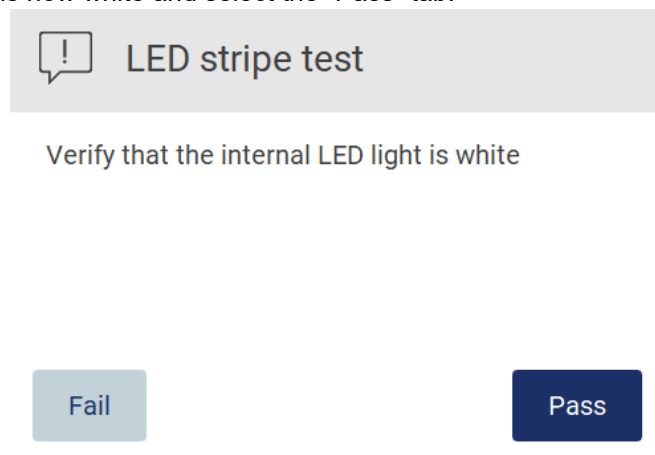
Picture 8.2-7 Green LED

9. Check that the LED is blue and select the “Pass” tab.



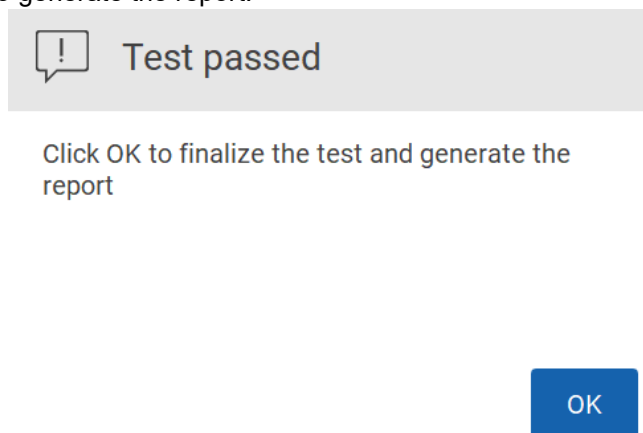
Picture 8.2-8 Blue LED

10. Check that the LED is now white and select the “Pass” tab.



Picture 8.2-9 White LED

11. Select the “OK” tab to generate the report.



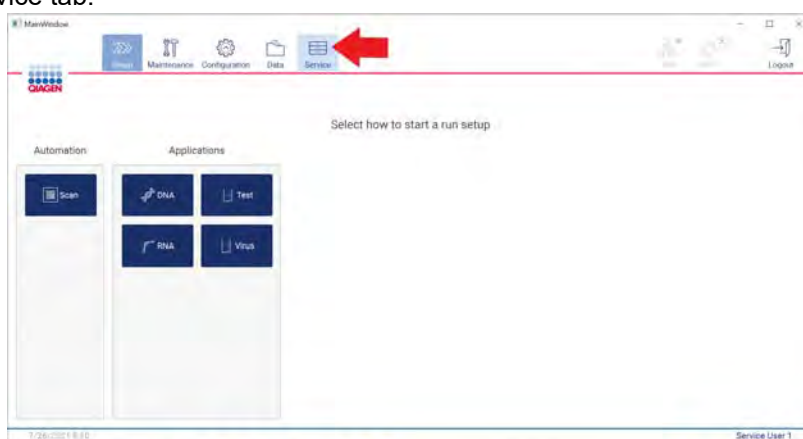
Picture 8.2-10 LED Report

8.3 VP2 - UV Light Test

Purpose	This test is to check the function of the UV light, which is used for decontamination.
Tool(s) Required	NA
Verification Time	1 minutes

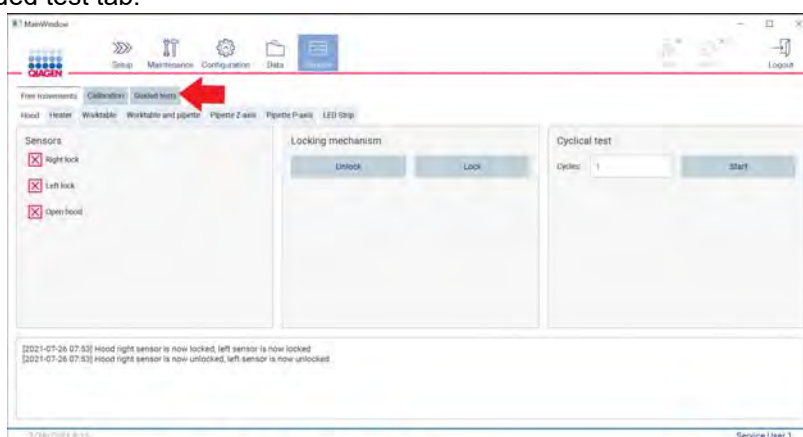
Table 8.3-1 VP2 UV Light Test

1. Log in to the EZ2 software as a service user.
2. Select the service tab.



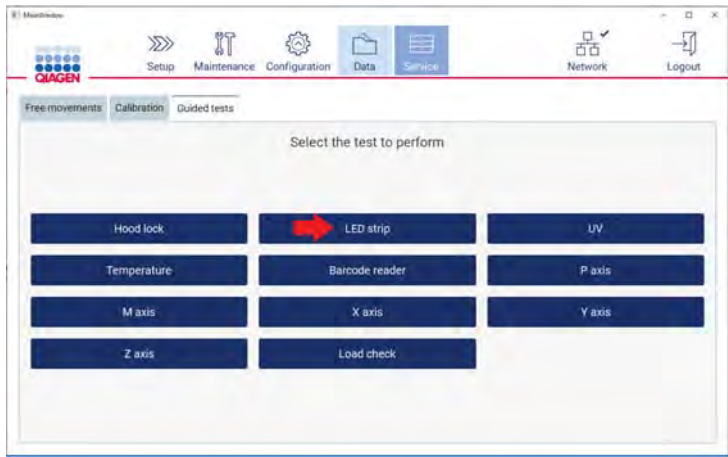
Picture 8.3-1 Service Tab

3. Select the guided test tab.



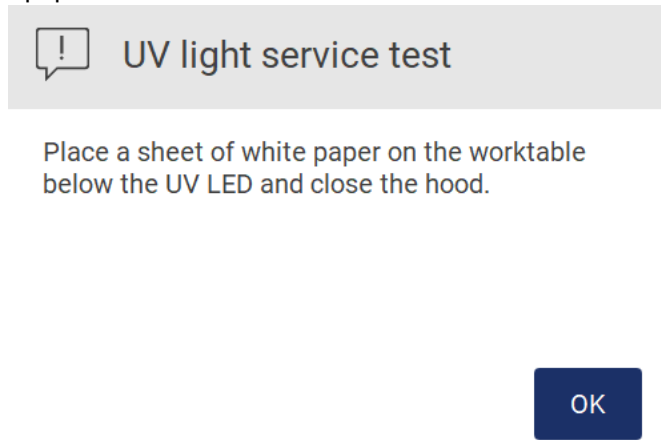
Picture 8.3-2 Guided Test Tab

4. Select the “UV” button.



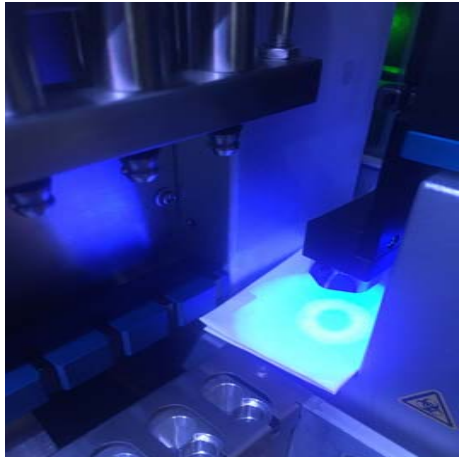
Picture 8.3-3 UV Tab

5. Place a white sheet of paper on the worktable under the UV LED and select the “OK” button



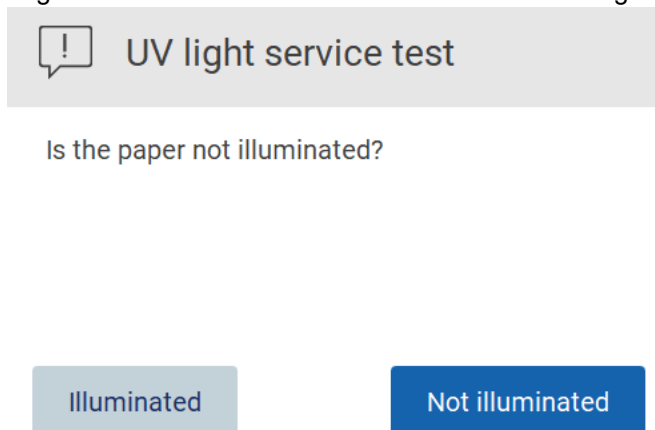
Picture 8.3-4 UV Test

6. Check that the UV light is working properly as seen in the picture below.



Picture 8.3-5 UV On Paper.

7. After checking the UV light select the “Illuminated” tab if the UV is working.



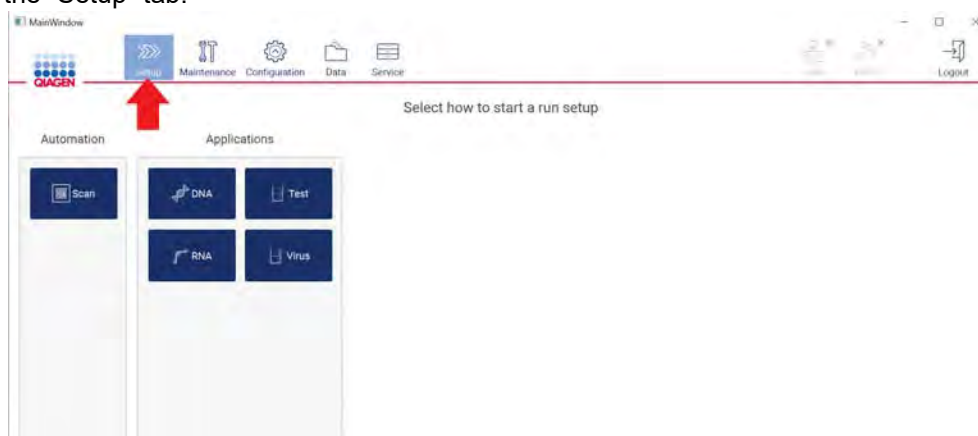
Picture 8.3-6 Illuminated

8.4 VP3 - Leakage Test

Purpose	This test is to check the tips are on correctly and that the O-rings and plungers in the pipette module are sealed and working correctly.
Item(s) Required	24 tips 24 tip holders Twenty Four (24) 2 ML tubes
Tool(s) Required	NA
Verification Time	7 minutes

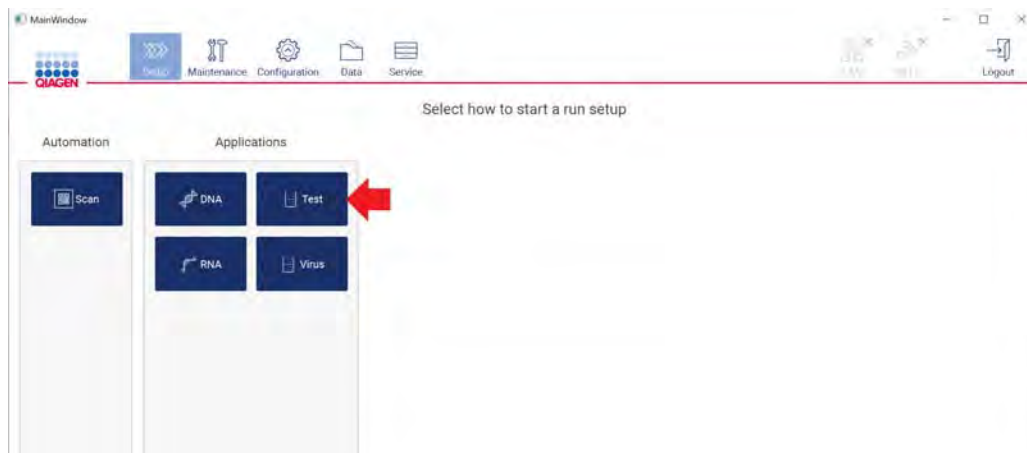
Table 8.4-1 VP3 Leakage Test

1. Log in to the EZ2 software as a service user.
2. Select the “Setup” tab.



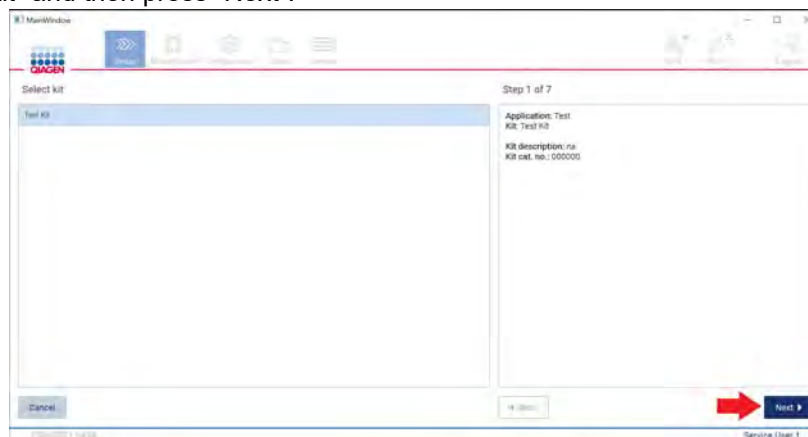
Picture 8.4-1 Setup

3. Select the “Test” button.



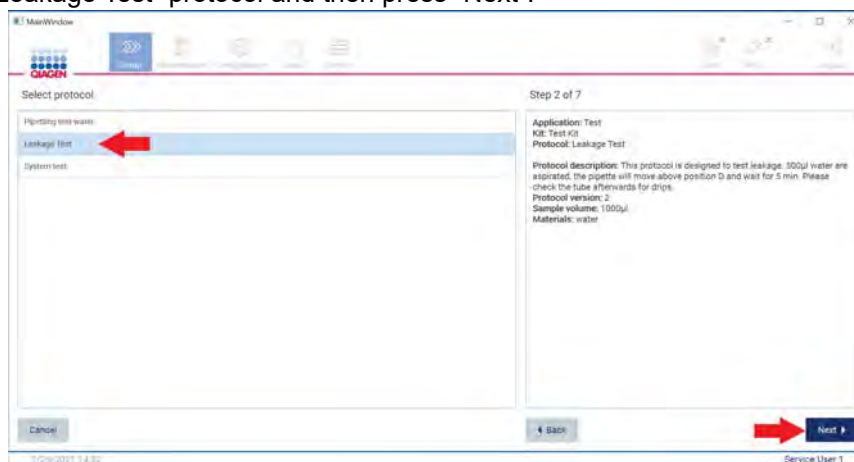
Picture 8.4-2 Test Applications

4. Select “Test Kit” and then press “Next”.



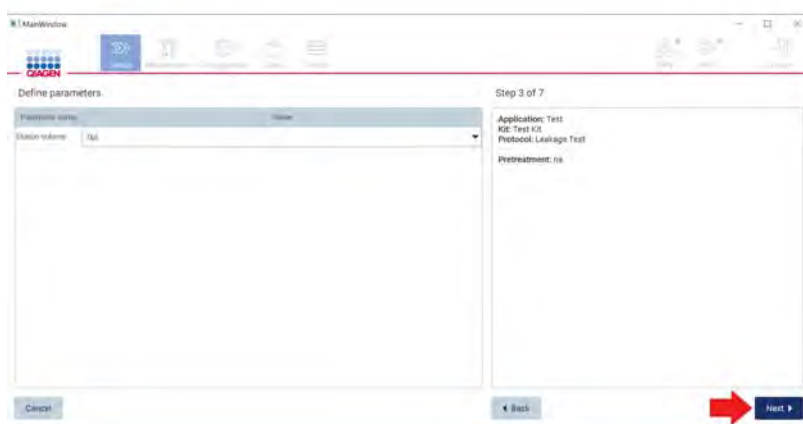
Picture 8.4-3 Test Kit

5. Select the “Leakage Test” protocol and then press “Next”.



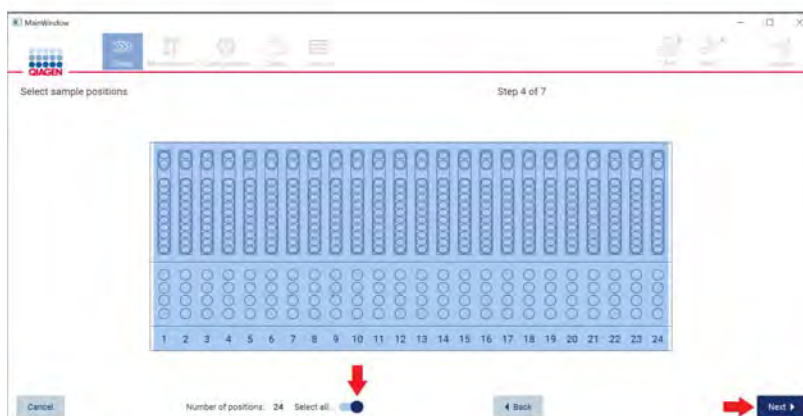
Picture 8.4-4 Leakage Test

6. Press the “Next” button.



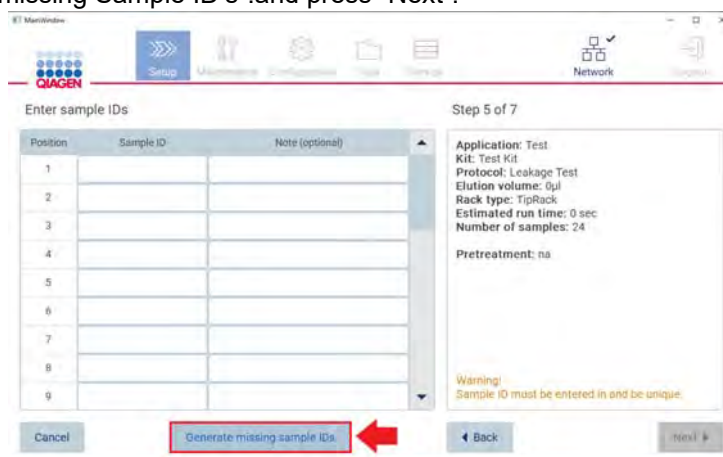
Picture 8.4-5 Elution

7. Use the slide button to select all the sample positions on the page and then press “Next”.



Picture 8.4-6 Sample Selected

8. Press “Generate missing Sample ID’s”.and press “Next”.



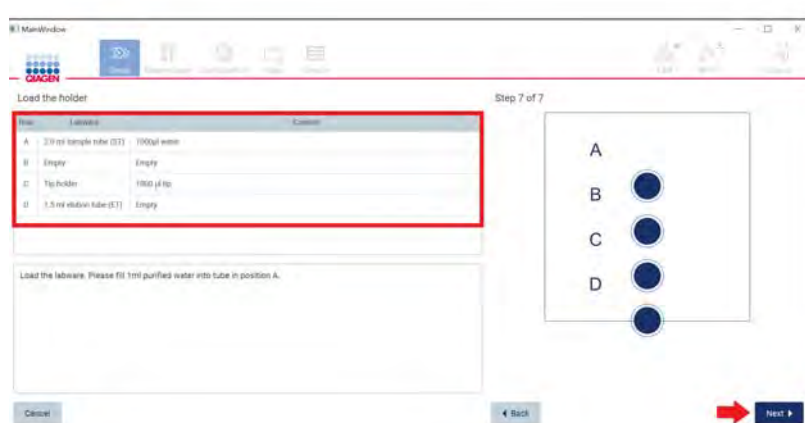
Picture 8.4-7 Sample ID

9. No cartridge is needed, select the “Next” tab on the cartridge page.



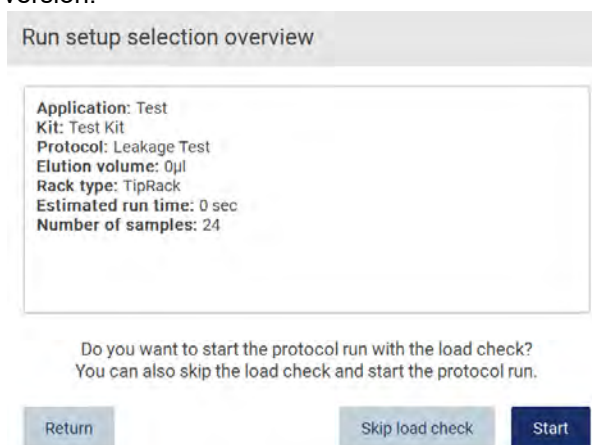
Picture 8.4-8 Cartridge

10. Load the positions as the page specifies and press “Next”.



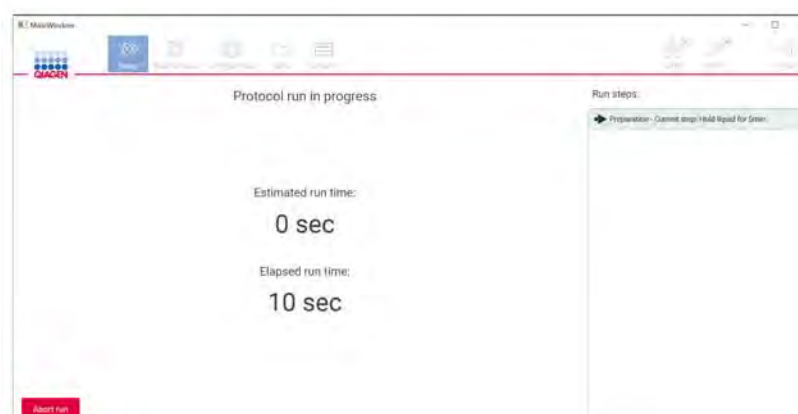
Picture 8.4-9 Loading

11. Select the “Skip load check” tab on the MDX or FX models of the EZ2 and the leakage test will start or press “Start” on the LS version.



Picture 8.4-10 Start Run

12. The protocol will run for 5 minutes and complete the test.



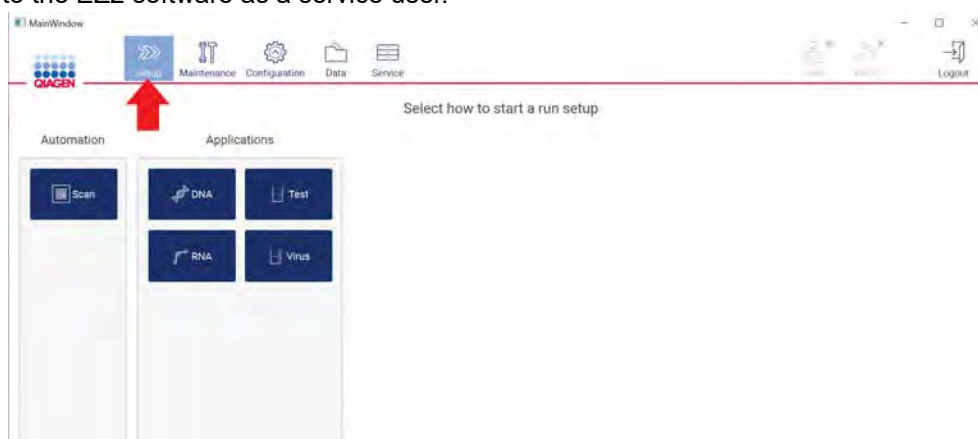
Picture 8.4-11 Leakage Timer

13. Check for leakage while the protocol is running.
14. Check the elution tubes are dry and that no liquid is in the bottom.
15. If no leakage occurred select the "Pass" button.

8.5 VP4 - Pipetting Test

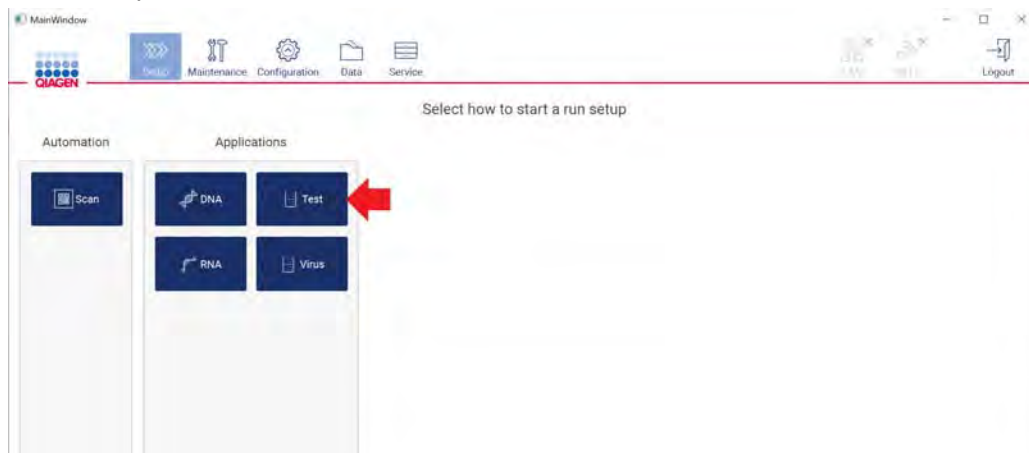
Purpose	This test is to check that all 24 pipette channels are pipetting the correct volume of fluid.
Item(s) Required	24 tips, 24 tip holders Twenty Four (24) 2 ML sample tubes with lids Twenty Four (24) 2 ML sample tubes
Tool(s) Required	Weight Scale
Verification Time	30 minutes

1. Log in to the EZ2 software as a service user.



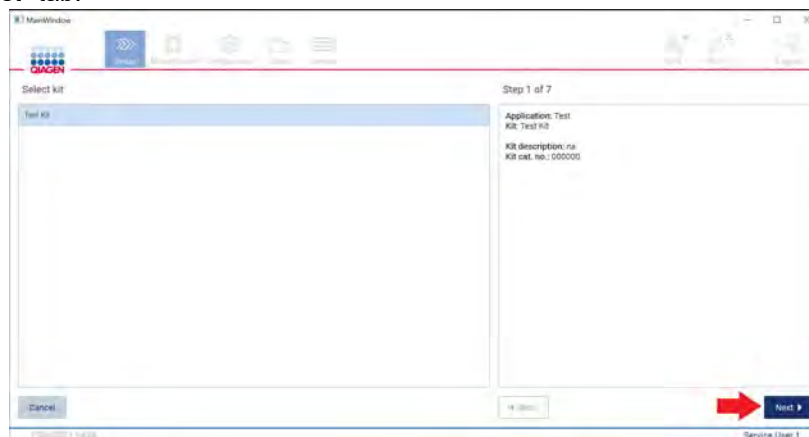
Picture 8.5-1 Setup

2. Select the “Setup” button.



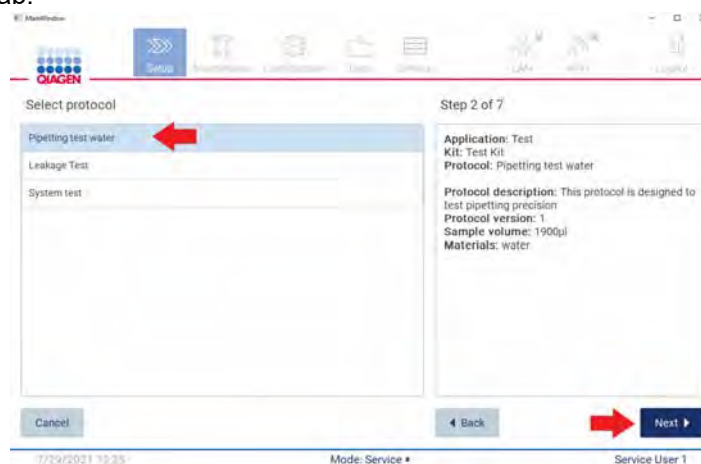
Picture 8.5-2 Test Applications

3. Select the “Test” tab.



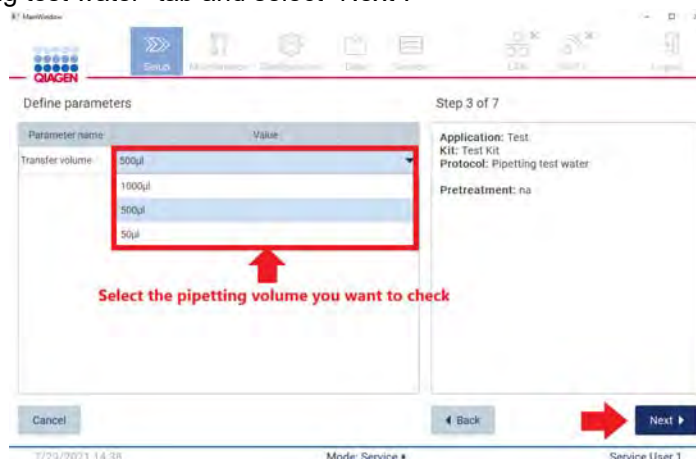
Picture 8.5-3 Test Kit

4. Select the “Next” tab.



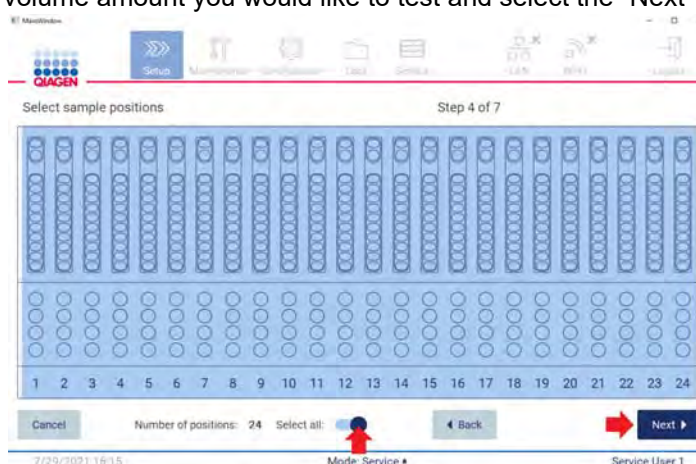
Picture 8.5-4 Pipette Test

5. Select the “Pipetting test water” tab and select “Next”.



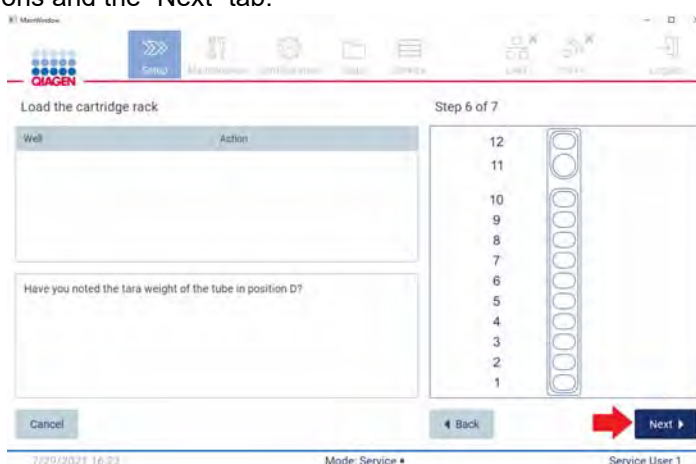
Picture 8.5-5 Pipetting Volume Amount

6. To verify the pipetting run all 3 different volume amounts the 1000, 500 and 50.
7. Select the transfer volume amount you would like to test and select the “Next” tab.



Picture 8.5-6 All Positions

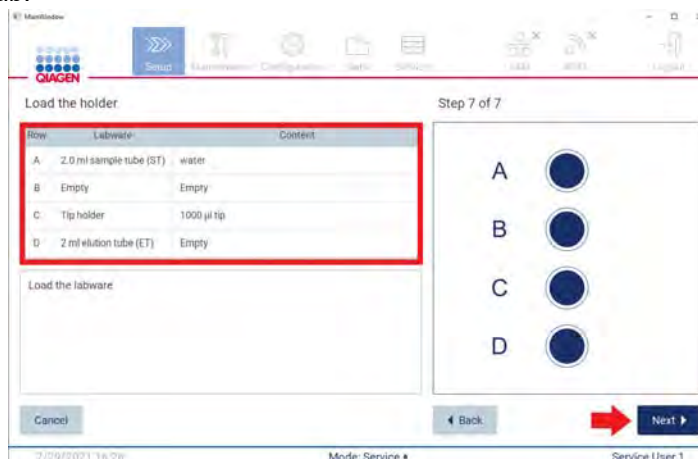
8. Select all the positions and the “Next” tab.



Picture 8.5-7 No Cartridge Needed

9. No cartridges will be needed for the pipetting test.

10. Select the “Next” tab.



Picture 8.5-8 Load Positions

11. Load a 2 mL tube in position A and fill it with water.

12. B position is empty.

13. Place a tip holder and tip in position C.



Picture 8.5-9 Number Tubes and Lids

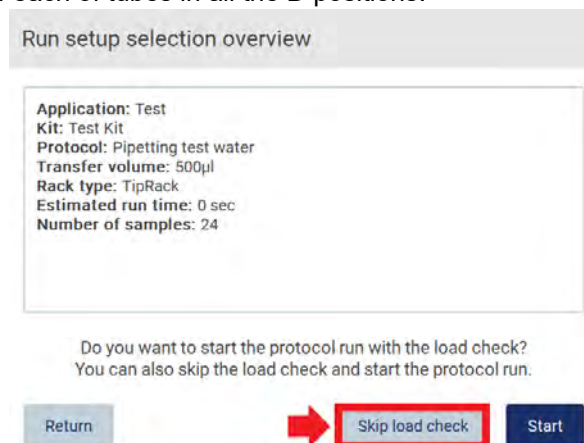
14. Number the tubes from 1-24 that go in position D.



Picture 8.5-10 Weighing Tubes

15. Use tubes with lids and weigh the 2 mL empty tubes that go in position D.

16. Document the weight of each of tubes in all the D positions.



Picture 8.5-11 Skip Load Check

17. Close the hood and select "Skip load check" on the MDX and FX version, and the pipette test will start immediately. If the EZ2 is a LS model just select "Start".
18. After the pipetting test has finished weigh each tube in position D and subtract the weight of the empty tube documented in step 14.
19. Enter the documented weights of each tube on the Excel spreadsheet.

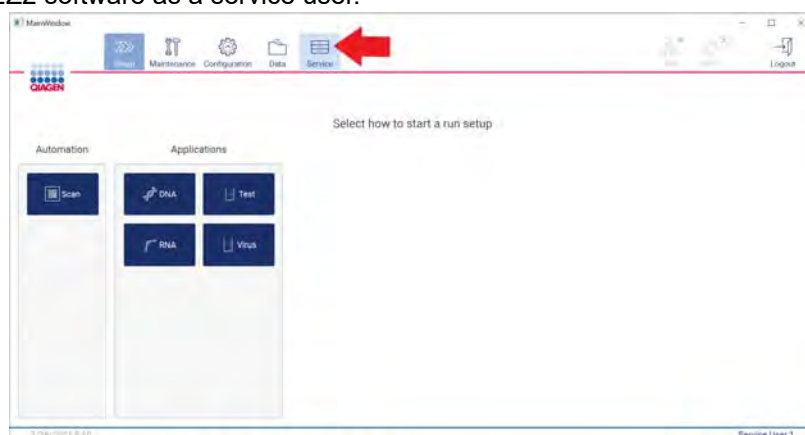
8.6 VP5 - Temperature Test

Purpose	This test is to verify that the heater module is reaching the correct temperature.
Tool(s) Required	Temperature Meter, Temperature Probe
Verification Time	24 minutes

Table 8.6-1 VP5 Temperature Test

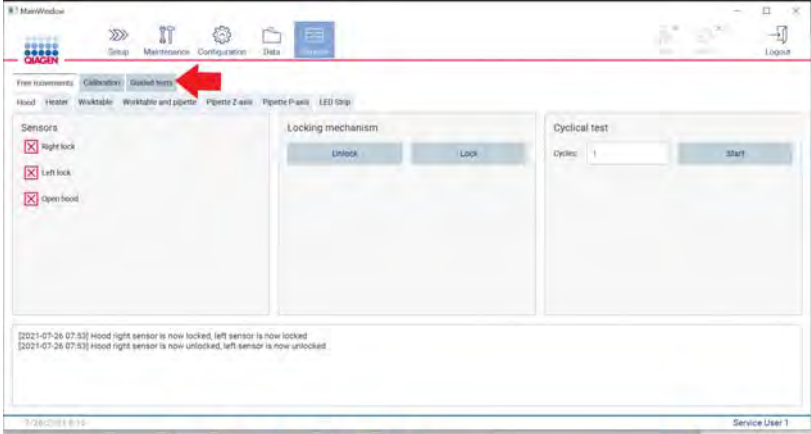
WARNING 	Hot surface The heating system can reach temperatures of up to 95°C (203°F). Avoid touching the heating system when it is hot. w12
---	---

1. Log in to the EZ2 software as a service user.



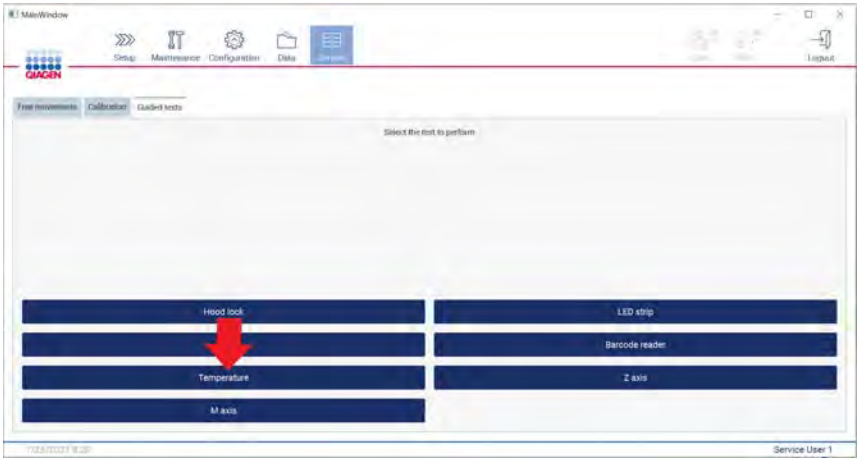
Picture 8.6-1 Service Tab

2. Select the service tab.



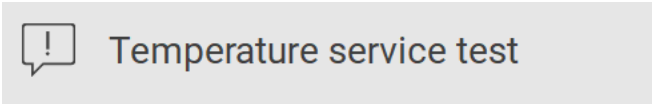
Picture 8.6-2 Guided Test Tab

3. Select the guided test tab.

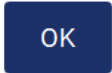


Picture 8.6-3 Temperature Test

4. Select the "Temperature" tab.



Close the hood.



Picture 8.6-4 Close The Hood

5. Close the hood and select the "OK" tab.

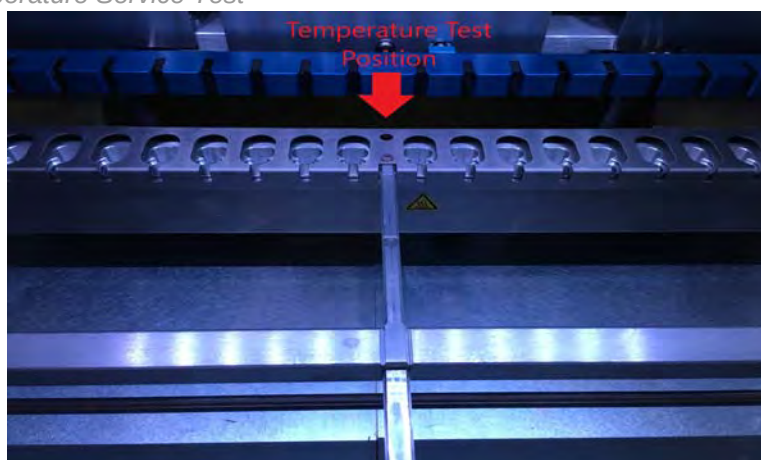


Temperature service test

Open the hood. Load the 200 µl water into the testing hole in the heating block.

OK

Picture 8.6-5 Temperature Service Test



6. Temperature Hole Position
7. Load 200 micro-liters in the temperature testing hole position.
8. Select the "OK" tab.



Temperature service test

Insert the thermoprobe in the testing hole and close the hood.

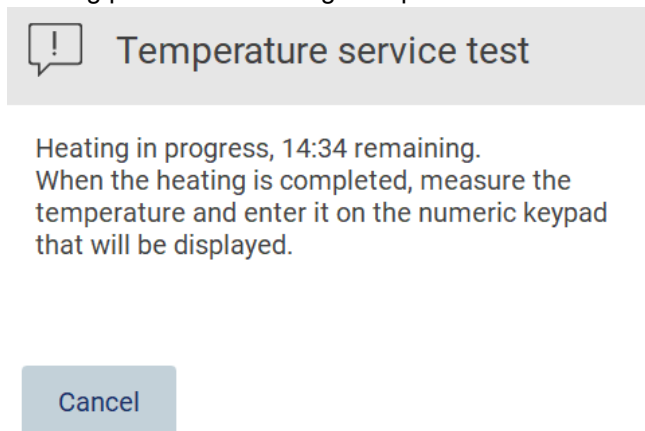
OK

Picture 8.6-6 Testing Probe



Picture 8.6-7 Temperature Probe Inserted

9. Insert the temperature testing probe in the testing hole position.



Picture 8.6-8 Temperature Waiting Time

10. The temperature test will begin counting down from 15 minutes to reach the correct temperature.



Picture 8.6-9 Enter Temperature

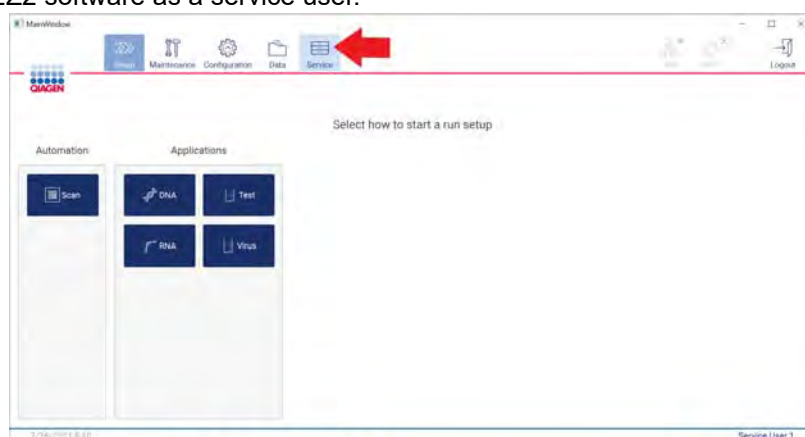
11. Once the countdown is completed enter the temperature in software and select the "Accept" tab.
12. The first temperature should be 60.0 degrees Celsius. Enter the temperature as 600.
13. Next the temperature test will count down from 8 minutes and enter the temperature the meter is now reading.
14. The second temperature should be 95 degrees Celsius, enter the temperature as 950 and select "Accept".
15. Ensure that the temperature test has passed and select "OK".
16. Open the hood and remove the temperature testing probe and clean the testing hole area.

8.7 VP6 - Hood Lock Test

Purpose	This test is to check that the hood is locked in the down position and can not open when the instrument is operating for the safety of the operator.
Tool(s) Required	NA
Verification Time	1 minutes

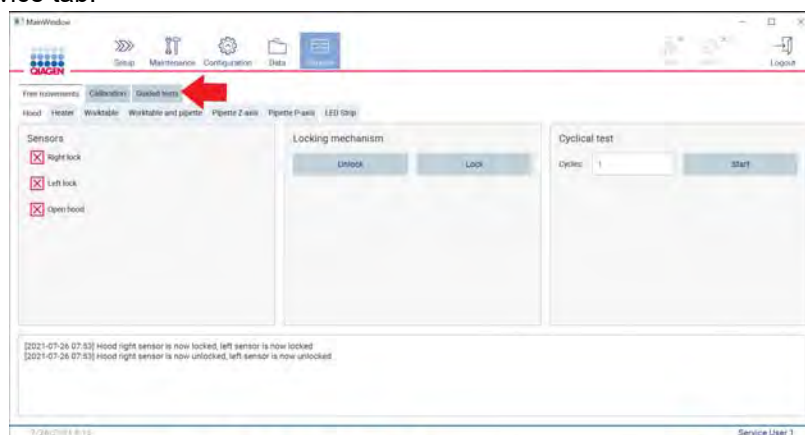
Table 8.7-1 VP6 Hood Lock Test

1. The rear cover will need to be in position while performing this test.
2. Log in to the EZ2 software as a service user.



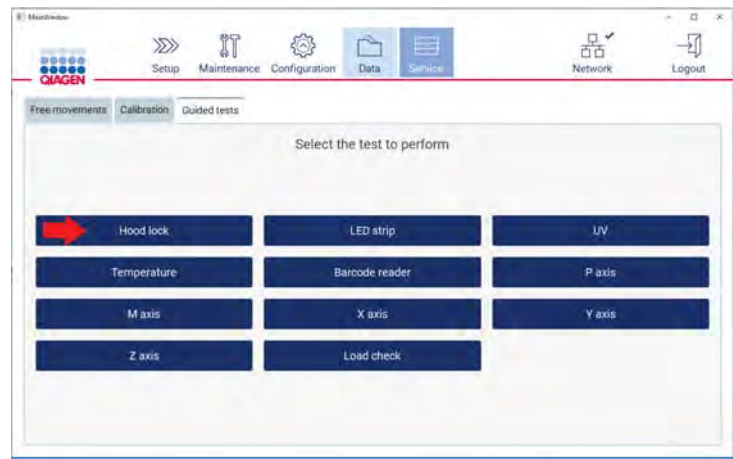
Picture 8.7-1 Service Tab

3. Select the service tab.



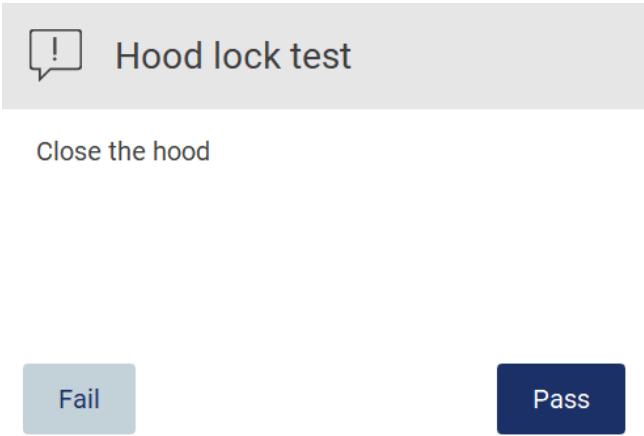
Picture 8.7-2 Guided Test Tab

4. Select the guided test tab.



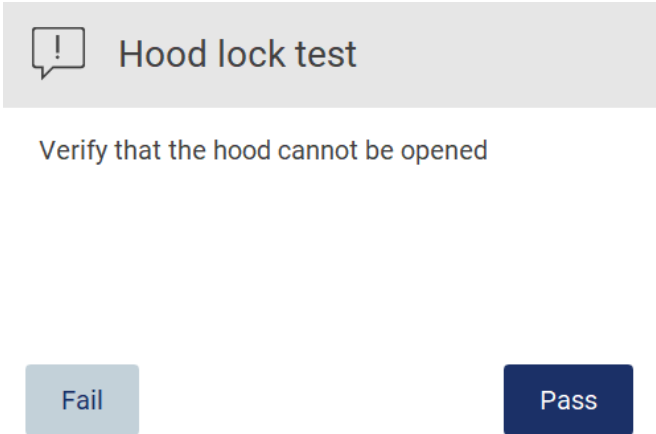
Picture 8.7-3 Hood Lock Tab

5. Select the “Hood Test” tab.



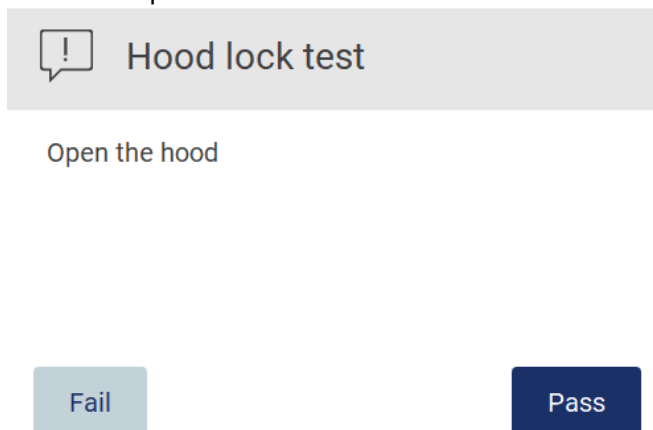
Picture 8.7-4 Close The Hood

6. Close the hood and select the “Pass” tab.



Picture 8.7-5 Hood Locked

7. Check that the hood can not be opened and select the "Pass" tab.



Picture 8.7-6 Open Hood

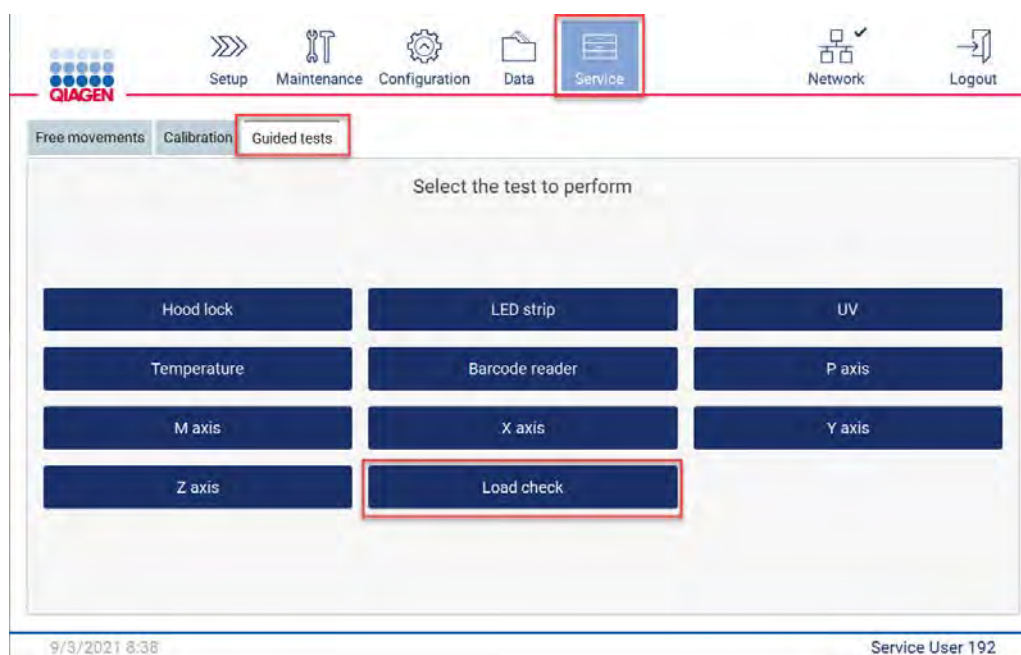
8. Verify that the hood can be opened and select the "Pass" tab.
9. Select the "OK" tab to generate the report.

8.8 VP7 - Load Check Test

Purpose	Confirm the load check process is able to successfully detect the loaded consumables, and identify loading errors.
Item(s) Required	Unpierced cartridge Pierced cartridge Three (3) 2mL Sample tubes with caps Two (2) 1.5mL Elution tubes with caps
Verification Time	8 Minutes

Table 8.8-1 VP7 Load Check Test

1. Log into the EZ2 system using the Service log in.
2. Press the Service environment button, and the Guided tests tab.

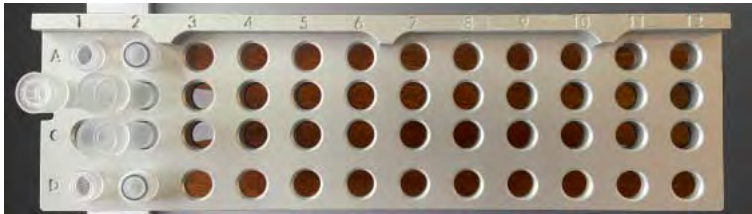


- 3. Press the Load check button to start the test.
- 4. Load the left cartridge rack according to the following table.



Picture 8.8-1 Load check cartridge rack

- 5. Load the left tip rack according to the following table.



Picture 8.8-2 Load check tip rack

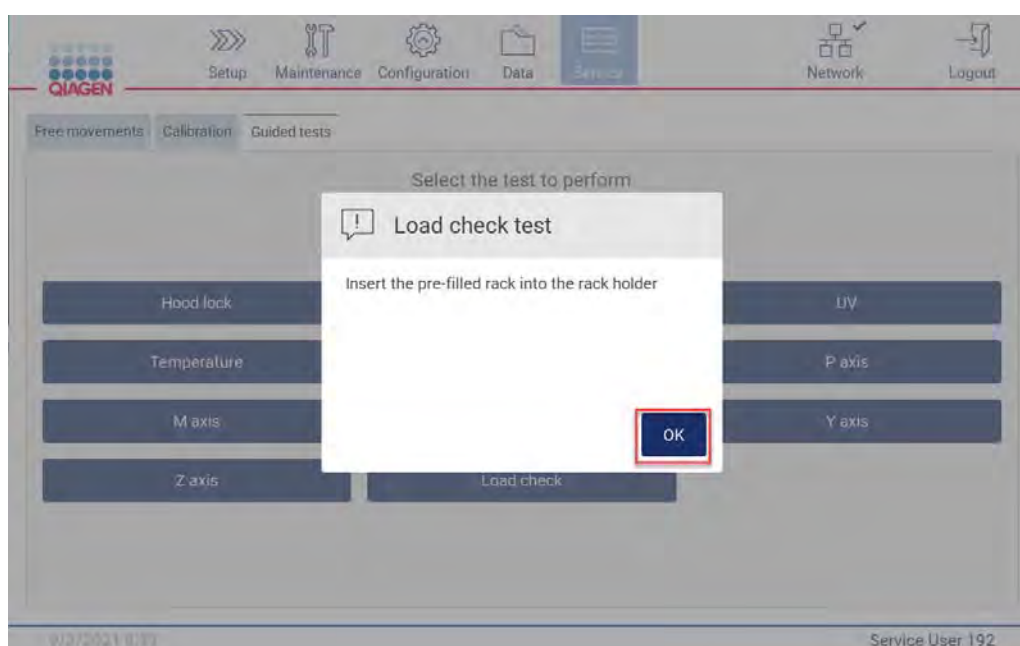
Rack	Position	Sample Columns		
		1	2	Remaining Positions
Cartridge Rack	Cartridge	Unpierced cartridge with QR code	Pierced cartridge with QR code	Empty
	Well 11	Empty	Capped 2mL Sample Tube	
Tip Rack	A	Uncapped 2mL Sample Tube	Capped 2mL Sample Tube	
	B	Empty	Tip holder without tip	
	C	Tip holder with tip	Tip holder without tip	
	D	Uncapped 1.5mL Elution Tube	Capped 1.5mL Elution Tube	

Table 8.8-2

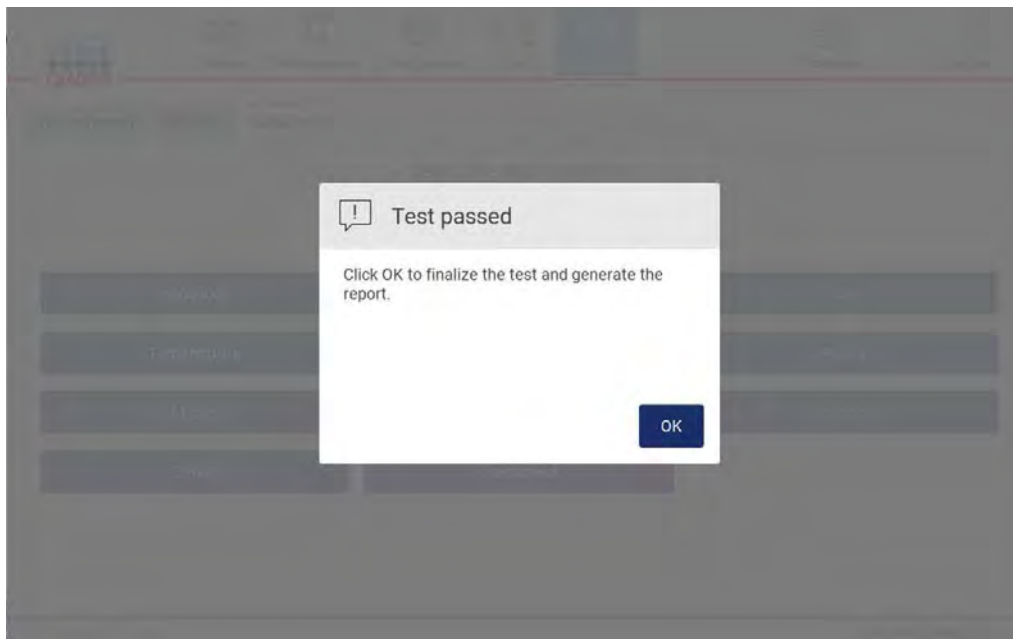
- 6. Insert the loaded cartridge and tip rack into the instrument.



Picture 8.8-3 Load check racks loaded into the EZ2



7. Press OK on the prompt that the prefilled racks are loaded.
8. Allow the Load check to complete.
9. After the Load check scans the racks, a prompt will inform the user the results of the QR code scan. Confirm the scanned results are consistent with the loaded cartridges.



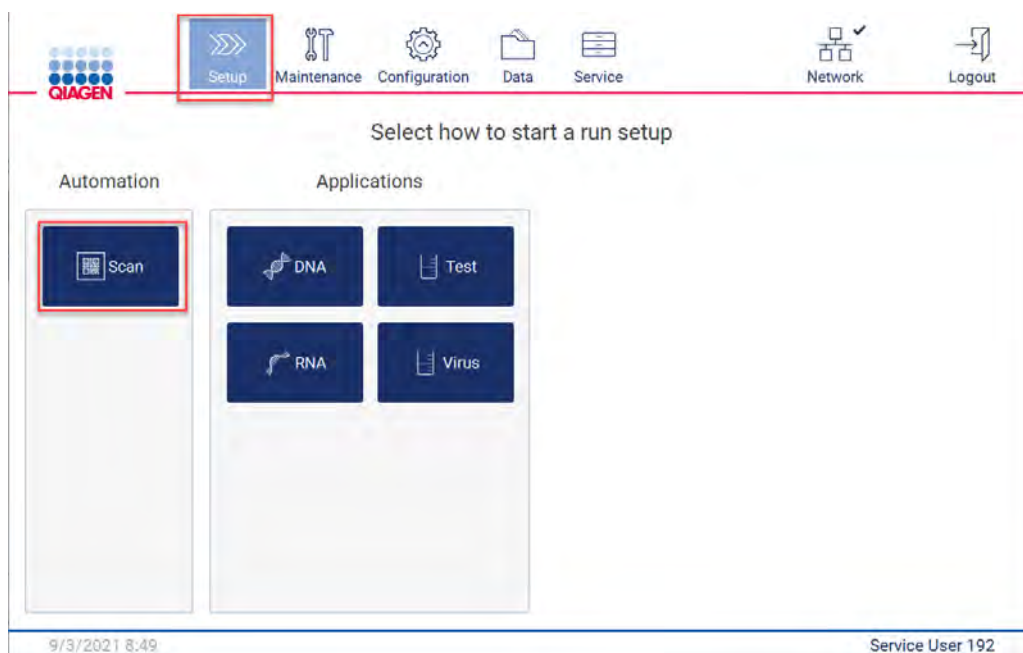
10. If the load check test was successful, there should be no error message and a prompt displayed indicating a successful test.
11. Press OK to generate the report.

8.9 VP8 - Workflow System Test

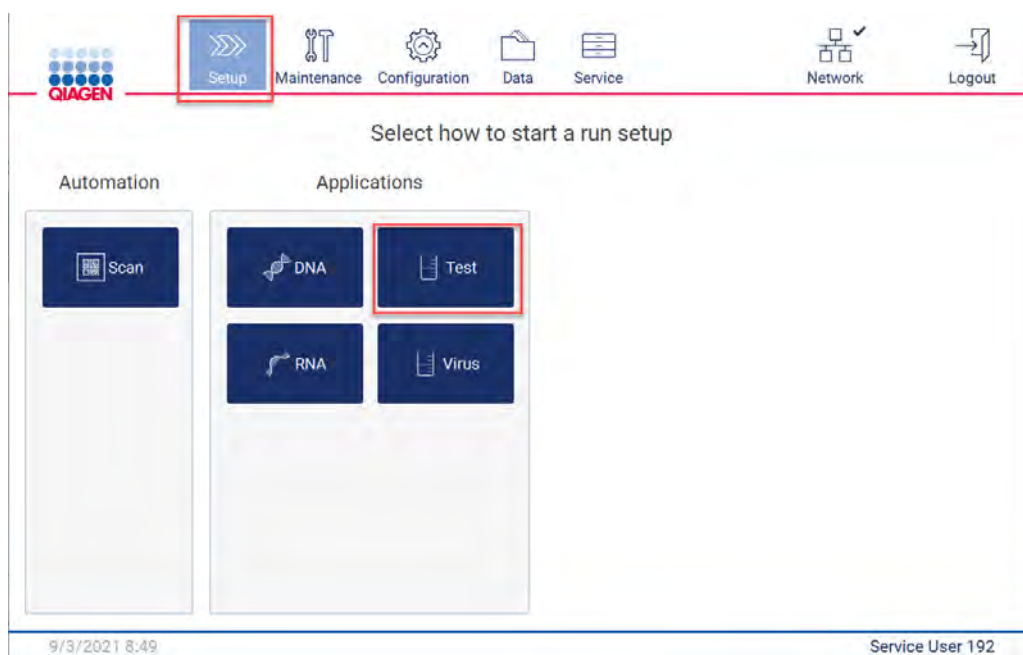
Purpose	This tests checks all steps of a typical protocol, to confirm that the instrument can successfully perform all steps of the workflow.
Tool(s) Required	Service Kit
Verification Time	25 minutes

Table 8.9-1 VP2 Workflow System Test

1. Log into the EZ2 system using the Service log in.
2. In the Setup environment, press the Scan button.



2.1.If there is no handheld barcode reader (EZ2 Connect LS), press the Test button under Applications.



2.2.Press Next after confirming Test kit is selected.

QIAGEN

Setup Maintenance Configuration Data Service Network Logout

Select kit

Step 1 of 7

Test Kit

Application: Test
Kit: Test Kit
Kit description: na
Kit cat. no.: 000000

Cancel Back Next

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3. Select System Test under the available protocols, and press Next.

QIAGEN

Setup Maintenance Configuration Data Service Network Logout

Select protocol

Step 2 of 7

Pipetting test water
Leakage Test
System test

Application: Test
Kit: Test Kit
Protocol: System test
Protocol description: This system test includes load check, please make sure to make a picture of load check results.
Protocol version: 2.2
Sample volume: 100µl
Materials: water

Cancel Back Next

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4. Confirm the elution volume, and press Next.

QIAGEN

Setup Maintenance Configuration Data Service Network Logout

Define parameters Step 3 of 7

Parameter name	Value
Elution volume	100µl

Application: Test
Kit: Test Kit
Protocol: System test
Pretreatment: na

Cancel Back Next

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5. Select all twenty-four positions with the toggle button. When all positions are active, press Next.

QIAGEN

Setup Maintenance Configuration Data Service Network Logout

Select sample positions Step 4 of 7

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24

Cancel Number of positions: 24 Select all: ☒ Back Next

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6. Press Generate missing sample IDs or enter the appropriate sample IDs and press Next.

Enter sample IDs

Step 5 of 7

Position	Sample ID	Note (optional)
1	2021-09-03_08-57_01	
2	2021-09-03_08-57_02	
3	2021-09-03_08-57_03	
4	2021-09-03_08-57_04	
5	2021-09-03_08-57_05	
6	2021-09-03_08-57_06	
7	2021-09-03_08-57_07	
8	2021-09-03_08-57_08	
9	2021-09-03_08-57_09	

Application: Test
 Kit: Test Kit
 Protocol: System test
 Elution volume: 100µl
 Rack type: TipRack
 Estimated run time: 4 min 36 sec
 Number of samples: 24
 Pretreatment: na

Cancel Generate missing sample IDs Back Next

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7. Prepare the cartridge racks with the QC water cartridges in all twenty-four (24) positions.
8. Place an uncapped 2.0mL screw cap Sample Tube in the well 11 heater position in all twenty-four cartridges.
9. After preparing the cartridge racks, insert the cartridge racks into the instrument.

Load the cartridge rack

Step 6 of 7

Well	Action
11	Load empty 2.0 ml sample tube (ST)

Prepare to take a picture of load check results!

Cancel Back Next

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10. Press Next once the cartridges, 2.0mL tube are loaded and the cartridge racks inserted into the instrument.
11. Prepare the tip racks with the labware in all 24 sample columns according to the following table.

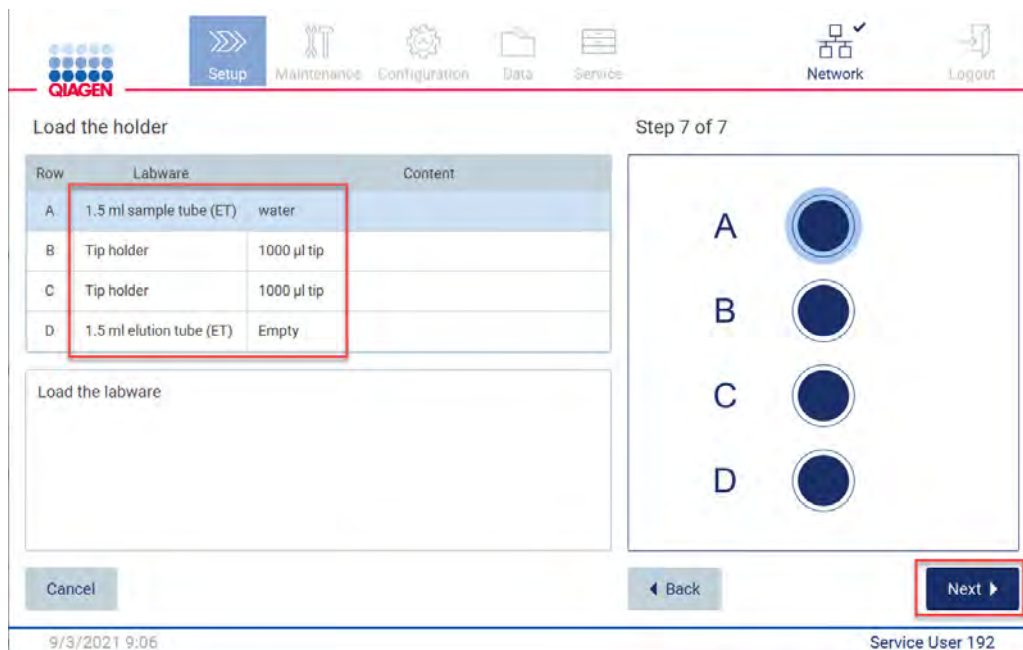
Row	Labware	Contents
A	1.5mL elution tube	xxxx water

Table 8.9-2

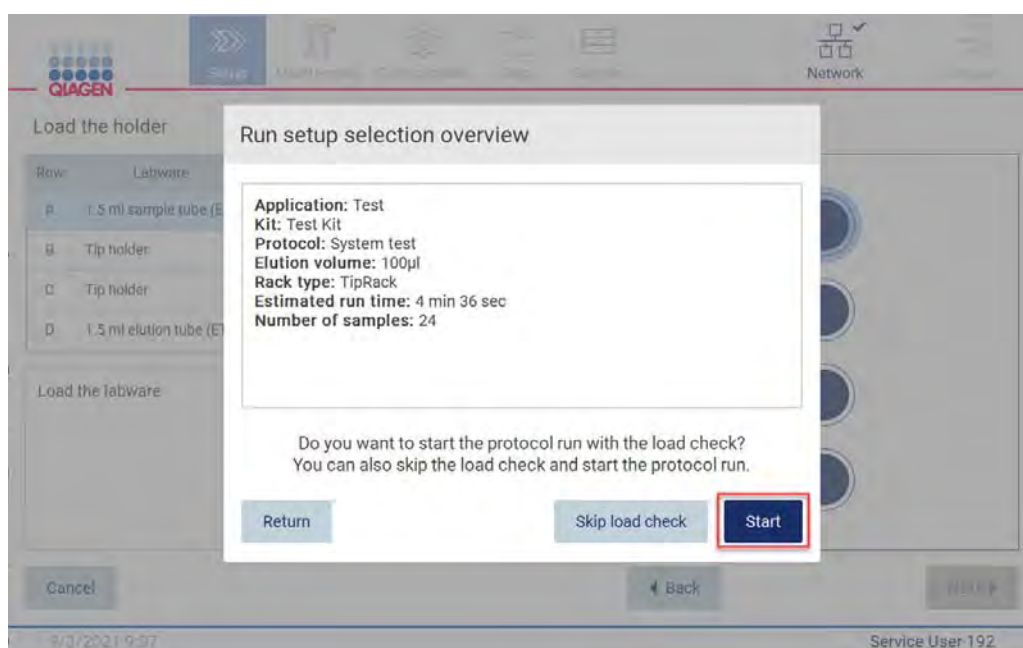
Row	Labware	Contents
B	Tip holder	Tip
C	Tip holder	Tip
D	1.5mL elution tube	Empty

Table 8.9-2

12. After loading the consumables in the tip racks, load the tip racks into the EZ2.
13. Once the tip racks are loaded and inserted into the EZ2, press Next.



14. Review the Run setup. If no changes are necessary, start the run by pressing Next. On Fx and MDx instruments, the load check must be performed.



15. During the run, observe the instrument.
 - Tip pick up should be smooth and complete.
 - The hood can not be opened.

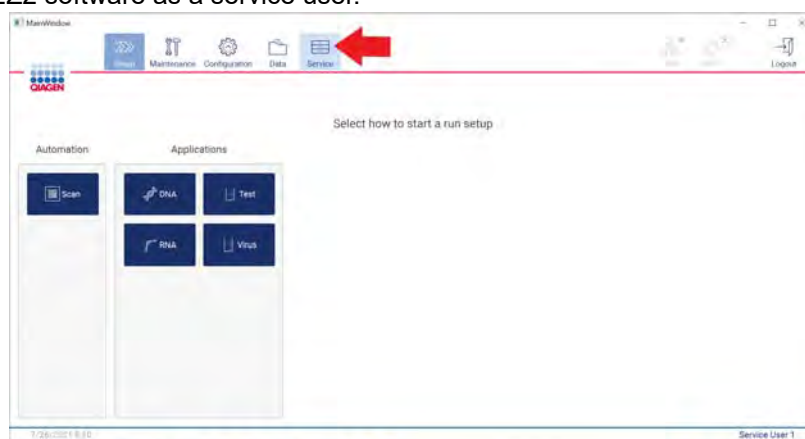
16. After the run completes successfully, check the LED strip is blinking. Confirm the volume in the elution tubes is approximately 100µL.
17. Clean the workdeck by removing used consumables. Perform the maintenance as prompted after the run completes.

8.10 VP9 - Barcode Reader Test

Purpose	This test is to check that the barcode ID is correctly read and translated to the software.
Tool(s) Required	list required tools
Verification Time	1 minutes

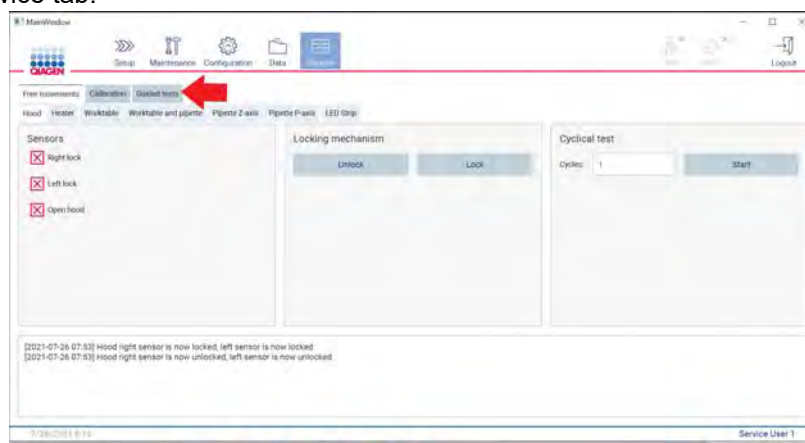
Table 8.10-1 VP9 Barcode Reader Test

1. Log in to the EZ2 software as a service user.



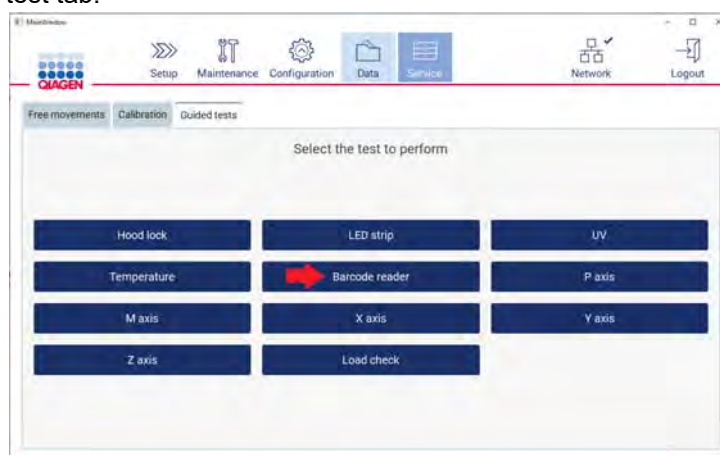
Picture 8.10-1 Service Tab

2. Select the service tab.



Picture 8.10-2 Guided Test Tab

3. Select the guided test tab.



Picture 8.10-3 Barcode Reader Test

4. Select the "Barcode Reader" tab.

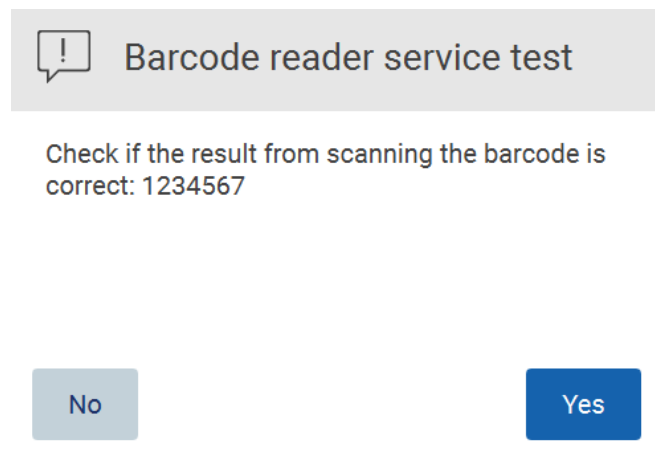


Picture 8.10-4 Barcode Scan Number

5. Use the handheld barcode scanner and scan the barcode below.



6.



Barcode reader service test

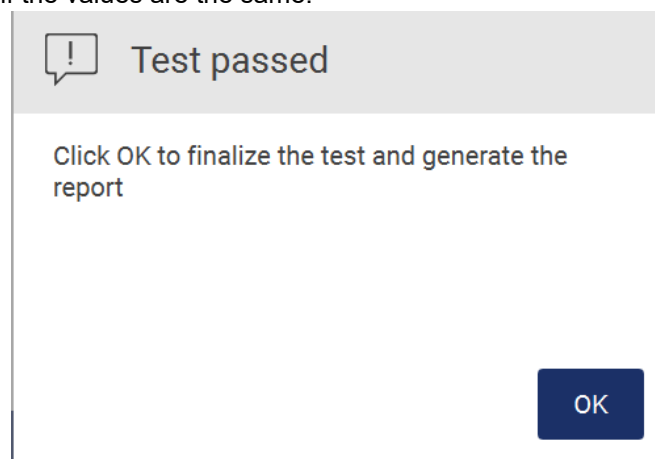
Check if the result from scanning the barcode is correct: 1234567

No Yes

Picture 8.10-5 Barcode Reading Verification

7. Verify that the result in the software is QIAGEN Test, consistent with the barcode scanned.

8. Select the “Yes” tab if the values are the same.



Test passed

Click OK to finalize the test and generate the report

OK

Picture 8.10-6 Barcode Testing Completed

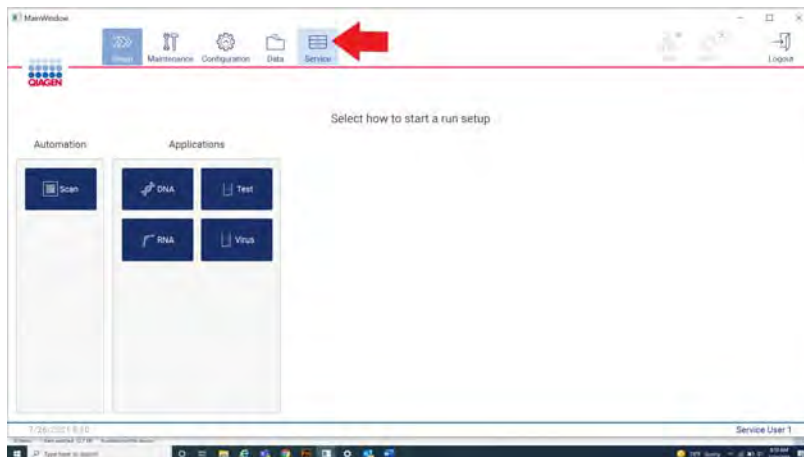
9. Select the “OK” tab to generate a report.

8.11 VP10 - Z Axis Movement Test

Purpose	This test is to check for step loss after parts have been replaced or calibrations have been completed.
Tool(s) Required	0.02- 1 mm Feeler Gauge Set
Verification Time	5 Minutes

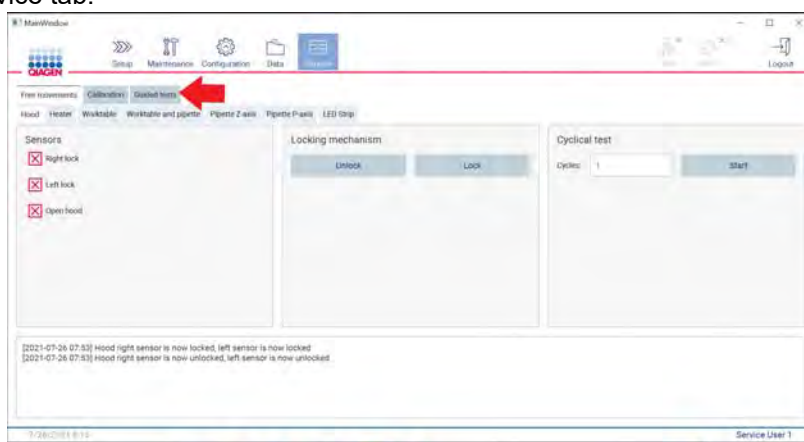
Table 8.11-1 VP10 Z Axis Movement Test

1. Log in to the EZ2 software as a service user.



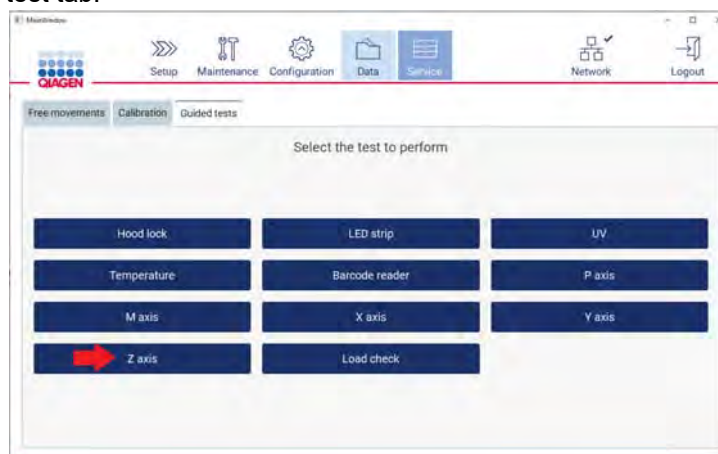
Picture 8.11-1 Service Tab

2. Select the service tab.



Picture 8.11-2 Guided Test Tab

3. Select the guided test tab.



Picture 8.11-3 Z Axis

4. Select the “Z Axis” tab.

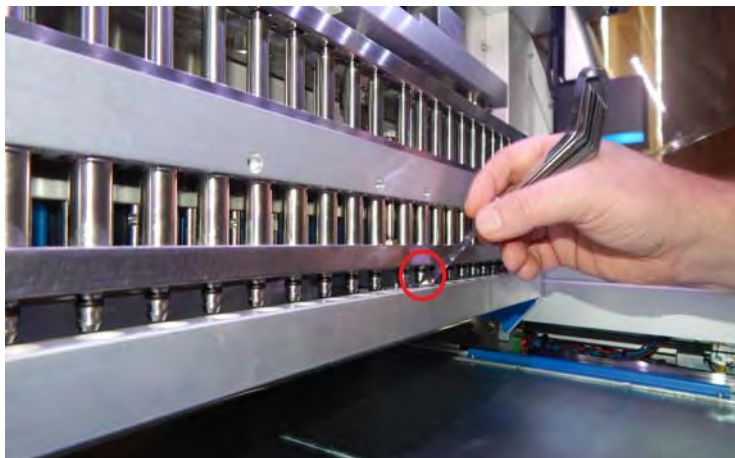


Z-Axis motor test

Measure the distance between the tip adaptor and the middle bar of the worktable.

OK

Picture 8.11-4 Measure



Picture 8.11-5 Feeler Gauge

5. Use a feeler gauge to check the gap between all the pipette and worktable positions.

Enter measured distance [μm]

7

8

9

4

5

6

1

2

3

0

Cancel

Accept

Picture 8.11-6 Enter 1st Measurement

6. Enter the smallest gap found with the feeler gauge.
7. The measurement is entered in as 1/10 of a millimeter.
8. Enter the first measurement in the software box and select the “Accept” tab.

9. The instrument will now move the Z to multiple positions and down again to be measured.

Enter measured distance [μm]

10

7 8 9

4 5 6

1 2 3

0

Cancel Accept

Picture 8.11-7 Enter 2nd Measurement

10. Measure the distance again and enter it in the software box then select the “Accept” tab.

Test passed

Click OK to finalize the test and generate the report

OK

Picture 8.11-8 Finalize

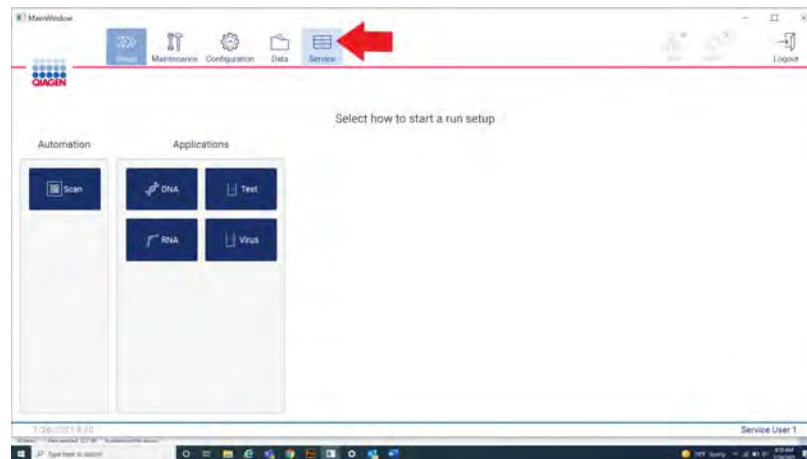
11. Select the “OK” tab to finalize the test and generate the report.

8.12 VP11 - M Axis Movement Test

Purpose	This test is to check for step loss after parts have been replaced or calibrations have been completed on the magnet module.
Item(s) Required	7 tips with tip holders
Tool(s) Required	list required tools
Verification Time	2 minutes

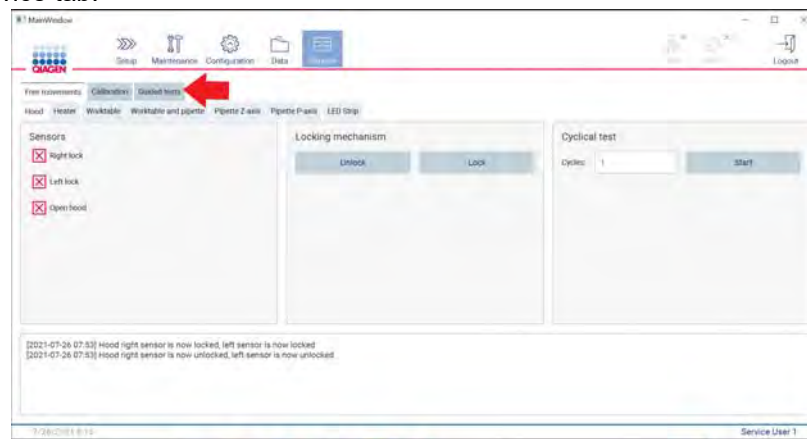
Table 8.12-1 VP11 M Axis Movement Test

1. Log in to the EZ2 software as a service user.



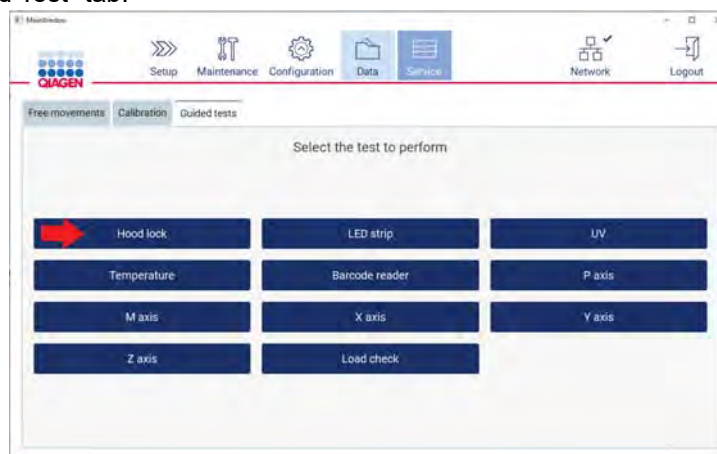
Picture 8.12-1 Service Tab

2. Select the service tab.



Picture 8.12-2 Guided Test Tab

3. Select the "Guided Test" tab.



Picture 8.12-3 M Axis

4. Select the “M axis” tab.



Picture 8.12-4 Tips

5. Open the hood and place tips in 1,4,8,12,16,20 and 24 in position C of the sample rack.



M-Axis motor test

Verify that there is no or almost no gap between the magnet and the tip.

Fail

Pass

Picture 8.12-5 1st Verification



Picture 8.12-6 Check Gap

6. Check the 7 positions, if the magnet has no gap or almost no gap from the tip select the “Pass” tab.

7. The instrument will now move the M axis to many positions to check for step loss.



M-Axis motor test

Verify that the gap between the magnet and the tip does not differ significantly from the one measured before.

Fail

Pass

Picture 8.12-7 2nd Verification

8. Verify that the magnet has little to no gap and select the "Pass" tab.



Test passed

Click OK to finalize the test and generate the report

OK

Picture 8.12-8 Finalize

9. Select the "OK" tab to generate the report.

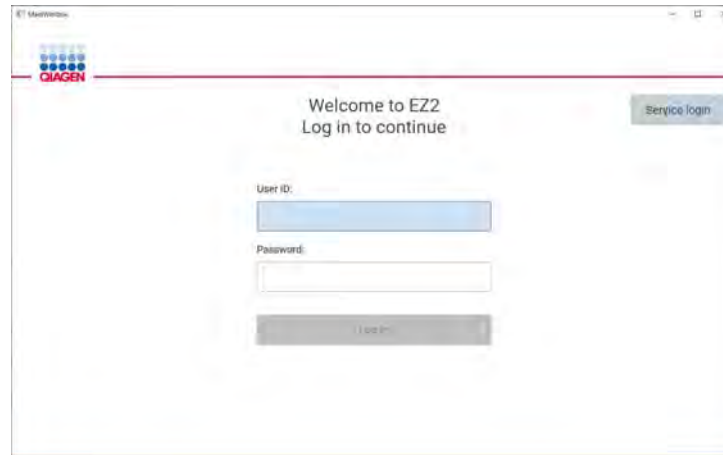
8.13 VP12 - System Initialization Test

Purpose	This test is to verify after service on the instrument is completed that the instrument functions correctly with no errors.
Tool(s) Required	NA
Verification Time	3 minutes

Table 8.13-1 VP12 System Initialization Test

1. Close the hood.
2. Turn the power on the EZ2 Connect.
3. Verify the system boots up.

- Log in to the EZ2 software as a service user with PIN.



Picture 8.13-1 Log In Screen

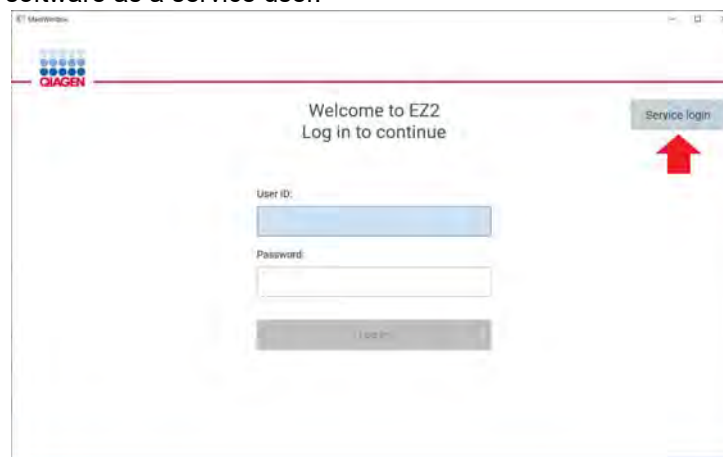
- Verify that the system initializes with no errors.
- After logging in to the software and the system initialized, test is complete.

8.14 VP13 - Service Switch

Purpose	This test is to check that the service switch is only active when the rear cover has been removed.
Tool(s) Required	T10
Verification Time	1 minutes

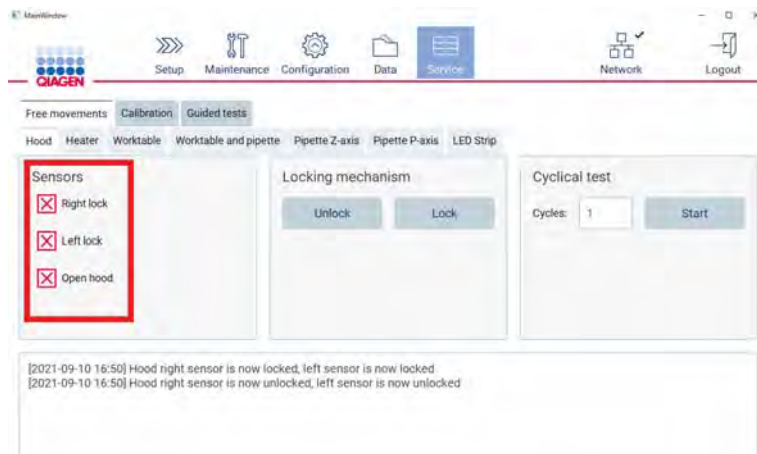
Table 8.14-1 VP13 Service Switch

- Remove the rear cover per [RP1 - Rear Cover - Removal Procedure](#).
- Plug in the EZ2 Connect and turn the power on.
- Log in to the EZ2 software as a service user.



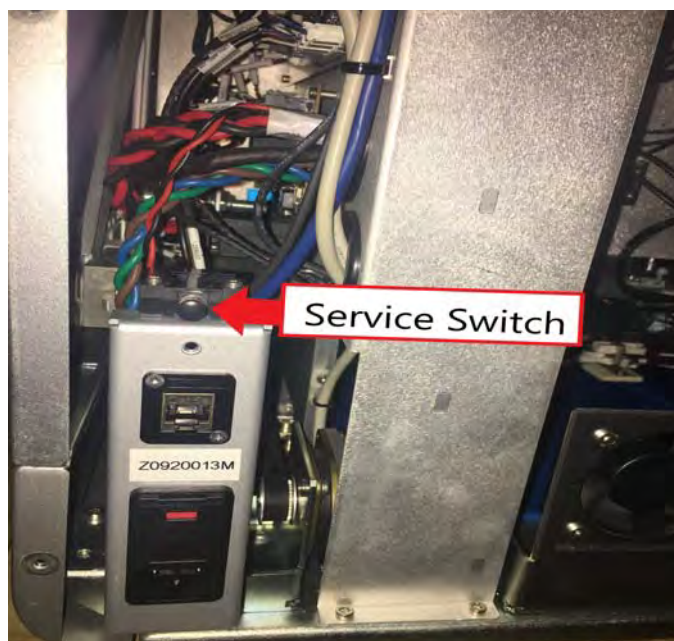
Picture 8.14-1 Service Log In

4. Select the service environment.



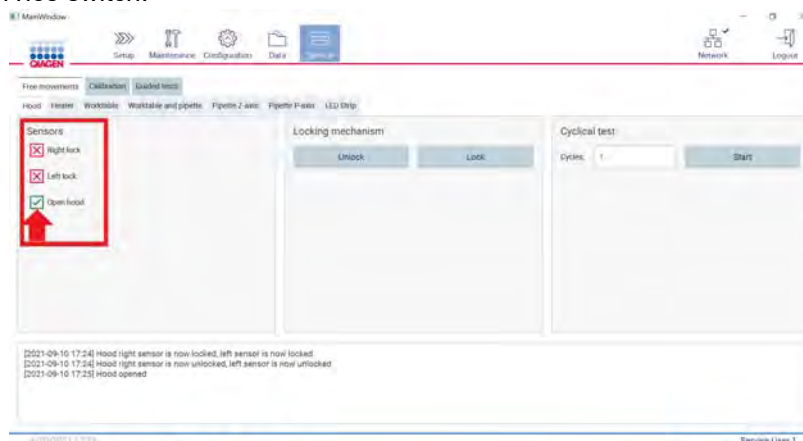
Picture 8.14-2 Rear Cover Off

5. Select the **"Free Movements"** environment.
6. All 3 hood sensors will have a red X next to the lock position.
7. Open the hood.



Picture 8.14-3 Service Switch

8. Push in the service switch.



9. The hood open sensor will now show a green check mark in the service environment.
10. The verification of the service switch is now complete.

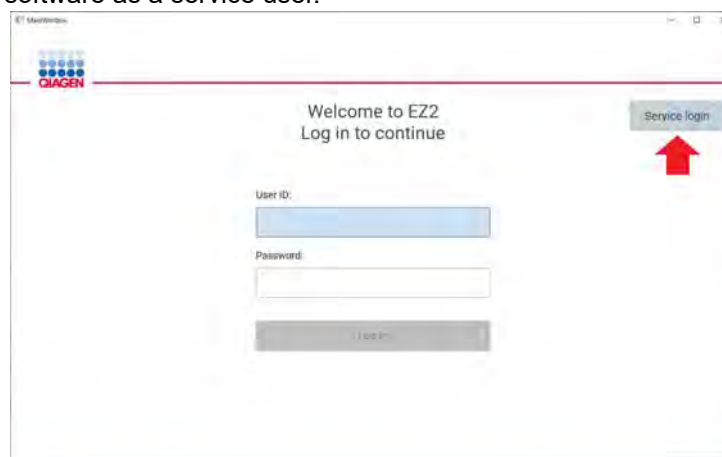
11. Replace the rear cover per [RP1 - Rear Cover - Replacement Procedure](#).

8.15 VP14 - Fan Test

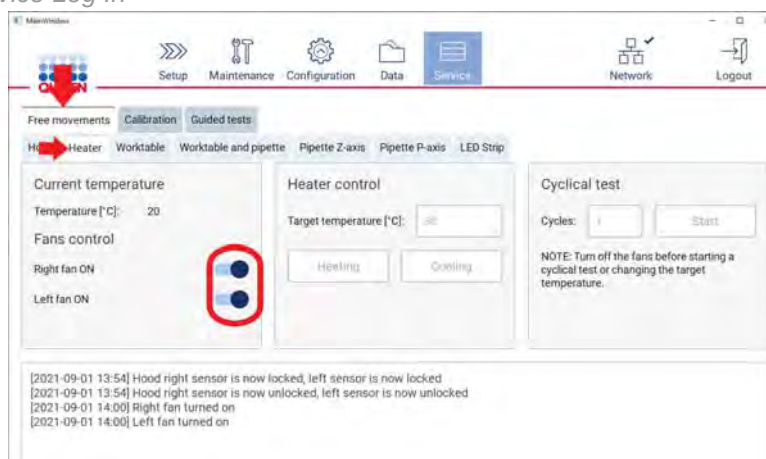
Purpose	This test is to verify that the fans are functional after being replaced or repaired.
Tool(s) Required	NA
Verification Time	1 minutes

Table 8.15-1 VP14 Fan Test

1. Log in to the EZ2 software as a service user.



Picture 8.15-1 Service Log In



Picture 8.15-2 Fan Test

2. In the service environment select the “Free movements” tab.
3. Select the “Heater” tab.

- To turn on the fans select the 2 buttons in the software.



Picture 8.15-3 Fans

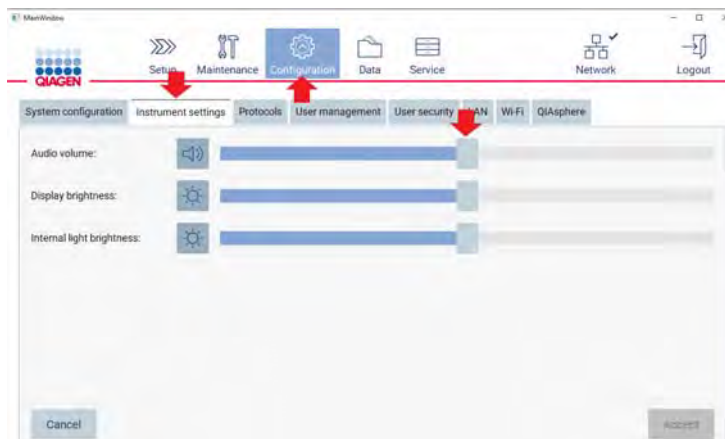
- Place your hand in front of the fans to check if they are operating correctly.
- Select the software buttons again to turn them off.

8.16 VP15 - Speaker Test

Purpose	This test is to verify the speaker is working after it is replaced or repaired.
Tool(s) Required	NA
Verification Time	1 minutes

Table 8.16-1 VP15 Speaker Test

- Log in to the EZ2 software as a service user.



Picture 8.16-1 Speaker Test

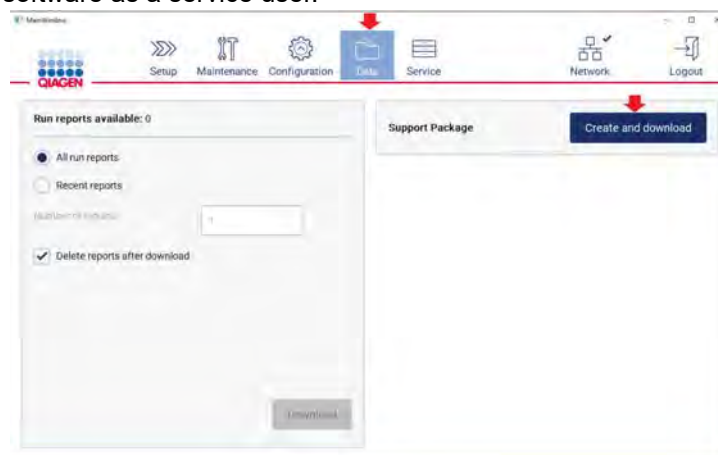
- In the configuration environment select "instrument Settings".
- Move the audio volume button to right above the halfway point.
- After moving the slider bar, an audible sound should be heard.

8.17 VP16 - USB Port Test

Purpose	This test is to check for step loss after parts have been replaced or calibrations have been completed.
Tool(s) Required	USB Stick
Verification Time	1 minutes

Table 8.17-1 VP16 USB Port Test

1. Log in to the EZ2 software as a service user.



Picture 8.17-1 USB Test

2. Select the Data environment.
3. Insert a USB stick in the instrument USB port(s) that need to be tested.
4. Select "Create and Download" button under the Support Package area of the Data environment.
5. Load the USB stick on a laptop and check to see if a support package was successfully downloaded.

Name	Status	Date modified	Type	Size
 EZ2-Z0920013M-20210619042528981.zip	 R	6/19/2021 4:25 AM	Compressed (zipp...	3,169 KB

Picture 8.17-2 Support Package

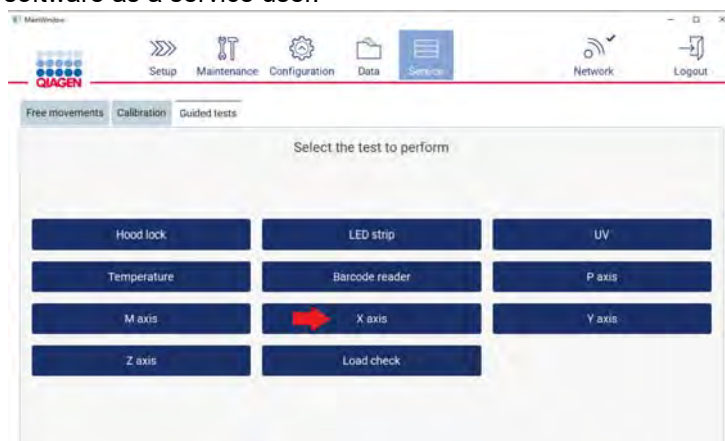
6. The support package is named in a manner similar to the picture above.

8.18 VP17 - X Axis

Purpose	This test is to check for step loss after parts have been replaced or calibrations have been completed.
Tool(s) Required	Metric Ruler
Verification Time	3 minutes

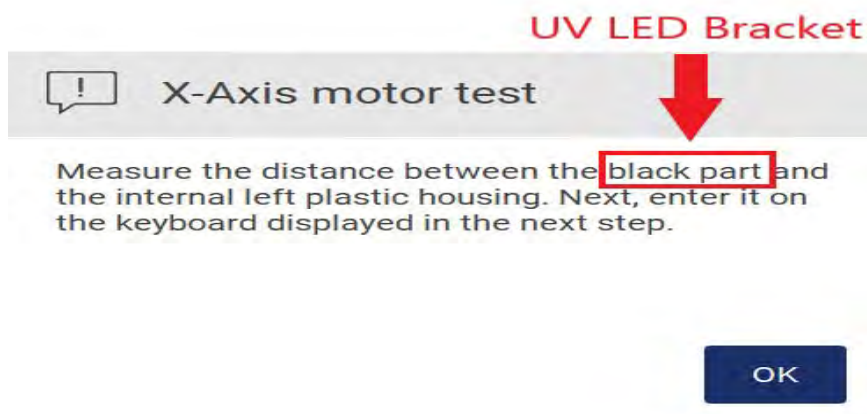
Table 8.18-1 VP17 X Axis

1. Log in to the EZ2 software as a service user.



Picture 8.18-1 X Axis Verification

2. In the service environment select "Guided tests".
3. Select the X axis tab under the guided tests.



Picture 8.18-2 X Axis Measurement

4. The UV LED housing will move to the left of the instrument.



Picture 8.18-3 X Axis Measurement

5. Measure the distance between the left plastic cover and the UV LED bracket.
6. Enter the measurement as 1/10 mm. The measurement in the picture is 15 mm, entered in the software box it will be 150.

Enter measured distance [μm]

150

7 8 9

4 5 6

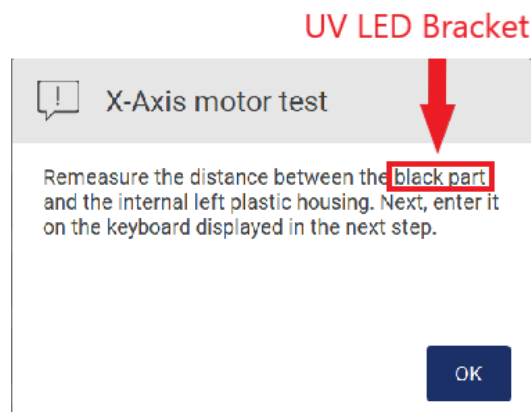
1 2 3

0

Cancel Accept

Picture 8.18-4 Enter measurement

7. Enter the measurement in the software box.



Picture 8.18-5 X Verification

8. The arm with the UV LED will go through a range of movements and stop again to be measured.

Picture 8.18-6 Second Measurement

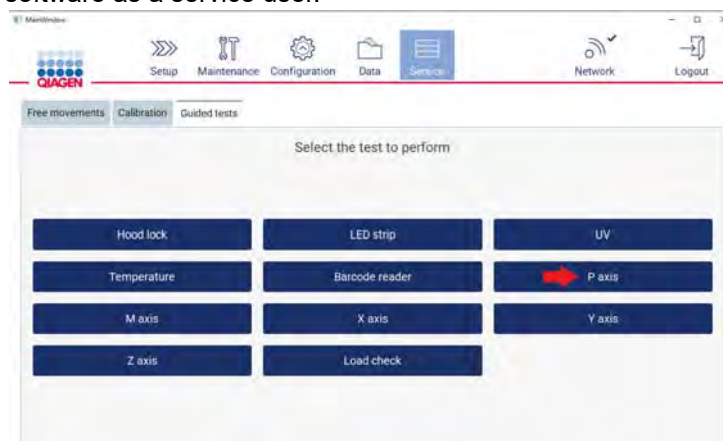
9. Measure the distance from the UV LED bracket to the left inside plastic cover again and enter the measurement in the software box.
10. If the 2 measurements are within the passing range the software will give a message that the test passed and press “OK” to complete the test.

8.19 VP18 - P Axis

Purpose	This test is to check for step loss after parts have been replaced or calibrations have been completed.
Tool(s) Required	0.02 - 1 mm Feeler Gauge Set
Verification Time	3 minutes

Table 8.19-1 VP18 P Axis

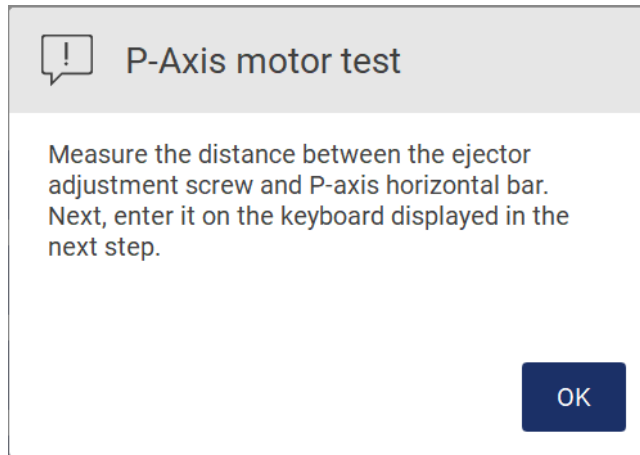
1. Log in to the EZ2 software as a service user.



Picture 8.19-1 Speaker Test

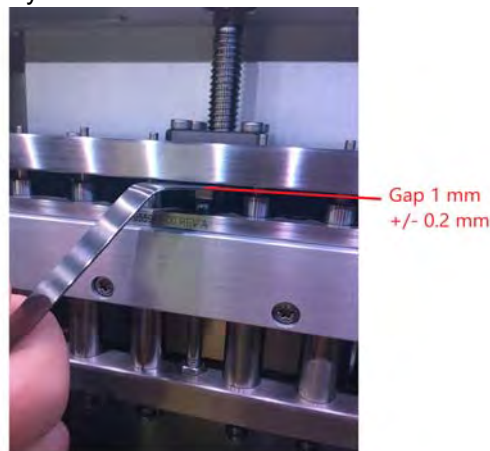
2. In the service environment select “Guided test”.

3. Select the P axis tab.



Picture 8.19-2 P Axis Check

4. Wait to press the "OK" button until you have measured the P axis.



Picture 8.19-3 P Axis Verification

5. Check the gap from the top of the adjustment screw to the bottom of the holding bar.
6. The gap should be 1 mm +/- 0.2 mm.



Picture 8.19-4 P Axis 3 Positions

7. Check all 3 positions of the P axis screws and select "OK" on the screen.

Enter measured distance [μm]

1| ✕ ⏮

7	8	9
4	5	6
1	2	3
0		

Cancel Accept

Picture 8.19-5 P Axis Measurement

8. When entering the measurement it will be in 1/10 of a millimeter.
 9. Enter the measurement of the feeler gauge in the box.
 10. Select "Accept".

! **P-Axis motor test**

Remeasure the distance between the ejector adjustment screw and P-axis horizontal bar. Next, enter it on the keyboard displayed in the next step.

OK

Picture 8.19-6 Repeat Measurement

11. Repeat steps 5-7 again.

Enter measured distance [μm]

1| ✕ ⏮

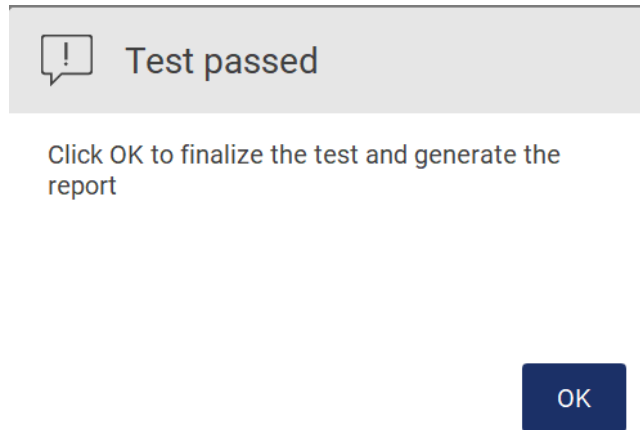
7	8	9
4	5	6
1	2	3
0		

Cancel Accept

Picture 8.19-7 Final Measurement

12. Enter the measurement with of feeler gauge in the box.

13. Select the Accept tab.



Picture 8.19-8 P Axis Verification Completed

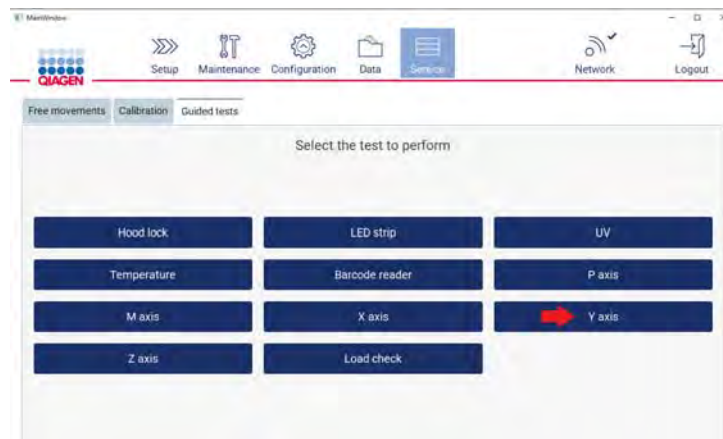
14. Select the "OK" if the testing passed to complete the P axis verification.

8.20 VP19 - Y Axis

Purpose	This test is to check for step loss after parts have been replaced or calibrations have been completed.
Tool(s) Required	0.02 - 1 mm Feeler Gauge Set
Verification Time	4 minutes

Table 8.20-1 VP19 Y Axis

1. Log in to the EZ2 software as a service user.



Picture 8.20-1 Speaker Test

2. In the service environment select "Guided tests".



Y-Axis motor test

Measure the distance between the tip adaptor and the front surface of the worktable bar. Next, enter it on the keyboard displayed in the next step.

OK

Picture 8.20-2 Y Axis Verification

3. Select the Y axis tab, then select OK.



Picture 8.20-3 Y Axis 1st Measurement

4. Use a 0.02 mm feeler gauge to check the distance between the tip adaptor and the front of the middle bar on all positions.

Enter measured distance [μm]

1|

7 8 9

4 5 6

1 2 3

0

Cancel Accept

Picture 8.20-4 Enter Measurement

5. Enter the measurement of the smallest gap in the software box and select the Accept tab.

6. The Y axis worktable will now go through a series of movements and stop to be measured again.



Y-Axis motor test

Remeasure the distance between the tip adaptor and the front surface of the worktable bar. Next, enter it on the keyboard displayed in the next step.

OK

Picture 8.20-5 2nd Measurement

7. Measure the distance between the tip adaptor and the front of the worktable bar.

Enter measured distance [μm]

1|

✕

⏮

7	8	9
4	5	6
1	2	3
0		

Cancel

Accept

Picture 8.20-6 Enter Measurement

8. The software measurement is in increments of 1/10 mm.
9. Enter the measurement in the software box and select Accept.



Test passed

Click OK to finalize the test and generate the report

OK

Picture 8.20-7 Y Verification Completed

10. Select the OK if the test passed to complete the verification.

9 Preventative Maintenance Guideline

9.1 Introduction

The objectives of the EZ2 preventative maintenance procedure are:

- Ensure the reliability and functionality of the system
- Perform a complete system inspection to ensure maximum system performance
- Upgrade or update the system according to the upgrade policies stated in the relevant instrument specific ISI notes
- Determining the customer's level of satisfaction with the instrument

This document is the guideline for performing the annual preventative maintenance for the EZ2.

9.2 Terms

In the PM procedure in chapter Table 9.4-1 we use the term *inspection*. An inspection includes the following tasks:

- look for damages
- look for corrosion
- look for dirt, precipitations from buffers, etc.

9.3 Requirements

All preventative maintenance (PM) procedures outlined in this guideline shall only be performed by QIAGEN authorized personnel.

INFORMATION 	Ensure all required material is on-site before starting the PM. It is strongly recommended to have all tools and other equipment available for the service visit.
---	--

9.3.1 Protective material

In order to prevent possible exposure to biological hazards, make sure to utilize appropriate hand, eye, and body protection.

Material	Comment
Eye protection	Safety glasses
Hand protection	Gloves (rubber, nitrile, etc.)
Body protection	Lab coat

Table 9.3-1 Protective material

9.3.2 Useful Documents

Document	Comment
EZ2 Connect User Manual	latest version for use with EZ2 Connect HB-2907 EZ2 Connect MDx User Manual HB-2908 EZ2 Connect LS User Manual
Service manual ⁺	latest revision SM-5000-0047, QFS, QTS, EZ2

Tab. 9.3-2 Useful documents

9.3.3 Software

Software	Comment
SOW-xxxx	EZ2 Connect Software
GPS Tool	Tool for converting xml reports into html

Tab. 9.3-3 PM Software

9.3.4 Tools

Tools	Comment
Feeler Gauge	0.02mm - 1.0mm
Torx Key Set	Range T5 - T40
Hex Key Set	Range HW2 - HW7
Hex Wrench	Size 3.0
Pliers	Combination, Long Nose, Cutting Pliers
Forceps / Tweezers	Bent Ends
Thermometer, incl. probe min. resolution 0.1°C min. accuracy +/- 0.6°C	e.g. Testo 926, with T needle probe (Testo order numbers: 0560 9261, 0628 0030): • resolution: 0.1°C • accuracy: +/- 0.5°C (device and probe) • calibration: according to manufacturer's prescriptions
Scale	WXSS205DU, DualRange (Mettler-Toledo): • resolution: 0.01 mg/0.1 mg • accuracy: +/- 0.03 mg (at 10 g) • accessories (container, evaporation shield) • calibration: according to manufacturer's prescriptions, and prior use (CarePac), respectively • Example scale only, any scale that meets the resolution and accuracy can be used.

Table 9.3-4 Tools

9.3.5 Special equipment

Tools	Comment

Table 9.3-5 Special equipment

9.3.6 Kits and consumables

SAP	Part Name	Comment
9027240	Installation Kit	• QC Cartridges • Tips and tip holders • Elution tubes • Sample tubes • Q-Card

Table 9.3-6 Kits and consumables

9.3.7 Parts

SAP	Part Name	Comment
9027085	PM Kit	Contains 48 o-rings and 24 d-rings
9027154	Air Filter	If needed

Table 9.3-7 Parts

9.3.8 Lubricants

SAP	Material	Comment
9027073	Grease Silicone, EZ2	Enough for multiple PM visits
9027075	Grease, LG2, EZ2	Enough for multiple PM visits
TBD	SuperLube, EZ2	Enough for multiple PM visits

Table 9.3-8 Lubricants

9.3.9 Cleaning Material

Material	Comment
Isopropyl Alcohol (IPA)	Contains minimum of 70% Isopropyl Alcohol (2-propanol)
Lint-free tissue or equivalent cleaning material	N/A
Long cotton or foam swabs	Recommended rod length minimum of 100mm.

Table 9.3-9 Cleaning Material

9.4 PM schedule

The approximate duration of the PM procedure is 5 hours. This time may vary, depending on the condition of the EZ2.

Subject	Task	Time
Cleaning	Clean the instrument	2.5 hours
Calibrations	Perform teaching and calibrations	0.5 hours
Verification	Run Verification Procedures	1.5 hours
Report	Fill in the PM visit record and sign off	0.5 hours
TOTAL		5 hours

Table 9.4-1 PM Timeline

9.5 Pre-Visit Actions

9.5.1 Contact Customer - Gather information

1. Discuss any problems and possible needs with the customer.

9.5.2 Check Salesforce/CRM for Implementation of ISIs

1. Check if all ISIs with upgrade class I or II have been implemented.
2. Review all other ISI's and determine implementation requirement.
3. If an ISI has not been implemented, prepare to perform at the PM and update SFDC/CRM after completion.

9.6 PM Procedure

9.6.1 General inspection

- Check that the EZ2 is clean and well maintained
- If required, re-train the customer on the maintenance procedure described in the EZ2 User Manual
- The latest user manuals are available at QIAGEN.com.

9.6.2 Cover disassembly

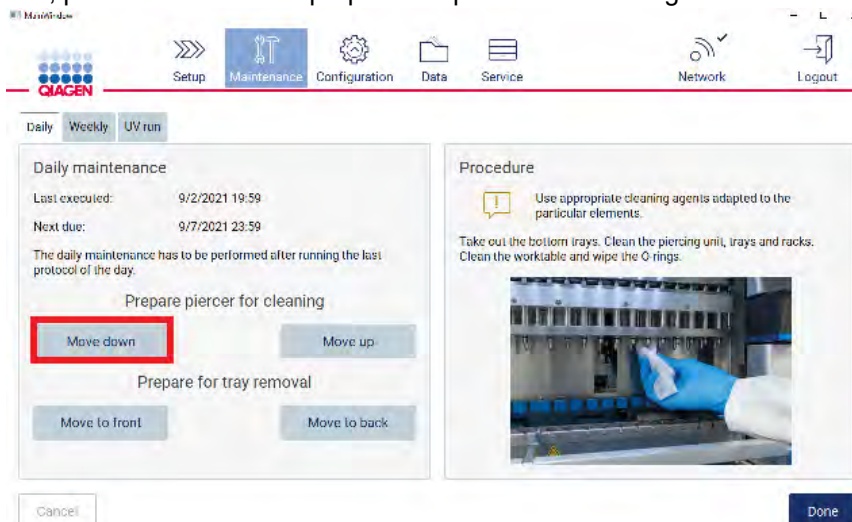
1. Remove the rear cover [see RP1 - Rear Cover - Removal Procedure](#).
2. Next remove the main cover; [see RP2 - Main Cover - Removal Procedure](#).
 - There is no need to remove the hood and top front cover to perform the PM procedure.

9.6.3 Perform Daily Maintenance

WARNING

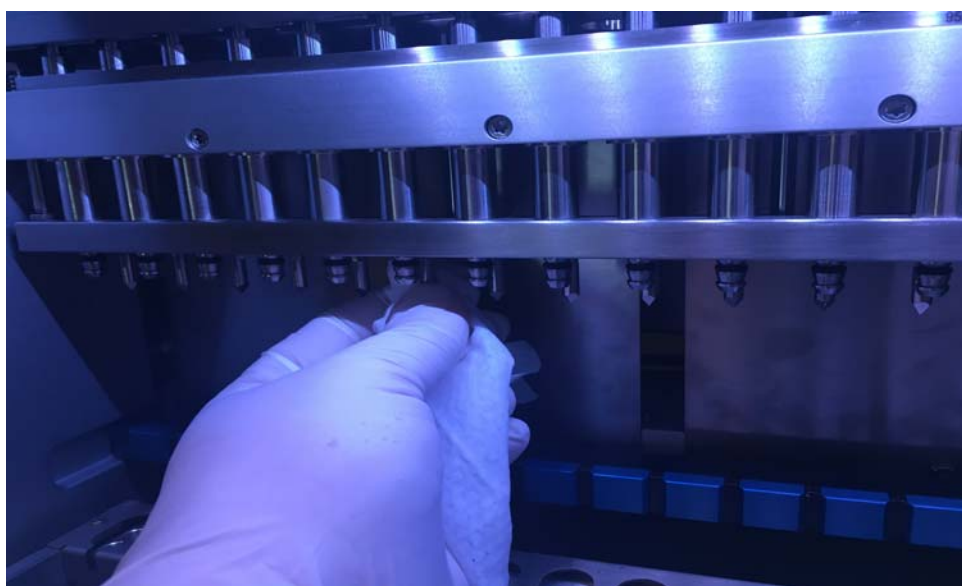

Be careful when cleaning the **piercing unit** - the spikes are very **sharp** and can be **contaminated**.

1. Log into the software and press the icon for the Maintenance Environment.
2. On the Daily tab, press Move down to prepare the piercer for cleaning.



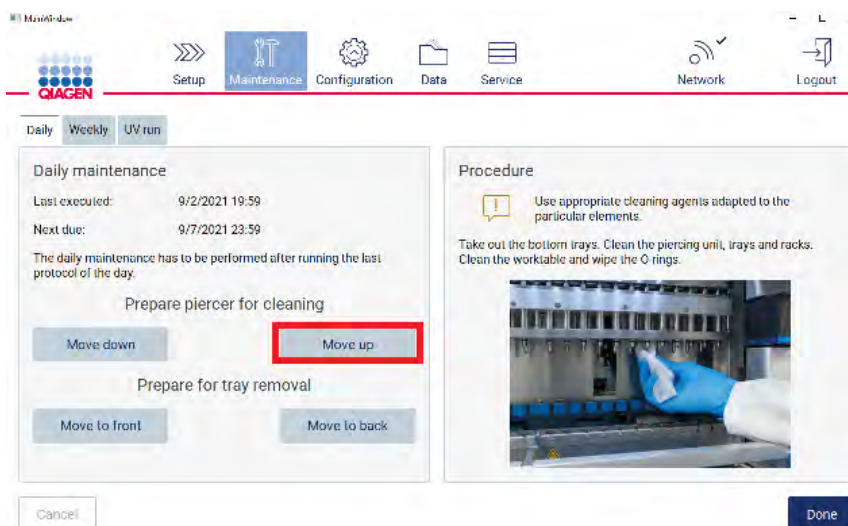
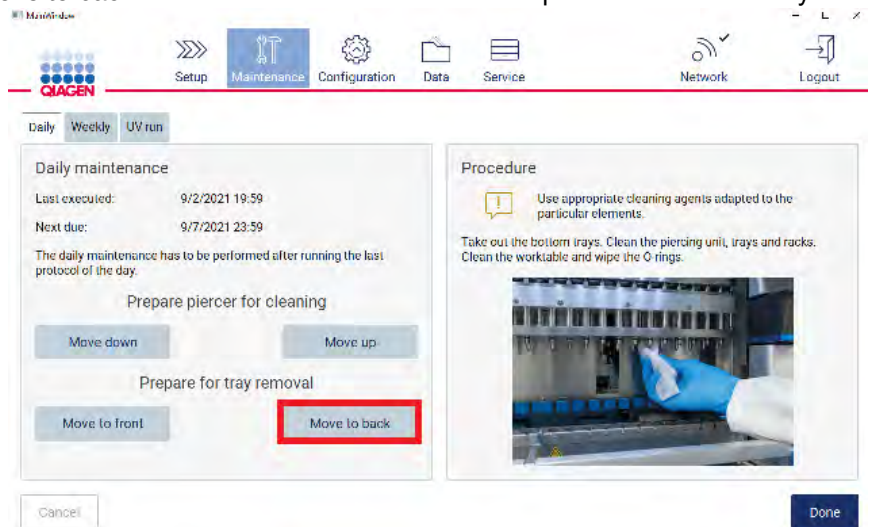
Picture 9.6-1 Piercer Down

3. Wearing gloves, wet a Kimwipe or similar with Isopropyl Alcohol or equivalent disinfectant, wipe down each piercing pin of the piercing unit.
4. Ensure each pin is completely clean and free of debris. Replace any piercing pin that can't be properly cleaned.
5. If the piercing unit does not appear to be appropriately maintained, train the customer regarding the maintenance procedures in the User Manual.



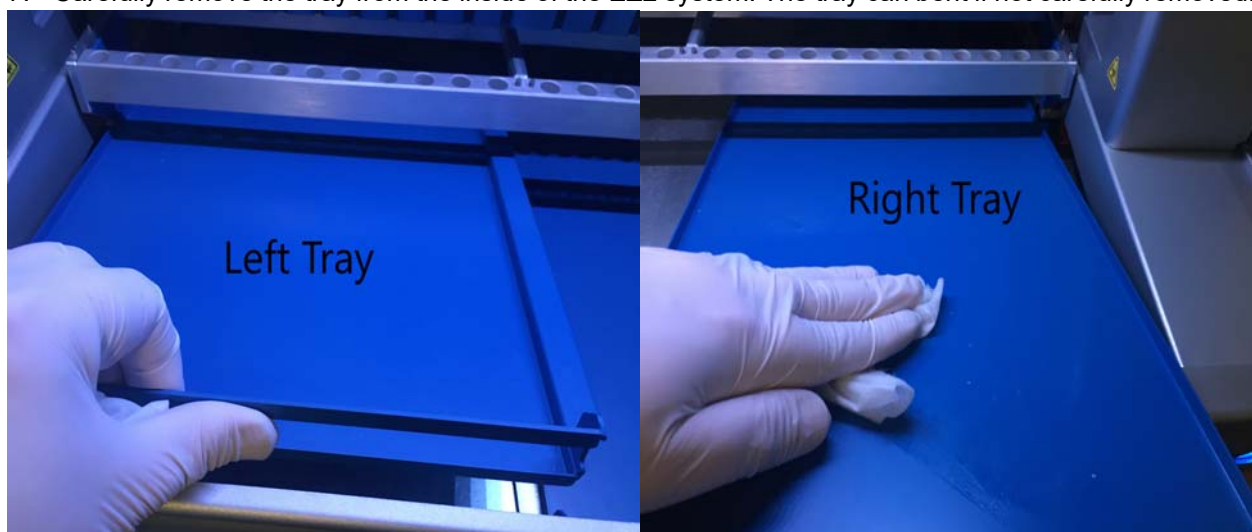
Picture 9.6-2 Clean the piercing unit

6. After the piercing unit is clean, press Move up to return the piercing unit back to the home position, then press the *Move to back* button to move the workdeck to a position where the trays can be removed.



Picture 9.6-3 Prepare for tray removal

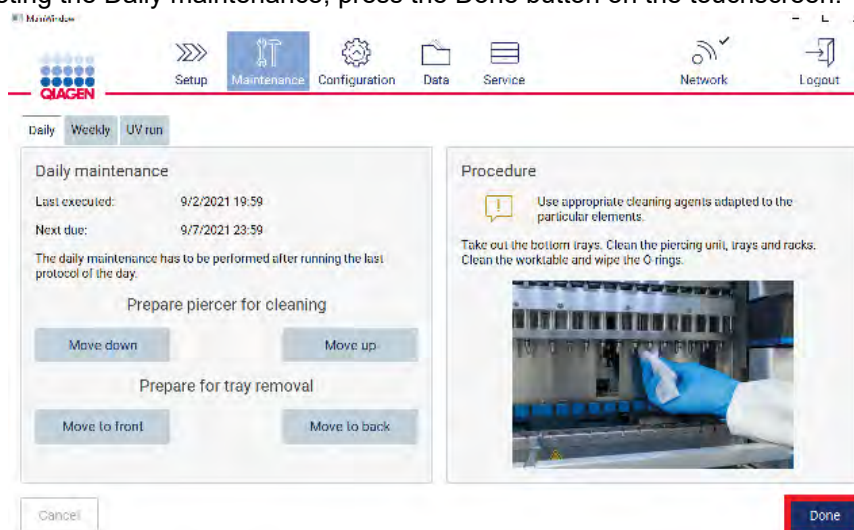
7. Carefully remove the tray from the inside of the EZ2 system. The tray can bent if not carefully removed.



Picture 9.6-4 Tray Removal

8. Clean the trays with Isopropyl Alcohol or equivalent disinfectant.

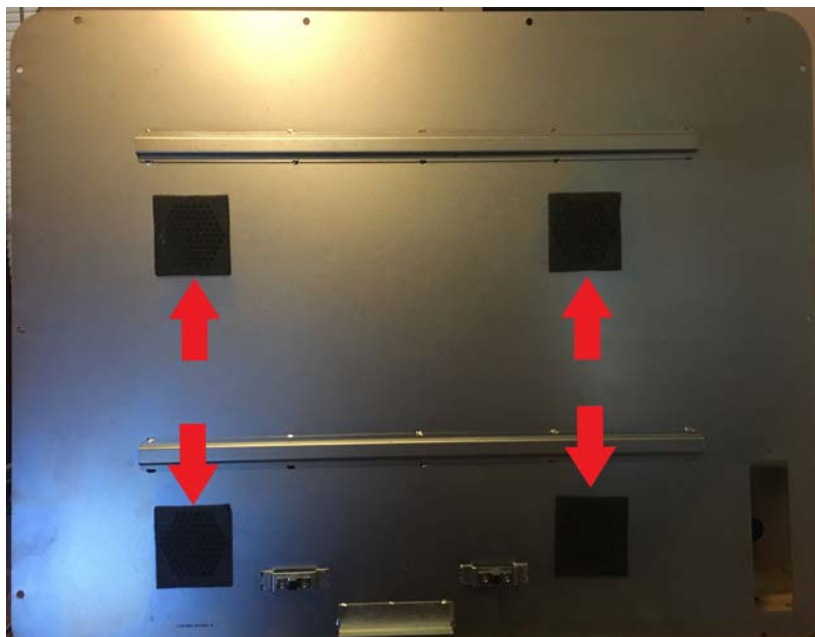
9. Carefully return the trays back into the instrument, ensuring that the trays are installed on the appropriate side.
10. After completing the Daily maintenance, press the Done button on the touchscreen.



Picture 9.6-5 Daily maintenance complete

9.6.4 Fan Filters

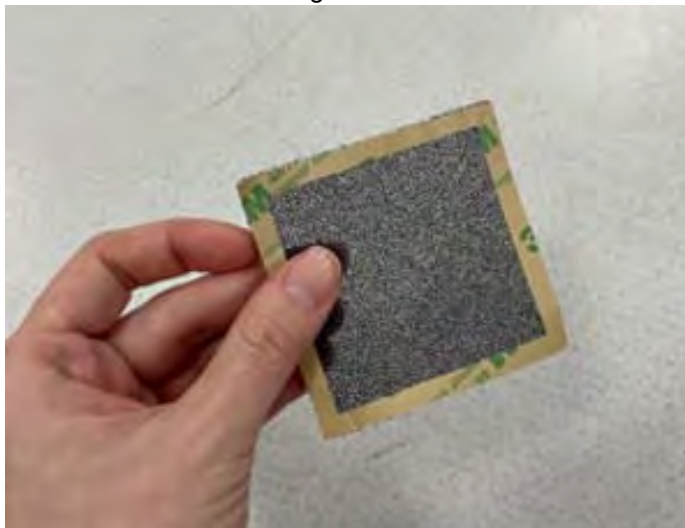
1. Replacing the fan filter is optional during the PM. Only replace the filters when they are damaged or dirty.
2. The fan filters are located on inside of the rear cover of the instrument.



Picture 9.6-6 Fan Filters

3. To remove the filters scrape off the old filters adhesive.

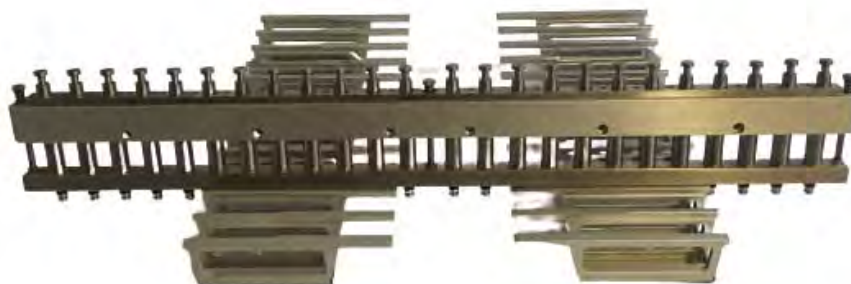
4. To replace the filters stick the adhesive side against the rear cover in the correct positions.



Picture 9.6-7 New Filter

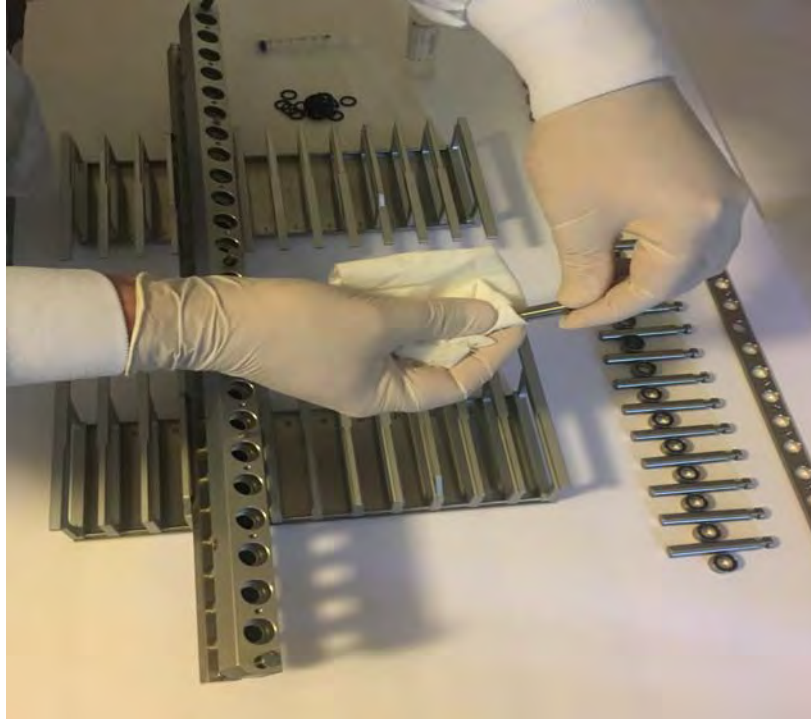
9.6.5 Disassemble & Clean Pipette Unit

1. Remove the pipette unit per [RP37 - Pipette Module - Removal Procedure](#).
2. It may help to use the cartridge rack to hold the plunger unit.



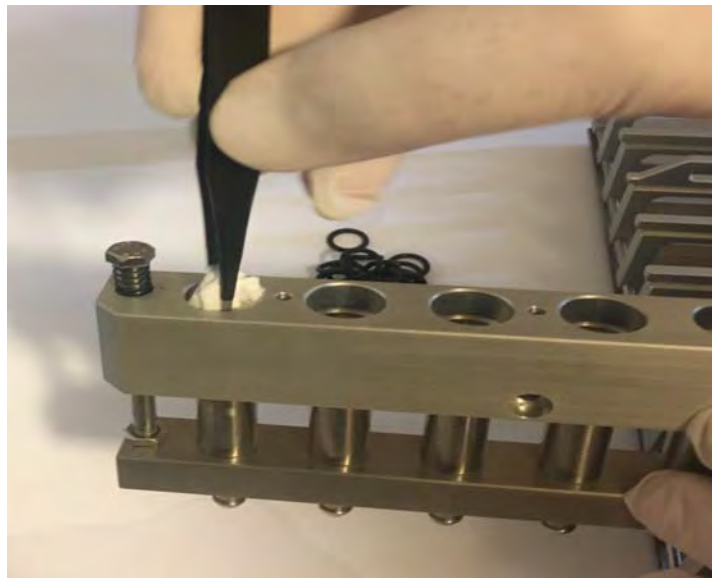
Picture 9.6-8 Plunger unit "holder"

3. Remove the plungers, collars and O-rings per [RP38 - Plunger/Collar/O-ring - Removal Procedure](#).

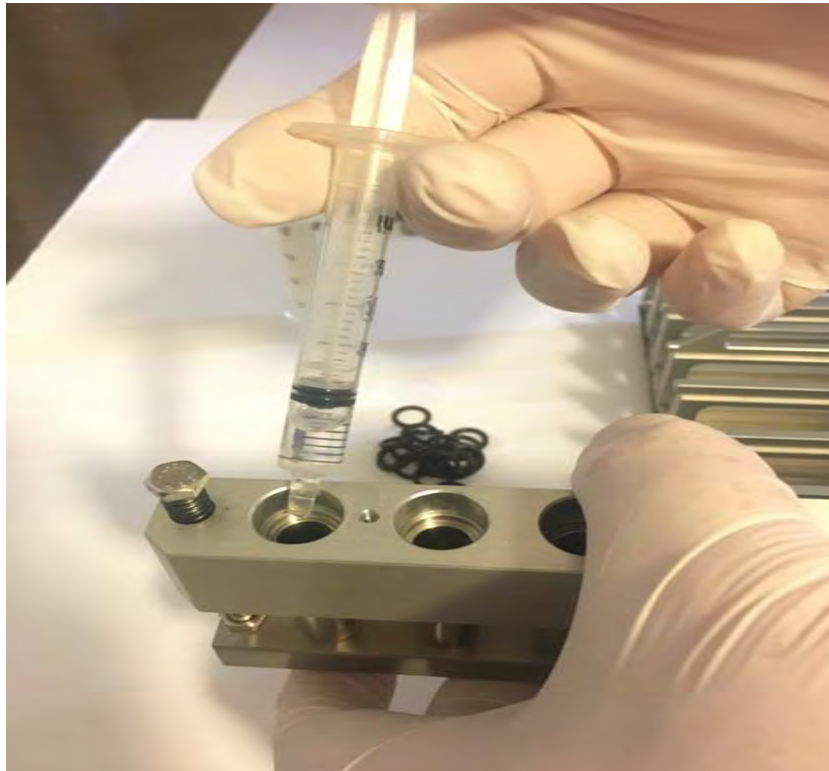


Picture 9.6-9 Plunger Dismantled

4. Keep the plungers and collars in order in the positions that they were removed from.
5. Use a lint free tissue moistened with 70% Isopropyl alcohol and clean the plungers, collars and plunger channels.

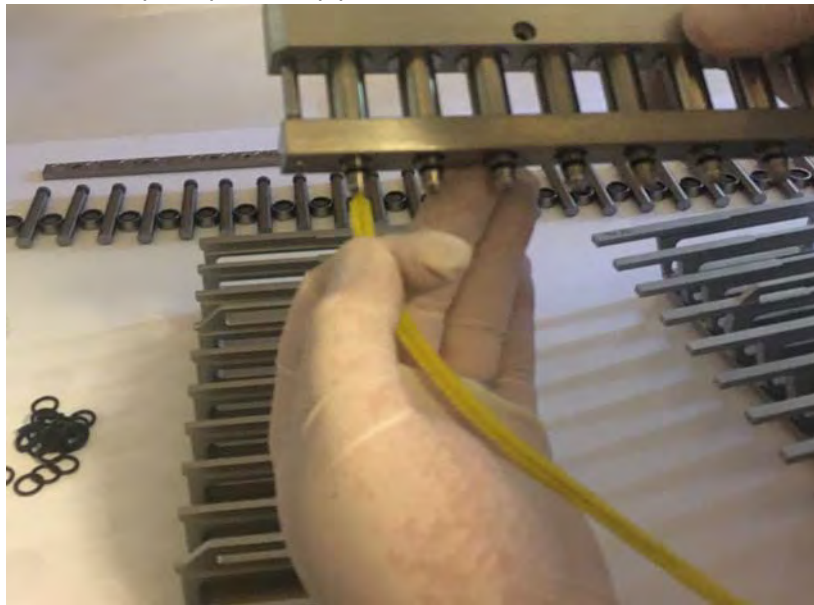


6. Place your finger over the end of the tip adapter and use a syringe to insert alcohol in the plunger housing.



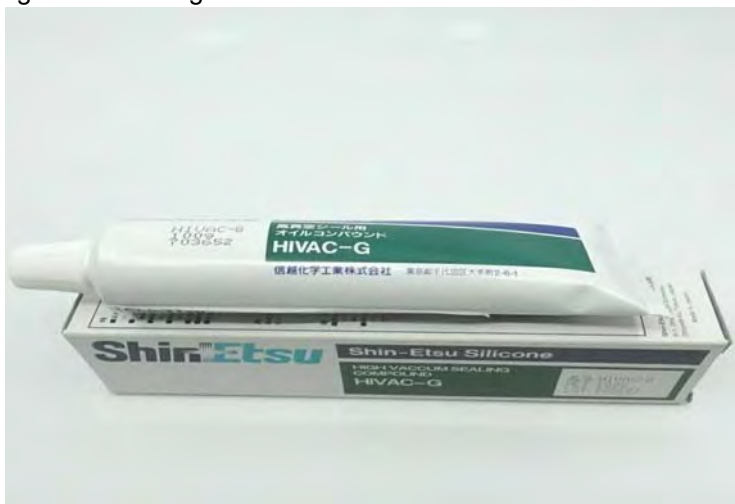
Picture 9.6-10 Plunger Housing Cleaning

7. Clean the inside of the plunger housing with forceps and a Kimwipe.
8. Clean the inside of the tip adapter with pipe cleaners.



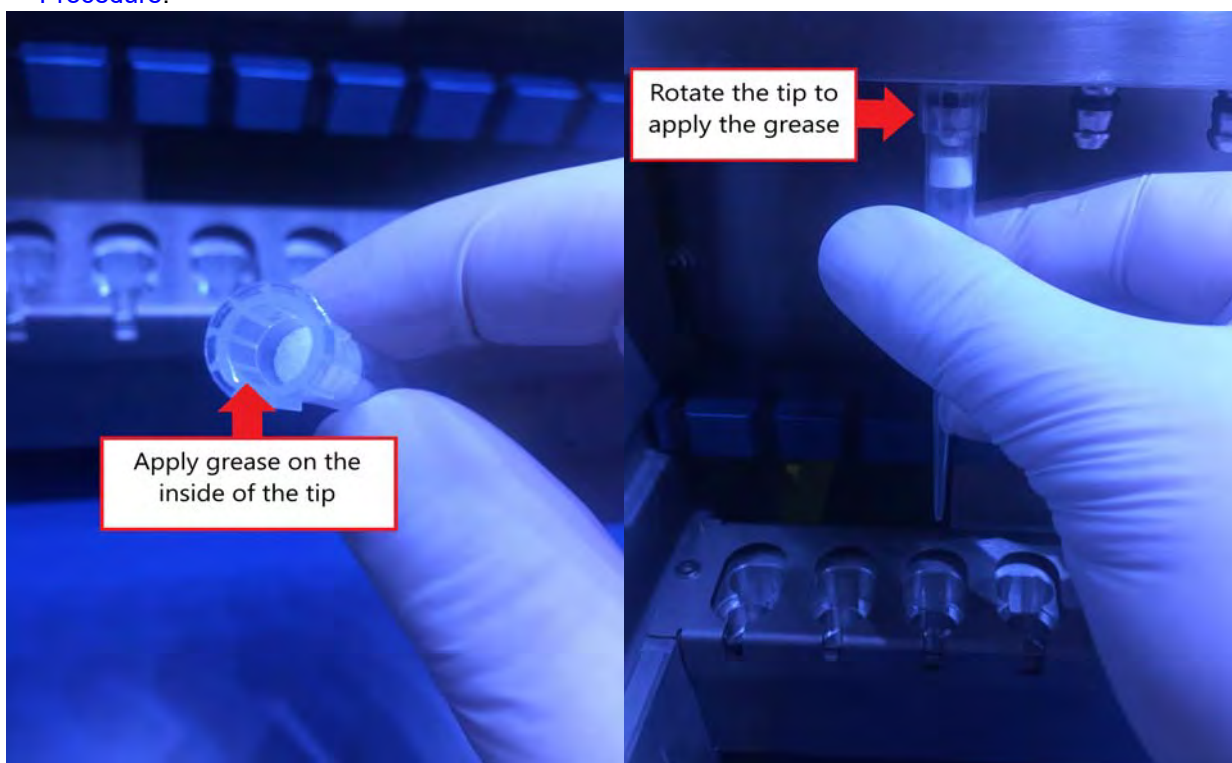
Picture 9.6-11 Cleaning Tip Adapters

9. Lubricate the O-rings with silicon grease..



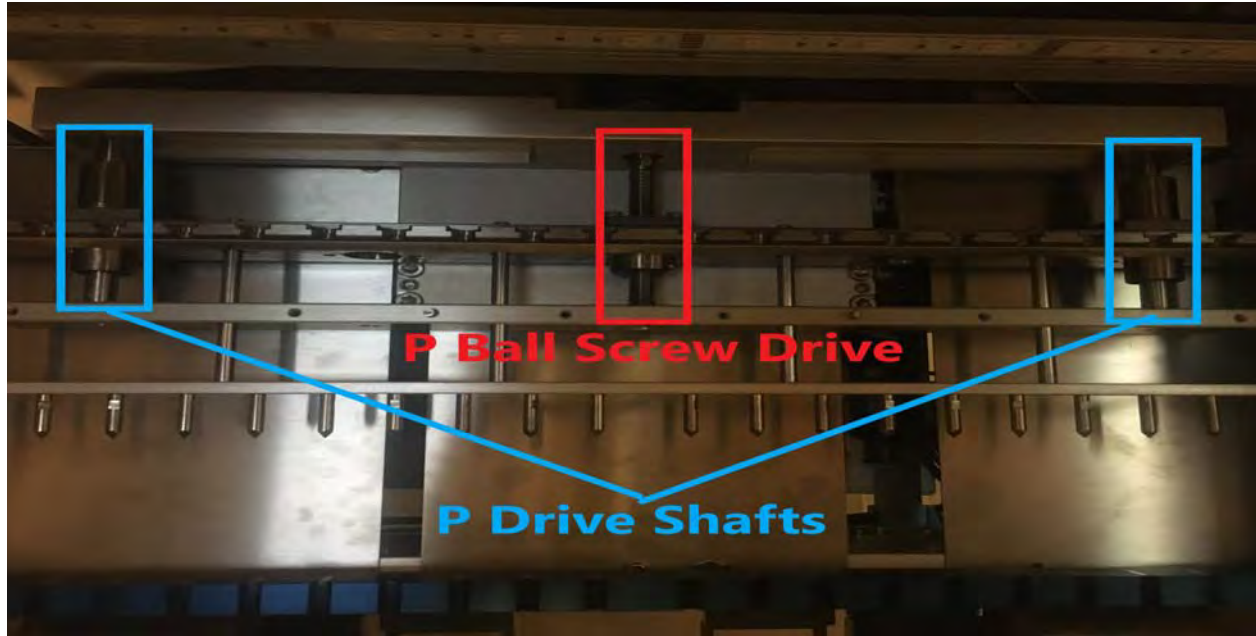
Picture 9.6-12 Silicon Grease Part Number 9027073

10. Put the O-rings, collars and plungers back in the pipette unit per [RP38 - Plunger/Collar/O-ring - Replacement Procedure](#).
11. Remove the D-rings on the pipette tip adapters per [RP39 - D-ring - Removal Procedure](#).
12. Replace the D-rings on the pipette tip adapter and apply silicon grease per [RP39 - D-ring - Replacement Procedure](#).



Picture 9.6-13 Greasing the D-rings

9.6.6 P Ball Screw Thread & Shafts



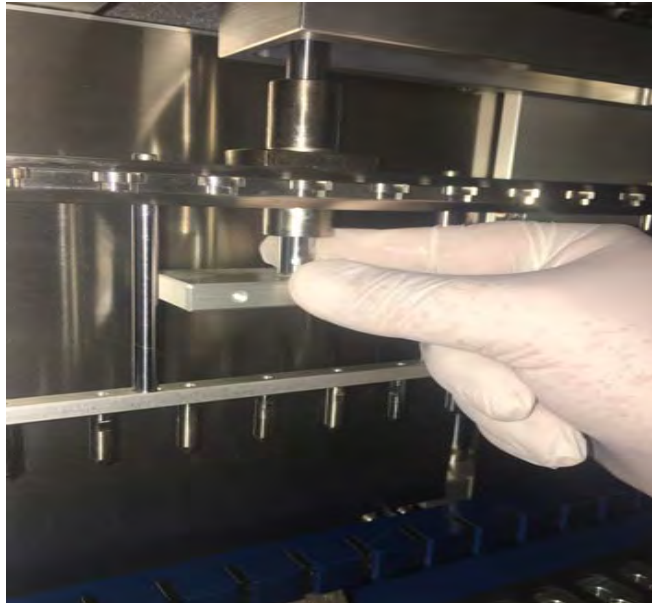
Picture 9.6-14 P Axis

1. Clean the P ball screw thread and shafts with a lint free cloth or Kimwipe.



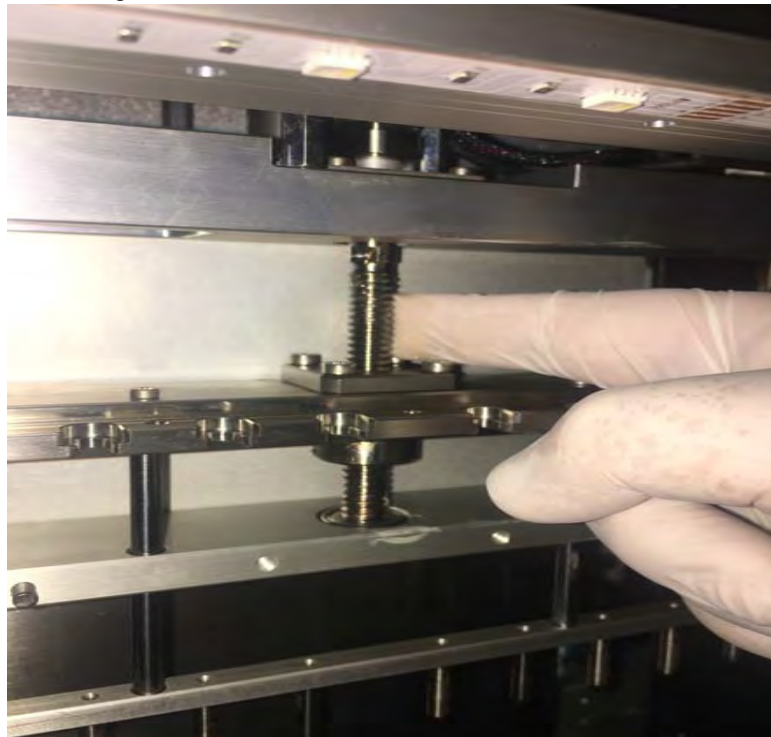
Picture 9.6-15 LG2 Grease 9027075

2. Apply LG2 grease, part number 9027075 on the fingertips of your gloved hand and lightly grease the P ball screw threading.



Picture 9.6-16 P Ball drive

3. Use the LG2 grease and grease the P shafts.



Picture 9.6-17 P-drive shaft

4. Install the pipette module per [RP37 - Pipette Module- Replacement Procedure](#).

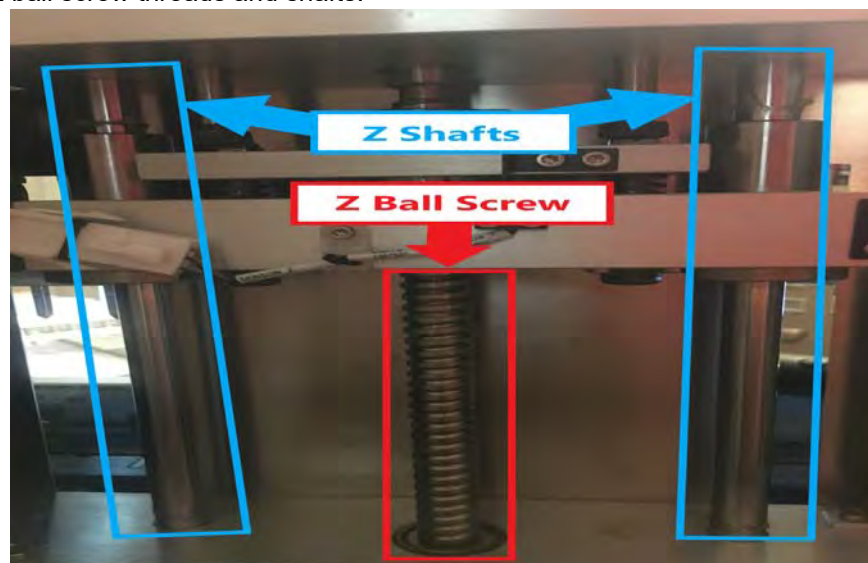
9.6.7 Z Ball Screw Thread & Shafts

1. Remove the 2 (T10) screws and open the Z access panel at the back of the instrument.



Picture 9.6-18 Z Access Panel

2. Clean the Z ball screw threads and shafts.



Picture 9.6-19 Z Ball Drive Cleaning

3. Place some LG2 grease on the end of your gloved fingers and lightly grease the 2 shafts and Z drive.

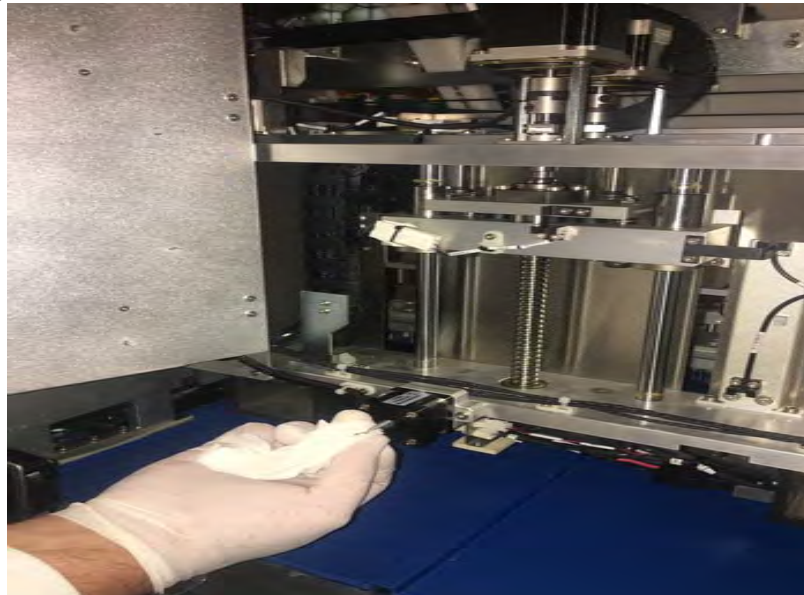


Picture 9.6-20 Grease Z Ball Threads and Shafts

4. Do not over grease the Z shafts or ball screw drive.

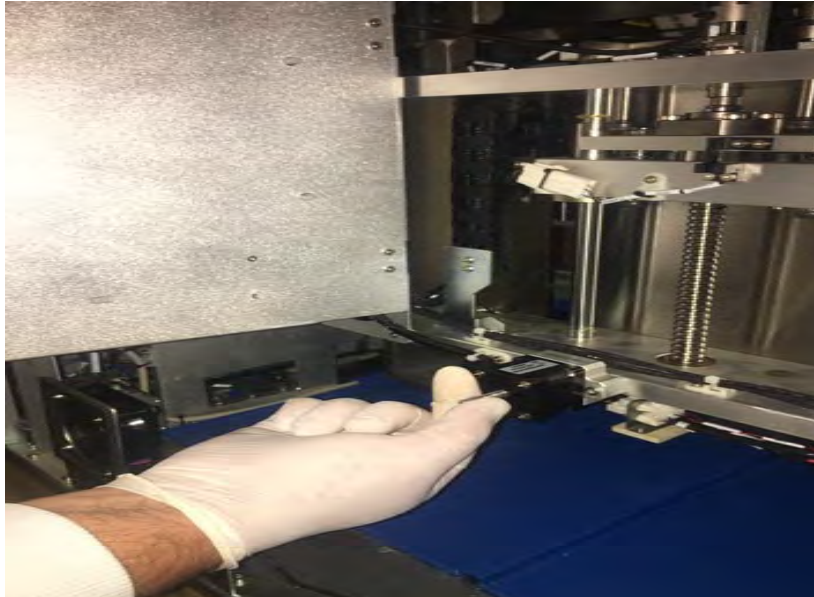
9.6.8 Magnet Unit Maintenance

1. Clean the magnet motor screw drive with a lint free cloth.



Picture 9.6-21 Clean Magnet Drive

2. Apply a light coat of LG2 grease on the fingertips of your gloved hand and apply a light coating on the magnet motor screw drive.



Picture 9.6-22 Grease M Motor Drive

3. Do not over grease the magnet motor screw drive.

9.6.9 X Axis Maintenance

1. Clean the X guide rail with a lint free cloth or Kimwipe, then move the UV LED arm to the left and clean the X guide rail where the arm was positioned.



Picture 9.6-23 X-axis cleaning

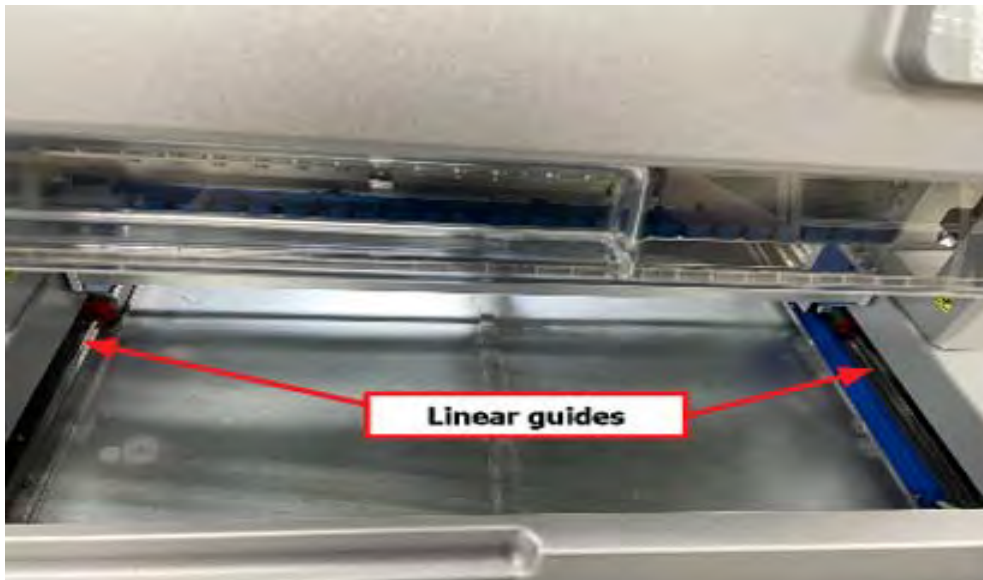
2. Do not grease the X guide rail.

3. Lubricate the outside and inside of the X pulley shaft flange bearing on each side of the pulley lightly with LG2 grease.



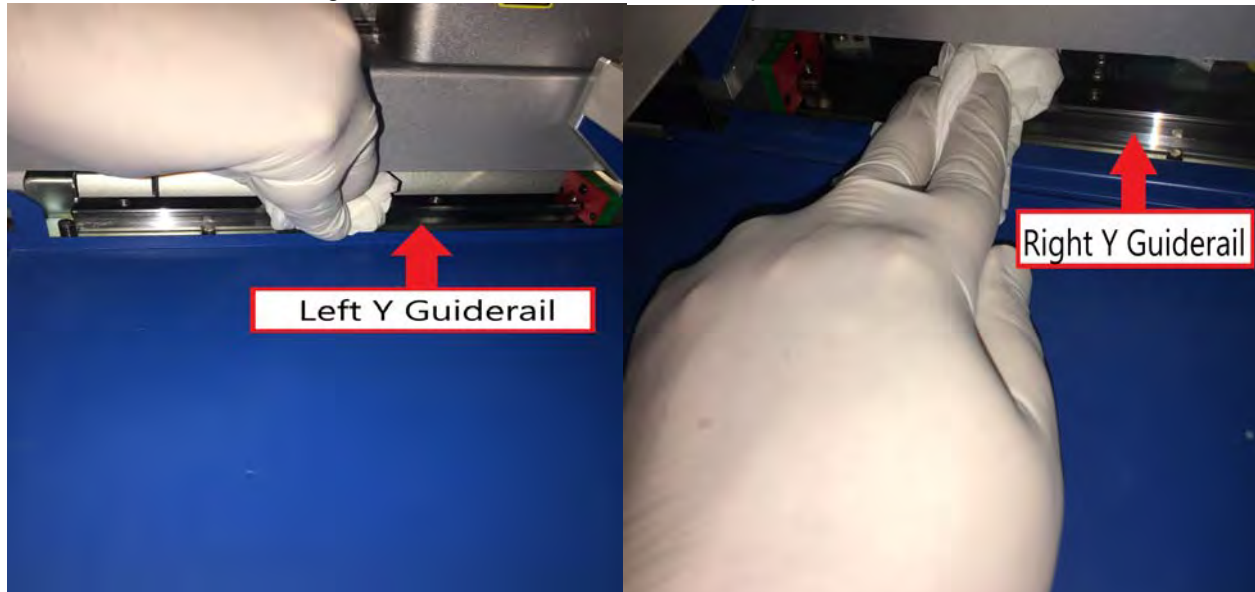
Picture 9.6-24 Lubricate X Pulley

9.6.10 Y Axis Maintenance



Picture 9.6-25 Y Axis Linear Guides

1. Clean the Y axis linear guides with a lint free cloth or Kimwipe.



Picture 9.6-26 Cleaning Y Linear Guides

2. Apply LG2 grease on the fingertips of your gloved hand and grease the Y linear guides.



Picture 9.6-27 Grease Y Linear Guides.

3. Do not over grease the linear guides.

9.6.11 Camera Maintenance (MDX and Fx Only)

1. Only clean the lens portion of the camera if needed, do not get the LED board wet that surrounds the camera lens.



Picture 9.6-28 Move Camera

2. With the instrument off, move the camera out of the home position to the center of the instrument.
3. Use a lint free cloth or Kimwipe with a small amount of alcohol to clean the lens portion of the camera.



Picture 9.6-29 Camera Lens

9.7 Operational Checks

9.7.1 Z Axis Calibration

1. Perform [CP3 Pipette Z-axis Calibration](#).

9.7.2 Y Axis Worktable Calibration

1. Perform [CP4 Worktable Calibration](#).

9.7.3 Pipette P Axis Calibration

1. Perform [CP5 Pipette P-Axis Calibration](#).

9.7.4 Magnet Calibration

1. Perform [CP6 Magnet Calibration](#).

9.7.5 Camera Calibration (MDX and Fx Only)

1. Perform [CP1 Camera Calibration](#).

9.7.6 Camera Exposure Calibration (MDX and Fx Only)

1. Perform [CP2 Camera Exposure Calibration](#).

9.7.7 Hood Lock Test

1. Perform [VP6 - Hood Lock Test](#).

9.7.8 LED Strip Test

1. Perform [VP1 - LED Strip Test](#).

9.7.9 Barcode Reader Test

1. Perform [VP9 - Barcode Reader Test](#).

9.7.10 Load Check Test (MDX and Fx Only)

1. Perform [VP7 - Load Check Test](#).

9.7.11 Temperature Test

1. Perform [VP5 - Temperature Test](#).

9.7.12 Leakage Test

1. Perform [VP3 - Leakage Test](#).

9.7.13 Pipetting Test

1. Perform [VP4 - Pipetting Test](#).

9.7.14 UV Light Test

1. Perform [VP2 - UV Light Test](#).

9.7.15 System Workflow Test

1. Perform [VP8 - Workflow System Test](#).

9.8 Call Closing/Record Keeping

1. Complete the PMVR, provide a copy to the client and upload a copy to the appropriate location in SFDC/CRM.
2. Download a Support Package and upload the support package to the appropriate location in SFDC/CRM.
3. Mark any ISI notes that were implemented during the PM as complete in SFDC/CRM.
4. Close the service call/work order according to procedure.

10 Software and Software Tools

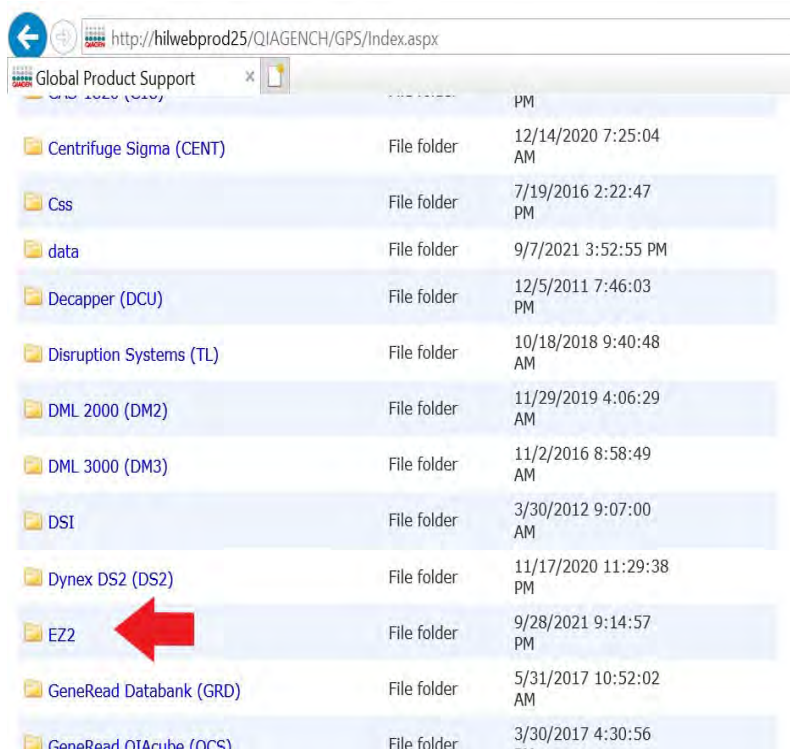
This Chapter is divided into different sections: User Software and Service Tools.

10.0.1 General Information

- The EZ2 Connect is shipped with the latest version of the User Software.
- The EZ2 Connect User Software runs on a touchscreen device that is integrated in the EZ2 Connect system.
- The EZ2 Connect User Software is composed of the operating system, the controller component, the operator GUI component, the technician GUI, and the Service Software.
- The operating system is Linux® based system with 1 GB flash memory.
- The operator GUI (user mode) is the interface between the user and the instrument.
- The technician GUI (R&D mode) provides additional features for running the instrument and for performing tests. It is accessible only to service personal who have been assigned a PIN code.
- The Service Software is run on the GUI that is able to run calibrations and verifications.
- The EZ2 Connect User Software can be installed or updated via the 3 USB ports on the instrument.

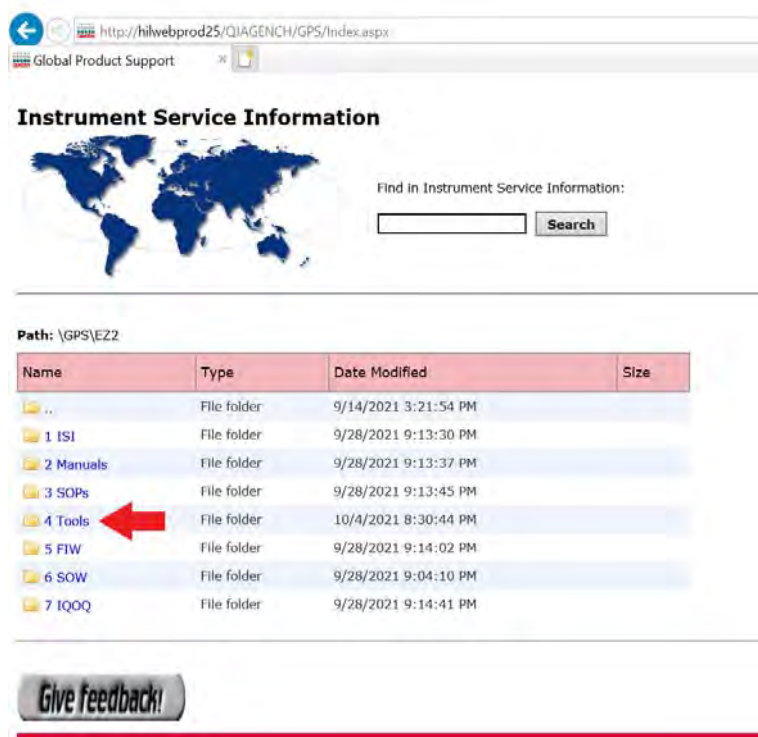
10.0.2 Log In As Service

1. To log in as a service user for the EZ2 Connect, visit the GPS portal.
2. Find and click on the EZ2 folder.



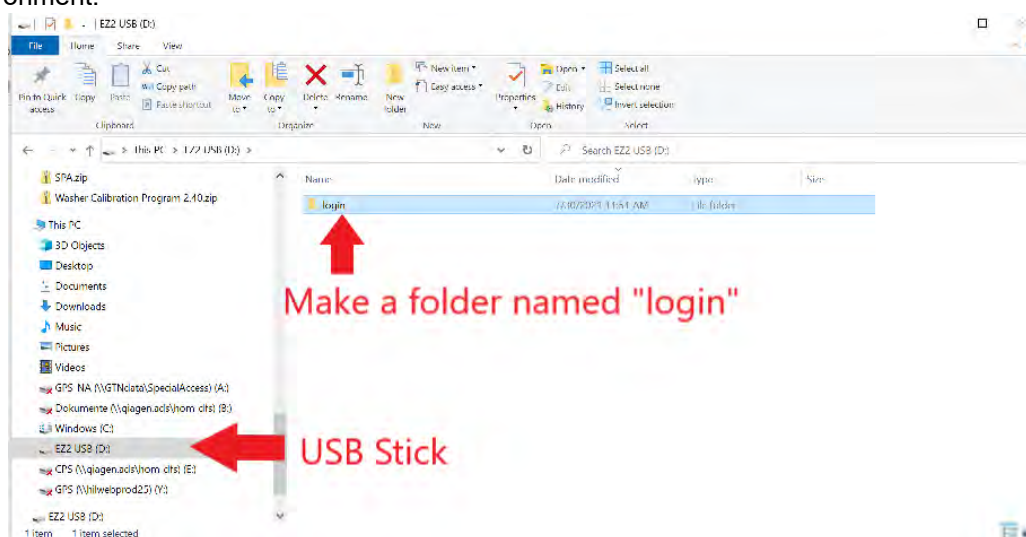
Picture 10.0-1 GPS EZ2 Folder

3. Inside the EZ2 folder find the **Tools** folder.



Picture 10.0-2 EZ2 Tools Folder

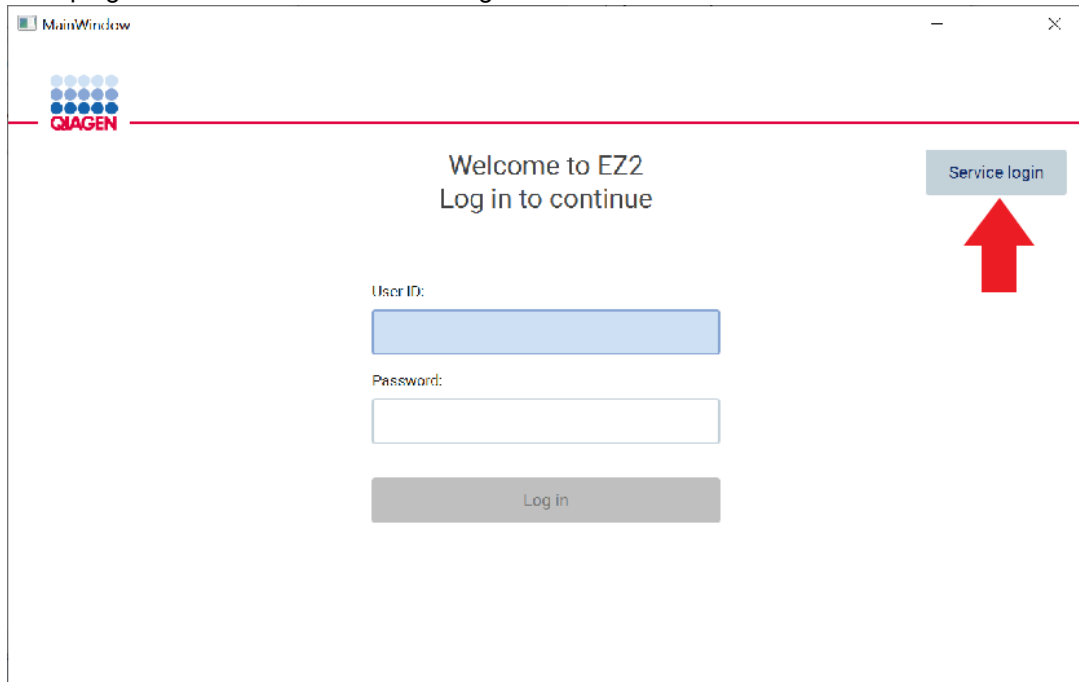
4. Find the **login** folder and copy it to a USB stick, or make a folder named **login** on the USB stick.
 5. The **login** folder will have a file named **service**, this is the key that lets you enter the service environment.



Picture 10.0-3 Login Folder

6. Plug the USB stick in one of the 3 USB ports on the instrument.

7. You can plug in the USB stick before starting the instrument or after it has initialized.

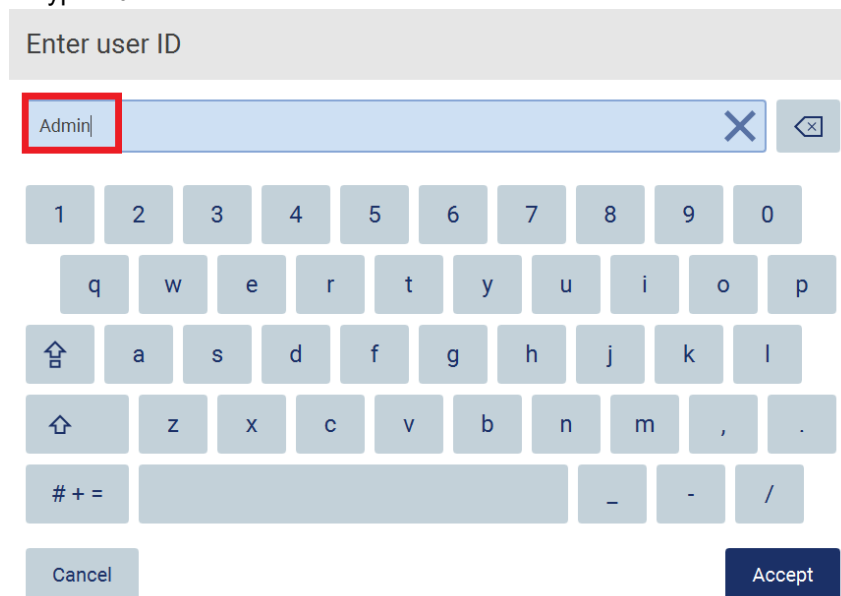


Picture 10.0-4 Service Login

8. Press the **Service Login** button on the screen.
9. A PIN code must be entered to run the service software, no user name is needed.
10. You are now logged in as a service user and all of the service environment is available.

10.0.3 Log In As The Administrator

1. To log in as the Administrator, no USB stick is needed.
2. For the user ID type **Admin**.



Picture 10.0-5 Admin Login

3. Type **Admin** for the password.

Enter password

Admin

1234567890

qwertyuiop

⌵asdgfhjkl

⌴zxcvbnm,.

#+= _ - /

CancelAccept

Picture 10.0-6 Admin Password

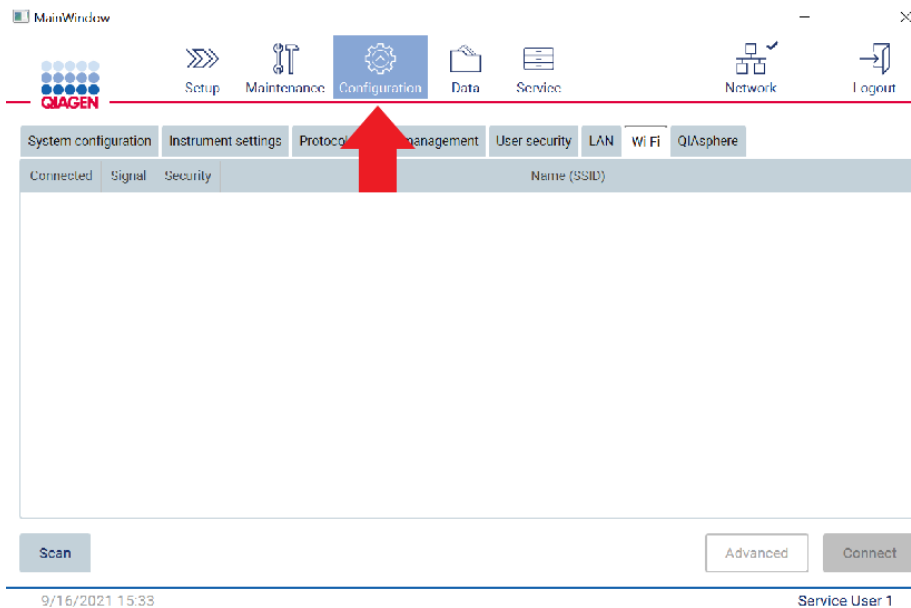
- 4. When logged in as the **Administrator** in IVD mode you can update the software and update protocols.
- 5. You will not be able to see or use the **Service** environment for calibrations or verification testing when signing in as admin.
- 6. All users will be able to create a report log, support package and audit trail.

10.0.4 Installation and Update Instrument Software

Installation, update or re-installation of the EZ2 Connect instrument software can be performed by the administrator or service personnel.

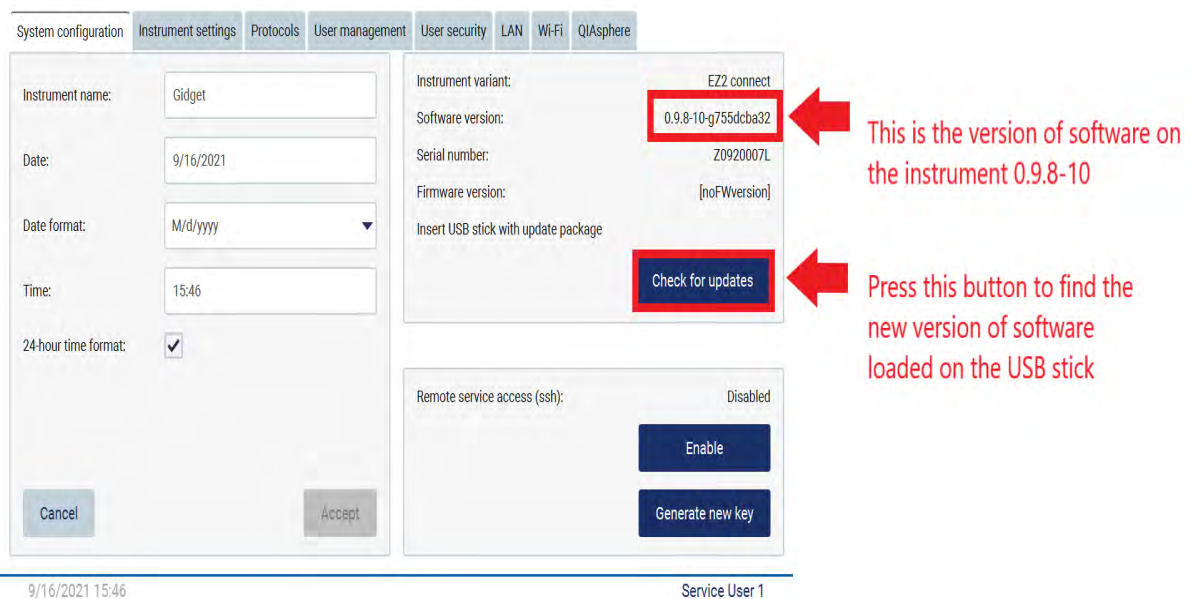
INFORMATION 	Before updating/installing the instrument software package, ensure to download all run reports and the support package to the USB drive.
---	--

1. Log in as a service user or administrator.



Picture 10.0-7 Configuration

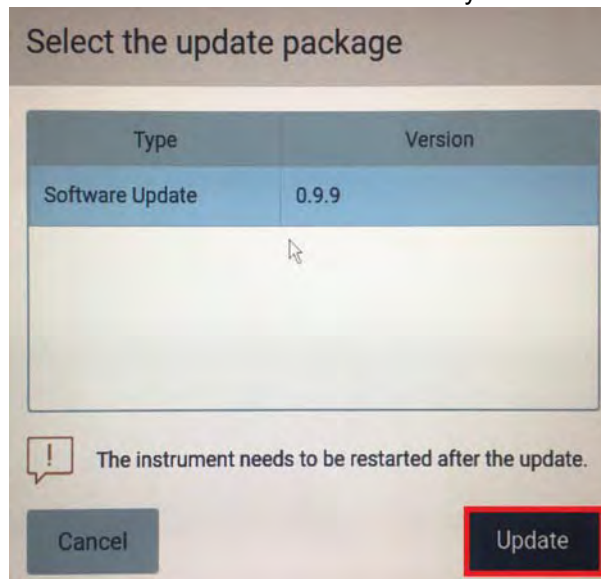
2. Press the **Configuration** button on the screen.
3. Remove the USB stick and load the new version of software on the USB stick, it does not need to be in a folder.



Picture 10.0-8 Check For Updates

4. Insert the USB stick in one of the 3 USB ports on the EZ2 Connect.
5. Press the **Check for Updates** button.

- The new software you loaded on the USB stick will be found by the software.

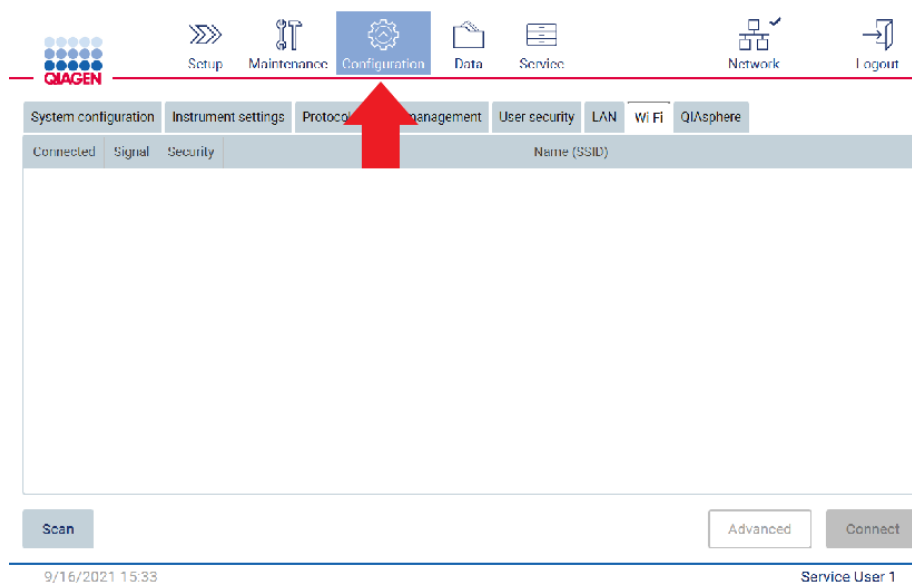


Picture 10.0-9 Update

- Press the new software you would like to load and press **Update**.
- The software may take a while to load, some updates may take over an hour.
- Follow the instruction on the screen and leave the USB stick in place till you see the log in screen again.

10.0.5 Protocol Updates

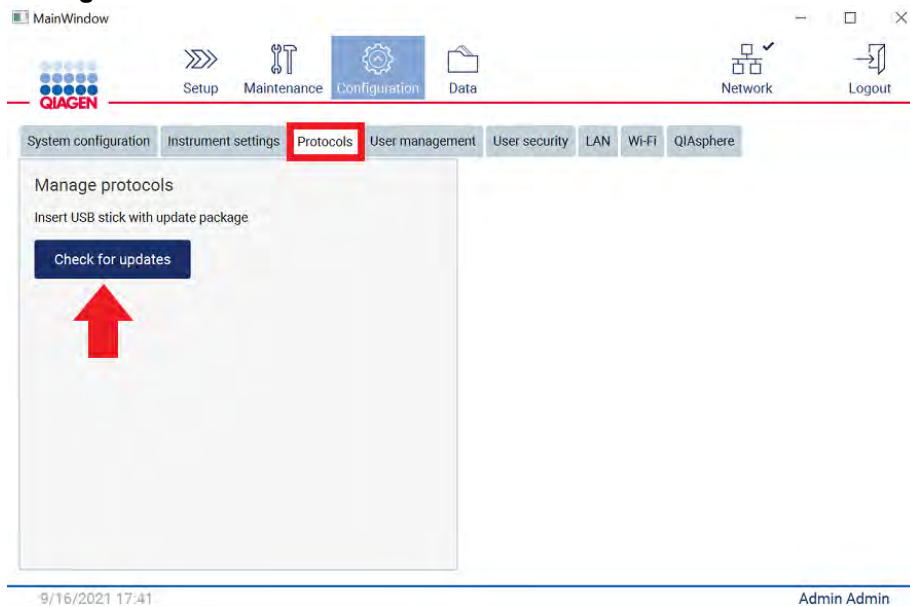
- Log in as a service user or administrator.



Picture 10.0-10 Configuration

- Load the protocol update from your computer on the USB stick.
- Insert the USB stick in one of the 3 USB port positions.

4. Select the **Configuration** environment.



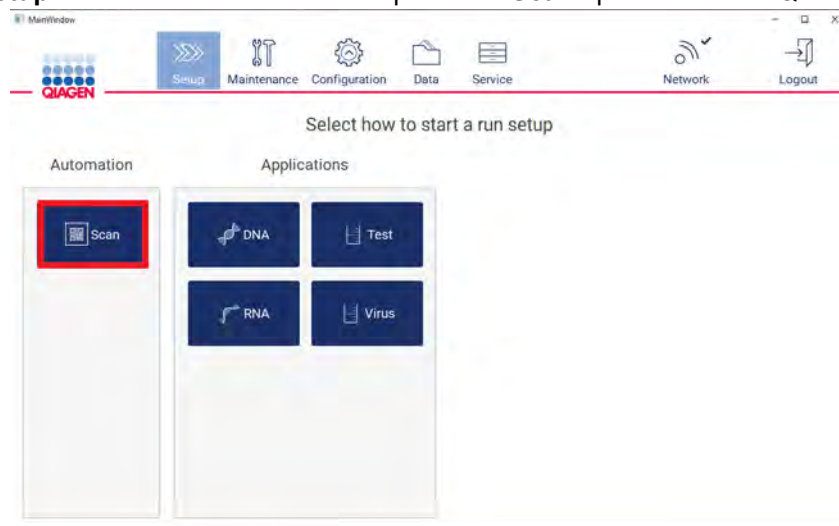
Picture 10.0-11 Protocol Updates

5. Select the **Protocols** tab.
6. Press the **Check for Updates** button.
7. Select the protocol update you would like to load.
8. Press the **Update** button.

10.1 User Software

10.1.1 Setup

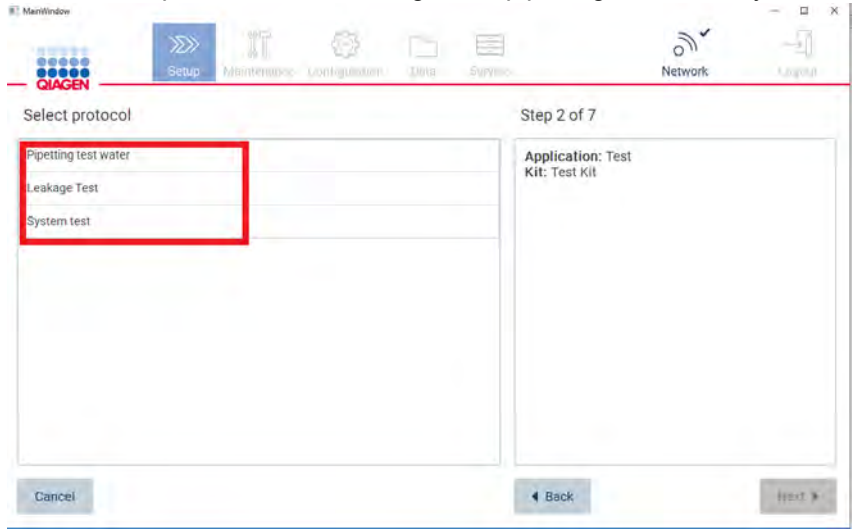
1. Under the **Setup** environment of the software press the **Scan** option to scan a Q-card.



Picture 10.1-1 Scan Q-card

2. In the **Setup** environment the customer can select the protocol they would like to run.

3. Service will use the **Test** option to run the leakage test, pipetting test or the system workflow test.



Picture 10.1-2 System Test

4. To run one or all of these test visit the [Verification Procedures \(VP\)](#) chapter for information on how to operate each test.

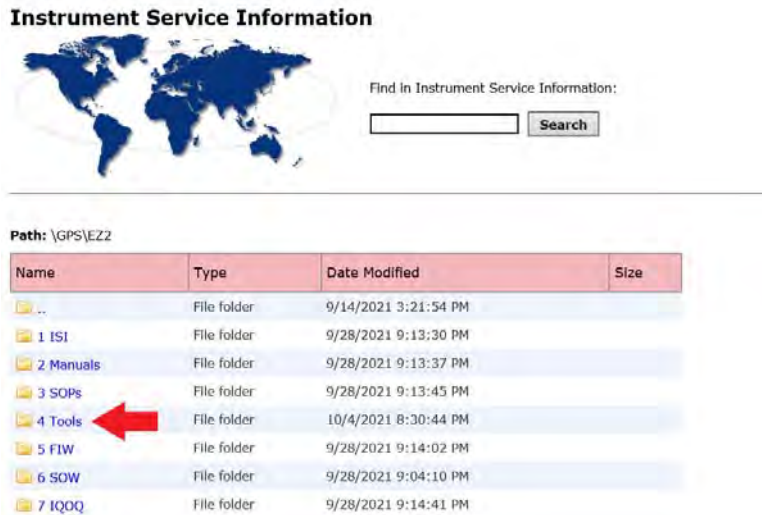
10.2 Software for Service and Troubleshooting

10.2.1 Creating A Support Package

- How to create a support package is located here: [Creating A Support Package](#).

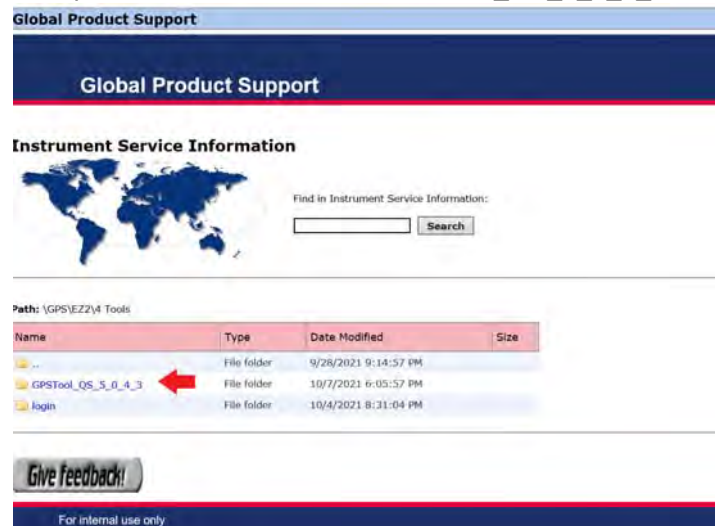
10.2.2 GPS Tool

- The GPS Tool is used to translate support package files to a readable document.
- Go to the GPS website and find the EZ2 folder and select the **Tools** folder.



Picture 10.2-1 Tools Folder

3. Once the tool folder is open find the folder named **GPSTool_QS_5_0_4_3** and click on the folder.



Picture 10.2-2 GPS Tool

4. Click the **GPS tool** folder.



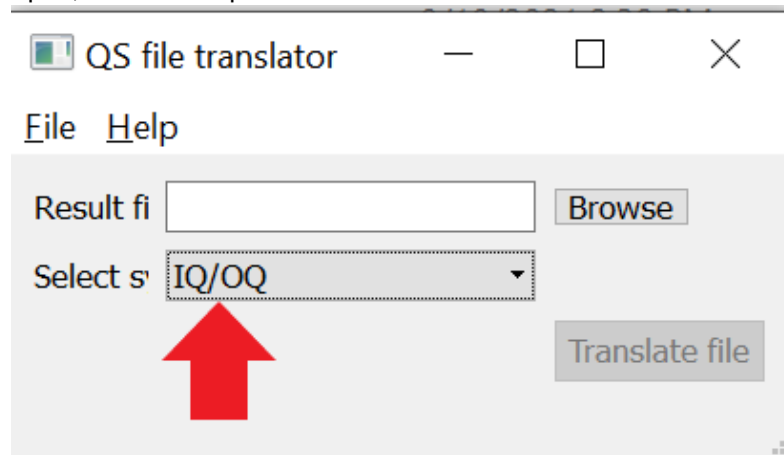
Picture 10.2-3 Open GPS Tool Folder

5. Download the **GPSTool.exe** to your computer.
 6. Open the **GPSTool.exe** translator program.

Name	Date modified	Type	Size
Data	10/7/2021 11:07 AM	File folder	
GPSTool.exe	8/18/2021 2:39 PM	Application	76 KB
libxml2.dll	8/18/2021 2:39 PM	Application extension	870 KB
libxslt.dll	8/18/2021 2:39 PM	Application extension	141 KB
log4cplus.dll	8/18/2021 2:39 PM	Application extension	274 KB
qlibxml.dll	8/18/2021 2:39 PM	Application extension	13 KB
qLog.dll	8/18/2021 2:39 PM	Application extension	41 KB
QtCore4.dll	8/18/2021 2:39 PM	Application extension	2,568 KB
QtGui4.dll	8/18/2021 2:39 PM	Application extension	8,119 KB
quazip.dll	8/18/2021 2:39 PM	Application extension	46 KB
qZip.dll	8/18/2021 2:39 PM	Application extension	37 KB

Picture 10.2-4 GPSTool.exe

7. The tool is now open, click the drop down arrow and Select **IQ/OQ**.

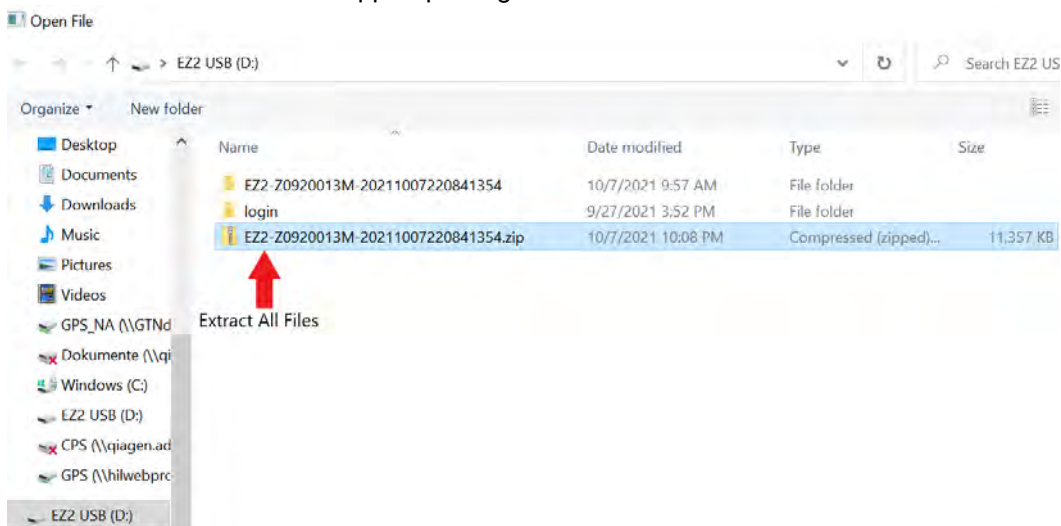


Picture 10.2-5 Select IQ/OQ

8. Click on the **Browse** button.

9. Find the support package file on your USB stick and right click.

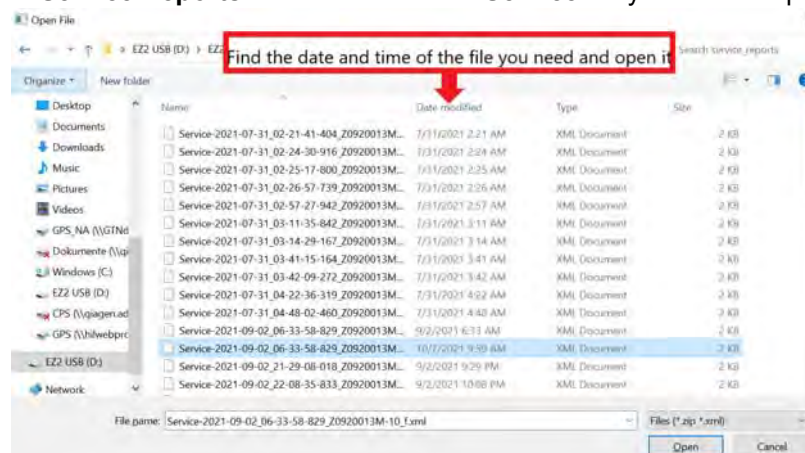
10. **Extract all files** on the support package.



Picture 10.2-6 Unzip the Support Package

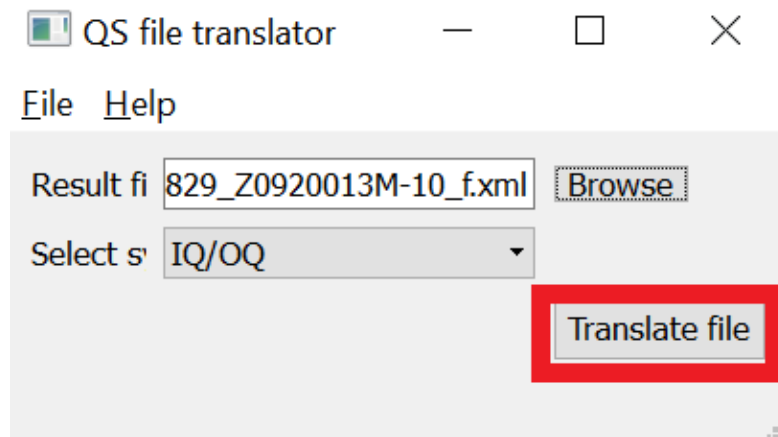
11. Open the unzipped support package and click on the **Reports** file.

12. Next click on the **Service Reports** file and click on the **Service** file you want to open.



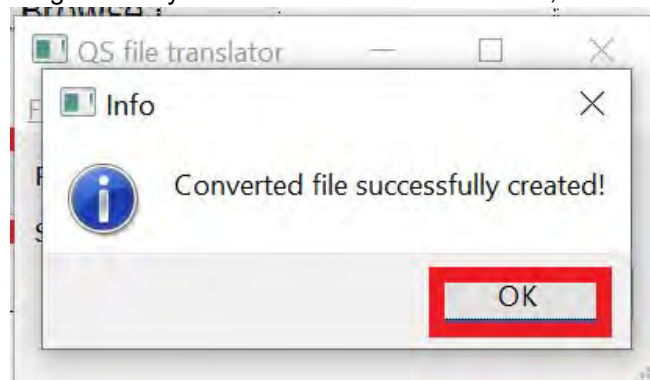
Picture 10.2-7 Service File

13. Once you have the file selected you want to open, click on the **Translate file** button.



Picture 10.2-8 Translate File

14. You will receive a message that says the files have been converted, click the **Yes** button.



Picture 10.2-9 Converted

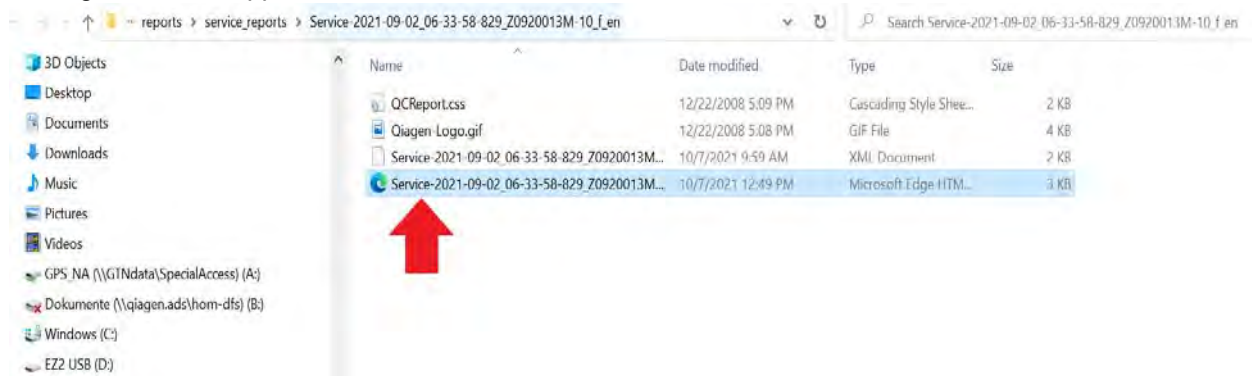
15. Return to the **Report** folder, then **Service Reports**.

16. Now you will see a new zipped folder.

Name	Date modified	Type	Size
Service-2021-07-31_02-21-41-404_Z0920013M...	7/31/2021 2:21 AM	XML Document	2 KB
Service-2021-07-31_02-24-30-916_Z0920013M...	7/31/2021 2:24 AM	XML Document	2 KB
Service-2021-07-31_02-25-17-800_Z0920013M...	7/31/2021 2:25 AM	XML Document	2 KB
Service-2021-07-31_02-26-57-739_Z0920013M...	7/31/2021 2:26 AM	XML Document	2 KB
Service-2021-07-31_02-57-27-942_Z0920013M...	7/31/2021 2:57 AM	XML Document	2 KB
Service-2021-07-31_03-11-35-842_Z0920013M...	7/31/2021 3:11 AM	XML Document	2 KB
Service-2021-07-31_03-14-29-167_Z0920013M...	7/31/2021 3:14 AM	XML Document	2 KB
Service-2021-07-31_03-41-15-164_Z0920013M...	7/31/2021 3:41 AM	XML Document	2 KB
Service-2021-07-31_03-42-09-272_Z0920013M...	7/31/2021 3:42 AM	XML Document	2 KB
Service-2021-07-31_04-22-36-319_Z0920013M...	7/31/2021 4:22 AM	XML Document	2 KB
Service-2021-07-31_04-48-02-460_Z0920013M...	7/31/2021 4:48 AM	XML Document	2 KB
Service-2021-09-02_06-33-58-829_Z0920013M...	10/7/2021 12:49 PM	Compressed (zipped)...	9 KB
Service-2021-09-02_06-33-58-829_Z0920013M...	10/7/2021 12:49 PM	XML Document	2 KB
Service-2021-09-02_21-29-08-018_Z0920013M...	9/2/2021 9:29 PM	XML Document	2 KB
Service-2021-09-02_22-08-35-833_Z0920013M...	9/2/2021 10:08 PM	XML Document	2 KB
Service-2021-09-02_22-30-08-032_Z0920013M...	9/2/2021 10:30 PM	XML Document	2 KB

17. New Zipped Service Folder

18. Right click the zipped folder and extract all the files.



Picture 10.2-10 Readable File

19. Open the .HTM file to view the readable file.

Test Report for OQ

9/2/2021 11:16 AM

Serial Number: Z0920013M

Application Version: 0.9.8

Instrument: Gadget

Overall Result: **FAILED**

Failed Test IDs: 10

Tested by QIAGEN: **Approved by system owner:**

Printed name: _____ Date: _____

Signature: _____ Signature: _____

Module Load check: **FAILED**

Result: **FAILED**

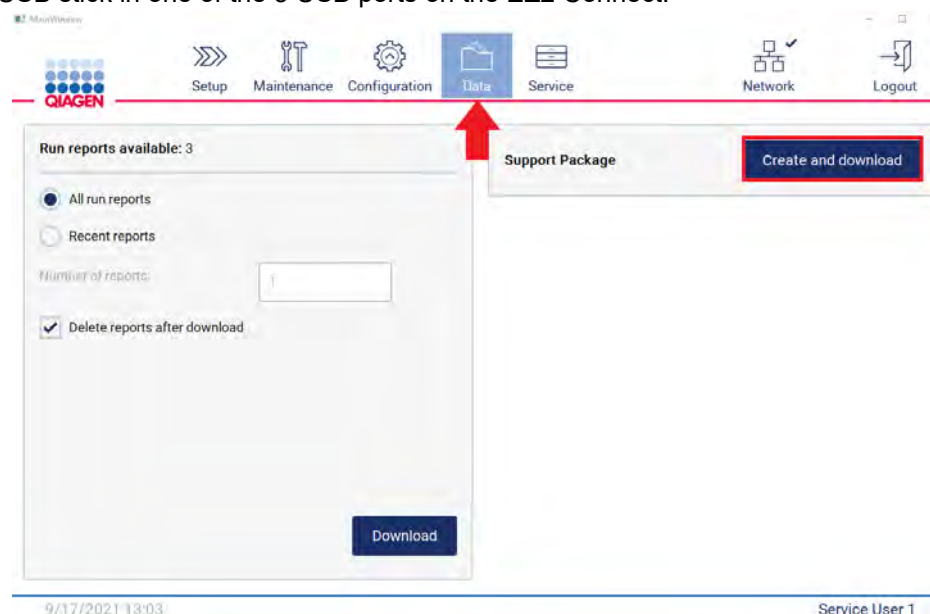
Test IDs: 10

Picture 10.2-11 Test Report

20. The test report is in a printable version. The report covers the testing that has been completed with a passed or failed marking on the report.

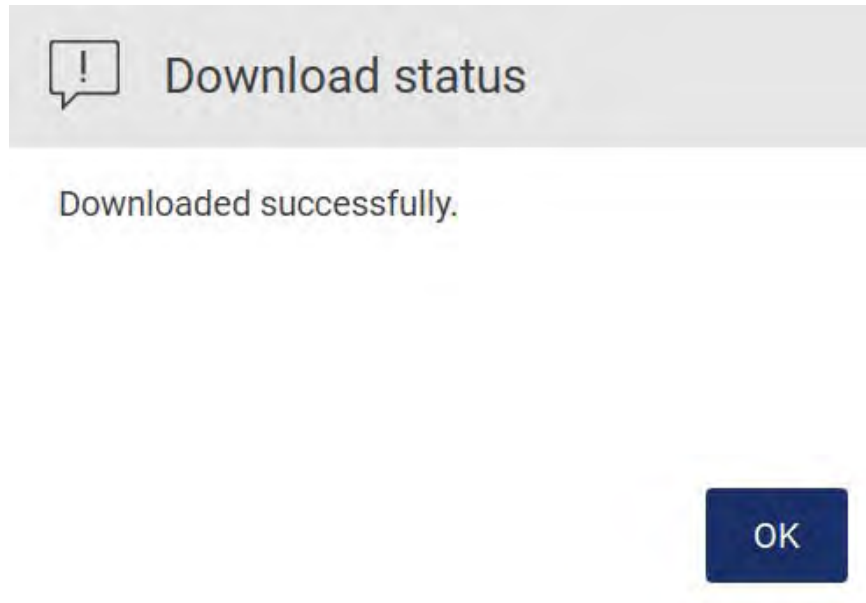
10.2.3 Downloading Reports

1. Log in to the software as service or administrator.
2. Insert a USB stick in one of the 3 USB ports on the EZ2 Connect.



Picture 10.2-12 Data Environment

3. Select the **Data** environment tab.
4. At the top of the screen you will see how many reports are available.
5. Select **All run reports** or **Recent reports**.
6. Uncheck **Delete reports after download**.
7. Press the **Download** button.



Picture 10.2-13 Download Status

8. Press the **OK** button.
9. Remove the USB stick from the instrument and plug it in to your computer.

10.2.4 Run Reports Content

1. An EZ2 run report is created by the software application after a run is completed, aborted or if it fails, after the user taps the **Finish** button on the screen that is shown after a run ends.

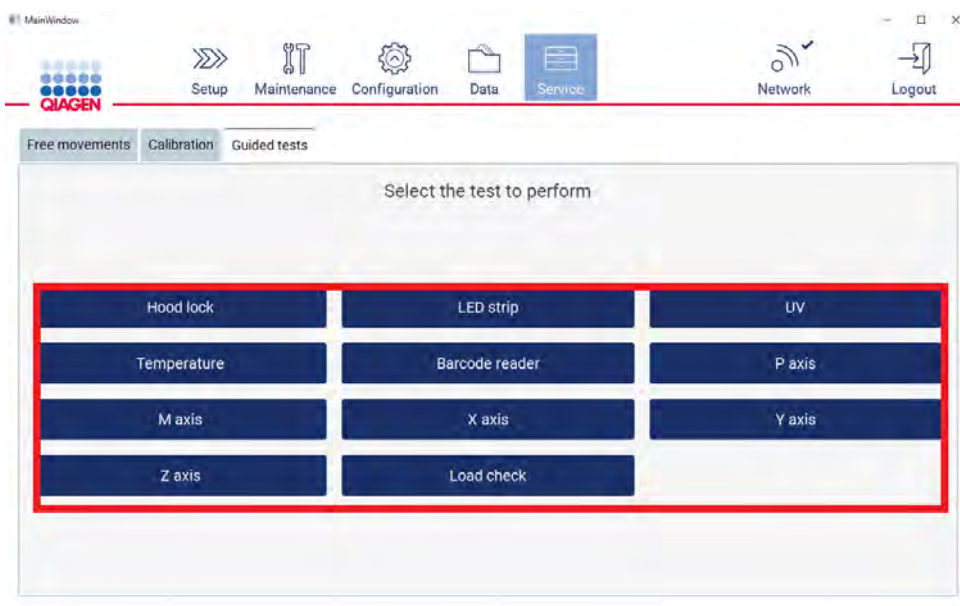
Each run report is saved in two formats - PDF and XML. Both formats include the same information, that is:

- The name of the user that was logged in when the run was started.
- The length of the run.
- The time when the run was started and when it ended.
- The name of the application that the protocol represents.
- The name of the kit.
- The name of the protocol.
- The name of the run report file, which includes the date when the run ended and the serial number of the instrument.
- The status of the run, which indicates if the run was completed successfully, if it failed, or if it was aborted.
- The status of the cleanup procedure that is done automatically after a run ends.
- The version number of the protocol.
- The protocol parameters selected during the run setup process.
- Information about the samples - their positions, names and any notes that were added by the user.

10.3 Service Software Environment

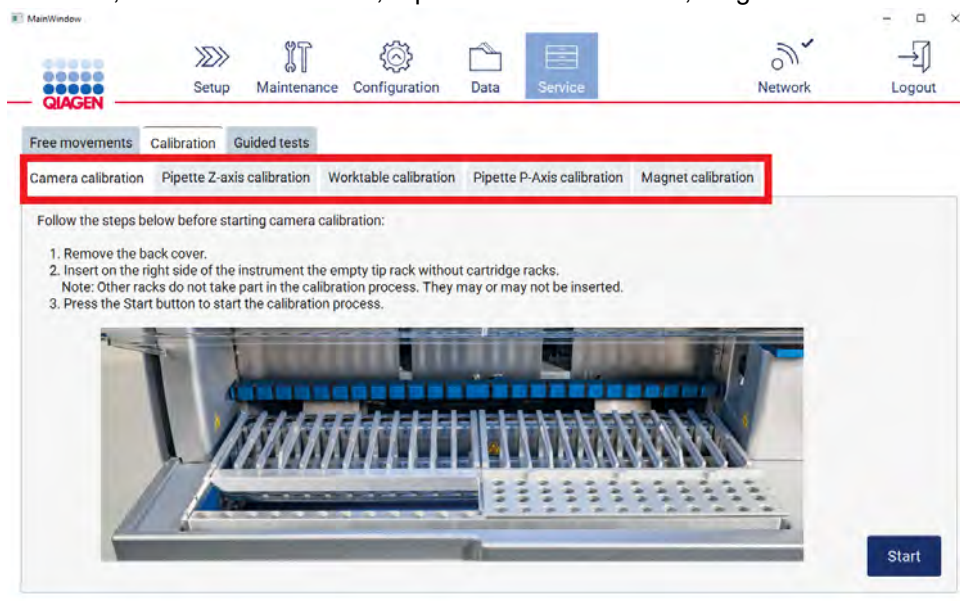
1. In the service environment you will find 3 categories: calibrations, guided test and free movements.

2. Under the **Guided Test** section, you will find verifications for these 11 categories: M-axis, Y-axis, Z-axis, P-axis, X-axis, LED test, Barcode reader test, Temperature test, Load check test, Hood lock test, UV test.



Picture 10.3-1 Verifications

3. For more information about the guided test see [Verification Procedures \(VP\)](#).
 4. In the **Calibrations** section of the software you will find these 5 categories: Camera calibration, Pipette Z-axis calibration, Worktable calibration, Pipette P-axis calibration, Magnet Calibration.

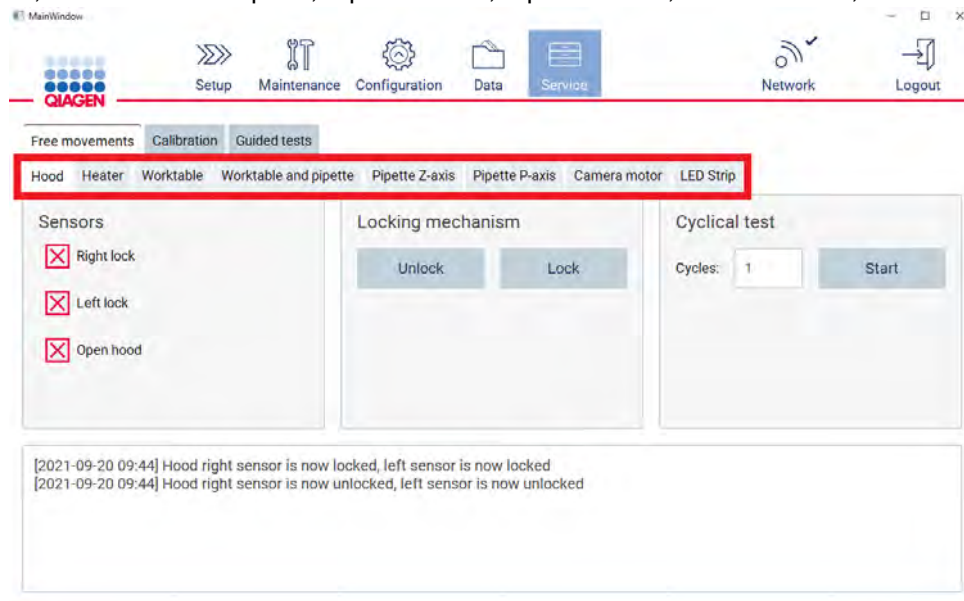


Picture 10.3-2 Calibrations

5. See [Calibration Procedures \(CP\)](#) for further details about available adjustments/calibrations.

10.3.1 Free Movements

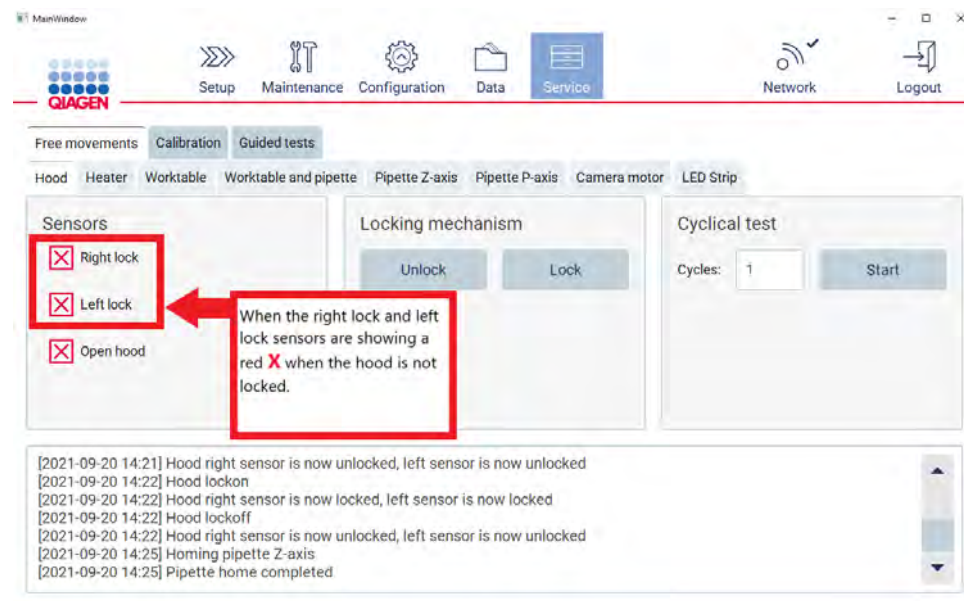
1. In the **Free Movements** section of the software you will find these 8 categories: Hood, Heater, Worktable, Worktable and Pipette, Pipette Z-axis, Pipette P-axis, Camera Motor, LED Strip.



Picture 10.3-3 Free Movements

10.3.1.1 Hood

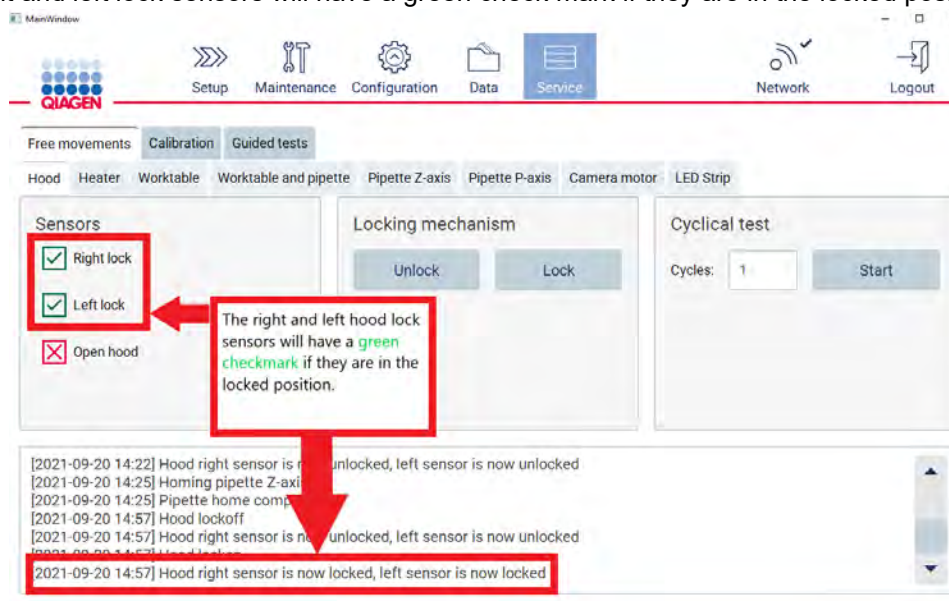
2. Under the **Hood** section of the software, the program is able to lock and unlock the hood.
3. The locking and unlocking can also be ran in the **Cycle Test** section. Enter the number of cycles to lock and unlock the hood in the text box. The hood section shows if the sensors are in a locked or unlocked position.



Picture 10.3-1 Hood Not Locked

4. The right and left lock sensors will have a red X if they are in the unlocked position.

5. The right and left lock sensors will have a green check mark if they are in the locked position.

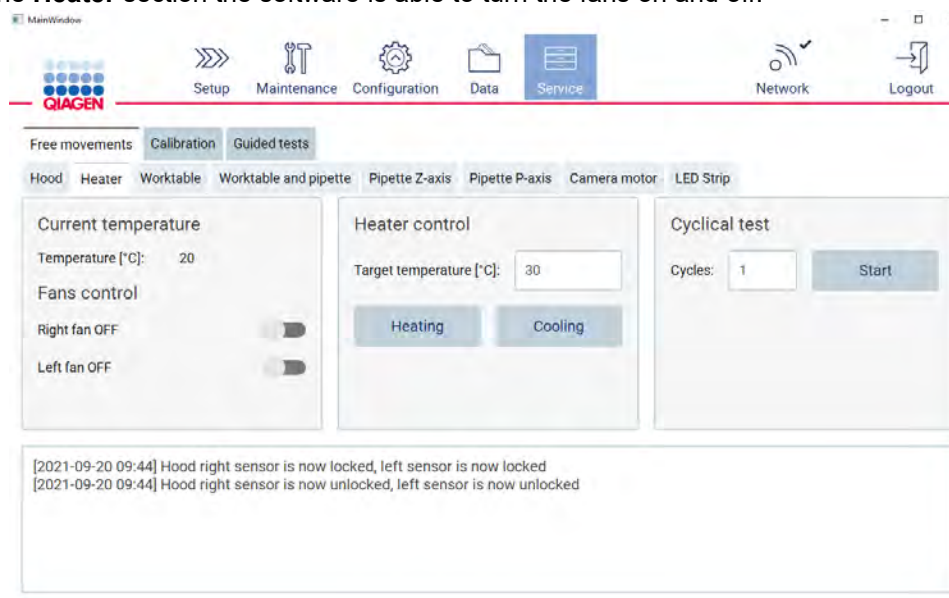


Picture 10.3-2 Locked Left & Right Sensors

6. The text at the bottom of the page will indicate last action of the sensors.
 7. The hood open sensor will only work when the rear cover is on or the service switch has been pressed in.
 8. The hood open sensor will have a red X when closed and a green check mark when the hood is open.

10.3.1.2 Heater

9. Under the **Heater** section the software is able to turn the fans on and off.



Picture 10.3-1 Heater

10. In the **Heater** section you can enter a temperature in the **Target Temperature** box. Press the **X** in the box to enter the temperature you want to reach.

Enter temperature [°C]

30|  

Input range: 30 - 110 °C

7	8	9
4	5	6
1	2	3
0		

Cancel Accept

Enter temperature [°C]

60|   

Input range: 30 - 110 °C

7	8	9
4	5	6
1	2	3
0		

Cancel Accept

Picture 10.3-2 Enter Temp Target

11. Check that the temperature heats to the target temperature you have entered with a thermometer and temperature probe.



Picture 10.3-3 Temperature Meter

12. To reach the target temperature it can take up to 15 minutes.



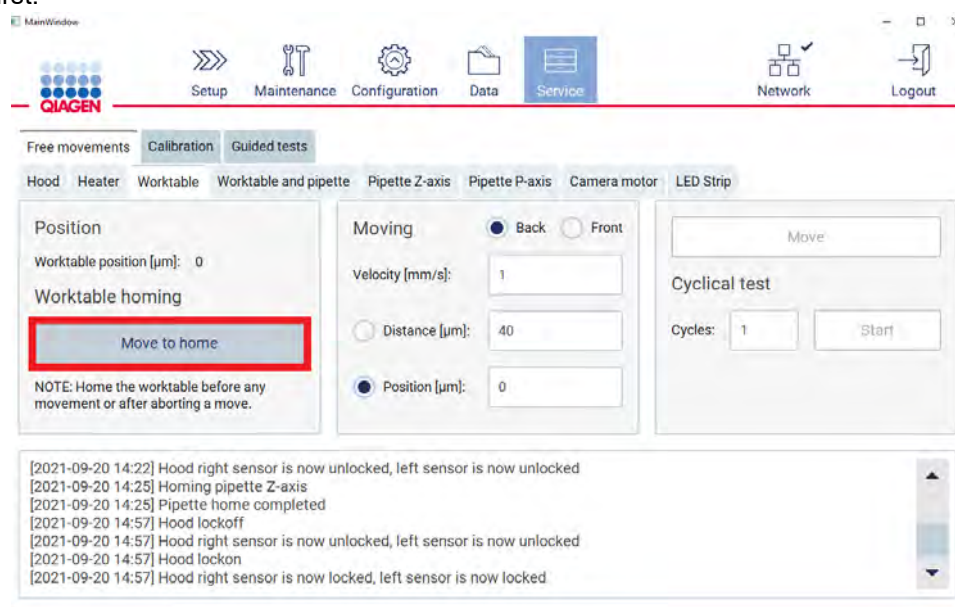
Picture 10.3-4 Background Temp

13. In the background of the software page the temperature will also show the temperature climbing to the target range.

14. When the temperature reaches the target press the **Stop** button to complete this test.

10.3.1.3 Worktable

1. Before you can use the worktable free movements portion of the software you must press the **Move to home** first.

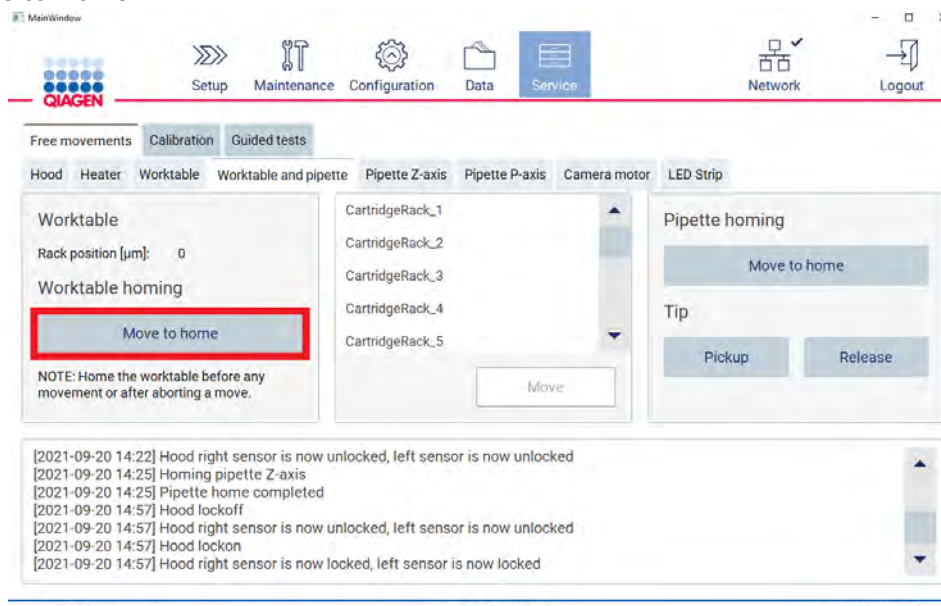


Picture 10.3-1 Home Worktable

2. After homing the worktable, change the **Velocity** 30.
3. The **Cyclical test** will move the worktable to the rear Y home and to the front Y home as one cycle.
4. Change the **Cyclical test** to how many you would like to run.
5. If this test is stopped before it is completed, the system will error and you will need to restart the instrument before anything else can be checked or tested.

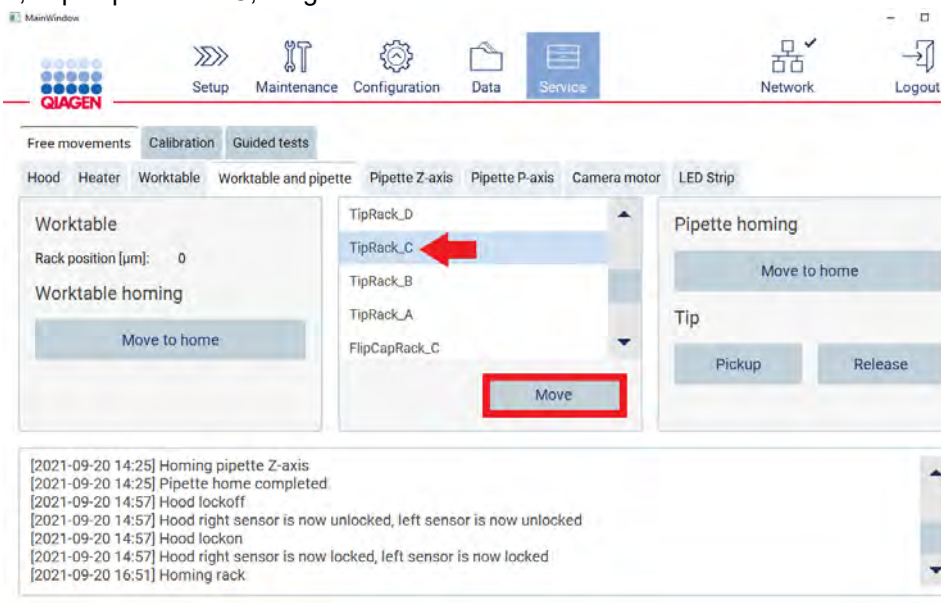
10.3.1.4 Worktable and Pipette

- Before you can use the worktable and pipette free movements portion of the software you must press the **Move to Home** first.



Picture 10.3-1 Move to Home

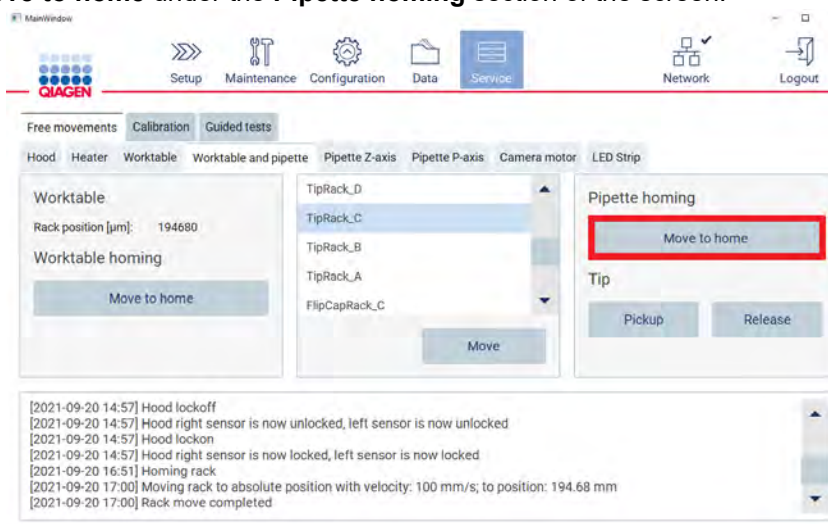
- After homing the worktable, you can select a worktable position to move to, Cartridge racks 1-12, Tip racks A-D, Flip cap racks A-C, Large Volume racks A-D.



Picture 10.3-2 Move to TipRack

- Load the tip rack position you would like to test for tip pick up.
- After you have loaded the tip rack press the **Move** button, this will move the tip rack C position under the pipette module.

5. Press the **Move to home** under the **Pipette homing** section of the screen.

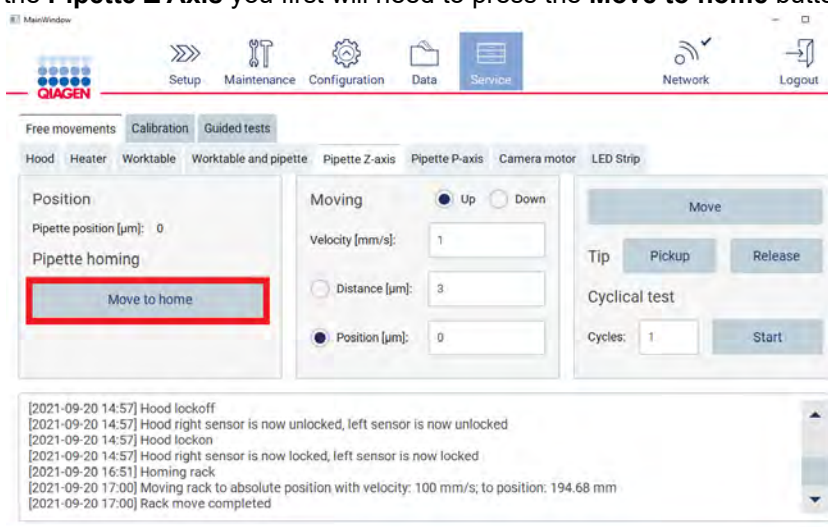


Picture 10.3-3 Move Pipette Home

6. Press the **Pickup** button to pick up the tips and press the **Release** button to remove the tips.

10.3.1.5 Pipette Z Axis

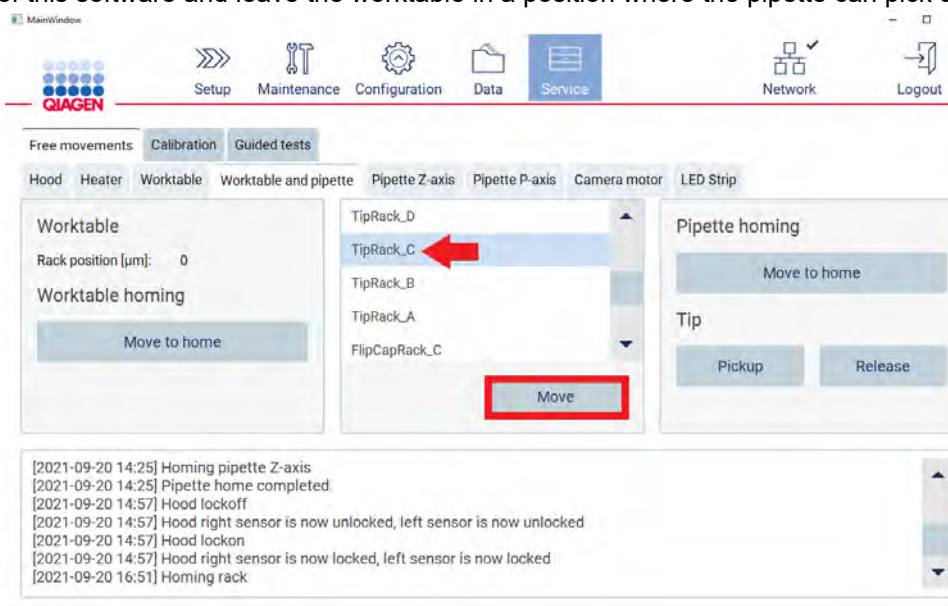
1. Before using the **Pipette Z Axis** you first will need to press the **Move to home** button.



Picture 10.3-1 Home Pipette

2. Change the **Velocity** to 30, this will speed up the movement of the Z for the pipette.
3. You can use the **Up** and **Down** to move the pipette the way you want to go but on the first movement it should be down because the pipette is in the up home position.

- if you would like to pick up and release the tips you will need to return to the **Worktable and Pipette** section of this software and leave the worktable in a position where the pipette can pick up tips.

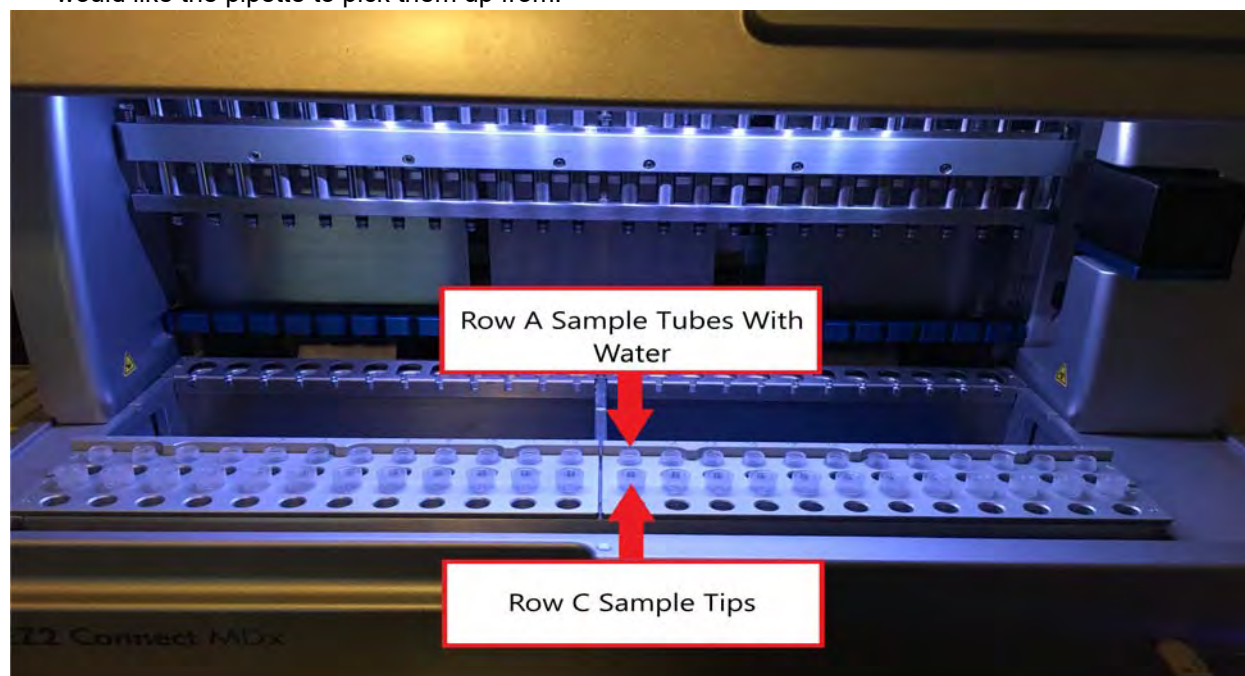


Picture 10.3-2 Move Worktable

- After the worktable is moved to the correct tip position you can now use the tip **Pickup** and **Release** buttons.

10.3.1.6 Pipette P Axis

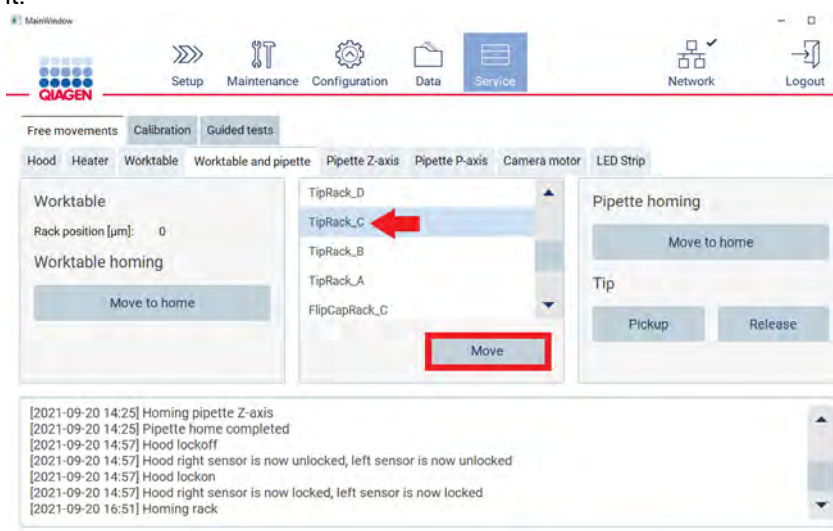
- To use the **Pipette P Axis** free movement to pick up fluid you will need to return to the **Worktable and Pipette** section of the free movements software page.
- Load the sample tube in the position you would like to pick fluid up from and load tips in the position you would like the pipette to pick them up from.



Picture 10.3-1 Sample Tubes & Tips

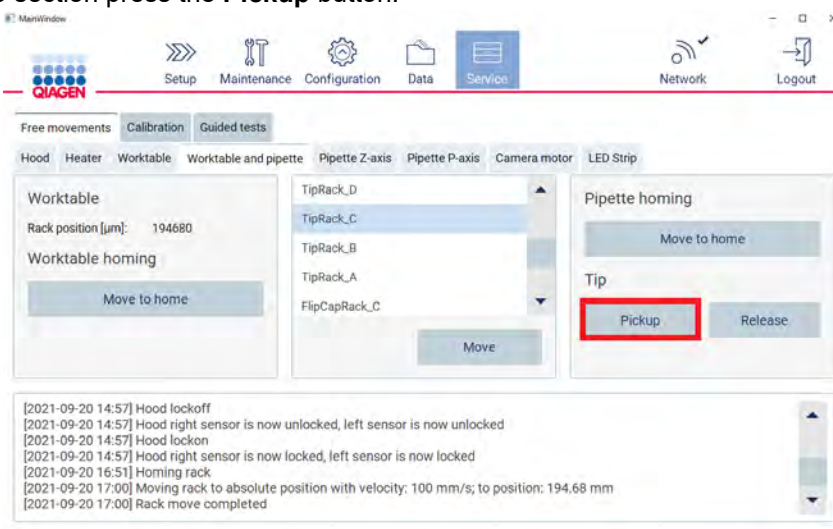
- In the picture above I have loaded sample tips in row C and sample tubes filled with water in row A.
- Return to the **Worktable and Pipette** section on the free movements software page.

- Press the **Move to home** button on the worktable section, then select **TipRack-C** and press the **Move** button below it.



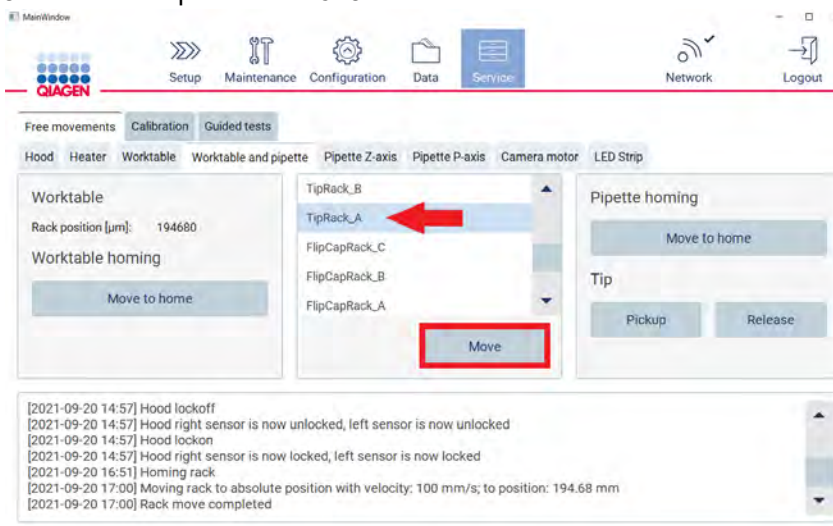
Picture 10.3-2 Worktable Move to Row C

- Press the **Move to home** under the Pipette homing section.
- Under the Tip section press the **Pickup** button.



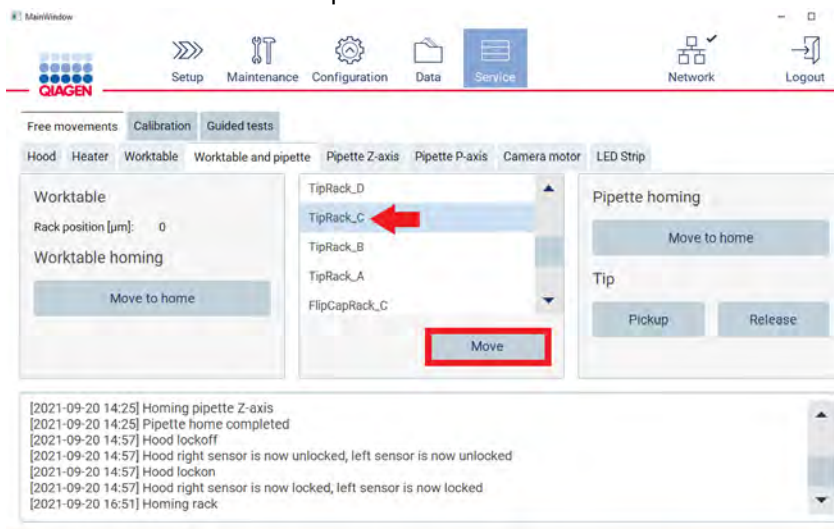
Picture 10.3-3

- The tips should now be on the pipette.
- Select **TipRack-A** and then press the **Move** button.



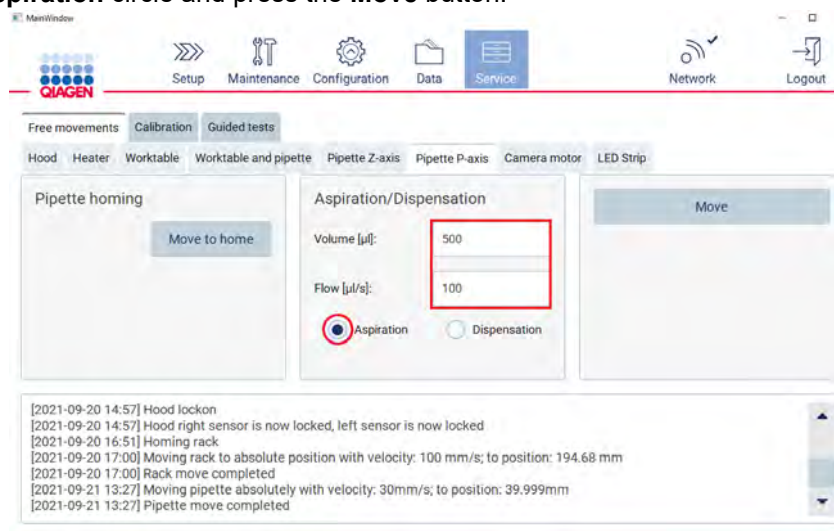
Picture 10.3-4 Move to Row A

10. Select the **Pipette Z axis** tab under free movements.
11. Press the **Down** option.
12. Type in the **Position** text box 39999 and press the **Move** button.



Picture 10.3-5 Tips in Fluid

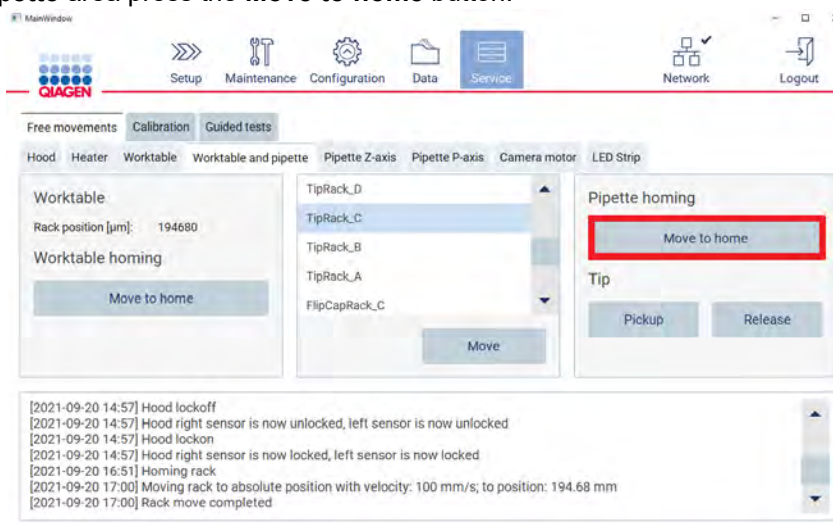
13. The tips should all be in the sample tube in row A in fluid.
14. Select the **Pipette P axis** tab under the free movements software page.
15. Change the volume amount to 500 and change the flow amount to 100.
16. Select the **Aspiration** circle and press the **Move** button.



Picture 10.3-6 Aspirate Fluid

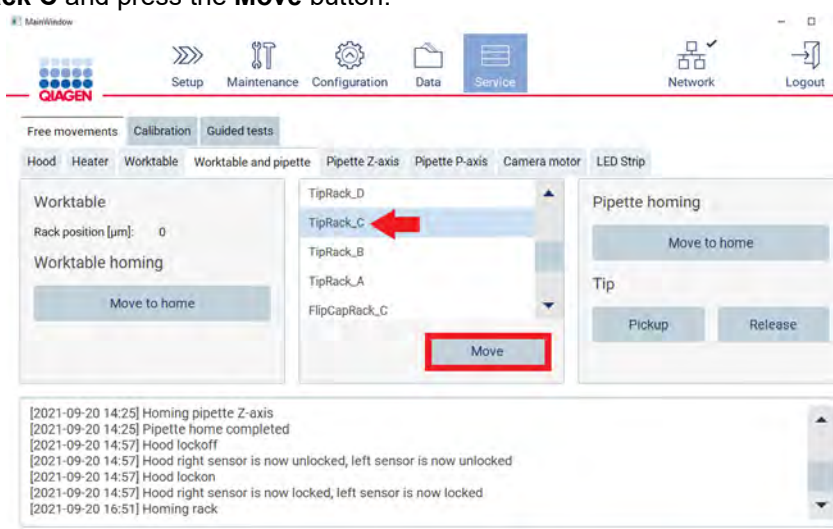
17. To empty the tip select the **Dispensation** circle and press the **Move** button.
18. To remove the tips return to the **Worktable and Pipette** tab on the free movements software page.

19. Under the **Pipette** area press the **Move to home** button.



Picture 10.3-7 Pipette Home

20. Select **TipRack C** and press the **Move** button.

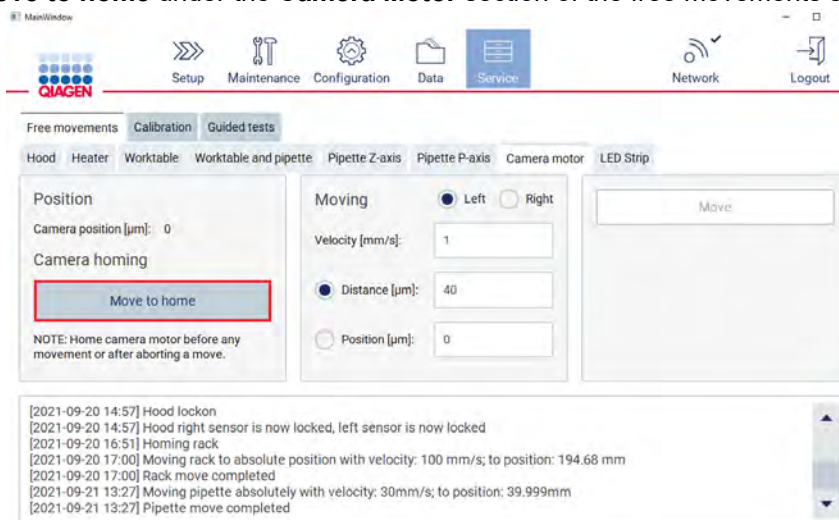


Picture 10.3-8 Tip Position C

21. Select the **Release** button and the pipette will push the tips off in the tip holders.

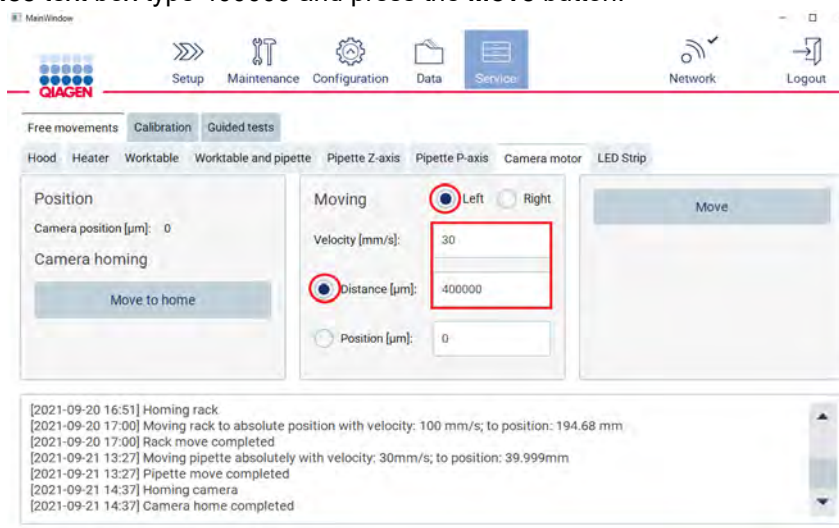
10.3.1.7 Camera Motor

1. Press the **Move to home** under the **Camera Motor** section of the free movements software.



Picture 10.3-1 Home Camera

2. The camera will need to move left after homing the camera.
3. Select the **Left** circle in the **Moving** section.
4. In the **Velocity** text box type **30**.
5. In the **Distance** text box type 400000 and press the **Move** button.



Picture 10.3-2 Move Camera

6. The camera will move 3/4 of the way across the work area.
7. Press the **Move to home** button again when testing is finished.

10.3.1.8 LED Strip

1. Select the **LED Strip** tab of the free movements page of the software.
2. Under the **LED Strip** it only has 2 options, to turn on the LED or to turn off the LED.

11 Troubleshooting

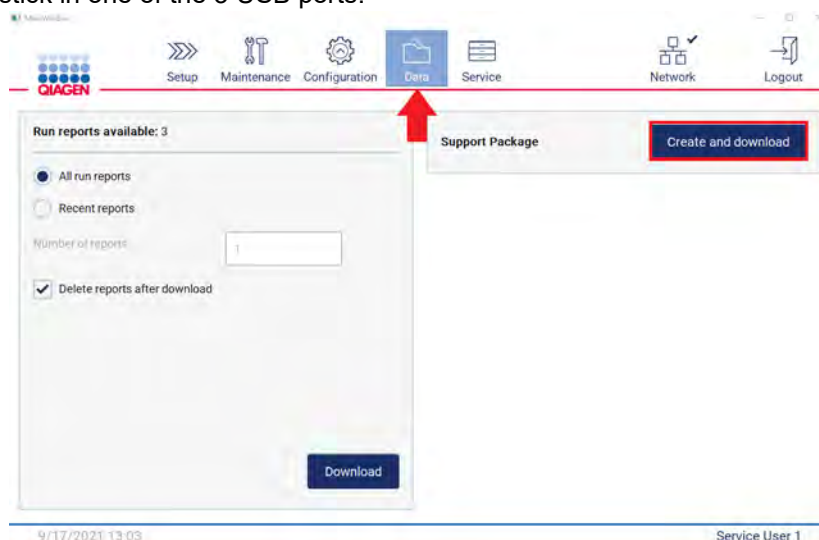
11.1 Troubleshooting Guide

A an Excel-based troubleshooting guide is available, it is: SM-5000-0047-A02, QFS, QTS, EZ2 Connect Troubleshooting Guide. This can be downloaded from Master Control.

11.2 Creating A Support Package

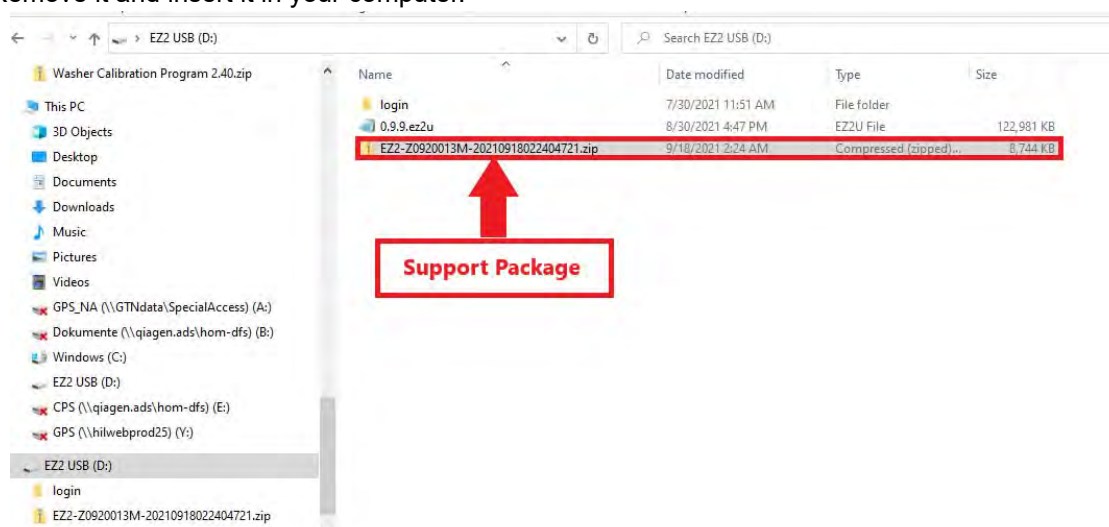
The Support Package can be saved as a compressed file (zip file) on a USB stick.

1. Log in to the software as service or administrator.
2. Insert a USB stick in one of the 3 USB ports.



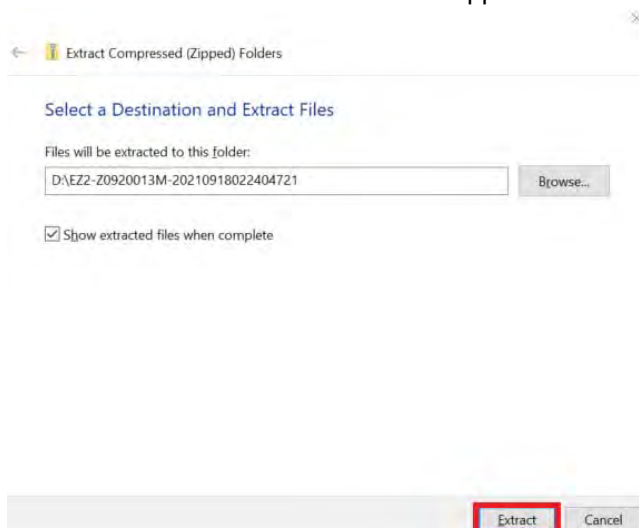
Picture 11.2-1 Data Environment

3. Select the **Data** environment on the software screen.
4. Press the **Create and Download** button on the screen.
5. Once the support package has finished downloading to the USB stick, the software give a message that the download is complete and press the **OK** button.
6. Remove it and insert it in your computer.



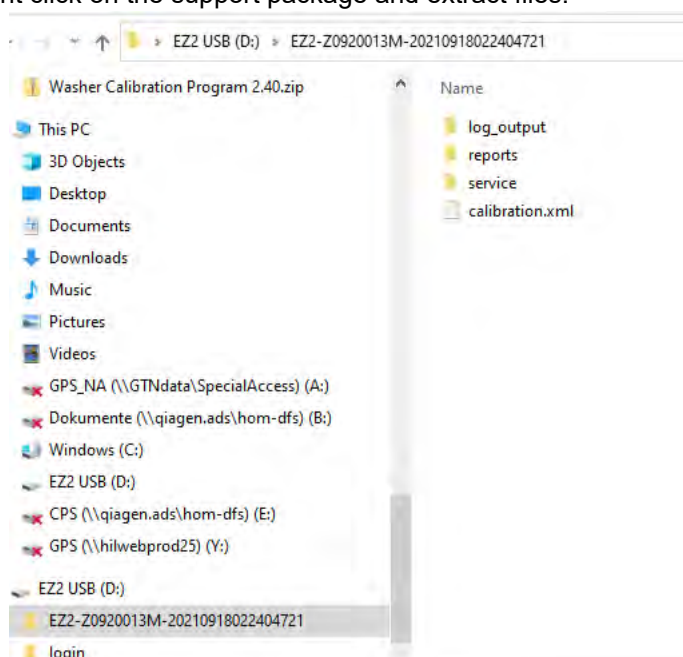
Picture 11.2-2 Support Package

7. The support package on the USB stick should look like the zipped folder in the picture above.



Picture 11.2-3 Extract Files

8. Right click on the support package and extract files.



Picture 11.2-4 Support Folders

9. After unzipping the support package, 3 support folders and 1 file will be available.
10. The **Log output** folder will contain error logs, and trace logs from past runs.
11. The **Reports** folder will contain information about protocols that have been ran on the instrument and service report files.
12. The **Service** folder will contain audit trails.
13. The **Calibration** file will contain the last calibration positions stored on the EZ2 Connect.

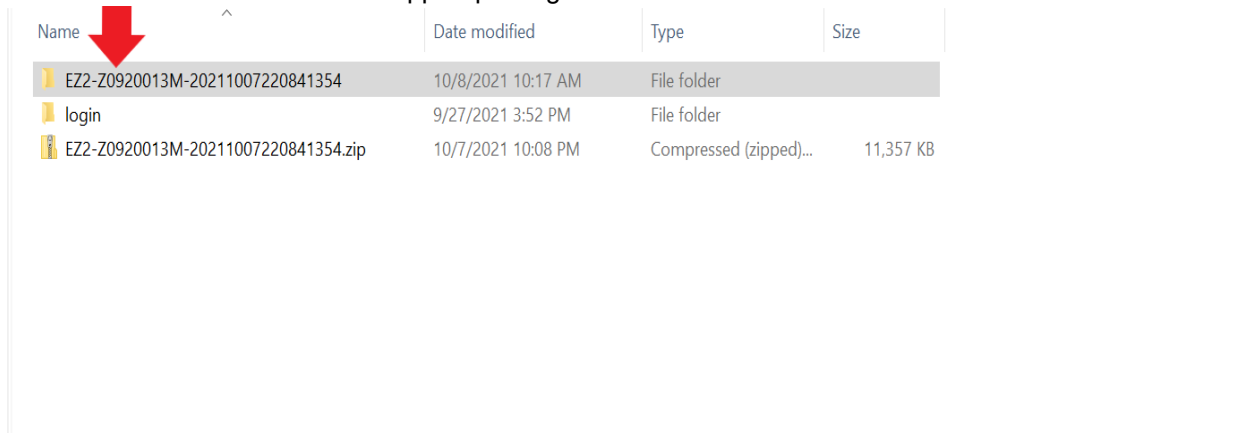
11.2.1 Log File Analysis

- A log file analysis tool is in development, in the meantime, we will provide guidance on how to find key information in the log files.

11.2.2 Viewing Trace Log Files

1. Trace log files are in the log folder of the support package.
2. The trace log files are best viewed using the free program called Notepad ++. We advise you to download and install Notepad ++ for log file analysis.

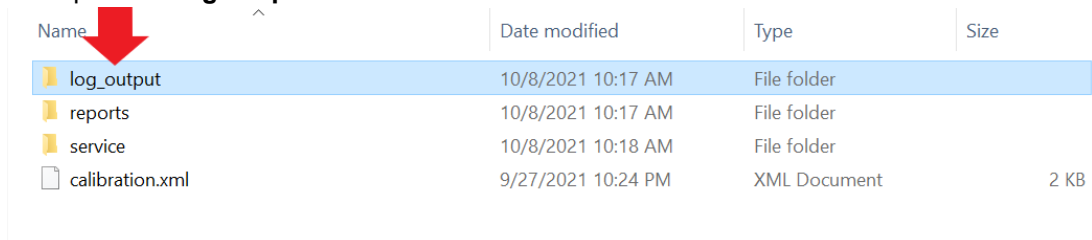
3. Notepad++ available from <http://notepad-plus-plus.org/>
4. Extract all of the files in the support package.



Name	Date modified	Type	Size
EZ2-Z0920013M-20211007220841354	10/8/2021 10:17 AM	File folder	
login	9/27/2021 3:52 PM	File folder	
EZ2-Z0920013M-20211007220841354.zip	10/7/2021 10:08 PM	Compressed (zipped)...	11,357 KB

Picture 11.2-5 Unzipped Support Package

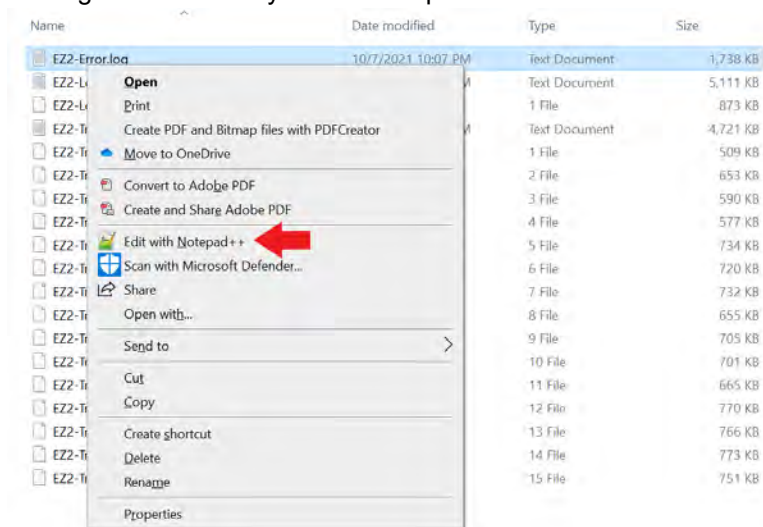
5. Open the unzipped support package.
6. Open the **Log Output** folder.



Name	Date modified	Type	Size
log_output	10/8/2021 10:17 AM	File folder	
reports	10/8/2021 10:17 AM	File folder	
service	10/8/2021 10:18 AM	File folder	
calibration.xml	9/27/2021 10:24 PM	XML Document	2 KB

Picture 11.2-6 Log Output Folder

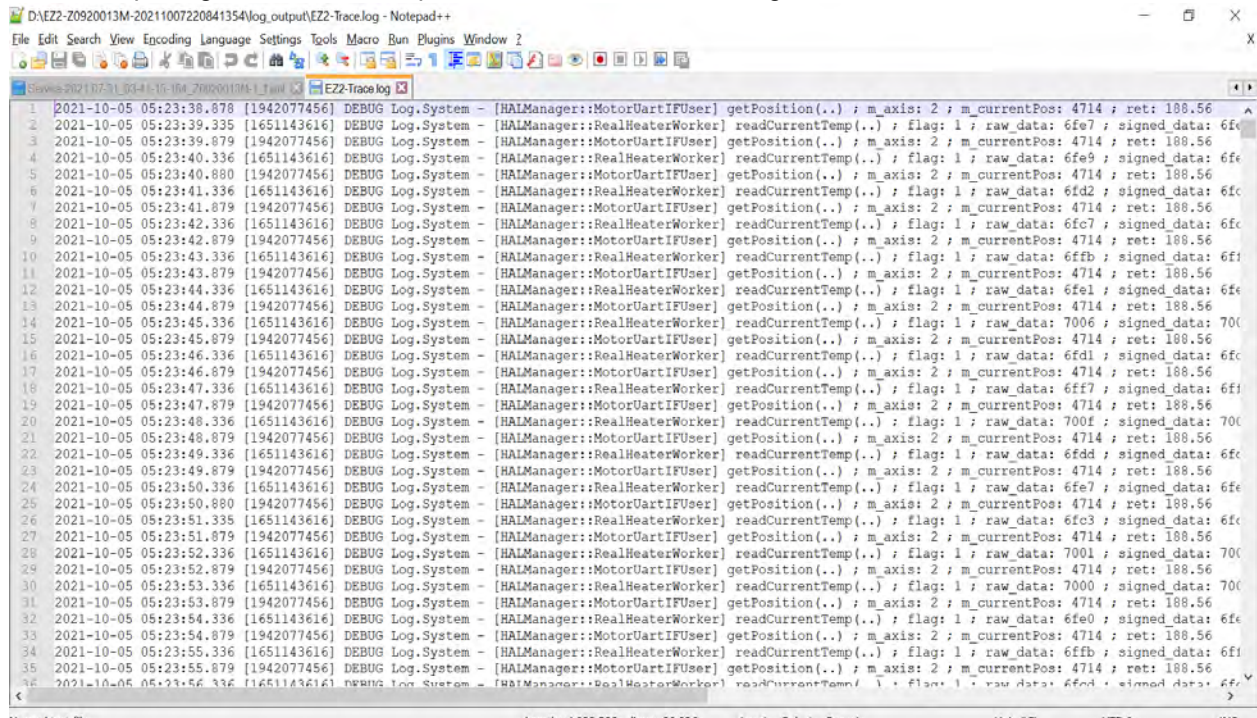
7. Right click the file you want to open and use the **Edit with Notepad ++** to open the file.



Name	Date modified	Type	Size
EZ2-Error.log	10/7/2021 10:07 PM	Text Document	1,738 KB
EZ2-L...		Text Document	5,111 KB
EZ2-L...		1 File	873 KB
EZ2-Tr...		Text Document	4,721 KB
EZ2-Tr...		1 File	509 KB
EZ2-Tr...		2 File	653 KB
EZ2-Tr...		3 File	590 KB
EZ2-Tr...		4 File	577 KB
EZ2-Tr...		5 File	734 KB
EZ2-Tr...		6 File	720 KB
EZ2-Tr...		7 File	732 KB
EZ2-Tr...		8 File	655 KB
EZ2-Tr...		9 File	705 KB
EZ2-Tr...		10 File	701 KB
EZ2-Tr...		11 File	665 KB
EZ2-Tr...		12 File	770 KB
EZ2-Tr...		13 File	766 KB
EZ2-Tr...		14 File	773 KB
EZ2-Tr...		15 File	751 KB

Picture 11.2-7 Use Notepad ++

8. After opening the file with Notepad ++, the files will be in a logical and readable format.



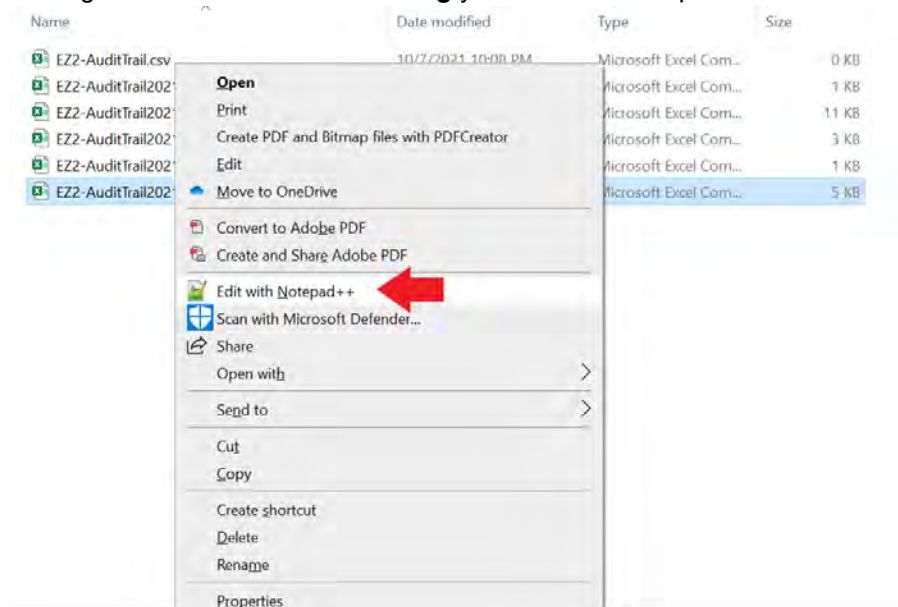
Picture 11.2-8 Readable Trace Log Files

9. Open the **Service** file in the support package, the audit trail logs can be opened with Notepad ++.

Name	Date modified	Type	Size
log_outp	10/8/2021 10:17 AM	File folder	
reports	10/8/2021 10:17 AM	File folder	
service	10/8/2021 10:18 AM	File folder	
calibration.xml	9/27/2021 10:24 PM	XML Document	2 KB

Picture 11.2-9 Service File

10. Right click on the **Audit Trail Log** you would like to open and use the **Notepad ++** program.



Picture 11.2-10 Open AuditTrail File with Notepad ++

11. The audit trail file should now be in a readable format as in the picture below.

```

1 Date,Time,User ID,Affected User ID,Sample ID,Message,
2 2021-09-27,21:13:48,,,SW version: 0.9.9,
3 2021-09-27,21:56:22,,,MDx mode: Service,
4 2021-09-27,21:56:47,Service User 4,,,Service user logged in,
5 2021-09-27,21:56:48,Service User 4,,,FW version: 3.16,
6 2021-09-27,21:56:50,Service User 4,,,Risk of injury is accepted by service user,
7 2021-09-27,21:57:04,Service User 4,,,Maintenance task started; due date: DailyMaintenance;2021-09-15T23:59:59,,
8 2021-09-27,21:58:33,Service User 4,,,Maintenance task done; due date: DailyMaintenance;2021-09-15T23:59:59,,
9 2021-09-27,21:58:58,Service User 4,,,Maintenance task started; due date: ManualUVRun,,
10 2021-09-27,22:00:43,Service User 4,,,Maintenance task aborted; due date: ManualUVRun,,
11 2021-09-27,22:01:30,Service User 4,,,Calibration started. Type: CameraExposure,,
12 2021-09-27,22:02:51,Service User 4,,,Calibration finished. Type : CameraExposure,,
13 2021-09-27,22:03:21,Service User 4,,,Calibration started. Type: CameraCalibration,,
14 2021-09-27,22:03:53,Service User 4,,,Calibration started. Type: CameraCalibration,,
15 2021-09-27,22:04:39,Service User 4,,,Calibration started. Type: CameraCalibration,,
16 NOT READING FILE
17 2021-09-27,22:09:48,,,MDx mode: Service,
18 2021-09-27,22:09:59,Service User 4,,,Service user logged in,
19 2021-09-27,22:10:00,Service User 4,,,FW version: 3.16,
20 2021-09-27,22:10:02,Service User 4,,,Risk of injury is accepted by service user,
21 2021-09-27,22:10:27,Service User 4,,,Calibration started. Type: CameraCalibration,,
22 2021-09-27,22:11:51,Service User 4,,,Calibration finished. Type : CameraCalibration,,
23 2021-09-27,22:12:08,Service User 4,,,Calibration started. Type: ZCalibration,,
24 2021-09-27,22:17:06,Service User 4,,,Calibration finished. Type : ZCalibration,,
25 2021-09-27,22:17:13,Service User 4,,,Calibration started. Type: YCalibration,,
26 2021-09-27,22:20:20,Service User 4,,,Calibration finished. Type : YCalibration,,
27 2021-09-27,22:20:25,Service User 4,,,Calibration started. Type: PCalibration,,
28 2021-09-27,22:21:11,Service User 4,,,Calibration finished. Type : PCalibration,,
29 2021-09-27,22:22:05,Service User 4,,,Calibration started. Type: MCalibration,,
30 2021-09-27,22:24:11,Service User 4,,,Calibration finished. Type : MCalibration,,
31 2021-09-28,02:37:37,,,SW version: 0.9.9,
32 2021-09-28,02:42:29,,,MDx mode: Service,
33 2021-09-28,02:42:38,Service User 4,,,Service user logged in,
34 2021-09-28,02:42:39,Service User 4,,,FW version: 3.16,
35 2021-09-28,02:42:41,Service User 4,,,Risk of injury is accepted by service user,
36 2021-09-28,02:42:48,Service User 4,,,Maintenance task started; due date: DailyMaintenance,,

```

Picture 11.2-11 Readable Audit Trail File

12. The Notepad ++ will also work for reading the calibration file in the support package.

```

1 <?xml version='1.0' encoding='utf-8'?>
2 <Calibrations version="0.9.7">
3   <Calibration Id="Default">
4     <Geometry>
5       <LocatingPoints>
6         <Point Name="CameraOrigin">
7           <X>5279</X>
8           <Y>8382</Y>
9           <Z>0</Z>
10          <M>0</M>
11          <P>0</P>
12        </Point>
13        <Point Name="PiercerOffset">
14          <X>0</X>
15          <Y>14000</Y>
16          <Z>0</Z>
17          <M>0</M>
18          <P>0</P>
19        </Point>
20        <Point Name="MagnetOn">
21          <X>0</X>
22          <Y>0</Y>
23          <Z>38500</Z>
24          <M>20919</M>
25          <P>0</P>
26        </Point>
27        <Point Name="RackHolderSlot">
28          <X>0</X>
29          <Y>77775</Y>
30          <Z>21228</Z>
31          <M>0</M>
32          <P>0</P>
33        </Point>
34        <Point Name="RackHolderPinRear">
35          <X>0</X>
36          <Y>180575</Y>
37          <Z>25277</Z>

```

Picture 11.2-12 Calibration File

11.2.3 GPS Tool

1. The GPS Tool is used to translate support package files to a readable document. How to use the GPS Tool is located here: [GPS Tool](#).

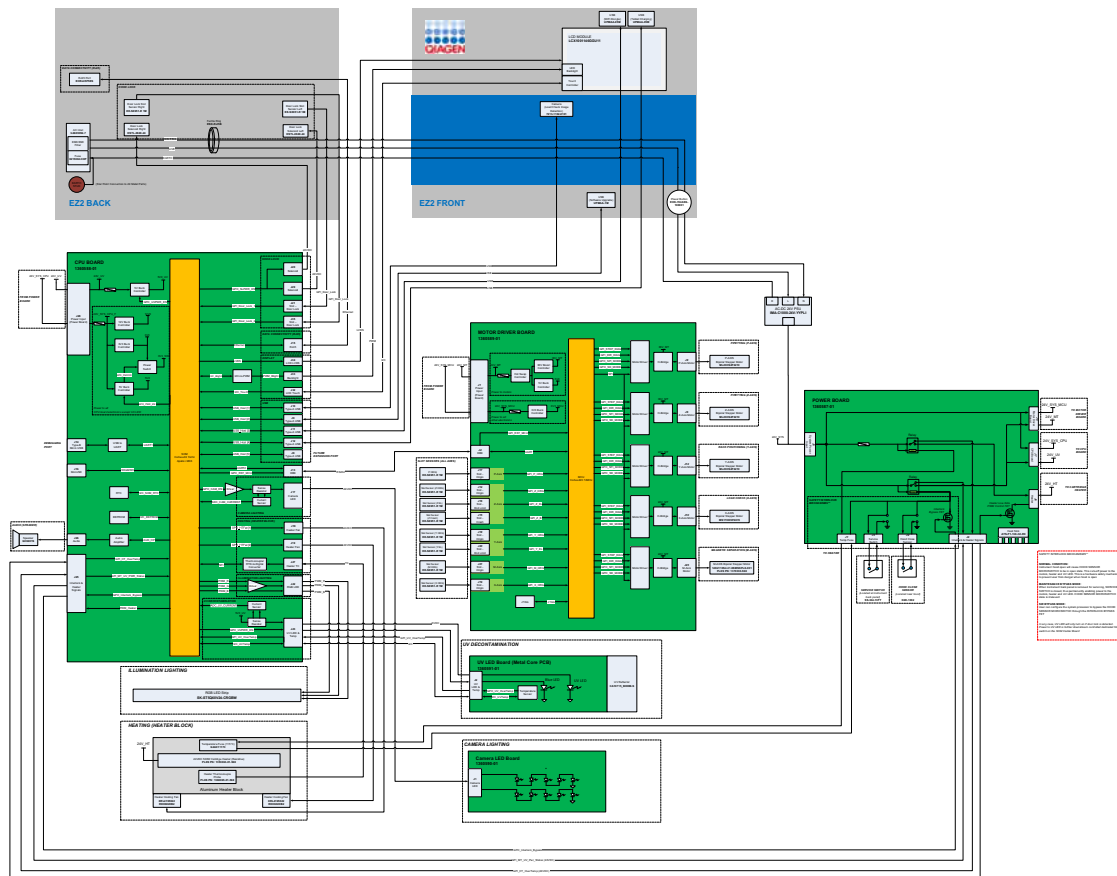
12 Spare Parts

12.1 Spare Part List

The spare parts list can be found in the excel sheet SM-5000-0047-A01, QFS, QTS, EZ2 Spare Parts.

13 Electronic Schematics

Title : Hardware Architecture Revision : C



14 ISI Notes

14.1 Purpose

The Instrument Service Information (ISI) note is an instrument history file containing important and urgent information about EZ2. The ISI note is a way to keep everyone up to date on the changes to EZ2 between service manual releases.

14.1.1 Procedure

Global Product Support is responsible for creating and distributing ISI notes. All members of the QIAGEN support organization are requested to send comments and suggestions about EZ2 that may result in the creation of an ISI note.

The following conditions must be fulfilled to justify the creation of an ISI-Note:

- Important information that will reduce operating expenses
- Hardware/Software bug(s) affecting the system reliability

14.1.2 Format

The ISI note will always be delivered as a PDF, and will be available from the document control system.

14.2 Implemented ISI-Notes

This section lists ISI notes that have been integrated into this manual.

ISI No.	Subject	Rev.	Upg. Class	See Page	Relevant
1		A	III		
3		A	IV		
4		A	III		
5		A	n.a.		

Table 14.2-1 Implemented ISI-Notes

14.3 Actual attached ISI-Notes

This section allows you to keep track of new ISI notes released after the publication of this manual.

ISI No.	Subject	N° of pages	Related to Chapter	Revision of ISI

Table 14.3-1 Actual attached ISI-Notes